Dell EMC SRDF/TimeFinder Manager for IBM i

Version 9.2 and later

Product Guide

REV 01

DELL EMC

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PREFACE

As part of an effort to improve its product lines, Dell Technologies periodically releases revisions of its software and hardware. Therefore, some functions described in this document might not be supported by all versions of the software or hardware currently in use. The product release notes provide the most up-to-date information on product features.

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Product information

Dell EMC SRDF/TimeFinder Manager for IBM i is a set of utilities that provides an IBM i interface to PowerMax, VMAX All Flash, VMAX3, and VMAX arrays. The product includes a Remote Data Facility (SRDF) control facility and a Dell EMC TimeFinder control facility. See "Standard features description" on page 22 for more detail.

SRDF/TimeFinder Manager also provides extended features for the IBM Independent Auxiliary Storage Pools (IASP) functionality. Display and assign multiple TimeFinder BCV devices. See "Extended features description" on page 304 for more detail.

Purpose

This product guide shows how to:

- Configure, and use the Dell EMC SRDF/TimeFinder Manager for IBM i
- Use the IBM Independent Auxiliary Storage Pools (IASPs) which are a set of switchable or private auxiliary disk pools.

Audience

This guide is for the IBM i security officer and operator, who control the SRDF and TimeFinder environment using IBM i host software.

Readers of this guide are expected to be familiar with the following:

- IBM i operating environments
- PowerMax, VMAX All Flash, VMAX3 Family, and VMAX Family arrays operating environment
- SRDF and TimeFinder operation

Related documentation

The following documents provide additional information about SRDF/TimeFinder Manager and Solutions Enabler:

- *Dell EMC Solutions Enabler, VSS Provider, and SMI-S Provider Release Notes* — Describe new features and any known limitations.
- *Dell EMC Solutions Enabler Installation and Configuration Guide* — Provides host-specific installation instructions.
- *Dell EMC Solutions Enabler CLI Reference* — Documents the SYMCLI commands, daemons, error codes and option file parameters provided with the Solutions Enabler man pages.
- *Dell EMC Solutions Enabler SRDF Family CLI User Guide* — Shows how to configure and manage SRDF environments using SYMCLI commands.
- *Dell EMC Solutions Enabler TimeFinder SnapVX CLI User Guide* — Shows how to configure and manage TimeFinder SnapVX environments using SYMCLI commands.
- *Dell EMC Solutions Enabler SRM CLI User Guide* — Provides Storage Resource Management (SRM) information related to various data objects and data handling facilities.
- *SRDF Interfamily Connectivity Information* — Defines the versions of PowerMaxOS, HYPERMAX OS and Enginuity that make up valid configurations for SRDF remote replication.
- *Dell EMC PowerMax Family Security Configuration Guide*— Describes how to configure PowerMax Family security settings.
- *Dell EMC VMAX All Flash and VMAX 3 Family Security Configuration Guide* — Describes how to configure VMAX All Flash and VMAX Family security settings.

The following documents provide additional hardware platform information:

- *Dell EMC PowerMax Family Product Guide* — Contains information relating to PowerMaxOS on the PowerMax and VMAX All Flash arrays.
- *Dell EMC VMAX All Flash Product Guide* — Contains information related to the VMAX All Flash arrays.
- *Dell EMC VMAX 3 Family Product Guide* — Contains information related to the VMAX 3 arrays.
- *Dell EMC PowerMAXOS 5978 Release Notes for Dell EMC PowerMax and VMAX All Flash*— Describes new features and any known limitations.
- *HYPERMAX OS for VMAX All Flash and VMAX3 Family Release Notes* — Describes new features and any known limitations.
- E-Lab™ Interoperability Navigator (ELN) — Provides a web-based interoperability and solution search portal. You can find the ELN at https://elabnavigator.EMC.com.

Typographical Conventions used in this document

This document uses the following type style conventions:

Normal	Used in running (nonprocedural) text for: • Names of interface elements, such as names of windows, dialog boxes, buttons, fields, and menus • Names of resources, attributes, pools, Boolean expressions, buttons, DQL statements, keywords, clauses, environment variables, functions, and utilities • URLs, pathnames, filenames, directory names, computer names, links, groups, service keys, file systems, and notifications
Bold	Used in running (nonprocedural) text for names of commands, daemons, options, programs, processes, services, applications, utilities, kernels, notifications, system calls, and man pages Used in procedures for: • Names of interface elements, such as names of windows, dialog boxes, buttons, fields, and menus • What the user specifically selects, clicks, presses, or types
Italic	Used in all text (including procedures) for: • Full titles of publications referenced in text • Emphasis, for example, a new term • Variables
`Courier`	Used for: • System output, such as an error message or script • URLs, complete paths, filenames, prompts, and syntax when shown outside of running text
`Courier bold`	Used for specific user input, such as commands
`Courier italic`	Used in procedures for: • Variables on the command line • User input variables
< >	Angle brackets enclose parameter or variable values supplied by the user
[ ]	Square brackets enclose optional values
|	Vertical bar indicates alternate selections — the bar means “or”
{ }	Braces enclose content that the user must specify, such as x or y or z
...	Ellipses indicate nonessential information omitted from the example

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PART 1
SRDF/TimeFinder Manager Standard features

Part 1 of this product guide describes the menus, configuration for SRDF/TimeFinder Manager, and shows how to control TimeFinder and SRDF replication sessions. It contains the following chapters and appendixes:

Chapter 1, "Standard Features Introduction"

Contains an overview of Dell EMC SRDF/TimeFinder Manager for IBM i and introduces the components that it supports for the SRDF family and TimeFinder family of replication solutions.

Chapter 2, "Configuration and Control Menus"

Describes the SRDF/TimeFinder Manager configuration and control menus.

Chapter 3, "Operational Features, Settings, and Displays"

Shows how to set up, maintain, and display the SRDF/TimeFinder Manager environment.

Chapter 4, "Configuring the Environment"

Shows the configuration steps that allow for control of SRDF and TimeFinder replication environments with IBM i hosts.

Chapter 5, "Controlling TimeFinder"

Shows how to control TimeFinder operations using the SRDF/TimeFinder Manager Control Menu.

Chapter 6, "Controlling SRDF"

Shows how to control SRDF links, including how to enable and disable consistency protection for SRDF/A and set the RDF mode.

Appendix A, "Standard Features Commands Reference"

Contains reference information, syntax, and examples for control commands.

Appendix B, "Symmetrix Integration Toolkit"

Lists the supported Solutions Enabler CLI commands that are packaged with the SRDF/TimeFinder Manager software in the Integration Toolkit, and provides some examples.

Appendix C, "Standard Features Messages"

Lists the messages that SRDF/TimeFinder Manager components may produce.

Appendix D, "Miscellaneous Device Information"

Describes miscellaneous device information such as device pairing compatibility and variable LUN support.

Appendix E, "Troubleshooting"

Provides some troubleshooting information for SRDF/TimeFinder Manager.

CHAPTER 1
Standard Features Introduction

This chapter provides an overview of SRDF/TimeFinder Manager for IBM i and introduces the components that it supports for the SRDF family and TimeFinder family of replication solutions.

This chapter also introduces the switchable Independent Auxiliary Storage Pool (IASP) functionality that SRDF/TimeFinder Manager supports for disaster recovery and backup operations in combined IBM i and array environments.

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Standard features description

SRDF/TimeFinder Manager is a set of host-based utilities that provides an IBM i interface to Remote Data Facility (SRDF) and TimeFinder. The SRDF/TimeFinder interface enables you to configure and control SRDF or TimeFinder operations on PowerMax, VMAX All Flash, VMAX3, VMAX, and DMX arrays attached to IBM i hosts. These operations are:

- Run TimeFinder commands to establish and split TimeFinder BCV devices where BCVs are used for local mirroring, or are used as the source device in multi-hop replication solutions.
- Create point-in-time copies of full volumes or individual data sets using TimeFinder/Clone copy sessions.
- Create logical point-in-time snapshots of images for:
  - Arrays running PowerMaxOS 5978 using TimeFinder SnapVX.
  - Arrays running HYPERMAX OS 5977 using TimeFinder SnapVX.
  - Arrays running Enginuity 5876, using TimeFinder/VP Snap.
  - Arrays, running Enginuity 5876 or earlier, using TimeFinder/Snap.

  Note: TimeFinder/Snap is not available on VMAX 10K arrays or on arrays running PowerMaxOS 5978 or HYPERMAX OS 5977.
- Run SRDF commands to perform establish and split operations for local replication solutions, or for extended-distance replication solutions, such as SRDF/Asynchronous (SRDF/A).
- SRDF/TimeFinder Manager provides support for the IBM i host environment, such as multipath, boot from SAN, and independent ASP (IASP) functionality.

Supported arrays and array features

SRDF/TimeFinder Manager is used in environments that contain PowerMax, VMAX All Flash, VMAX3, and VMAX arrays.

The following rules and requirements apply to VMAX 10K arrays and arrays running PowerMaxOS 5978 or HYPERMAX OS 5977:

- D910 device emulation only. Emulation for 2107 devices is not available.
- Thin devices only. Thick devices are not available.
- Thin (bound) gatekeeper devices only.

Note: Meta devices are available on VMAX 10K arrays but not on arrays running PowerMaxOS 5978 or HYPERMAX OS 5977.

Enginuity, HYPERMAX OS, and PowerMaxOS requirements

The version of PowerMaxOS, HYPERMAX OS, or Enginuity that you need depends on the type of storage array:

Table 1 PowerMaxOS, HYPERMAX OS, and Enginuity requirements

Storage array	Required version of PowerMaxOS, HYPERMAX OS, or Enginuity
PowerMax Family	PowerMaxOS 5978
VMAX All Flash Family	PowerMaxOS 5978 or HYPERMAX OS 5977
VMAX3 Family	HYPERMAX OS 5977
VMAX Family	Enginuity 5876

TimeFinder functionality

The TimeFinder family of local replication software allows users to non-disruptively create and manage point-in-time copies of data to allow operational processes, such as backup, reporting, and application testing to be performed independent of the source application, and to maximize service levels without impacting performance or availability.

TimeFinder technologies available on PowerMaxOS 5978 or HYPERMAX OS 5977

TimeFinder SnapVX is the TimeFinder technology available on arrays running PowerMaxOS 5978 or HYPERMAX OS 5977. TimeFinder Mirror and Clone commands can be run on PowerMaxOS 5978 or HYPERMAX OS 5977, but emulations use SnapVX as the underlying technology. When running emulation sessions, devices cannot be a SnapVX source or a link target; and when devices are SnapVX sources or link targets they cannot be used for emulation sessions. TimeFinder Snap sessions are not supported, therefore, there is no emulation for this local replication type.

TimeFinder SnapVX

SnapVX creates point-in-time copies directly in the Storage Resource Pool (SRP) of the source device, thus eliminating the concepts of target devices and source/target pairing. If point-in-time data is needed, linked targets are created. The maximum number of SnapVX snapshots for each source device is 256.

Storage Resource Pools

All point-in-time data saved in the SRP of the source device is limited by the TF_reserve_capacity as defined in the SRP. Once this limit is reached no more allocations are allowed for snapshot data. Point-in-time data is saved in the SRP using a shared allocation mechanism. For multiple snapshots that are saving the same tracks, only one track is saved in the SRP.

Snapshot identification

Snapshots have a user-supplied name. The same name may be used for multiple devices and/or multiple snapshots. For multiple snapshots a generation number is produced to differentiate point-in-time copies of the same name on the same devices. Snapshots and their associated generation numbers are ordered by a timestamp. The most recent snapshot is generation 0 and older snapshots are consecutive increasing positive numbers. As snapshots are established and terminated, the generation numbers are reassigned by snapshot timestamps so there are no gaps in the generation numbers.

The snapshot names are assigned by the user at the time the snapshot is established. After that, snapshots are uniquely identified for subsequent commands by name, source device, and generation number. If the generation number is omitted, generation 0 is assumed. Snapshots are established on source devices only. The point-in-time copy data is saved in the SRP of the source device.

Snapshot Termination and Restore

Snapshots can only be terminated when there are no links. Snapshots may be set for automatic termination with an optional “Time To Live” setting specified in days. Snapshots automatically terminate when the Time to Live has expired (plus or minus 15 seconds), but only if there are no links. If the Time to Live has expired on a snapshot with one or more links, the snapshot is terminated when the last link has been unlinked.

Restore to the original source device is supported and is differential by default. If the original source device is a link target of another snapshot, the link must be fully copied prior to the restore.

Failed Snapshots

A snapshot can fail if no more allocations can be made from the SRP for the snapshot or if a corruption occurs on the source device. To recover a “Failed” snapshot terminate and re-establish it.

Consistent Snapshots

By default all snapshots are consistent. Depending on the state of the devices at the time of the snapshot, Local ECA or SRDF/A is used to ensure that I/O is paused while the snapshot is created.

The following rules apply to the source devices, when the snapshot is created, in order to ensure that the snapshot is consistent:

- A mixture of R1 and R2 devices is not allowed.
- A mixture of R1 and non-RDF devices is allowed. The R1 devices can be a mixture of RW and NR on the RDF link.
- A mixture of R2 and non-RDF devices is allowed if the R2 devices are NR on the RDF link.
- A mixture of R1 and R21 devices is allowed if the R1->R21 is NR on the RDF link.
- A mixture of R2 and R21 devices is allowed:
  - if the R1->R2 and the R1->R21 are all RW on the RDF link.
  - if the R1->R2 and the R1->R21 are all NR on the RDF link.

Note: When SnapVX is used remotely and there are several SRDF groups in the configuration, SRDF mode for the links should be synchronized or links should be split.

If the source devices do not meet these criteria the snapshot cannot be created.

Managing Point-in-time data

SnapVX defaults to only having the source device participating in the sessions. If point-in-time data is needed, the user creates one or more links from the snapshot session to one or more target devices. Links can be created in Copy mode for a permanent copy on the target device, or in NoCopy mode for temporary use. Linked point-in-time copies can be mapped to and made visible to a host. Point-in-time copies are identified by a user-supplied name and device.

In SRDF/TimeFinder Manager pairing is done from a device file.

Linking rules

- The target device must be of equal size as or larger than the source device.
- Maximum number of linked targets per source device is 1024.
- Host writes to a linked target device do not change the point-in-time copy, so only the linked target device contains a modified point-in-time copy. To restore a modified point-in-time copy to the original source device, a snapshot from the target device must be established followed by a Copy mode link of that snapshot to the source device. When the copy is complete both the link and the target device's snapshot may be removed.
- A relink command automatically unlinks an existing link to a snapshot and links to a new selected snapshot. When the relink is executed the copy is differential between the original linked snapshot and the newly linked snapshot.
- The relink command is also used to relink to the same snapshot. This has the effect of refreshing the point-in-time copy on the link target when it's been modified by host writes.
- A link to a target that has a snapshot may not be removed until the link is fully copied. If the link is a NoCopy, it must first be changed to Copy and then removed when fully copied.

- When a device is used as a link target, all of its previous track allocations are deallocated. Any pre-existing data on the target is lost during a link or relink. When a link with Copy mode is issued and the data is fully copied; the point-in-time data remains on the target device when an unlink is issued.
- When a link with NoCopy mode or Copy mode is issued, but the data is not fully copied, after the unlink the integrity of the data is uncertain.
- Devices with persistent track allocations cannot be used as links targets.
- Linked devices must have all tracks defined before being used as a source for another snapshot.
- A link can fail if:
  - Allocations are no longer available from the SRP associated with the link device. If more space is added to the SRP, it may be possible to recover from a failed link. Otherwise, the link must be terminated and re-established.
  - A corruption occurs on the link device, the failed link must be terminated and re-established.

TimeFinder technologies supported for Enginuity 5876 or earlier

TimeFinder/Clone, TimeFinder/Snap, and TimeFinder/Mirror are supported on arrays running Enginuity 5876 or earlier. Except where noted these technologies are referred to as “TimeFinder” in this document.

TimeFinder base products

- **TimeFinder/Clone** — Provides clone copy sessions that create point-in-time copies of full volumes or individual data sets. TimeFinder/Clone enables users to make copies of data simultaneously on multiple target devices from a single source device. The data is available to the target host immediately upon activation, even if the copy process has not completed. Data may be copied from a single source device to as many as 16 target devices. Both source and target devices can be either a standard device or a BCV device.
- **TimeFinder/Snap** — Provides snap copy sessions that create economical, pointer-based replicas simultaneously on multiple target devices from a single source device where only the pre-images of changed data are written. TimeFinder/Snap enables users to configure special devices called virtual devices (VDEVs) and save area devices (SAVE devices).

  Note: TimeFinder/Snap is not supported on VMAX 10K arrays.

- **TimeFinder/Mirror** — Provides full physical copies of data from a standard device, which are online for regular I/O operations from the host. Data is copied to BCV devices as a mirror of the primary data. Data is available to the target host after devices are split.

  Note: For Enginuity version 5876, TimeFinder/Mirror native operations are no longer available. However, TimeFinder/Mirror commands are still available using the TimeFinder/Clone Emulation feature. TimeFinder/Clone Emulation enables TimeFinder/Mirror scripts that were developed and deployed on previous generation DMX arrays to run on the VMAX Family arrays. Clone Emulation is the default for RAID 5/RAID 6-protected BCVs. TimeFinder/Mirror operations work on DMX arrays running Enginuity version 5773.

SRDF functionality

Within the SRDF family of software, SRDF/TimeFinder Manager provides:

- SRDF Cascaded replication
- SRDF Extended Distance Protection (SRDF/EDP)
- SRDF Reserve Capacity with Transmit Idle
- SRSF/A Delta Set Extension (DSE)
- SRDF/Synchronous (SRDF/S)
- SRDF/Asynchronous (SRDF/A)
- Concurrent SRDF/S
- Concurrent SRDF/A
- SRDF/A Multi-Cycle
- SRDF/Metro

Note: SRDF/S and SRDF/A are two of the SRDF base products. Except where noted, SRDF/S and SRDF/A are called SRDF in this guide.

For detailed information on configuring SRDF for IBM i see Chapter 6, page 165.

Ensuring data integrity

SRDF/TimeFinder Manager prevents suspension of the SRDF links under conditions that can result in inconsistent data on the target volumes.

Internal mechanisms on the array maintain the current status of every *track* stored in the system. In an SRDF configuration, a track status is *invalid* until the data received from a host write request is written to both the source and target arrays.

In synchronous mode, the array responds to the host with access to the source (R1) device on a write operation only after the array containing the target (R2) device acknowledges that it has received and checked the data.

When the command to suspend the SRDF links is executed, SRDF/TimeFinder Manager checks for invalid tracks. There should be little or no write activity from the host, because normal operation is to perform one of the following:

- Bring the source IBM i host to a restricted state using the STMENDSBS command.
- Hold all jobs and job queues before suspending SRDF links.
- For IASP operations, vary off the ASP before a split operation.

Note: For V6R1 and higher, you can use the Change ASP Activity command (CHGASPACT) to quiesce an IASP if it is varied on and the split can be run without varying it off. The CHGASPACT command suspends initiations of new transactions and operations before a split operation, writes changes to auxiliary storage before the split and then resumes initiations of new transactions and operations after split.

These processes ensure that one or more target volumes contain a point-in-time frozen image that exactly matches the image of the source storage device at the time the links are suspended.

The *Dell EMC SRDF Introduction* contains information on SRDF link states, such as SyncInProg, Synchronized, and Split.

Multiple SRDF configurations

System storage can span across multiple source arrays with a like number of target arrays providing mirror copies. SRDF/TimeFinder Manager ensures that the frozen image is consistent across all arrays before a Suspend or Split operation is performed. This ensures that no invalid tracks are present on any of the target arrays when the frozen image is created. Figure 1 shows a system where a single image spans two arrays.

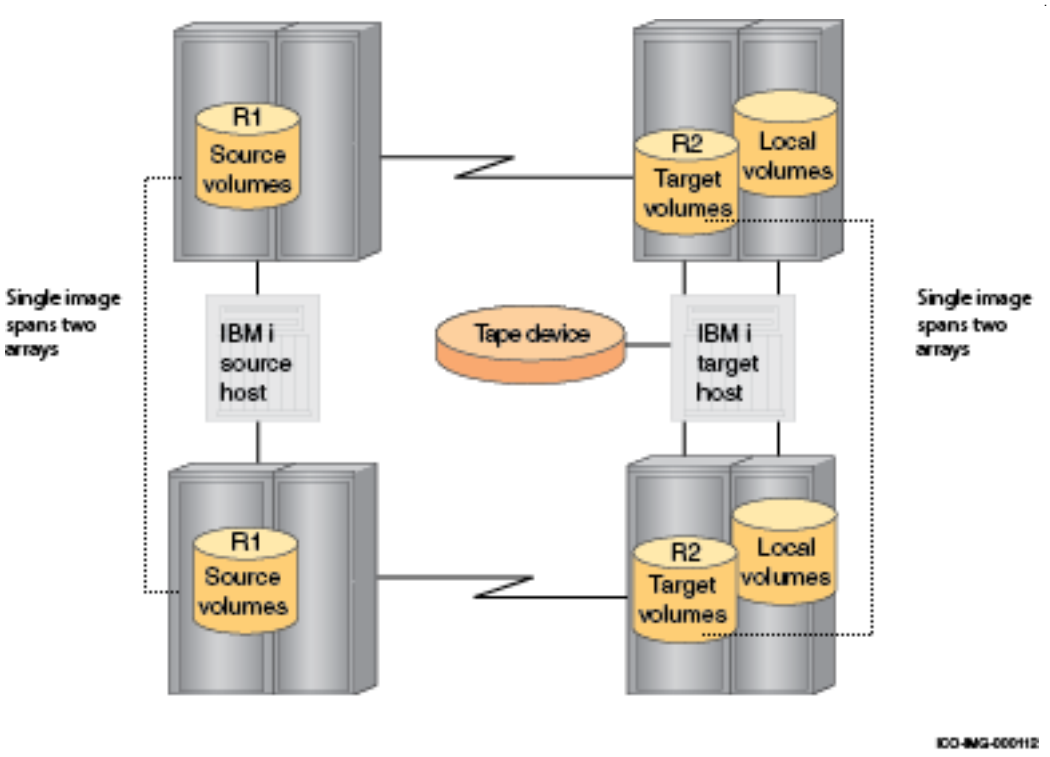


Figure 1 Multiple arrays — single image

Note that both source (R1) and target (R2) volumes shown in Figure 1 may also be part of a TimeFinder BCV pair (not shown).

Supported features for arrays running PowerMaxOS 5978 or HYPERMAX OS 5977

For SRDF/TimeFinder Manager, and IBM i operating system version 6.1.1 or higher with device type D910-099, automated FAST control is available on arrays that run PowerMaxOS 5978 or HYPERMAX OS 5977. The automated FAST handles all thin device binding and allocation, and manages the movement of application data across all of the storage capacity defined in a Storage Resource Pool (SRP). Data movement is based on gathered performance data and a performance Service Level Priority (SLP) defined for the application.

Supported features for arrays running Enginuity 5876

The following features are available on arrays running Enginuity 5876 with SRDF/TimeFinder Manager, and IBM i operating system version 6.1.1 or higher with device type D910-099:

- FAST VP
- VLUN VP Migration
- Virtual Provisioning

For a list of all minimum requirements for D910 support refer to "D910 device support (thin devices)" on page 294.

These features are available using native IBM i Solutions Enabler CLI commands, and are also available from open systems hosts using Solutions Enabler CLI commands, Unisphere for PowerMax, or Unisphere for VMAX.

Virtual Provisioning is also available through SRDF/TimeFinder Manager commands and screens.

FAST VP

FAST VP allows for automated policy-driven storage tiering that performs dynamic non-disruptive movement of data to meet performance and capacity demands for applications. FAST can move data among three tiers Flash, Fibre Channel, or SATA drives depending on the demand for the data. FAST VP also allows for movement of data at the sub-LUN level. FAST VP is configured using IBM i Solutions Enabler CLI commands.

Refer to "Supported Solutions Enabler CLI commands" on page 262 for specific Solutions Enabler CLI commands for managing FAST VP.

VLUN VP Migration

VLUN VP Migration enables transparent, non-disruptive data mobility for standard volumes between storage tiers and RAID protection schemes while providing constant data availability and protection. VLUN VP Migration can be used to migrate standard devices and meta devices of any emulation — FBA, CKD, and IBM i. VLUN VP Migration can be configured using IBM i Solutions Enabler CLI commands.

Refer to "Supported Solutions Enabler CLI commands" on page 262 for specific Solutions Enabler CLI commands for managing VLUN VP Migration.

Virtual Provisioning

In Virtual Provisioning, thin devices present to the host disk configurations with inflated capacity. Thin devices can be created with an inflated capacity, because the actual storage space for the data written to thin devices is on DATA devices that are located in a thin pool. When more storage space is needed the thin device can provision more storage from the pool without disrupting the host application. Virtual Provisioning for IBM i is managed through the SRDF/TimeFinder Manager commands and screens. Refer to "Working with Virtual Provisioning™ and thin pools" on page 114 for details on Virtual Provisioning configuration.

Note: If using Solutions Enabler CLI commands to create FAST policies, create tiers, and migrate data, refer to the latest *Dell EMC Solutions Enabler CLI Reference* for details on how to use the `symfast`, `symtier`, and `symmigrate` commands. If using Unisphere for PowerMax or Unisphere for VMAX, the online help has details on how to perform FAST. For more information on FAST VP and VLUN VP Migration operations refer to *Dell EMC Solutions Enabler Array Controls and Management CLI User Guide*.

Thin device compression

Data within a thin device can be compressed to save space and uncompressed when needed. Data can either be compressed or uncompressed manually, or can be compressed by FAST as part of its processing. Refer to "Working with Virtual Provisioning™ and thin pools" on page 114 for details on thin device compression configuration and management using SRDF/TimeFinder Manager screens, or through the command "CTLPOOL" on page 215.

Thin device migration

Virtual LUN VP (virtual pools) migration allows thin device allocations to be moved between pools. For a given list of devices, the data tracks are moved from their current pool to a target pool. Thin device migration is only allowed for locally attached arrays.

Two types of migration are supported:

- Full migration — Full migration migrates all of the thin device's allocations. After the migration, if the thin device is not already bound to the target pool, it is automatically bound to that pool.
- Pool-level migration — Pool-level differs from the full migration of a thin device in that not all the thin device allocations are migrated, just the allocations that are in the source pool. Any allocations in pools other than the source pool are not migrated to the target pool. After the migration, device binding does not change to the target pool. If rebinding the thin device to the target pool is desired, it is the user's responsibility to rebind the device to the target pool.

Prior to a migration, the thin devices, source pool, and target pool need to be identified.

Refer to "Working with Virtual Provisioning™ and thin pools" on page 114 for details on thin device migration configuration and management using SRDF/TimeFinder Manager screens, or through the command "CTLPOOL" on page 215.

Supported operations and configurations

Because SRDF/TimeFinder Manager is an IBM i interface to array functionality, the following operations can be performed and configurations are supported:

- Display and assign multiple TimeFinder BCV devices.
- Run TimeFinder commands to establish and split TimeFinder BCV devices used in replication solutions that vary in complexity, such as where BCVs are used for local mirroring or are used as the source device in multihop replication solutions.
- Run SRDF commands to suspend and resume SRDF Remote Link Directors and perform establish and split operations for local replication solutions or for extended-distance replication solutions, such as SRDF/A.
- Use TimeFinder/Clone Emulation in configurations containing RAID 5/RAID 6-protected BCVs. The following devices do not take mirror positions of the standard device:
  - Standard devices paired with RAID 5/RAID 6-protected BCV devices
  - Mixed-BCV configurations — some BCV devices are RAID 5/RAID 6 protected, and some BCV devices are not RAID 5/RAID 6 protected
  - RAID 5/RAID 6-protected BCVs used as R1 devices (BCV/R1)
  - Remote RAID 5/RAID 6-protected BCVs
- Create, activate, and recreate TimeFinder/Snap virtual copy sessions or TimeFinder/Clone copy sessions for creating logical point-in-time snapshots of images for arrays.

  Note: Recreate functionality, for TimeFinder/Snap virtual copy sessions, is only available in environments running Enginuity 5876 or higher.

- Use IASPs to present a target device with an IASP image to another partition or host for business continuance (BC) processes.
- Establish SnapVX snapshots and link point-in-time data to target devices if needed.
- Concurrent RDF.
- Multi-hop configurations that work with combinations of TimeFinder and SRDF.

TimeFinder/Consistency Group

TimeFinder/Consistency Group (TimeFinder /CG) enables SRDF/TimeFinder Manager to split local or remote BCVs, clones, and snap images (Figure 2 on page 32) while the source system is operational. This feature is referred to as Split While Active (SPLITWA). “CTLACT” provides further detail on SPLITWA.

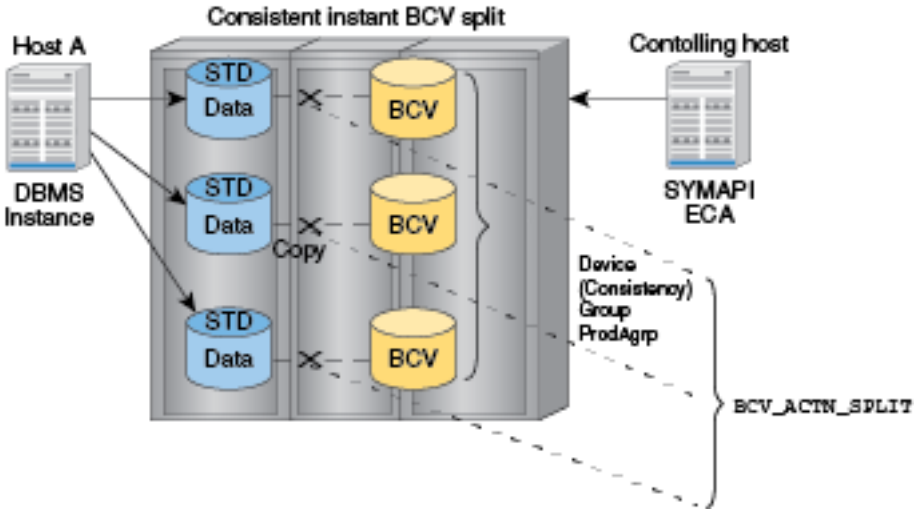


Figure 2 TimeFinder/CG

TimeFinder/CG features are as follows:

- Ensures dependent write consistency for all devices in the step that was selected through the SRDF/TimeFinder Manager menu or from the CTLACT command.
- SPLITWA operations do not require that the IBM i host be in a restricted state or that Hold Jobs operation be performed.
- The image that SPLITWA creates can be IPL'd on another partition or host. Data in main memory at the time of the split is not available on the split image.

TimeFinder/CG prerequisites are as follows:

- SRDF/TimeFinder Manager client and the controlling host must have a gatekeeper configured on each locally attached array. For detailed information on array gatekeeper management, refer to the *Dell EMC Solutions Enabler Installation and Configuration Guide*.
- Separate license key, and the `netcnfg` file must be configured on the client using the CRTNETCNFG command. This ensures that correct information (node name, IP address, port, and security level) points to the remote server. Only non secure connections are supported for IBM i.

Note: When using this split type, the data in main memory is not included in the image. Only use SPLITWA when the host and applications are in a quiesced state. If that is unacceptable to your business needs, use other operating system functions like journaling and database commitment control to get an image up to the last transaction.

Note: When SPLITWA is used remotely and there are several SRDF groups in the configuration, SRDF mode for the links should be synchronized or links should be split.

Supported TimeFinder features for cascaded sessions

SRDF/TimeFinder Manager supports the following TimeFinder features for cascaded sessions:

- Recreate of Clone session with Snap or VP Snap session off Clone target
- Restore from TimeFinder cascaded Snap or VP Snap to a fully copied Clone
- Incremental restore of Clone target to active Snap or VP Snap source device
- Time Finder/Snap or VP Snap off SRDF/A R2 in SRDF multi-site topologies

Recreate of Clone session with Snap or VP Snap session off Clone target

This feature permits recreate of Clone sessions when the target device is the source of a cascaded Snap or VP Snap session. The subsequent activation of the recreated session is only allowed after the first pre-copy is complete. Following the recreate operation when the state of the Clone target is Not Ready.

Restore from TimeFinder cascaded Snap or VP Snap to a fully copied Clone

Note: This feature is supported only for IASP configurations.

This feature permits cascaded Clone/Snap or Clone/VP Snap sessions to copy data back from Snap devices to Clone devices, as shown in Figure 3.

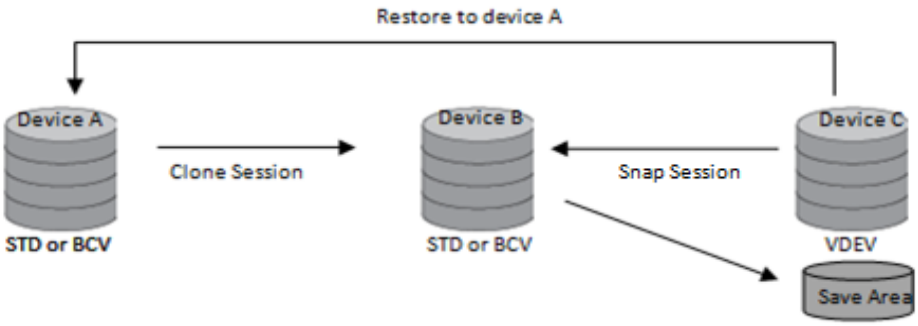


Figure 3 Restore Snap session to fully copied Clone

Rules and restrictions for restoring snap session to Clone

- In a cascaded session where A/B pair is a Clone session and B/C pair is a Snap session:
  - If the AB pair is in the Copied or Split state and the BC pair is in the CopyOnWrite or Copied state, incremental restore is allowed for the BC pair. Following the restore, the state of the VDEV device is Not Ready.
  - If the AB pair is in the Copied or Split state and BC pair in the Restored state, incremental restore is allowed for the AB pair.

- Once both the AB and the BC pair sessions are restored, only the terminate operation is allowed on either pair. For the AB pair this is termination of the entire session, for BC pair this is termination of the persistent restore session. Once the AB pair is terminated, all operations for a non-cascaded session are allowed for the remaining BC pair. Once the BC pair persistent restore is terminated, all operations for a cascaded session are allowed for both the AB and BC pairs.

- In a cascaded session where the AB pair is a Clone Emulation session and the BC pair is a Snap session:
  - If the AB pair is in the Split state and BC in the CopyOnWrite or Copied state, an incremental restore is allowed for the BC pair. Following the restore the state of the VDEV device is Not Ready.
  - If the AB pair is in the Split state and the BC pair is in the Restored state, a restore is not allowed for the AB pair until the persistent restore session is terminated for the BC pair. This restriction allows for compatibility with native TimeFinder.
  - When the BC pair is restored, the only operations allowed are cancelling the AB pair session or terminating the BC pair restore session.
- In a cascaded session where the AB pair is a Clone session and the BC pair is a VP Snap session:
  - If the AB pair is in the Copied or Split state and BC in the CopyOnWrite or Copied state, an incremental restore is allowed for the BC pair. Following the restore the state of the target device is Not Ready.
  - If the AB pair is in the Copied or Split state and the BC pair is in the Restored state, an incremental restore is allowed for the AB pair.
  - Once both the AB and the BC pair sessions are restored only the terminate operation is allowed on either pair. For the AB pair the entire session terminates, for BC pair the persistent restore session terminates. Once the AB pair is terminated, all operations for a non-cascaded session are allowed for the remaining BC pair. Once the BC pair persistent restore is terminated, all operation for a cascaded session are allowed for both the AB and BC pairs.
- In a cascaded session where the AB pair is a Clone Emulation session, and the BC pair is a VP Snap session:
  - Cascading from TF Clone Emulation targets is not supported.
- Session cancellation/termination operations:
  - For the AB pair, the Clone terminate operation is allowed when the pair is in any state.
  - For the BC pair, the Clone terminate or Snap terminate operation is allowed when the pair is in any state.
  - For the AB pair, the Clone Emulation cancel operation is only allowed when the pair is in the Split state.

Incremental restore of Clone target to active Snap or VP Snap source device

This feature permits incremental restore of Clone target to source devices with active Snap or VP Snap sessions. Pair state for Snap or VP Snap session should be in the CopyOnWrite or Copied state before the restore.

Note: This feature is supported only for IASP configurations.

Time Finder/Snap or VP Snap off SRDF/A R2 in SRDF multi-site topologies

Note: For SRDF/TimeFinder Manager, this feature does not include SRDF/Star, as SRDF/TimeFinder Manager does not support SRDF/Star configurations.

This feature permits TimeFinder Snaps and Clones off the R2 device of an R21 to an R2 SRDF device pair operating in asynchronous SRDF mode.

Also required for this feature is support for SRDF/A group-level and device-level write pacing for R21 to R2 SRDF device pairs in cascaded configurations, which are supported in the following Enginuity versions:

- Group-level write pacing, whether on demand or using autostart, the R21 array requires Enginuity 5876.159.102 or later and the R2 array requires Enginuity 5876 or later.
- Device-level write pacing, whether on demand or using autostart, the R21 array requires Enginuity 5876.159.102 and the R2 requires Enginuity 5876 or later.

The following rules and restrictions apply for this feature:

- The R21 array must run Enginuity 5876.159.102 or later.
- The R2 array must run Enginuity 5876 or later.
- If the R21 to R2 SRDF/A session is active, the SRDF pair states cannot be TransIdle, and SRDF/A device-level write pacing must be activated and supported for the R21 to an R2 SRDF/A session. The R21 device must also be pace-capable.
- If the R21 to R2 SRDF/A session is not active, SRDF/A device-level write pacing must be configured with autostart enabled for the SRDF group that contains the R1 mirror of the R21 device.

Load Source Mirroring

The use of remote load source mirroring (RLSM) is supported to provide protection for the SAN boot load source device. With IBM i 6.1 and later operating system on the host, multipathing of the load source is available. This is the preferred method for load source protection with the i 6.1 and later operating systems, and it is supported on PowerMax, VMAX All Flash, and VMAX arrays. For more information, refer to the *Dell EMC Host Connectivity Guide for IBM i*.

Note: The following information is only provided as an overview and example. Dell EMC recommends that you use updated PTFs and follow all IBM i and OS/400 documentation and technical support procedures for RLSM and other storage changes. The IBM Redbook, *iSeries in Storage Area Networks: A Guide to Implementing FC Disk and Tape with iSeries* (SG 24-6220-01) contains more detail on RLSM.

- Supported device types for RLSM:

  6718-50 to 2105-A82
  4326-50 to 2105-A85
  4327-50 to 2105-A84
  4328-50 to 2107-A86

- Unsupported device type for RLSM:

  6717-50 to 2105-A81

The mirroring process can take 15 minutes to several hours, depending on the server and the number of devices in the configuration.

The IBM backup and recovery documentation contains detailed information about how to recover systems with mirrored protection.

RLSM with TimeFinder and SRDF

In TimeFinder and SRDF configurations, each load source device and its mirror has its own BCV or R2 device. Both devices are processed for control actions, such as establish and split operations.

The disk configuration status (in SST) for both the load source and its mirror should be Active for normal operations. If the mirror device status is other than Active (that is suspended, resuming, or unknown), a warning message is written to the job log during the Split control action processing and when the split operation is performed. If the load source device status is other than Active, an error message is issued, and the split operation stops.

Required 2107/2105 emulation configuration for RLSM

The remote load source mirror device must be presented to the host as an unprotected device (210x-A8x), as protection is provided by the use of the host level mirroring.

It is recommended that all other devices be presented to the host as protected devices (210x-A0x).

210x-Axx emulations are external disk emulations based on the IBM ESS/DSS storage subsystems. IBM recognizes the disks as being connected through Fibre Channel disk controllers and not within the IBM server racks.

Descriptions of the emulation types are as follows:

- 210x-A8x emulation is reported to the host as an unprotected disk which allows it to be used for host level mirroring. Any protection internal to the array, which may have local protection (2-way mirror and/or remote protection (SRDF)), does not affect the "unprotected" state of the 210x-A8x disk.
- 210x-A0x emulation is reported to the host as a parity-protected disk which cannot be used for host level mirroring. Any protection internal to the array, which may have local protection (2-way mirror and/or remote protection (SRDF)), does not affect the "parity-protected" state of the 210x-A8x disk.

To configure remote load source mirroring, the "mirror" device must be an unprotected type (210x-A8x) and it must be added to the ASP. Using the IBM DST (Dedicated Service Tool - with options available during a manual IPL) enable the load source mirroring option, and then select the option to start mirrored protection. Follow the on-screen instructions and the system will IPL. Once the operating system starts, the mirroring begins during the storage management recovery phase of the IPL. For complete details on configuring and running RLSM refer to the IBM documentation.

SRDF/TimeFinder terminology and restrictions

SRDF/TimeFinder Manager activates and provides access to the array's pre-configured BCV, clone or snap devices, SnapVX control operations, and SRDF link control. These operations are controlled using menu-driven interfaces or a command line interface.

SRDF/TimeFinder Manager concepts and terminology used in the menus and command line interface are described below.

Terminology

SRDF/TimeFinder Manager uses the concept of an *image* as the high-level controlling object, and the concepts of *paths* and *steps* for representing how data on the source device gets copied to one or more target devices.

Images

An image is a set of *auxiliary storage pools (*ASPs) that can be operated on simultaneously. The types of images are as follows:

- System image — A combination of the system ASP and all user ASPs. SRDF/TimeFinder Manager represents a System image on screens as SYSTEM.
- IASP image — An image that contains the following:
  - A primary IASP and all secondary IASPs, if any, that are associated with the primary IASP group. The image name of the *ASP group* is the name of the primary IASP.
  - A user defined file system (UDFS) IASP. The image name of the UDFS is the name of the disk pool of that UDFS.

  See "SRDF/TimeFinder Manager Extended Features" on page 301 for more information on IASP functionality.
- NON_CONFIG image — A nonconfigured unit, or any user ASP that is not part of the SYSTEM image. SRDF/TimeFinder Manager does not control or process the devices in the NON_CONFIG image.

Paths and steps

SRDF/TimeFinder Manager uses the following concepts to configure, monitor, and control the array configuration:

- A *path* consists of steps, which describe how data on devices of the source image gets to devices of the target image.
- Each *step* is a description of the replication mechanism used to copy data from one device to another device.

  SRDF/TimeFinder Manager supports paths containing up to five steps. Steps in the path are predefined and are identified by their position (1 through 5) in the path. Step numbers are used with control commands to identify which steps in a path are to be processed.

  Example of paths and steps:
  - Configuration type — BCV/R1 (STD → BCV/R1 → R2)
  - Path name — BCVR1
  - Steps on path— /BCV1/RDF1

  Refer to "Configuring paths" on page 82 for more information.

Usage Restrictions

SRDF/TimeFinder Manager has the following usage restrictions.

User profile restriction

SRDF/TimeFinder Manager must be installed under the QSECOFR user profile as follows:

- All objects are owned by the security officer.
- All users must belong to the *SECOFR user class and have *ALLOBJ, *JOBCTL, and *SERVICE special authority.
- Users that do not belong to the *SECOFR user class with *ALLOBJ authority must have the following special authorities for the flowing objects to run SRDF/TimeFinder Manager menu options and commands:

  QSYS/QCSTCTL(*SRVPGM)
  QSYS/QCSTCRG1(*SRVPGM)
  QSYS/QCSTCRG4(*SRVPGM)

  To satisfy this requirement, system administrators must grant users *USE access to the above objects. The command EDTOBJAUT can be used for this purpose.
- Users need *IOSYSCFG and *JOBCTL authority in their user profile or inherited from a group that the user belongs to.
- To use commands such as DISCOVER, STMENDSBS, and STMSTRSBS a user must belong to the QSECOFR class and have *ALLOBJ, *JOBCTL, and *SERVICE special authority either in their profile or inherited from a group the user belongs to.

Single instance restriction

The Work with Images and Work with Pools screens do not lock the SYMAPI database file, which allows multiple jobs to access SRDF/TimeFinder Manager screens at the same time. In addition, the STATUS action does not lock the SYMAPI database file. However, the single instance restriction does apply to SRDF/TimeFinder Manager control actions, where only one control action can be run at a time.

CHAPTER 2
Configuration and Control Menus

This chapter describes the commonly used SRDF/TimeFinder Manager menus. It also introduces running SRDF/TimeFinder Manager commands from remote IBM systems.

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- Navigating SRDF/TimeFinder Manager 42
- Accessing the main menu 42
- Dell EMC SRDF/TimeFinder Manager for IBM i main menu 43
- Control Menu 44
- Configuration by Image screen 47
- Work with Associated Paths screen 48
- Disk Units screen 50
- List of Pools screen 56
- Running remote commands 57

Overview

The following screens, detailed in this chapter, are accessed from the main menu and are used to control SRDF/TimeFinder Manager actions:

- The Control Menu — Performs control actions for all supported SRDF and TimeFinder operations and other system-control operations.
- Configuration by Image screen — Performs operations at the image and step level.
- Work with Associated Paths screen — Performs operations at the path level.
- Disk Units screen — Performs operations at the device level.
- List of the Pools screen — Displays existing pools and performs control actions for all supported actions on pools.

Navigating SRDF/TimeFinder Manager

When working in SRDF/TimeFinder Manager screens, the following actions return to a previous screen or exit from SRDF/TimeFinder Manager screens:

- Press **F5** to refresh the information on SRDF/TimeFinder Manager screens.
- Press **F3** to return to the Configuration by Image screen.
- Press **F12** to display the previous screen.
- Repeatedly press **F3** or **F12** until control returns to the IBM i Main Menu.

Accessing the main menu

To access the Dell EMC SRDF/TimeFinder Manager for IBM i main menu:

1. On the source IBM i host, ensure that the product library EMCCTL is present in the library list by entering:

   ```
   ADDLIBLE EMCCTL
   ```

   Note: EMCCTL must be at the top of the library list to ensure proper execution of SRDF/TimeFinder Manager operations.

2. At the command prompt on the system console, enter:

   ```
   GO MAINCTL
   ```

   The Dell EMC SRDF/TimeFinder Manager for IBM i main menu (Figure 4 on page 43) appears.

Dell EMC SRDF/TimeFinder Manager for IBM i main menu

The Dell EMC SRDF/TimeFinder Manager for IBM i main menu is an IBM i interface to access array functionality.

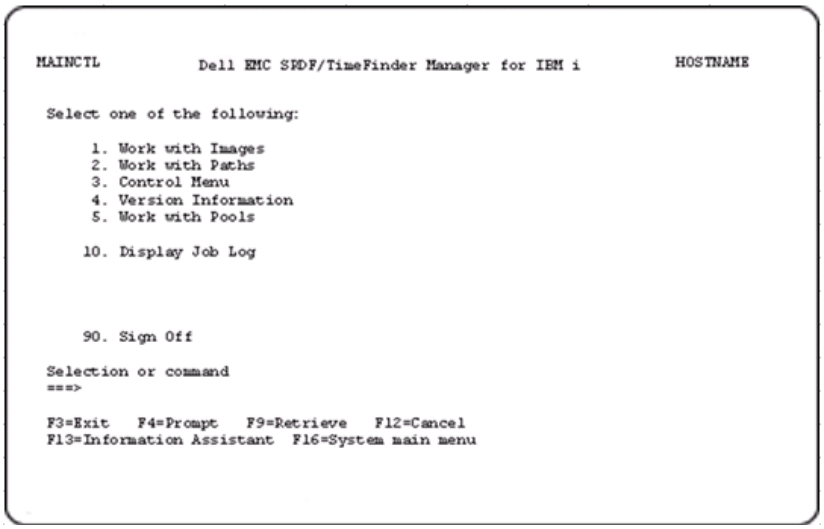


Figure 4 Dell EMC SRDF/TimeFinder Manager for IBM i main menu

Table 2 describes the options on the Dell EMC SRDF/TimeFinder Manager for IBM i main menu and lists where to find more information on each option.

Table 2 Dell EMC SRDF/TimeFinder Manager for IBM i main menu options (page 1 of 2)

Main menu option	Description	Where described
1. Work with Images	Displays the Configuration by Image screen used to add or remove ASPs from images or display other screens that enable you to associate and work with paths.	"Configuration by Image screen" on page 47
2. Work with Paths	Displays the Work with Available Paths screen that enables you to add new paths to the path file. It also enables you to delete a path from the path file and from the image file if the path has configured steps.	"Configuring paths" on page 82
3. Control Menu	Displays the Control Menu which contains options to execute SRDF, TimeFinder, or IASP control action commands, control SRDF groups, configure pools, and control other system level actions.	"Control Menu" on page 44
4. Version Information	Displays current program version information for SRDF/TimeFinder Manager software components.	"Viewing version information" on page 77.

Table 2 Dell EMC SRDF/TimeFinder Manager for IBM i main menu options (page 2 of 2)

5. Work with Pools	Displays the List of the Pools screen where you can create or delete save or data pools, as well as work with thin devices (bind, rebind, or unbind thin devices to the data pool), compress or uncompress thin device data, and migrate thin devices (full or pool level). Also enables access to other screens that allow you to enable, disable, add, or remove save or data devices in the pool. For arrays running HYPERMAX OS 5977, only freeall and rename actions are supported.	"List of the Pools screen" on page 56.
10. Display Job Log	Displays the current job log.	This is an OS/400 function; it is not documented in this guide.
90. Sign Off	Exits SRDF/TimeFinder Manager and signs off from the IBM i host.	

Control Menu

Use the Control Menu to:

- Perform SRDF, TimeFinder, or IASP control actions against devices in a step.
- Independent ASP control
- Discover arrays and host configuration detail.
- Toggle the SRDF mode type.
- Bring an IBM i host to a restricted state.
- Start subsystems.
- Control Daemons.
- Control device pools whose devices are configured for SRDF/A DSE, SNAP, or Virtual Provisioning operations.
- List SRDF groups.
- Control Dynamic SRDF Groups.
- Set SRDF groups.
- Create and delete RDF pairs.
- Create, delete, and display SRDF/Metro vWitness information.
- Add and remove node information.

To access the Control Menu, select option **3. Control Menu** from the Dell EMC SRDF/TimeFinder Manager for IBM i main menu (Figure 4 on page 43).

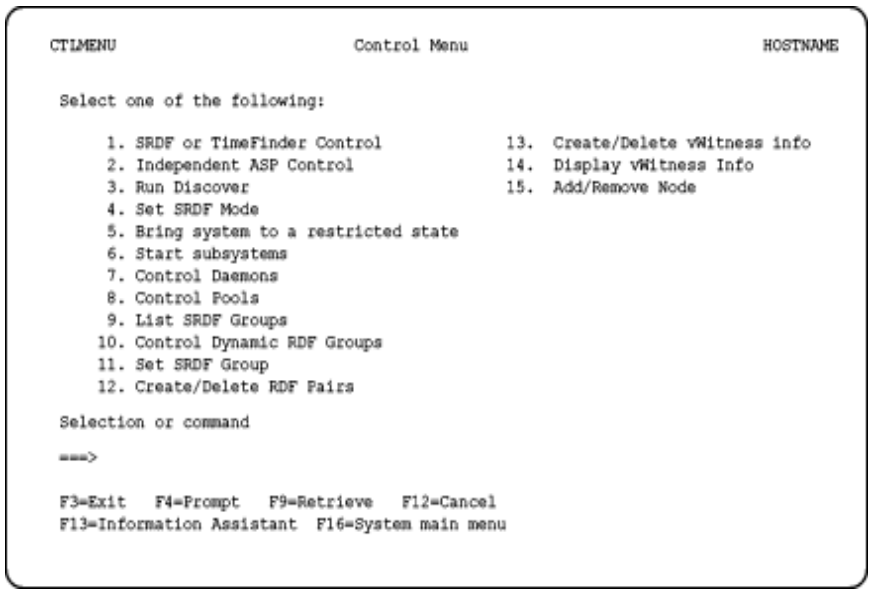


Figure 5 Control Menu

Table 3 provides a brief description of the Control Menu options and lists where each option is discussed in greater detail.

Table 3 Control Menu options (page 1 of 2)

Control Menu option	Description	Where discussed
1. SRDF or TimeFinder Control	Displays the Dell EMC SRDF or TimeFinder Control (CTLACT) screen for controlling devices in an image/path/step for SRDF and TimeFinder operations.	"" on page 165 "Controlling TimeFinder" on page 135 "CTLACT" on page 206
2. Independent ASP Control	Displays the Dell EMC Independent ASP Control (IASPCTL) screen, for processing IASP control actions.	"SRDF/TimeFinder Manager Extended Features" on page 301
3. Run Discover	Displays the Run Discover Process (DISCOVER) screen, which executes the DISCOVER command.	"Discovering array and host configuration details" on page 63 "DISCOVER" on page 227
4. Set SRDF Mode	Displays the Set SRDF Mode (SETRDFMOD) screen for setting the SRDF mode (synchronous, asynchronous, or Adaptive Copy disk, Adaptive Copy write pending mode) to use for a selected image/path/step.	"SETRDFMOD" on page 250
5. Bring system to a restricted state	Puts an IBM i host in a restricted state and ends the subsystems, except the subsystem running the Solutions Enabler daemon jobs. The two daemon jobs STORAPID and STORGNSD must be running to execute the control commands.	"STMENDSBS" on page 252
6. Start subsystems	Starts subsystems by executing the program defined in the SYSTEM value QSTRUPPGM.	"STMSTRSBS" on page 255

Table 3 Control Menu options (page 2 of 2)

Control Menu option	Description	Where discussed
7. Control Daemons	Displays the Control Dell EMC daemon process (STORDAEMON) screen for starting or stopping the SYMAPI server from the client (target) IBM i host.	"Working with Solutions Enabler daemons" on page 72
8. Control Pools	Displays the Control Pool (CTLPOOL) screen for controlling SRDF/A DSE or SNAP SAVE pools, or Virtual Provisioning thin pools.	"CTLPOOL" on page 215
9. List SRDF groups	Displays the List of the RDF Groups screen that lists all SRDF groups. It also shows the number of seconds that data on the R2 is behind the data on the R1 device and if the RDF group participates in a Star configuration.	"Listing and monitoring SRDF groups" on page 196
10. Control Dynamic RDF Groups	Displays the Control SRDF Groups screen for adding, modifying, and removing SRDF groups	"CTLRDFGRP" on page 221
11. Set SRDF group	Displays the Set SRDF Group (SETRDFGRP) screen for controlling attribute settings for an RDF group.	"SETRDFGRP" on page 245
12. Create/Delete RDF Pairs	Displays the Create/Delete RDF Pairs (CRTDLTRDFP) screen for creating or deleting dynamic SRDF pairs.	See "RDF group/device compatibility" on page 166 for more information on RDF group/device pairing. "CRTDLTRDFP" on page 203
13. Create/Delete vWitness info	Displays the vWitness Control Action (CTLVWIT) screen for adding, deleting, enabling, and disabling SRDF/Metro vWitness instances.	See "Maintain vWitness information" on page 192 and "CTLVWIT" on page 225
14. Display vWitness info	Displays the Display vWitness Information (DSPVWIT) screen for viewing the SRDF/Metro vWitness information.	See "View vWitness information" on page 193 and "DSPVWIT" on page 229
15. Add/Remove Node Information	Configure details of remote nodes, such as the back-up or disaster recovery node, so that they can be monitored and the operator alerted should any of them become unavailable.	See "Configuring node information" on page 65 and "STORDAEMON" on page 256

Configuration by Image screen

Use the Configuration by Image screen to perform image and step level operations.

To access the Configuration by Image screen, select option **1. Work with Images** from the Dell EMC SRDF/TimeFinder Manager for IBM i main menu (Figure 4 on page 43).

```
CTLDIMG            Symmetrix Configuration by Image          HOSTNAME

Type option, press Enter.
1=Add ASP to image 2=Work with paths   4=Remove ASP from image
5=Work with disk units                 8=Associate path

                          # of     Devices
Opt Image         ASP     Units    Type
 __ SYSTEM                  22     Symmetrix
 __                  1      10     Symmetrix
 __                  5      12     Symmetrix
 __ IBMIASP                  2     Non_Symmetrix
 __                 33       2     Non_Symmetrix
 __ PIASP2                   5     Symmetrix
 __                144       3     Symmetrix
 __                145       2     Symmetrix
 __ NON_CONFIG              18     Mixed
 __                  3      13     Non_Symmetrix
 __                         5     Symmetrix


 F3=Exit    F9=Command Entry    F10=Collapse/Show ASP   F12=Cancel
```

Figure 6 Configuration by Image screen

Press **F10** to display the information at the image level (Figure 7).

```
CTLDIMG                Symmetrix Configuration by Image          HOSTNAME

Type option, press Enter.
1=Add ASP to image 2=Work with paths   4=Remove ASP from image
5=Work with disk units                 8=Associate path

                        # of   Devices
Opt Image       ASP     Units  Type
 __ SYSTEM                22   Symmetrix
 __ IBMIASP                2   Non_Symmetrix
 __ PIASP2                 5   Symmetrix
 __ NON_CONFIG            18   Mixed


F3=Exit   F9=Command Entry     F10=Collapse/Show ASP   F12=Cancel
```

Figure 7 Configuration by Image screen — collapsed

Press **F10** again to display information at the ASP level (Figure 6).

Table 4 describes the fields on the Configuration by Image screen.

Table 4 Fields for the Configuration by Image screen

Field	Description
Opt	Entry field for configuration action options.
Image	**SYSTEM —** Displays the system ASP (ASP 1) and all basic user ASPs (2-32) configured on the system. **NAME** — Displays the name of the IASP image, when IASP images (ASPs 33-255) are present in the configuration. **NON_CONFIG** — Displays non-configured units (devices) and any user ASPs not part of the SYSTEM image.
ASP	ASP number.
# of Units	Total number of devices in the ASP or image, depending upon which screen appears.
Devices Type	Displays the device types in the configuration: • **Symmetrix** — Contains only Symmetrix devices. • **Non_Symmetrix** — Contains only non-Symmetrix devices. • **Mixed** — Contains both Symmetrix and non-Symmetrix devices.

Work with Associated Paths screen

Use the Work with Associated Paths screen to perform operations on paths.

To access Work with Associated Paths screen, select option **2. Work with Paths** from the Configuration by Image screen (Figure 6 on page 47).

```
CTLPATH                Work with Associated Paths                    HOSTNAME
 IMAGE= SYSTEM
 Type option, press Enter.
  4=Remove path  5=Work with disk units  6=Pair   7=Unpair
 11=IASP view   14=Establish/Create     15=Split/Activate    16=Configure IASP
 17=Add IOA     18=Remove IOA           19=Cancel/Terminate  20=Link/Relink
 21=Unlink      22=Work with Snapshots  23=Change Link Mode  24=Restore
                                         Invalids/ Path/
Opt Path Name  Step    Status            % Copied  Snapshot                 Gen
    CASCSVX            Configured                  /SVX1/SVX1/SVX1/SVX1
               SVX1    Copied                100% first_step              0
               SVX1    Restored              100% second_step             0
               SVX1    NoSnapshots             0%
               SVX1    NotPaired             N/A

    LSNAPVX1           Configured                  /SVX1
               SVX1    CopyInProg             55% End_of_Quarter          1


                                                                       Bottom
 F3=Exit   F5=Refresh   F6=Add Path   F9=Command Entry   F12=Cancel
```

Figure 8 Work with Associated Paths screen — overview

Table 5 describes the fields on the Work with Associated Paths screen.

Table 5 Fields for the Work with Associated Paths screen

Field	Description
Opt	The entry field for configuration action options.
Path Name	The name of the path used to move data on devices of the source image to devices of the target image.
Step	The description of how data is copied from source devices to target devices.
Status[a]	Identifies the status of the selected path.
Invalids/%Copied	The number of invalid tracks remaining during SRDF or TimeFinder synchronization operations, or percent of copy completion for TimeFinder/Clone or TimeFinder/Snap operations.
Path/Snapshot	Description of steps in the path associated with the selected image. Snapshot name is displayed if target devices for the SVXn are linked to the snapshot, or source devices restored from the snapshot and target devices are not linked.
Gen	For SVXn step, if target devices are linked to the snapshot or source devices restored from the snapshot and target devices are not linked then the generation number is displayed.

a. Possible step status for SnapVX :

- NotConfigured – step is not paired and no snapshots for source devices in the step exist.
- NotPaired – step is not paired and at least one snapshot for source devices in the step exists.
- Established – devices are paired, snapshots exist but are not linked. Step has snapshots, is paired, but not linked.
- NoSnapshots – step is paired and no snapshots for source devices in the step exist.
- If the step is paired, at least one snapshot for source devices in the step exists but target devices are not linked, the status of the first Snapshot would be presented. If snapshot has restores, the status would be Restored or RestoreInProg.
- RestNotPaired – step is not paired and source devices are restored.
- Status of the link – If target devices are linked.

Disk Units screen

Use the Disk Units screen to perform operations at the device level. The Disk Units screen can be displayed from either the array view or IBM i view.

To access the Disk Units screen, select option **5=Work with disk units** for the desired step in the Work with Associated Paths screen (Figure 8 on page 48).

The Disk Units screen for SnapVX type of step Figure 11 on page 55 displays a different header than all other type of steps.

```
CTLUNITS                          Disk Units                              HOSTNAME

 Image SYSTEM       ASP *ALL   Path SRDFMETRO     Step number  1       Step type RDFM1
 RDF mode Active               Consistency protection N/A
 Clone Emulation N/A           BCV Configuration N/A      Target Type N/A
 Snap Pool Name N/A            Clone Type N/A
 5=Show details    6=Pair    7=Unpair   8=Cancel
    Unit      Source  Target  Source       Target                 Invalids/   Bias
Opt #    ASP  Dev.    Dev.    Symmetrix    Symmetrix    Status    % Copied
      1   1  0012E   00112   000196701855 000196701854 Suspended       0      T
      2   1  0012F   00113   000196701855 000196701854 Suspended       0      T
      3   1  00130   00114   000196701855 000196701854 Suspended       0      T
      4   1  00131   00115   000196701855 000196701854 Suspended       0      T

                                                                          Bottom
F3=Exit  F5=Refresh  F9=Command Entry  F10=Mapping View  F11=IBM i View  F12=Cancel
```

Figure 9 Disk Units screen

```
CTLUNITO                      Disk Units IBM i View                       HOSTNAME

 Image SYSTEM      ASP *ALL  Path N/A         Step number N/A   Step type N/A

 Unit       Serial      Resource    Disk  Disk   I/O      I/O
 #     ASP  Number      Name        Type  Model  Adapter  Bus  Ctl  Dev
    1    1  50-8A333    DD038       2107  0A85      2       0    1   10
    8    1  50-62333    DD011       2107  0A84      2       0    1    2
    9    1  50-66333    DD010       2107  0A84      2       0    1    3
   10    1  50-6A333    DD009       2107  0A84      2       0    1    4
   13    1  50-8E333    DD034       2107  0A85      2       0    1   11
   14    1  50-6E333    DD036       2107  0A84      2       0    1   12
   15    1  50-76333    DD037       2107  0A84      2       0    1    5
   17    3  50-7A333    DD013       2107  0A84      2       0    1    6
   18    3  50-72333    DD014       2107  0A84      2       0    1   13
   19    3  50-7E333    DD012       2107  0A84      2       0    1    7
   20    3  50-86333    DD032       2107  0A84      2       0    1    9
   23    3  50-82333    DD033       2107  0A84      2       0    1    8


                                                                      Bottom
 F3=Exit   F9=Command Entry   F11=Symmetrix View   F12=Cancel
```

Figure 10 IBM i View

Table 6 describes the fields on the Disk Units screen for both the array and IBM i views.

Table 6 Fields for the Disk Units screen (page 1 of 4)

Field	Description
Array view	
RDF mode	Type of SRDF/A mode for the RDF devices in a step. Possible values are: • **N/A** — The device is not RDF capable. • **Active** — The selected image is part of a SRDF/Metro configuration. • **Synchronous** — The mode for the selected image is set to Synchronous. • **Asynchronous** — The mode for the selected image is set to Asynchronous. • **Adaptive Copy** — The mode for the selected image is set to adaptive copy. Possible values are Adaptive Copy Write Pending Mode, Adaptive Copy Disk Mode, Adaptive Copy Mixed Mode.
Consistency Protection	Indicates whether or not consistency protection is enabled for SRDF/A. Possible values are: • **N/A** — The device is not in SRDF/A mode. • **Enabled** — The device is running in asynchronous mode and consistency protection is enabled. • **Disabled** — The device is running in asynchronous mode and consistency protection is disabled.
Clone emulation	Indicates the clone for the step. Possible values are: • **N/A** — The BCV clone emulation is not enabled for the step. • **Enabled** — There are clone sessions for all pairs of BCV devices. • **Mixed** — Not all BCV devices in the step are clone enabled.
BCV Configuration	Indicates the configuration type for the BCV device in a step. Possible values are: • **N/A** — Not all devices in the BCV step are configured or the step type is not BCV. • **Standard** — All BCV devices in the step for the selected path of the system image are configured as additional mirrors of a standard array device. • **RAID5/6** — The BCV devices in the step for the selected path of the system image are configured for RAID 5 or RAID 6 or both RAID 5 and RAID 6 protections. • **Mixed** — Devices have mixed BCV protections — Some devices are standard BCVs; some devices are RAID 5 and RAID 6-protected BCVs.
Target Type	Indicates the type of target device for a Clone (NCL*n*) step. Possible values are: • **BCV** — All target devices are thick BCV devices. • **Standard** — All target devices are standard thick devices. • **Mixed** — Target devices are a mix of BCV, standard, and thin devices. • **TDEV** — All target devices are standard thin devices (allowed for VP Snap). • **BCV_TDEV** — All target devices are thin BCV devices (allowed for VP Snap). • **TDEV RDF** — All target devices are RDF thin devices. • **TDEV BCV/RDF** — All target devices are RDF BCV thin devices. • **TDEV BCV/STD** — Target devices are a mix of standard and BCV Thin devices (allowed for VP Snap). • **N/A** — This is not a Clone (NCL*n*) step.
Snap Pool Name	Indicates the pool name used for the SNAP session.
Clone Type	Indicates the Clone type of Clone type of the step • **Regular** — Clone type is a normal type • **VP Snap** — Clone type is a Virtual Space Efficient type • **Mixed** — Clone type is mixed, regular and VP Snap • **N/A** — indicates other types of the step
OPT	Entry field for configuration action options.

Table 6 Fields for the Disk Units screen (page 2 of 4)

Field	Description
Unit #	Number assigned by the system to identify a specific disk. **0** — Non-configured devices.
ASP	ASP number in the current configuration. **0** — Non configured devices.
Source Device	Array device name of the source device.
Target Device	Array device name of the target device.
Source Symmetrix	Array serial number for the source device.
Target Symmetrix	Array serial number for the target devices (only if present). If BCV, Clone, or Snap devices are present, the serial number is the same as the source array.
Status of the RDF pair	Status for the selected SRDF pair (if relevant). Possible values are: • **N/A** — Device is not SRDF capable. • **Consistent** — The R2 mirrors of SRDF/A devices are in a Consistent state, which signifies the normal state of operation for device pairs operating in asynchronous mode. • **Failed Over** — R1 is Not Ready or Write Disabled, and the operation is failed over to the R2 device. • **Invalid** — The default state when another RDF state applies, such as when the combination of R1, R2, and RDF link states and statuses do not match any other pair state. An Invalid status can occur if there is a problem at the disk director level. • **Partitioned** — The SYMAPI is currently unable to communicate through the corresponding RDF path to the remote array. A Partitioned status may apply to devices within an RA group. • **R1 UpdInProg** — The R1 is currently Not Ready or Write Disabled to the host. There are invalid local (R1) tracks on the source side and the link is Ready or Write Disabled. • **R1 Updated** — The R1 is currently Not Ready or Write Disabled to the host. There are no local invalid tracks on the R1 side and the link is Ready or Write Disabled. • **Synchronized** — The R1 and R2 are currently in a synchronized state and have identical data. • **SyncInProg** — Synchronization is currently in progress between the R1 and the R2. There are existing invalid tracks between the two pairs and the logical link between both sides of an RDF pair is up. During an Establish operation, data is copied from the standard device to the RDF device until both devices contain identical data. • **Split** — SRDF links were established and are currently split. The R1 and the R2 are currently Ready to their hosts, but the link is Not Ready or Write Disabled. • **Suspended** — The RDF links have been suspended and are Not Ready or Write Disabled. If the R1 is Ready while the links are suspended, any I/O will accumulate as invalid tracks owed to the R2 device. • **TransIdle** — The RDF group is operating in SRDF/A Transmit Idle mode. The SRDF/A session cannot push data in the transmit cycle across the link because the link is down.

Table 6 Fields for the Disk Units screen (page 3 of 4)

Field	Description
Status for BCV or Clone enabled pair	Status for the selected BCV or Clone enabled pair (if relevant). Possible values are: • **N/A** — Device is not paired. • **Invalid** — If the device is a meta head, and the members are not in a consistent state, then the overall status of the meta device is mixed and invalid. • **NeverEstab** — The BCV device was paired but never established with a standard or R2 device, or Cancel is executed against this device pair. • **RestInProg** — When the restore action is executed, data is copied from the BCV or Clone to the standard device until both devices contain identical data. • **Restored** — The BCV or Clone enabled device and the standard device have identical data, although the data was originally on the BCV or Clone enabled device prior to being synchronized. Any changes to the standard device are also written to the BCV or Clone enabled device. The BCV or Clone enabled device is unavailable to the host for processing. • **Split** — The BCV or Clone enabled device is completely separated, or split, from the standard devices allowing each device to be accessed separately by the host. • **SplitBfrRes** — The split occurred when a BCV or Clone enabled device was being restored to a standard device. The BCV or Clone enabled device is separated from the standard device, although the standard is not completely synchronized. No restrictions. • **SplitBfrSyn** — The split occurred when a BCV or Clone enabled device was synchronizing. The BCV or Clone enabled device is separated from the standard device although the BCV or Clone enabled device is not completely synchronized. • **SplitInProg** — The BCV or Clone enabled devices are in the process of being separated or split from the standard devices. • **SplitNoInc** — The BCV or Clone enabled devices are completely separated or split from the standard devices but cannot be incrementally established or restored. No restrictions. • **Synchronized** — The BCV or Clone enabled devices and the standard device have identical data. Any changes to the standard device are also written to the BCV or Clone enabled device. When a step state is Synchronized, a Split operation can be performed. The BCV or Clone enabled device is unavailable to the host for processing. • **SyncInProg** — Synchronization is currently in progress between the standard or R2 device and the target device.

Table 6 Fields for the Disk Units screen (page 4 of 4)

Field	Description
Status for Snap or Clone pair	Status for the selected Snap or Clone device pair (if relevant). Possible values are: • **Copied** — The entire device has been written to and is in a fully copied state. • **CopyInProg** — When background copying, the state of the device pair is Copy In Progress. When the copy session is activated, background copying begins so that a full copy of the data will become available on the target device. • **CopyOnAccess** — After activating the copy session, only those tracks that have been written to the source or written to/read from the target will be copied to the target device. A full data copy to the target device will not occur unless all of the device tracks are accessed or written to while participating in the active session. • **CopyOnWrite** — Devices participating in the copy session have been activated. Data and pointers will be copied upon any first write to a track on the source device or any host write to the target device. • **Created** — A copy session between a source device and a target device has been created. • **CreateInProg** — A copy session between a source device and a target device is in progress. • **Failed** — The copy session has failed because the log device pool is full. • **Invalid** — Applies to a meta device only. The meta device states do not consistently match. • **Not Created** — The target device is not actively participating in a copy session. • **Precopy** — The precopy process keeps checking for new writes to be precopied to the target device until the copy session is activated. Once activated, the normal background copy mechanism starts and the precopy operation ends. Precopy can occur in the Created or Recreated state. • **RestInProg** — Data restore to the specified target device is in progress. • **Restored** — Data has been fully restored to the specified target device. • **TermInProg** — The copy session between a source device and a virtual device is in the process of terminating.
Invalids /%Copied	The number of invalid tracks remaining during SRDF or TimeFinder synchronization operations, or percent of copy completion for TimeFinder/Clone or TimeFinder/Snap operations.
Bias	SRDF/Metro configurations only: Indicates whether the source array is the bias side of a SRDF/Metro configuration. The field is blank when the source array is not on the bias side, contains S when the source array is the bias side, or contains T when the target array is the bias side.
IBM i View	
Serial Number	IBM i serial number.
Resource Name	IBM i resource name for the device.
Disk Type	The device type being emulated.
Disk Model	The model of the device being emulated.
I/O Adapter	The address of an adapter on an IOP.
I/O Bus	Address of a bus on an IOP.
Ctl	Address of a controller on an I/O bus.
Dev	Address of a storage device on a controller.

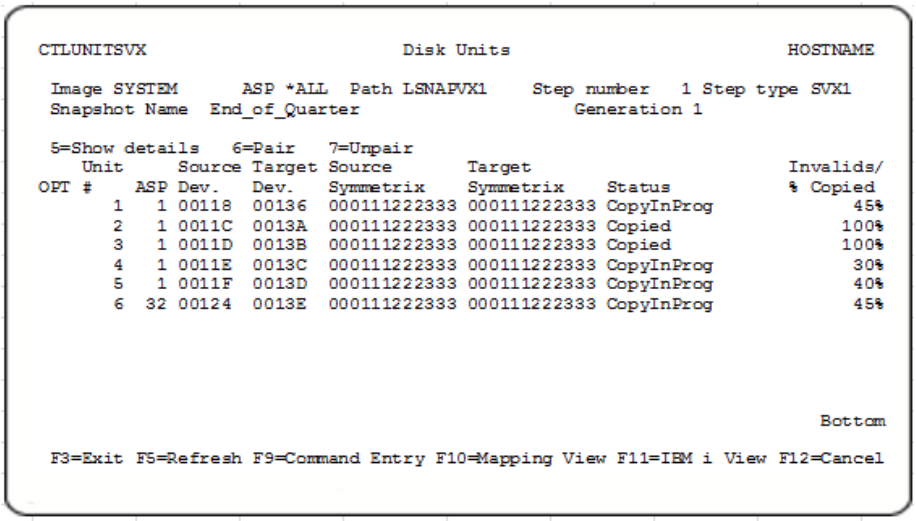


Figure 11 Disk Units screen — SVX type of step

Table 7 describes the fields on the Disk Units screen for SVX type of step.

Table 7 Fields for the Disk Units screen for SVX type of step (page 1 of 2)

Field	Description
Array view	
Snapshot Name	Name of the SnapVX snapshot.
Generation	Identifies the point-in-time copy when multiple snapshots are made for the same name on the same devices.
OPT	Entry field for configuration action options.
Unit #	Number assigned by the system to identify a specific disk. **0** — Non-configured devices.
ASP	ASP number in the current configuration. **0** — Non configured devices.
Source Device	Array device name of the source device.
Target Device	Array device name of the target device.
Source Symmetrix	Array serial number for the source device.

Table 7 Fields for the Disk Units screen for SVX type of step (page 2 of 2)

Field	Description
Target Symmetrix	Array serial number for the target devices (only if present). If target devices are present, the serial number is the same as the source array.
Status for SnapVX pair	Status for the SnapVX pair. Possible values are: • **N/A** — Source device is not paired and not restored. • **Linked** — Target device is linked to snapshot in NoCopy mode. • **Copied** — Target device is linked to snapshot in Copy mode when it is 100% copied. • **CopyInProg** — Target device is linked to snapshot in Copy mode and it is in transient state to Copied. • **RestoreInProg** — Restore from snapshot to source device is in progress and source device is not paired. • **Restored** — Source device is restored from the snapshot and source device is not paired. • **NotLinked** — Target device is not linked to the snapshot. • **Failed** — Link failed if no more allocations can be made from the SRP associated with the link device or if a corruption occurs on the link device.
Invalids/%Copied	Percent of the total device size that has been copied to the target when the COPY flag is set, or when the source device is restored from the snapshot.

List of Pools screen

Use the List of Pools screen to work with pools.

To access the List of Pools, select option **5. Work with Pools** from the EMC SRDF/TimeFinder Manager for IBM i main menu (Figure 4 on page 43).

```
CTLPOOLS                        List of the Pools                        HOSTNAME

 Type option, press Enter.
  5=Work with the Pool     4=Delete Pool  7=Rename Pool   8=Rebalance Thin Pool
 10=End Rebal Thin Pool  11=Change max subs  13=Compr Enable  14=Compr Disable
                    Pool           Emul      Pool  # of
 Opt  Symmetrix     Name           Type      Type  Devices    State     Compr
      000111222333  DEFAULT_POOL   N/A       SNAP  0          Enabled   N/A
      000111222333  d910           FBA       DATA  10         Enabled   Enabled
      000111222333  d910_bcv       FBA       DATA  10         Enabled   N/A
      000111222333  NON_POOLED     N/A       DATA  960        N/A       N/A
      000222333444  DEFAULT_POOL   N/A       SNAP  64         Enabled   N/A
      000222333444  mig6r6         FBA       DATA  256        Enabled   N/A
      000222333444  mig7r5         FBA       DATA  256        Enabled   N/A
      000222333444  tbcv           FBA       DATA  512        Enabled   N/A
                                                                        Bottom
 F3=Exit     F5=Refresh     F6=Create Pool     F7=Work With Unbound Thin Devices
 F9=Command Entry           F12=Cancel         F8=Work With Migration Sessions
```

Figure 12 List of the Pools screen

Table 8 describes the fields on the List of the Pools screen.

Table 8 Fields for the List of the Pools screen

Field	Description
Opt	The entry field for configuration action options.
Symmetrix	Array serial number
Pool Name	Name of the pool.
Emulation Type	Emulation type of the pool devices.
Pool Type	The type of device pools. Possible values are SNAP, DSE, or DATA
# of Devices	Number of devices in the pool.
State	Status of the devices in pool. Possible values are Enabled, Disabled, Balancing, or N/A
Compr	Compression status of the thin pool. Possible values Enabled, Disabled, Enabling, Disabling, or N/A. Field is only relevant for thin pools. For all other pools state is N/A.

Running remote commands

For SRDF/TimeFinder Manager versions earlier than 8.0, preparing to run SRDF/TimeFinder Manager commands from a remote IBM system requires several configuration steps, including starting the REXEC server on both the local and remote system. In order to simplify using this feature, SRDF/TimeFinder Manager 8.0 and later uses the STMRMTCMD command and executes remote commands through the STM server.

See “STMRMTCMD” on page 254 for more information on how to run the remote feature for STM commands.

CHAPTER 3
Operational Features, Settings, and Displays

This chapter shows how to set up, maintain, and display the SRDF/TimeFinder Manager environment.

- Setting user-configurable preferences 60
- Discovering array and host configuration details 63
- Configuring node information..... 65
- Configuring IFS files to support Solutions Enabler 70
- Working with Solutions Enabler daemons..... 72
- Working with the SYMAPI server 74
- Viewing version information 77

Setting user-configurable preferences

SRDF/TimeFinder Manager has the following user-configurable preferences:

- Suppress or allow pop-up messages ("Controlling pop-up messages")
- Set restricted state wait time ("Setting the restricted state wait time")
- Set the wait time value for SRDF/A ("Controlling the SRDF/A Split wait time")
- View values in the FLSHCFG data area ("Viewing the current values in the FLSHCFG data area")
- Reset the lockbox password ("Resetting the lockbox password")

Controlling pop-up messages

SRDF/TimeFinder Manager can determine how pop-up messages display. For example, during BCV Split or SRDF Split/Suspend operations, pop-up messages or screens display when additional action is required to proceed with the operation.

Pop-up messages display by default. To suppress pop-up messages and screens, use the CHGDTAARA command to set the value in the DSPMSG data area in the EMCCTL library to 1. When the value is set to **1**, all messages go into the job log during SRDF or TimeFinder operations, and no messages display even in interactive jobs. This only applies when the operation is initiated from the CTLACT command or option **1. SRDF or TimeFinder Control** on the Control Menu.

To allow pop-up messages and screens, use the CHGDTAARA command to set the value in the DSPMSG data area in the EMCCTL library to 0.

The default is to allow pop-up messages and screens.

Setting the restricted state wait time

SRDF/TimeFinder Manager provides an option on the Control Menu that sets the system to a restricted state. Depending on the configuration, the time it takes for a host to go into restricted state can vary. By default, SRDF/TimeFinder Manager waits 1800 seconds for control to return, after the action to bring the system to a restricted state is issued from the Control Menu. If the host cannot go into restricted state within the wait time, the option times out with error CTL0151. Use the WRKSBS command to verify the current status of the subsystems.

If additional restricted state wait time is required, use the CHGDTAARA command to set the appropriate value (in seconds) in the SBSWAIT data area in the EMCCTL library.

Controlling the SRDF/A Split wait time

In an SRDF/A configuration, during an SRDF Split/Suspend operation, the control program needs to wait at least two cycle times after the flush to ensure that all changes are made to the R2 devices, before the actual split occurs. This process usually takes two times the average SRDF/A cycle time, but could take longer depending upon the configuration. The control program waits for a maximum of 300 seconds (the default) before failing with an error.

SRDF/TimeFinder Manager can control the wait time for a Split/Suspend operation before failing with an error.

To control the Split wait time, set a value in the SRDFATIME data area using the CHGDTAARA command.

Viewing the current values in the FLSHCFG data area

A preference can be set to view the current values in the FLSHCFG data area.

To view the current values in the FLSHCFG data area, use the following command. If necessary, change EMCCTL to the library containing the data area.

```
DSPDTAARA EMCCTL/FLSHCFG
```

The Display Data Area screen (Figure 13) appears:

```
                          Display Data Area
                                                        System:   HOSTNAME
Data area . . . . . :   FLSH FG
  Library . . . . . :     EMCCTL
Type  . . . . . :   *CHAR
Length  . . . . . :   150
Text  . . . . . :

          Value
Offset     *...+....1....+....2....+....3....+....4....+....5
    0     'Suspend timeout (seconds)              000060     '
   50     'Timeout action End/Continue (E/C)      E          '
  100     'Option SUSPEND/F.RCW.RT (S/F)          S          '




                                                                Bottom
Press Enter to continue.

F3=Exit   F12=Cancel
```

Figure 13 Display Data Area screen with FLSHCFG data area with default values

You can modify three fields:

- **Suspend timeout (seconds)** — If the database Suspend operation does not complete within this time, the action defined in the following field occurs. The default value is 000060.
- **Timeout action End/Continue (E/C)**

  E — When a suspend timeout occurs, an escape message is written in the joblog, the Suspend operation is terminated, and the SRDF/TimeFinder Manager Split operation is terminated.

  C — When a suspend timeout occurs, a diagnostic message is written in the joblog, the Suspend operation continues, and the SRDF/TimeFinder Manager operation continues.

  The default value is **E**.

Note: If the Suspend operation fails, an escape message is written in the joblog. If operation is successful, a completion message is written to the joblog. When C is specified in the data area check the Additional Message Information.

- **Option SUSPEND/FRCWRT (S/F)**

  When performing a SPLITWA action in the CTLACTTGT command, the value of this field determines which mode the CTLACTTGT uses to run the CHGASPACT command:

  S — CHGASPACT runs in *SUSPEND mode.

  F — CHGASPACT runs in *FRCWRT mode.

  The default value is **S**.

  To change values in this data area, use the CHGDTAARA command. For example:

  - To change the Suspend timeout (seconds):

    ```
    CHGDTAARA DTAARA(EMCCTL/FLSHCFG (40 6)) VALUE('nnnnnn')
    ```

  - To change Timeout Action End/Continue (E/C):

    ```
    CHGDTAARA DTAARA(EMCCTL/FLSHCFG (90 1)) VALUE('E')
    ```

Resetting the lockbox password

The lockbox password for array accessibility is set to the default (*hostname*@SELockbox1) when the CRTSYMAPI and CRTSYMAPIC commands are run during SRDF/TimeFinder Manager installation. The password can be changed by running the following SYMCLI command:

```
symcfg –lockbox [–password <Password>]
   reset –ssv
   setpw [–new_password <NewPassword>]
```

Discovering array and host configuration details

The Discover function is used to perform the discovery process after changes to the host configuration or array configuration. However, by default, the SRDF/TimeFinder Manager Discover function is turned off and it does not automatically run the discovery process while viewing configuration details, performing control operations, or making configuration changes.

The discovery process is activated one of two ways:

- Changing the discover data area value in the EMCCTL library — default is set to discovery process off.
- Running the DISCOVER command.

Note: The discovery process run from the DISCOVER command is independent of the setting in the EMCCTL library.

Turning on the discovery process from the EMCCTL library

To turn on the discovery process, use the CHGDTAARA command to set the Discover data area in the EMCCTL library to 1. The Discover function remains on and the discovery process runs until the value is set to 0.

Running the DISCOVER command

The DISCOVER command performs a one-time discovery process, and does not enable automatic discovery. For subsequent configuration changes, unless the Discover data area value is set to 1, the command must be run again to update configuration files.

Run the discovery process as follows:

1. Select option **3. Control Menu** from the Dell EMC SRDF/TimeFinder Manager for IBM i main menu (Figure 4 on page 43).
2. Select option **3. Run Discover** from the Control Menu to display the Run Discover Process (DISCOVER) screen (Figure 14 on page 64).

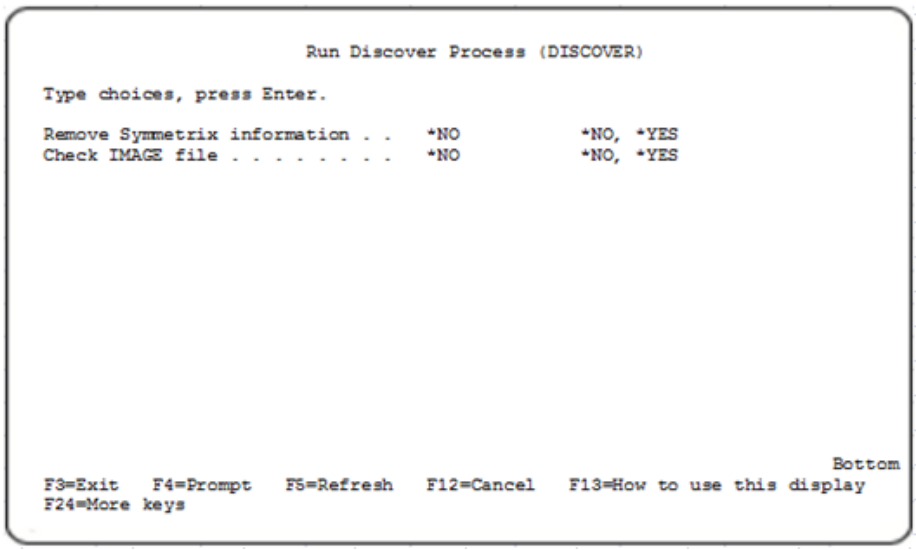


Figure 14 Run Discover Process (DISCOVER) screen

3. Enter the following required parameters:
   a. **Remove Symmetrix information**:
      - *__NO__ — Do not remove old array data from the SYMAPI database. *NO is the default.
      - *__YES__ — Remove previously discovered array information that is no longer valid from the SYMAPI database.

CAUTION

When the RDF link is down, do not run DISCOVER with Remove Symmetrix information set to *YES. Doing so can remove relevant R2 information, such as BCV or Clone pairing for the array from the SYMAPI database.

   b. **Check IMAGE file**:
      - *__NO__ — Do not verify information stored in the IMAGE file with current device information of the host. *NO is the default.
      - *__YES__ — Verify information stored in the IMAGE file with the current device information of the host and remove any old device information from the file.
4. Press **Enter** to start the discovery process.

Turning off the discovery process

To turn off the discovery process, use the CHGDTAARA command to set the Discover data area in the EMCCTL library to 0.

Configuring node information

SRDF/TimeFinder Manager can monitor remote nodes, for example those providing disaster recovery facilities. Should one of those nodes become unavailable for any reason, SRDF/TimeFinder Manager sends a message to the system operator of the IBM System i.

SRDF/TimeFinder Manager keeps a list of the nodes that it is to monitor. It also provides facilities for you to:

- Access the management facilities for node monitoring
- Add nodes to the list
- Remove nodes from the list
- Change the Notify User parameter
- Display the list of nodes
- Control the monitoring of the remote nodes

Accessing the node monitoring management facilities

To access the facilities for managing how SRDF/TimeFinder Manager monitors nodes:

1. Select option **3. Control Menu** from the Dell EMC SRDF/TimeFinder Manager for IBM i main menu (Figure 4 on page 43).
2. Select option **15. Add/Remove Node** from the Control menu to display the Display Node Information (NODEDETAIL) screen (Figure 15).

```
NODEDETAIL            Display Node Information                    HOSTNAME

Type option, press Enter.

4=Delete Node       5=Change Notify User

Opt   Node Name     Node IP             Node      Notify      Last
                                        Failed     User     Updated
-     CLOVIS        10.243.148.119        N         N      201804250300
-     STMDEV        10.243.148.141        Y         Y      201804250300
-     STMDEV2       10.243.148.140        N         Y      201804250300









F3=Exit F5=Refresh F6=Add Node F9=Command Entry F12=Cancel
```

Figure 15 The Display Node Information screen

For each node in the list, the screen shows:

- The name of the node
- The IP address of the node
- An indication of whether the node is unavailable: **Y** shows that the node is unavailable while **N** shows that the node is available.
- The date and time when the availability of the node was last checked.

Adding a node

To add a node to the list of monitored nodes:

1. Access the Display Node Information screen.
2. Press **F6** to display the Add Node Information screen:

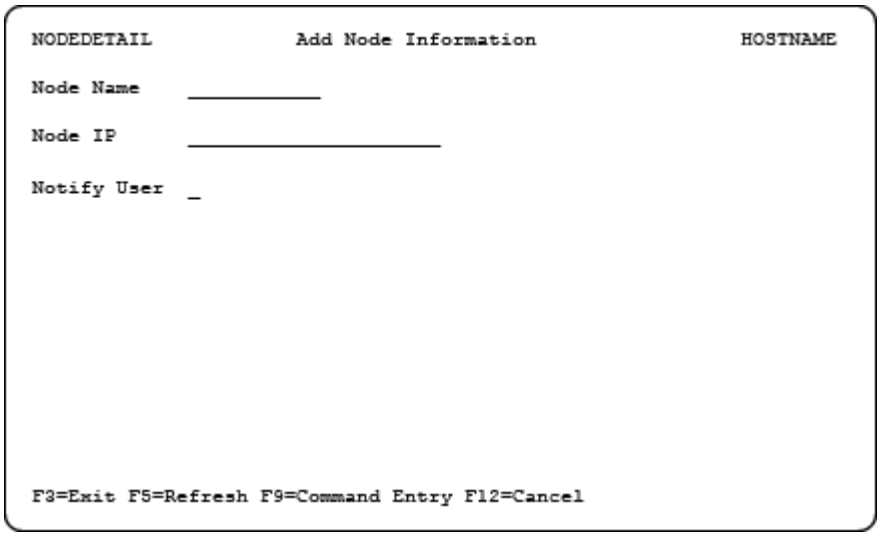


Figure 16 Add Node Information Screen

3. Enter values for the **Node Name** and **Node IP** parameters.
4. The **Notify User** parameter specifies whether the daemon periodically checks the node and notifies the user. Enter **Y** to enable periodic checking and notification of the node. Enter **N** for the daemon to check the node and notify the user only when the daemon starts up.
5. Press **Enter**.

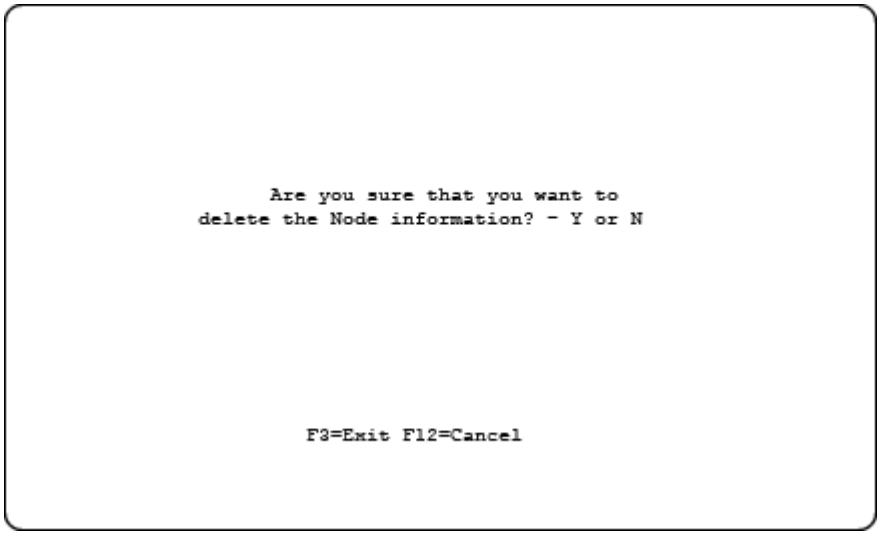
   The Display Node Information screen re-appears with the new entry included in the list.

Removing nodes

To remove nodes from the list of monitored nodes:

1. Access the Display Node Information screen.
2. Enter **4** in the **Opt** column for each node to remove from the list.
3. Press **Enter**.

   A confirmation screen appears:



4. Type **Y** and press **Enter**.

Changing the Notify User parameter

To change the setting of the Notify user parameter for one or more nodes:

1. Display the Access the Display Node Information screen.
2. Enter **5** in the **Opt** column for each node to remove from the list.
3. Press **Enter**.

Controlling the monitoring of remote nodes

A daemon process called NODEDAEMON carries out the monitoring of remote nodes and recording their status. You can start and stop the daemon process:

1. Select option **3. Control Menu** from the Dell EMC SRDF/TimeFinder Manager for IBM i main menu.
2. Select option **7. Control Daemons** from the Control Menu.

   The Control Dell EMC daemon process (STORDAEMON) screen (Figure 17) appears.

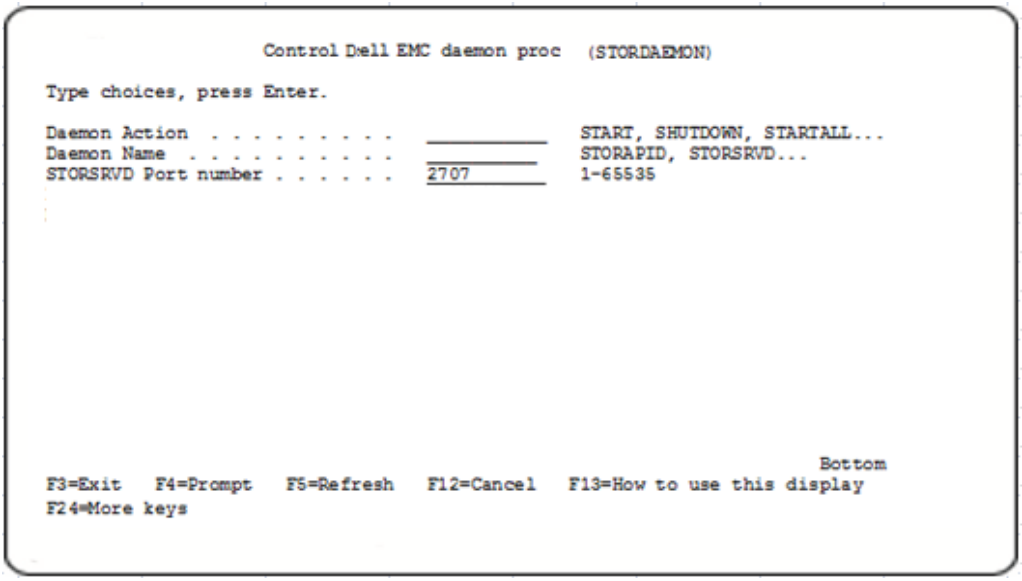


Figure 17 Control Dell EMC daemon proc (STORDAEMON) screen

3. In the **Daemon Action** field enter **STRNODEMON** (to start the daemon) or **ENDNODEMON** (to stop the daemon).
4. Press **Enter**.

Configuring IFS files to support Solutions Enabler

In order to support Solutions Enabler operations, the required files are installed to the IBM i Integrated File System (IFS) during SRDF/TimeFinder Manager installation. These files are installed in the following IFS directories:

- `/var/symapi/config` – Holds the Solutions Enabler configuration files, and an alias file that can be used to run CLIs in a QShell environment.
- `/usr/storapi/man/man1` – Holds the help files for the CLIs.
- `/usr/storapi/locales/en` – Holds the `error.properties` file that enables error reporting in different languages.

The installation of these Solutions Enabler files is performed by the CRTSYMAPI or CRTSYMAPIC program. When either of these programs are run, these files are restored from a save file in the EMCCTL or EMCCTLC library. The save files are called SYMAPICNFG, STORAPILOC and STORAPIMAN.

Modifying the Solutions Enabler configuration files

Once installed the SYMAPICNFG files require some modification for Solutions Enabler to function properly.

Modifying the readme configuration files

The following "README" files are templates can be renamed or copied to a file that does not include "README" in the file name. For example `README.daemon_options` file can be renamed to `daemon_options`.

- `/var/symapi/config/README.daemon_options` — The daemon_options file contains parameters that control the behavior of the various Solutions Enabler daemons. Each daemon reads this file at startup and applies any settings that apply. The daemon_options file is created by the CRTSYMAPI or CRTSYMAPIC program. For SRDF/TimeFinder Manager, the daemon options that control which subsystem the daemon is started in is set to the EMCCTL subsystem as a default. See "Working with Solutions Enabler daemons" on page 72 more for information. The parameters in this file can be modified by the user.
- `/var/symapi/config/README.daemon_users` — The daemon_users file lists the local users that are authorized to control and, in some cases, make use of the various Solution Enabler daemons. The parameters in this file can be modified by the user.
- `/var/symapi/config/README.options` — This options file contains behavior parameters that can be set to critically change the default behavior of SYMCLI operations, SYMAPI calls, and their control actions. It can be used to set certain global restrictions as well as customize and streamline command line coding to the user-specific environment. The options file is being set by the CRTSYMAPI file.

Note: Altering these files is not a prerequisite for normal use, and changing parameters in these files should only be attempted by experienced SYMCLI or SYMAPI users. Changes to these files may cause unwanted feature restrictions, or may cause the array environment to stop functioning.

- `/var/symapi/config/README.netcnfg` — This network service file maps a service name to a remote server host node name and/or internet address, and to a port number, to provide a TCP/IP connection for executing remote SYMAPI functions. The NETCNFG file is created by the CRTNETCNFG command. This file can be modified to list more than one service by defining one unique system per line.

Modifying the QShell configuration files

The following files require modification to allow aliases to function in QShell:

- `/var/symapi/config/alias/profile` — This file is a template used to modify the `/etc/profile` file.

  Copy the line from the `/var/symapi/config/alias/profile` file and paste it into the `/etc/profile` file. If `/etc/profile` does not exist then copy the `profile` file from the `/.../alias` directory to `/etc` directory.

  To set this feature at the user level:

  1. Copy the `profile` file from the `/etc` directory to the `/home/<`*user directory*`>`.
  2. Change the `profile` file name in the user directory to `.profile`. (add period in front of "profile").

- `/var/symapi/config/alias/qshrc` — This file holds setup information that is used by QShell. When running QShell, the `/etc/profile` file points to `qshrc` file to set the environmental values defined by this file.

  1. Verify that the `qshrc` file points to the default library EMCCTL or EMCCTLC, by checking that the following two lines are set as shown below:
     - `export PATH=$PATH:/QSYS.LIB/EMCCTL.LIB`
     - `liblist -af EMCCTL`
  2. If the lines in the `qshrc` file, shown above, do not point to the EMCCTL or EMCCTLC library, modify the lines accordingly.

Working with Solutions Enabler daemons

The STORDAEMON command allows for SRDF/TimeFinder Manager to start, stop, and query the Solutions Enabler daemons. Refer to "STORDAEMON" on page 256 for details on using this command.

The daemons that are managed are:

- **Storapid (Base)** — Provides centralized gatekeeper device management for all Solutions Enabler applications. It alleviates contention when there are limited gatekeeper resources available and also eliminates the need for every client to constantly select, open, lock, and ping for an available gatekeeper device for every online function.

  Additionally, the base daemon monitors Symmetrix External Locks (SEL) and Device External Locks (DEL), and releases any lock(s) held by a crashed application. The base daemon also eliminates the need for Solutions Enabler applications to run as root.

- **Storsrvd (Symapi)** — Provides the SYMAPI server, a multi-threaded program that listens for SYMAPI sessions and management requests initiated by the STORDAEMON command. The server also listens for management requests from the system operator console.

- **Storgnsd (GNS)** — Provides a global, distributed repository for DG and CG group definitions.

Configuring daemons using the daemons options file

There is a daemon options file used to control daemon behavior. At startup each daemon reads this file and applies all relevant settings. In most cases the default settings are sufficient. However, there are four options available that are specific to IBM i. The file is located in the IFS directory `/var/symapi/config` and must be named `daemon_options`.

The options specific to IBM i are:

- **ibmi_jobq_name = <*JOBD> | jobq name** — An example jobq name is EMCCTL. If the jobq name is not set, the default *JOBD is used. When *JOBD is used the job description defined in the current user profile is used to determine the job queue name. The second option does not need to be set if the library of named job description is in the library list. If the second option is specified and the first option is set to *JOBD, the second value is ignored.

- **ibmi_jobq_lib_name = <*LIBL> | Library name** — An example library name is EMCCTL. If the Library name is not set, the default value *LIBL is used.

  Note: For both of these options the name has a maximum character length of 10, and upper and lower case characters are allowed. If an option is longer than 10 characters an error message appears.

- **ibmi_allow_subsystem_start = <yes>|<no>** — The default value is yes. If the daemon jobs are not started yet, this value is used to control automatic startup of the subsystem that is being used for the Solutions Enabler daemons. If the subsystem cannot be started, the daemon jobs do not start and an error message is generated.

  Both Base and GNS daemon jobs are started when an SRDF/TimeFinder Manager or Solutions Enabler command is run on the host. By default SRDF/TimeFinder Manager uses the EMCCTL subsystem to run the daemon jobs. The subsystem defined (ibmi_jobq_name) in the daemon options file must be started manually if the daemon_option, **ibmi_allow_subsystem_start**,is set to **no**. If the daemon option, **ibmi_allow_subsystem_start**, is set to **yes** the subsystem and daemon jobs start automatically.

  If you want the daemon jobs run in the QBATCH subsystem, in the daemon options file set the option as follows:

  - **ibmi_jobq_name = qbatch**
  - **ibmi_jobq_lib_name = *libl**

  Note: Make sure that the subsystem and job queue entry is set to allow multiple jobs and multiple threads.

- **redirect_std_fds = enable | disable** – For IBM i, the default value for this option is **enable**. This option ensures that the standard output for the running daemon is not directed to the default spool file. This option does not need to be set in the `daemon_options` file if default is being used.

  When this option is set to **disable**, a spool file is created and all the output for the running daemon job is directed to a spool file. The spool file can become large when running for an extended period of time. Eventually the file reaches its maximum allowable size and an informational message is generated. When the message appears, the daemon job halts and all SRDF/TimeFinder Manager operations are impacted until the user replies to the message.

  Note: The daemon may stop running if this message is not replied to for a long period of time. If this happens, restart the deamon.

For more information on configuring Solutions Enabler daemons refer to the *Dell EMC Solutions Enabler Installation and Configuration Guide*.

Working with the SYMAPI server

The SYMAPI server is required for SPLITWA operations, or to run remote CLI commands on the controlling host. The controlling host can either be another IBM i host or a Windows host which has access to the array. It must also have IP connectivity from the IBM i host where SRDF/TimeFinder Manager is installed.

Configuring the SYMAPI server

When TimeFinder/CG is used in the replication solution, the `netcnfg` file on the client host must be configured. The `netcnfg` file contains the IP address of the SYMAPI server.

Prior to configuring the `netcnfg` file:

1. Confirm IP connectivity to the controlling host.
2. Use the IBM i CFGTCP command, to add an entry to the local host table on the IBM i host where SRDF/TimeFinder Manager is installed. Enter Option 10.

Configure the `netcnfg` file on the client host as follows:

1. On the source IBM i host, run the following command:

   CRTNETCNFG NODENAME (*nodename*) **NETWADDR** (*ip address*)

   You can specify the node name, the IP address, or both items.

   The Create SYMAPISVR netcnfg file (CRTNETCNFG) screen (see Figure 18) appears.

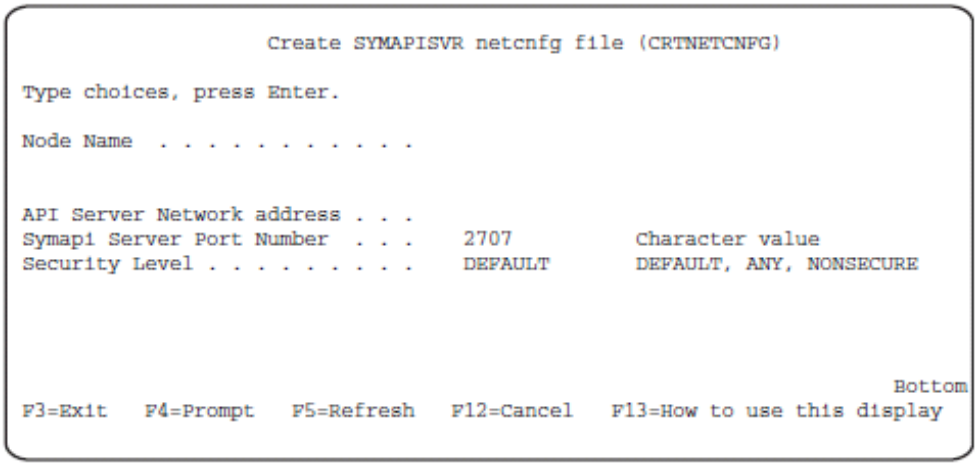


Figure 18 Create SYMAPISVR netconfg file (CRTNETCFG) screen

2. Enter the following information:
   - **API Server Network address —** The network address or node name of the API server that was entered into the local host table in step 2 .

     Note: When using a node name, make sure the node name also appears in the host table for the system or is listed in the DNS tables of the attached network.

- **SYMAPI Port Number** — The TCP/IP port number. The default value is **2707**.
- **Security Level** — The security connection level for the server: DEFAULT, ANY, NONSECURE, or SECURE.

Note: A SECURE connection is not supported for IBM i hosts.

3. Press **Enter**.

   An entry similar to the following is added to the `netcnfg` file:

```
SPLITWA_SERVICE - TCPIP - 10.123.456.789 2707 -
```

Starting the SYMAPI Server

To start the SYMAPI Server:

1. Select option **3. Control Menu** from the EMC SRDF/TimeFinder Manager for IBM i main menu.
2. Select option **7. Control Daemons** from the Control Menu.

   The Control Dell EMC daemon proc (STORDAEMON) screen (Figure 19) appears.

```
                    Control Dell EMC daemon proc  (STORDAEMON)

Type choices, press Enter.

Daemon Action  . . . . . .  _________   START, SHUTDOWN, STARTALL...
Daemon Name  . . . . . .    _________   STORAPID, STORSRVD...
STORSRVD Port number . . . . .  2707    1-65535

                                                                  Bottom
F3=Exit   F4=Prompt   F5=Refresh   F12=Cancel   F13=How to use this display
F24=More keys
```

Figure 19 Control Dell EMC daemon proc (STORDAEMON) screen

3. Enter the following information:
   - **Daemon Action** — START
   - **Daemon Name** — STORSRVD
   - **STORSRVD Port Numer** — Default is 2707.

Note: Refer to "STORDAEMON" on page 256 for more information.

Stopping the SYMAPI Server

To start the SYMAPI Server:

1. Select option **3. Control Menu** from the EMC SRDF/TimeFinder Manager for IBM i main menu.
2. Select option **7. Control Daemons** from the Control Menu.

   The Control Dell EMC daemon proc (STORDAEMON) screen (Figure 19) appears.
3. Enter the following information:
   - **Daemon Action** — SHUTDOWN
   - **Daemon Name** — STORSRVD

   Note: Refer to "STORDAEMON" on page 256 for more information.

Viewing version information

To view the current program version information:

1. On the source IBM i host, display the Dell EMC SRDF/TimeFinder Manager for IBM i main menu (see "Accessing the main menu" on page 42).
2. Select option **4. Version Information**.

   The Program Version Information screen (Figure 20) appears listing current program version information for the Base SRDF/TimeFinder Manager, I/O Driver, and SYMAPI.

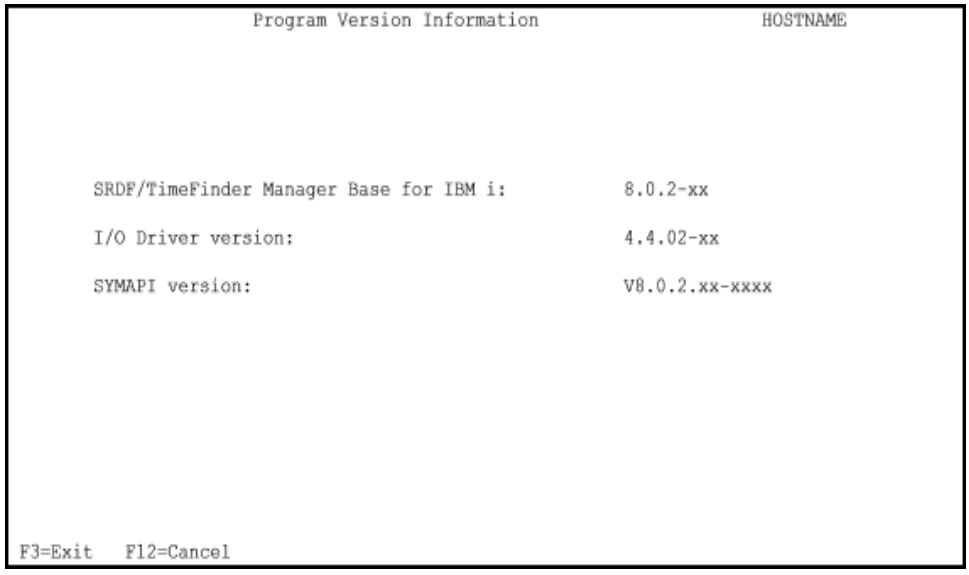


Figure 20 Program Version Information screen

Note: The version number of the operating environment appears on the Disk Units Details screen (Figure 45 on page 103).

CHAPTER 4
Configuring the Environment

This chapter describes the configuration steps that enable SRDF/TimeFinder Manager to control SRDF and TimeFinder replication environments with IBM i hosts.

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- Configuring images 80
- Configuring paths 82
- Working with paths 85
- Configuring device pairs 92
- Configuring device pools 105
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Configuration overview

The initial configuration consists of:

- Creating paths and associating paths with images
- Pairing devices for TimeFinder environments
- Configuring devices and device pools

If TimeFinder/CG is used in the replication solution, configure the `netcnfg` file on the client host as well. Refer to "Working with the SYMAPI server" on page 74 for a complete description of the procedure.

Carry out these configuration steps only once, after a new installation. However, if system storage resources are altered, such as when a new disk is added to an ASP using IBM i System Service Tools, repeat these configuration steps.

Configuring images

The procedure for adding ASPs to an image or removing ASPs from an image is similar. This operation allows ASPs to participate or be excluded from control operations. The removed ASPs display in the NON_CONFIG image on the Symmetrix Configuration by Image screen.

Adding ASPs to an image

To add ASPs to an image:

1. Select option **1. Work with Images** on the Dell EMC SRDF/TimeFinder Manager for IBM i main menu to display the Symmetrix Configuration by Image screen.
2. On the Symmetrix Configuration by Image screen, enter **1** in the **Opt** column for the ASP in the NON_CONFIG image pool to be added to the SYSTEM image (Figure 21 on page 81).

The Configuration by Image screen re-appears (Figure 21) with the ASP removed from the NON_CONFIG image and added to the ASP list of the SYSTEM image.

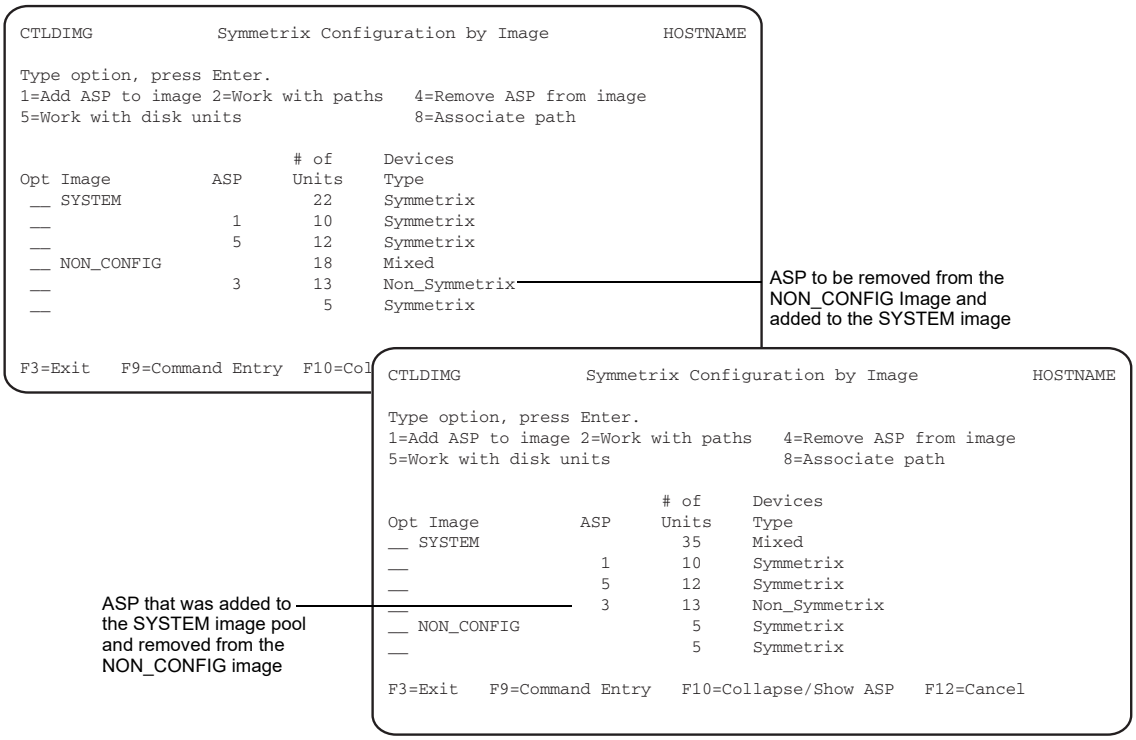


Figure 21 Adding an ASP to an image

Removing ASPs from an image

To remove ASPs from an image:

1. On the Configuration by Image screen, enter **4** in the **Opt** column of the ASP to be removed from the SYSTEM image (Figure 22).

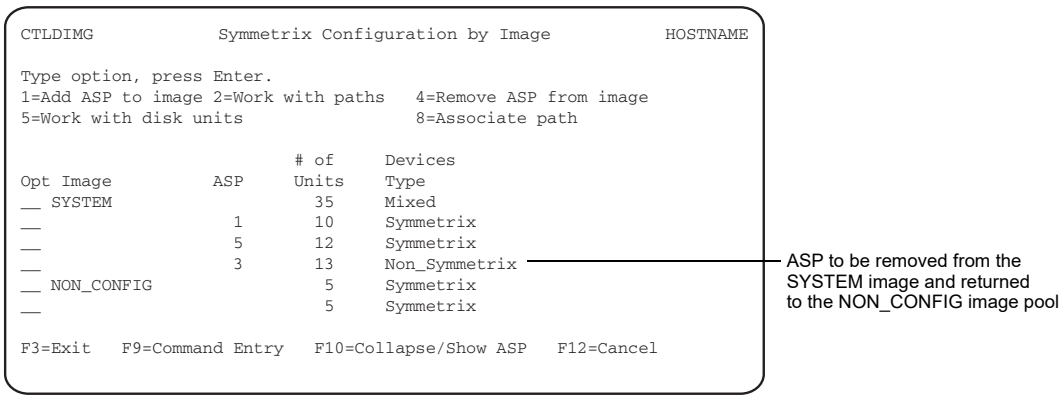


Figure 22 Removing an ASP from the SYSTEM image

The Confirm Remove ASP from the image screen (Figure 23) appears.

```
:.....................................:
:     Confirm Remove ASP from the image   :
:                                         :
:    Removing an ASP from the image will  :
:    erase all pairing information.       :
:    Do not remove an ASP from the image  :
:    until you have cancelled/terminated  :
:    any clone enabled or snap steps.     :
:                                         :
:    Remove an ASP?     Y or N            :
:                                         :
:    F3=Exit    F12=Cancel                :
:.....................................:
```

Figure 23 Confirm Remove ASP from the image screen

2. Enter **Y** to confirm the ASP removal from the SYSTEM image.

   The Configuration by Image screen (Figure 24) appears with the ASP added to the ASP list in the NON_CONFIG image pool.

ASP removed from the SYSTEM image and returned to the NON_CONFIG image

```
CTLDIMG            Symmetrix Configuration by Image            HOSTNAME

Type option, press Enter.
1=Add ASP to image 2=Work with paths   4=Remove ASP from image
5=Work with disk units                 8=Associate path

                                 # of     Devices
Opt Image          ASP           Units    Type
 __ SYSTEM                        22      Symmetrix
 __                  1            10      Symmetrix
 __                  5            12      Symmetrix
 __ NON_CONFIG                    18      Mixed
 __                  3            13      Non_Symmetrix
 __                               5       Symmetrix


F3=Exit   F9=Command Entry  F10=Collapse/Show ASP     F12=Cancel
```

Figure 24 Returning an ASP to the NON _CONFIG pool

When an ASP is removed, the target image for that ASP may not be synchronized with the source standard devices. This feature adds flexibility to SRDF and TimeFinder operations. For example, for system configurations where one or more ASPs containing non-Symmetrix devices are permitted, allowing operations to continue for some ASPs while the ineligible ASPs are removed. Note that the SRDF link status for the image must be in an eligible state.

Configuring paths

Paths can contain up to five steps. Steps in the path are defined during configuration and are identified by their positions (1 through 5) in the path. Step numbers are used with control commands to identify which steps in the path are to be processed (Table 9 on page 83).

- Each image can have multiple paths. With IASP images, only one step of a path can be presented to a target host at a time.
- A path cannot be associated with an image (IBMIASP) that contains non-Symmetrix devices.
- You cannot associate a path on the NON_CONFIG image.

You can display overall status for the path associated with an image and for each step in the path at any time during the configuration process. Refer to the "Work with Associated Paths screen" on page 48 for more information.

There are two types of paths:

- Predefined paths — Refer to Table 9 lists the predefined paths.
- Manually defined paths — Use when data needs to be moved in a manner that is not defined in an existing path. "Working with paths" on page 85 describes how to define a path.

An image that contains devices spread across multiple arrays must have the same configuration steps on all participating arrays.

Once a path has been configured, you can access the path by the image and path name to display full configuration details and separately control each step of the path. Controlling the path ensures that the replicated data gets from source devices to target devices according to the complexity of the replication solution.

Predefined and user-defined paths

Use Table 9 as a guide when associating paths in the configuration process.

Steps in the path are shown in the user interface screens with a forward slash (/) separator between steps, which represents how data gets from the host source devices (STDs) to the target devices.

A step in the path is identified by its position (1 through 5) in the path as shown in this example:

- Using YOURPATH3, the number of steps associated with the full path /BCV1/RDF1/BCV1/RDF1/BCV1 is 5:

```
/BCV1/RDF1/BCV1/RDF1/BCV1
  1    2     3    4     5
```

If a step in any path contains the same source and target devices on the same source and target arrays, the step type is the same in all paths for this combination. Unique combinations of source and target devices require unique step types in all paths.

Table 9 Predefined and user-defined configurations and paths (page 1 of 2)

Configuration type	Path name	Steps in the path
Predefined configurations and paths		
SRDF (STD/R1 → R2)	SRDF	/RDF1
Cascaded SRDF	CASCADRDF	/RDF1/RDF1
Concurrent Cascaded SRDF	CASCADRDF2	/RDF2/RDF1
Concurrent RDF	SRDF2	/RDF2
Local BCV (STD →BCV)	LOCALBCV	/BCV1
Local Clone Emulation (STD → BCV)	LOCALCLN	/CLN1
Local Clone (STD → BCV or STD)	LOCALNCL	/NCL1
Local Snap (STD → VDEV)	LOCALSNAP	/SNP1

Table 9 Predefined and user-defined configurations and paths (page 2 of 2)

Configuration type	Path name	Steps in the path
Local SnapVX (STD → BCV or STD	LOCALSNAPVX	/SVX1
Duplicate Snap (STD → VDEV g VDEV)	DUPSNAP	/SNP1/SNP2
Remote BCV (STD/R1 → R2 → BCV)	REMOTEBCV	/RDF1/BCV1
BCV/R1 (STD → BCV/R1 → R2)	BCVR1	/BCV1/RDF1
BRBCV (STD → BCV/R1 → R2 → BCV)	BRBCV	/BCV1/RDF1/BCV1
Remote Clone Emulation (STD/R1 → R2 → BCV)	REMOTECLN	/RDF1/CLN1
Remote Clone (STD/R1 → R2 → BCV or STD)	REMOTENCL	/RDF1/NCL1
Remote Snap (STD/R1 → R2 → VDEV)	REMOTESNAP	/RDF1/SNP1
Remote SnapVX (STD/R1 → R2 → BCV or STD)	REMOTESNAPVX	/RDF1/SVX1
SRDF/Metro	SRDFMETRO	/RDFM1
Remote Mirror using SRDF/Metro	METROBCV	/RDFM1/BCV1
Remote Clone with emulation using SRDF/Metro	METROCLN	/RDFM1/CLN1
Remote native Clone using SRDF/Metro	METRONCL	/RDFM1/NCL1
Remote SnapVX using SRDF/Metro	METROSVX	/RDFM1/SVX1
User-defined configurations and paths		
Multiple BCVs each with a unique path name and step	MULT_BCV1 MULT_BCV2 MULT_BCV3 MULT_BCV4	/BCV1 /BCV2 /BCV3 /BCV4
Concurrent RDF (RDF1 and RDF2) requires two path names	CONC_SRDF1 CONC_SRDF2	/RDF1 /RDF2
Concurrent BCV (BCV1 and BCV2) with unique steps	CONC_BCV1 CONC_BCV2	/BCV1 /BCV2
Multihop to third array: (STD/R1 → R2 → BCV/R1 → R2) (STD/R1 → R2 → BCV/R1 → R2 → BCV/R1) (STD → BCV/R1 → R2 →BCV/R1 → R2 → BCV)	YOURPATH1 YOURPATH2 YOURPATH3	/RDF1/BCV1/RDF1 /RDF1/BCV1/RDF1/BCV1 /BCV1/RDF1/BCV1/RDF1/BCV1
Predefined configurations and paths		
SRDF (STD/R1 → R2)	SRDF	/RDF1
Cascaded SRDF	CASCADRDF	/RDF1/RDF1

Step Types

Table 10 lists step types and the number of replication copies that SRDF/TimeFinder Manager allows for each step.

Table 10 Supported step types for SRDF and TimeFinder operations

Replication step type	Number of replication copies allowed	Description
RDF*n*	*n* = 1 or 2	Data is replicated by SRDF. Concurrent RDF can have up to two target devices.
BCV*nnnn*	Maximum number of TimeFinder copies allowed is limited by the version the array's operating environment.	Data is replicated by TimeFinder/Mirror.
CLN*nnnn*		Data is replicated by TimeFinder/Clone emulation.
NCL*nnnn*		Data is replicated by TimeFinder/Clone.
SNP*nnnn*		Data is replicated by TimeFinder/Snap.
SVX*nnnn*		Data is replicated by TimeFinder/SnapVX.

Working with paths

The following sections show how to work with paths.

Adding/creating paths

A path must be created when data needs to be moved in a way that is not already defined.

If a step in any path contains the same source and target devices on the same source and target array, the step type should be the same for all paths. A unique combination of the source and target devices must have a unique step type in all the paths.

To add a path, do one of the following:

Option 1 - add path

On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **2. Work with Paths** to display the Work with Available Paths screen (Figure 25).

```
CTLDPATH                 Work with Available Paths                  HOSTNAME

 Type option, press Enter.
  4=Delete path
 Opt       Path Name         Path
           BCVR1             /BCV1/RDF1
           BRBCV             /BCV1/RDF1/BCV1
           CASCADRDF         /RDF1/RDF1
           CASCADRDF2        /RDF2/RDF1
           DUPSNAP           /SNP1/SNP2
           LOCALBCV          /BCV1
           LOCALCLN          /CLN1
           LOCALNCL          /NCL1
           LOCALSNAP         /SNP1
           LOCALSVX          /SVX1
           REMOTEBCV         /RDF1/BCV1
           REMOTECLN         /RDF1/CLN1
           REMOTENCL         /RDF1/NCL1
           REMOTESNAP        /RDF1/SNP1
           REMOTESVX         /RDF1/SVX1
           SRDF              /RDF1
                                                                    More...
 F3=Exit   F6=Add Path   F12=Cancel
```

Figure 25 Work with Available Paths screen

Option 2 - add path

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **1. Work with Images** to display the Configuration by Image screen (Figure 6 on page 47).
2. Select option **2. Work with paths** on the Configuration by Image screen to display the Work with Associated Paths screen (Figure 26).

```
CTLPATH                Work with Associated Paths                    HOSTNAME
 IMAGE= SYSTEM
 Type option, press Enter.
  4=Remove path  5=Work with disk units  6=Pair   7=Unpair
 11=IASP view   14=Establish/Create     15=Split/Activate    16=Configure IASP
 17=Add IOA     18=Remove IOA           19=Cancel/Terminate  20=Link/Relink
 21=Unlink      22=Work with Snapshots  23=Change Link Mode  24=Restore
                                         Invalids/ Path/
Opt Path Name  Step    Status            % Copied  Snapshot                  Gen
    CASCSVX            Configured                  /SVX1/SVX1/SVX1/SVX1
                SVX1   Copied                100%  first_step                0
                SVX1   Restored              100%  second_step               0
                SVX1   NoSnapshots             0%
                SVX1   NotPaired              N/A

    LSNAPVX1           Configured                  /SVX1
                SVX1   CopyInProg             55%  End_of_Quarter             1

                                                                          Bottom
 F3=Exit   F5=Refresh   F6=Add Path   F9=Command Entry   F12=Cancel
```

Figure 26 Work with Associated Paths screen — add path

Option 3 - add path

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, (Figure 4 on page 43) select option **1. Work with Images**.

   The Configuration by Image screen (Figure 6 on page 47) appears.

2. Enter **8** (Associate path) in the **Opt** column of the image you want to associate a path with and press **Enter**.

   The Paths not Associated with the Image screen (Figure 27) appears.

```
SELPATH                Paths not Associated with the Image          HOSTNAME

 IMAGE= SYSTEM

 Type option, press Enter.
 1=Select path

 Opt  Path Name   Path
      BCVR1       /BCV1/RDF1
      BRBCV       /BCV1/RDF1/BCV1
      CASCADRDF   /RDF1/RDF1
      CASCADRDF2  /RDF2/RDF1
      DUPSNAP     /SNP1/SNP2
      LOCALBCV    /BCV1
      LOCALCLN    /CLN1
      REMOTEBCV   /RDF1/BCV1
      REMOTECLN   /RDF1/CLN1
      REMOTENCL   /RDF1/NCL1
      REMOTESNAP  /RDF1/SNP1
                                                                    More...

 F3=Exit    F6=Add Path    F12=Cancel
```

Figure 27 Paths not Associated with the Image screen

3. From any of these screens press **F6** (Add path).

The Paths defined on the System screen appears (Figure 28), which lists all the defined paths.

```
ADDPATH                    Paths Defined on the System              HOSTNAME

      Path Name   Path
      BCVR1       /BCV1/RDF1
      BRBCV       /BCV1/RDF1/BCV1
      CASCADRDF   /RDF1/RDF1
      CASCADRDF2  /RDF2/RDF1
      DUPSNAP     /SNP1/SNP2
      LOCALBCV    /BCV1
      LOCALCLN    /CLN1
      LOCALNCL    /NCL1
      LOCALSNAP   /SNP1
      LOCALSVX    /SVX1
      REMOTEBCV   /RDF1/BCV1
      REMOTECLN   /RDF1/CLN1
      REMOTENCL   /RDF1/NCL1
      REMOTESNAP  /RDF1/SNP1
      REMOTESVX   /RDF1/SVX1
      SRDF        /RDF1
                                                                   More...
 Enter Path Name:
 Enter Path:
 Enter=Saves Changes    F3=Exit    F12=Cancel
```

Figure 28 Paths Defined on the System screen

4. Provide the following information:
    a. **Enter Path Name** — Type the name associated with one or more steps to be used in the replication schema.
    b. **Enter Path** — Type the path to use to replicate data from a source device to a target device. Refer to "Working with paths" on page 85 for more information about paths.
5. Press **Enter** to save the changes.

Note: When you press **F6** (Add path) from the Work with Associated Paths screen, the new path is immediately associated with the image.

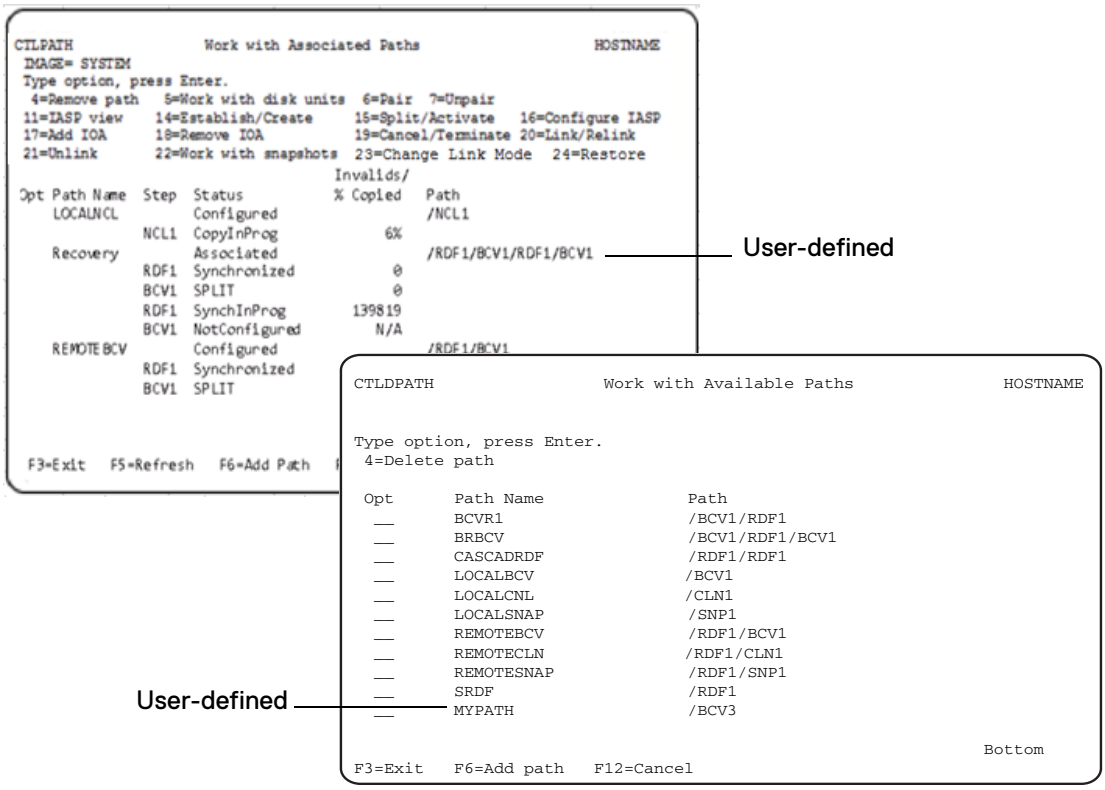


Figure 29 New paths added to screens

If additional paths are required, repeat step 1 through step 3.

Removing (disassociating) a path

To remove (disassociate) a path:

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **1. Work with Images** to display the Configuration by Image screen (Figure 6 on page 47).
2. On the Configuration by Image screen, select option **2. Work with paths** to display the Work with Associated Paths screen (Figure 25 on page 85).
3. On the Work with Associated Paths screen, enter **4** in the **Opt** column of the path to be removed from the image file.

   Depending on how the path is configured, the result is one of the following:

   - If the first step in a path is not configured, the path is disassociated from the image without warning.
   - If a path has configured steps, the Disassociating a Path screen (Figure 30 on page 90) prompts to disassociate the selected path or cancel the operation.

     If a path has a step in which a Snap or a Clone is not terminated or a clone-enabled BCV is not canceled, the path cannot be disassociated.

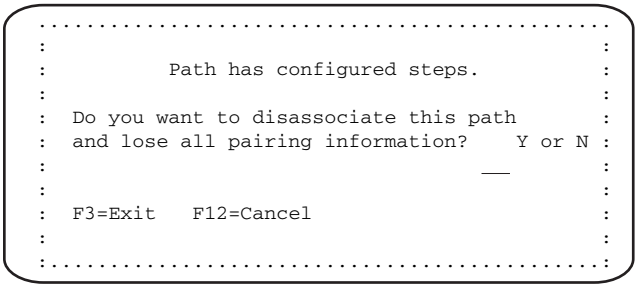


Figure 30 Disassociating a Path screen

4. Type **Y** and press **Enter**.

   The Work with Associated Paths screen appears stating that the path name selected has been disassociated from the image.

SRDF/TimeFinder Manager does not delete disassociated paths. These paths remain available on the Paths not Associated with an Image screen for another association with the same image or a different image.

To permanently delete a disassociated path, follow the steps in "Deleting a path."

Deleting a path

Delete a path as follows:

1. On the Work with Available Paths screen (Figure 25 on page 85), enter **4** in the **Opt** column of the path name to be removed.

   Depending on the status of the path, the result is one of the following:

   - If the path is associated with any image, the Deleting a Path Associated with an Image screen appears a prompt to delete the path and all its pairing information or cancel (Figure 31).

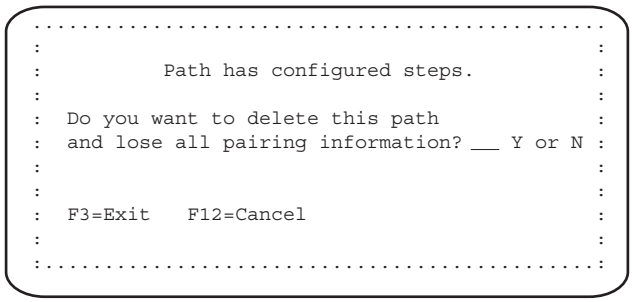


Figure 31 Deleting a Path Associated with an Image screen

- If the path is not associated with any image, Figure 32 appears a message confirming path deletion.

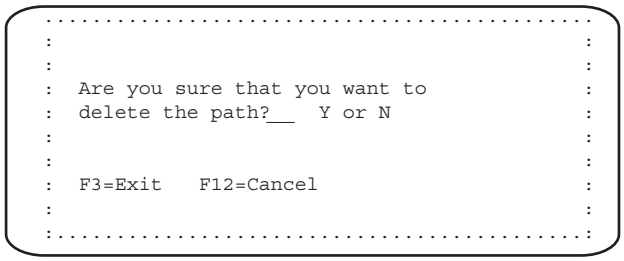


Figure 32 Deleting a Path Not Associated with an Image screen

2. Type **Y** and press **Enter**.

   The Work with Available Paths screen (Figure 25 on page 85) appears with the selected path deleted.

Associating paths with an image

The following steps describe how to associate paths with an image:

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, (Figure 4 on page 43) select option **1. Work with Images**.

   The Configuration by Image screen (Figure 6 on page 47) appears.

2. Enter **8** in the **Opt** column of the image you want to associate a path with and press **Enter**.

   The Paths not Associated with the Image screen (Figure 27 on page 87) appears. Enter option **1** in the **Opt** column of one or more names of the paths you want associated with the image, and press **Enter**.

   SRDF/TimeFinder Manager associates all paths you selected and displays the Configuration by Image screen (Figure 6 on page 47).

3. On the Configuration by Image screen, enter **2** in the **Opt** column of the image.

   The Work with Associated Paths screen (Figure 33 on page 92) appears, which contains all paths that have been associated with the image. In this example, the image is SYSTEM. Note that all ASPs in the image have the same paths available.

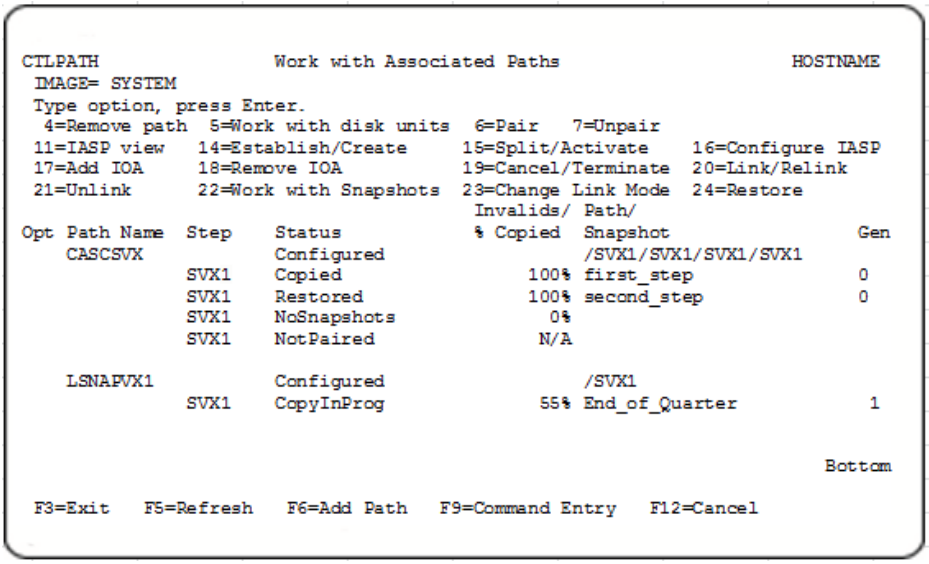


Figure 33 Work with Associated Paths screen — associating paths with image

Configuring device pairs

After an image is selected, and a path is associated with the image, create pairs of all BCV, Clone, or Snap devices used in the TimeFinder replication solution.

Pairing overview and rules for replication pairs

The BCV device pairing algorithm attempts to prevent the selection of a BCV device that is on the same physical drive as one of the mirrors of the standard device. If a BCV has two mirrors, only the first mirror is checked.

To search for the optimal BCV pair, SRDF/TimeFinder Manager uses the following selection criteria by priority:

1. If a BCV exists, that was the last BCV paired and established with the current standard device, SRDF/TimeFinder Manager selects this BCV device first if the BCV is not on the same physical drive. If not, the search continues. The best BCV pair to use is a device with the same emulation or a compatible emulation as the standard device that is not on the same physical disk as any of its mirrors.
2. If SRDF/TimeFinder Manager cannot find a compatible BCV device that is not on the same disk as one of the mirrors, it performs the pairing sends an informational message "CTL0083" to the job log. This message states the standard and BCV devices numbers.
3. If SRDF/TimeFinder Manager finds an optimal BCV device for a standard or R2 device, but this standard or R2 device was previously established with a different BCV device, it sends an informational message "CTL0009" or "CTL0083" to the job log .The message states the standard device is now paired with the BCV device but was previously paired with a different BCV device.

Dynamic SRDF pairing is available from SRDF/TimeFinder Manager as "Dynamic SRDF pairs" on page 176 describes.

Note: Except where noted, wherever BCVs are referenced in the pairing discussion, Snap devices can be used interchangeably.

Table 11 lists the configuration (pairing) action for each supported replication step type.

Table 11 Configuration actions for replication types

Replication step type	Description	Configuration (pairing) action
RDF1	Single SRDF (SRDF/S or SRDF/A) available for the source devices for this step.	If pairing R1-R2 devices is predefined - no user action is required. See "RDF group/device compatibility" on page 166 for more information on RDF group/device pairing.
RDF1, RDF2	Concurrent RDF	If SRDF/TimeFinder Manager performs automatic RDF group selection[a], no user action is required. See "RDF group/device compatibility" on page 166 for more information on RDF group/device pairing.
BCV*nnnn*[b]	TimeFinder/Mirror	Requires device pairing: automatic or manual.
CLN*nnnn*[b]	TimeFinder/Clone	Requires device pairing: automatic or manual.
NCL*nnnn*[b]	TimeFinder/Clone	Requires device pairing: automatic or manual.
SNP*nnnn*[b]	TimeFinder/Snap	Virtual devices are considered target devices. Requires device pairing: automatic or manual.
SVX*nnnn*[b]	TimeFinder/SnapVX	Copies are created directly in the Storage Resource Pool of the source device, no target device needed except for Link, Relink and Unlink actions. Pairing for the SVXn type of step is done only from a device file. Manual pairing at the device level is allowed.

a. When a path is associated with the image in a Concurrent RDF configuration, SRDF/TimeFinder Manager picks the first group in the list for all devices that belong to the image and the same RDF group. For example, if R1 devices are in RDF group 0B and 0E, /RDF1 will always pick RDF group 0B and /RDF2 will always pick 0E.

b. The maximum number of TimeFinder copies allowed is limited by the version of PowerMaxOS, HYPERMAX OS, or Enginuity.

BCV device pairing

BCV device pairing can be performed as follows:

- Automatically — When you are pairing by image, SRDF/TimeFinder Manager automatically makes selections as described in "Pairing overview and rules for replication pairs" on page 92. For example, when you perform BCV pairing, the Pair option attempts to find a matching available BCV device on the array and pairs the standard or R2 device with a BCV device for the specified image.
- Using a device file with source and target device pairs - A text file containing pairs of source and target devices. Refer to "Creating a device file" on page 100 for more information.
- Manually — You can individually select the BCV device from the All STD/BCV/SNAP Devices of Matching screen to pair with a standard device.

Using meta devices

When using meta devices, the BCV selection criteria above are applied at the meta device member level. SRDF/TimeFinder Manager does a single pass through all the devices in the image.

Note: If you attempted to unpair all the devices and pair them again, it is likely that SRDF/TimeFinder Manager creates the same device pairing. If this pairing is not sufficient, try manually swapping around several devices, which may create different pairings.

Note: For arrays running PowerMaxOS 5978 or HYPERMAX OS 5977, there is no meta device support. Creating RDF pairs between an array running Enginuity 5876 and an array running PowerMaxOS 5978 or HYPERMAX OS 5977, where the devices on the 5876 array are meta devices, is allowed provided the appropriate 5876 fixes are applied to the array.

TimeFinder/Clone device pairing

The source and target devices in a TimeFinder/Clone device pair can be either standard devices or BCV devices as long as they are all of the same size and type. Clone copies of striped or concatenated meta devices can also be created, but the source and target meta devices must have completely identical stripe count, stripe size, and capacity. Once it is activated, the copy can be instantly accessed by a target's host, even before the data is fully copied to the target device.

In environments running Enginuity 5876 or higher, the target device of a clone session can be used as the source for one or more clone sessions. This allows a clone operation to take place with a device that is already involved in a clone operation without ending the first clone session. This applies to clone target devices of both TimeFinder/Clone sessions and TimeFinder/Clone sessions.

For target devices that are BCV devices, automatic pairing is performed. For standard target devices, either the name of the file containing these devices must be provided or manual pairing must be performed.

For TimeFinder/Clone emulation sessions, source devices can only be standard devices and target devices can only be BCV devices.

Note: Do not mix standard and BCV devices for TimeFinder/Clone (NCL*n*) step device pairing, if planning to use TimeFinder Consistency Group ECA (SPLITWA) splits. Use either standard or BCV devices for the entire group of devices.

TimeFinder/Snap device pairing

A TimeFinder/Snap device pair consists of a standard, R1, R2, or a BCV device and a virtual device.

Duplicate snaps

A duplicate snap is a duplicate of a point-in-time copy of a virtual device that has been activated in a previous snap session. This second point-in-time copy session resides with the source device of the original snap session and is charged as part of the maximum number of sessions for that source device.

The duplicate snap is an actual copy of the virtual device to another virtual device.

In order to support duplicate snaps, SRDF/TimeFinder Manager supports paths such as /SNP1/SNP2.

A duplicate snap requires creation and activation, described as follows:

- Creation of a duplicate session.

  Following the creation of the duplicate snap, a virtual device to virtual device session is in the Created state. Although this is the same point-in-time as the original virtual device, the time stamp is not inherited from the original virtual device's session. The newly created session has its own time stamp indicating the actual time of creation.

  When the duplicate session is in the Created state, the original session between the standard device and the virtual device cannot be terminated until the duplicated session is activated. Also, duplicated sessions in the Created state are limited to a maximum of two sessions for any standard device.

- Activation of the duplicate session.

  Following the activation of the duplicate snap, the session reverts to a normal snap session between the source device of the original session and the duplicate virtual device, which may be recreated or terminated the same as any other snap session. Similar to the create operation; the activated session has a time stamp that reflects the time of the latest operation. Once activated, the normal snap rules for maximum number of sessions associated with a standard device apply.

  Following the activation of the duplicate snap, a new path with a step such as/SNP2 can be associated. This picks up existing pairing (source device and the virtual device that was used as duplicate snap). Operations can be performed on either the second step of the path /SNP1/SNP2 or first step of the /SNP2 path.

After terminating the existing copy session, virtual devices can again be used for duplicate snap (pairing for virtual device — virtual device is restored). The Disk Units screen and the Disk Units Details screen show the current status of the copy sessions.

SnapVX device pairing

Copies are created directly in the Storage Resource Pool of the source device, so no target device needed except for Link, Relink and Unlink actions. Pairing for this step type can only be done from a device file. See “Creating a device file” on page 100.

Pairing devices

Devices can be paired using the following methods:

- Automatic — SRDF/TimeFinder Manager attempts pairing all devices in the image.
- Manual — each target is individually selected and paired.
- Using a device file with source and target device pairs - A text file containing pairs of source and target devices. See “Creating a device file” on page 100 for more information.

Note: For SRDF pairing operations, static pairing occurs during initial array configuration, and dynamic pairing occurs while the array is in operation. For arrays running PowerMaxOS 5978 or HYPERMAX OS 5977, only dynamic SRDF groups are supported. To support dynamic SRDF groups, the “CTLRDFGRP” (Control Dynamic RDF Group) command is available.

Pairing Devices automatically

To set up device pairing:

1. On the Work with Associated Paths screen, enter **6** i9n the **Opt** column of the step of the associated path for the image. The TimeFinder Pairing Options screen appears (Figure 34).

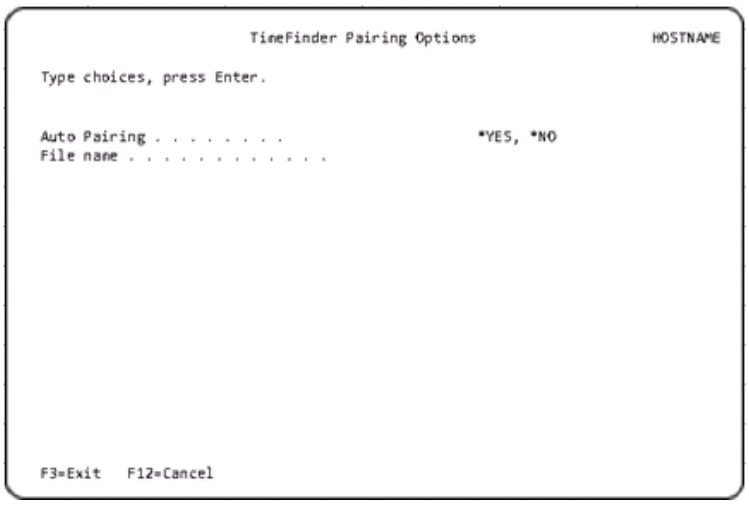


Figure 34 TimeFinder Pairing Options screen

2. Enter Pairing Option as follows:
   - YES — Automatic pairing is performed. For NCL*n* (TimeFinder/Clone) type of the step, target devices are chosen from available BCV devices.
   - NO — Automatic pairing is not selected. The name of the device file containing pairs of source and target devices must be provided.

   Note: See "Creating a device file" on page 100 before entering a device file name.

For SnapVX, the Device File Used for SnapVX Pairing screen appears.

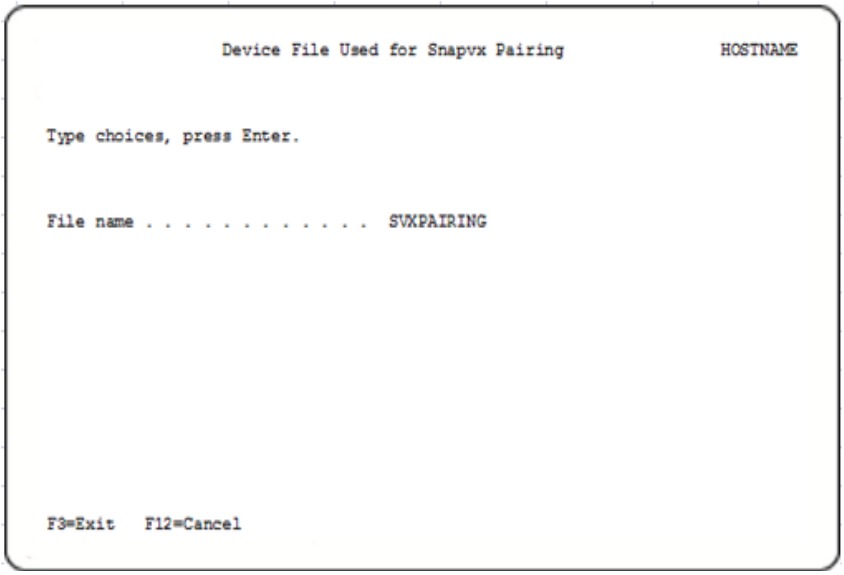


Figure 35 Device File used for SnapVX

For TimeFinder/Mirror and TimeFinder/Clone when automatic pairing is selected and standard, RAID 5, and RAID 6 BCVs exist in the configuration, the Type of the BCV devices screen appears (Figure 36).

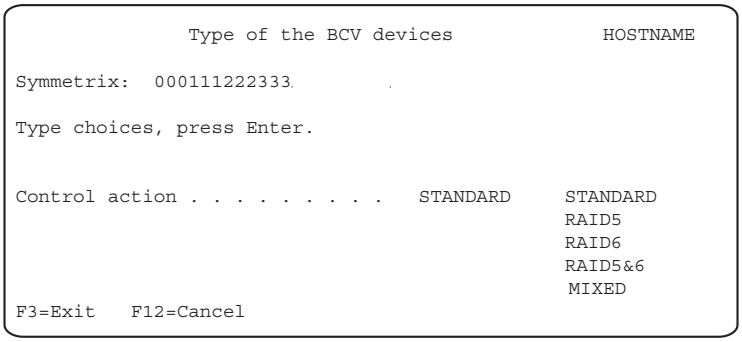


Figure 36 Type of the BCV devices screen

The Work with Associated Paths screen appears with the Status field updated (Figure 33 on page 92).

Pairing devices manually for BCV, Clone, Snap or SnapVX devices

Refer to "Miscellaneous Device Information" on page 293 for a list of compatible devices. Perform device pairing as follows:

1. On the Disk Units screen enter **6** in the **Opt** column of the device (Figure 37 on page 98).

Note: For the SnapVX type of step the Disk Unit screen contains different header information. (Figure 42 on page 102). Table 7 on page 55 defines the fields on the Disk Unit screen for SnapVX.

```
CTLUNITS                          Disk Units                              HOSTNAME

 Image SYSTEM      ASP *ALL  Path LOCALNCL    Step number   1   Step type NCL1
 RDF mode N/A                Consistency protection N/A
 Clone Emulation N/A         BCV Configuration N/A      Target Type BCV
 Snap Pool Name N/A          Clone Type Regular
 5=Show details   6=Pair   7=Unpair   8=Cancel
    Unit     Source Target Source       Target                          Invalids/
OPT #    ASP Dev.   Dev.   Symmetrix    Symmetrix    Status             % Copied
       1   1 008A   0046   000111222333 000111222333 PreCopy                 99%
       8   1 0062   0092   000111222333 000111222333 PreCopy                 99%
       8   1 0066   0096   000111222333 000111222333 PreCopy                 99%
      10   1 006A   009A   000111222333 000111222333 PreCopy                100%
      13   1 008E   004A   000111222333 000111222333 PreCopy                100%
 6    14   1 006E          000111222333              N/A                     N/A
      15   1 0076          000111222333              N/A                     N/A
      17   3 007A   00A6   000111222333 000111222333 PreCopy                100%
      18   3 0072   00AA   000111222333 000111222333 PreCopy                100%
      17   3 007E   00AE   000111222333 000111222333 PreCopy                100%
      18   3 0086   00B2   000111222333 000111222333 PreCopy                100%
                                                                          More…

 F3=Exit F5=Refresh F9=Command Entry F10=Mapping View F11=IBM i View F12=Cancel
```

Figure 37 Disk Units screen with Pair option for device not paired

For a step type of Clone and SnapVX, the Type of the target device screen appears (Figure 38).

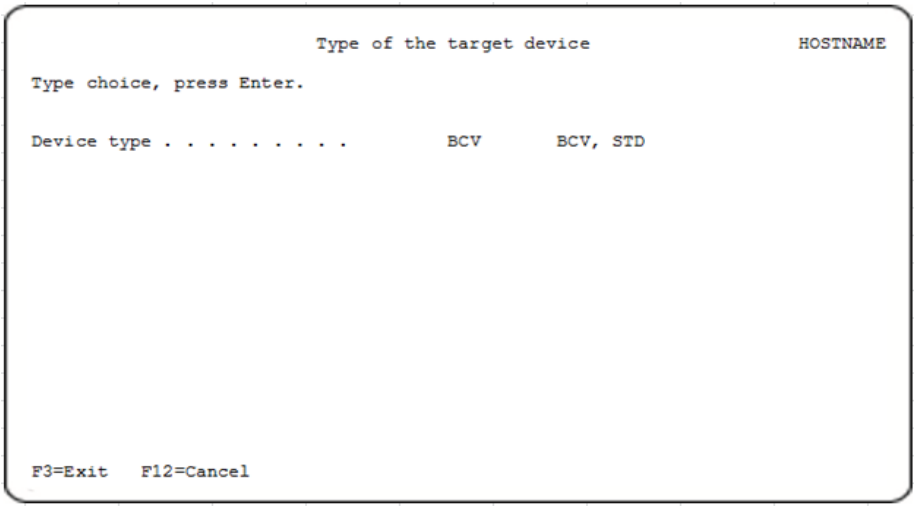


Figure 38 Type of the target device screen

2. Choose the target device type (Standard or BCV).
3. The All STD/BCV/SNAP Devices of matching screen appears (Figure 39).

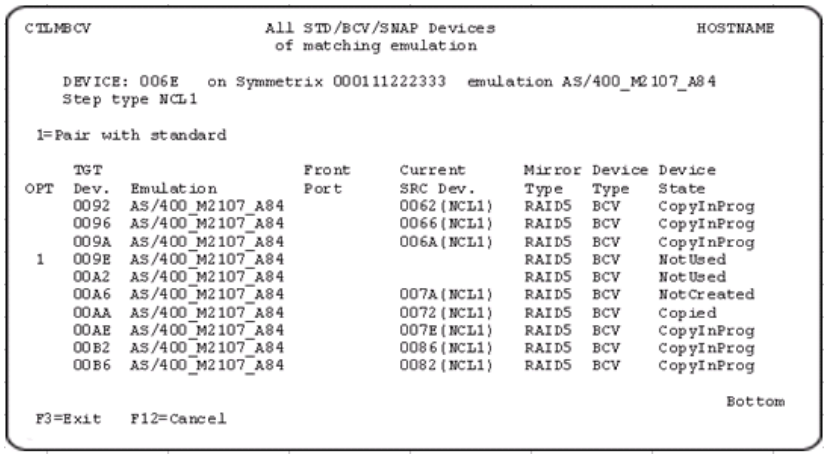


Figure 39 All STD/BCV/SNAP devices of matching screen

The All STD/BCV/SNAP devices of matching screen displays a list of the STD/BCV/SNAP devices that are of the same or compatible type as the source type.

4. Use Option **1**=Pair with Standard for pairing devices.

   The Disk Units screen appears with the device paired (Figure 40).

```
CTLUNITS                        Disk Units                              HOSTNAME

 Image SYSTEM      ASP *ALL  Path LOCALNCL    Step number   1   Step type NCL1
 RDF mode N/A                Consistency protection N/A
 Clone Emulation N/A         BCV Configuration N/A      Target Type BCV
 Snap Pool Name N/A          Clone Type Regular
 5=Show details   6=Pair   7=Unpair   8=Cancel
    Unit      Source Target Source       Target                          Invalids/
OPT #    ASP Dev.   Dev.   Symmetrix    Symmetrix    Status             % Copied
      1   1 008A   0046   000111222333 000111222333 PreCopy                 100%
      8   1 0062   0092   000111222333 000111222333 PreCopy                 100%
      9   1 0066   0096   000111222333 000111222333 PreCopy                 100%
     10   1 006A   009A   000111222333 000111222333 PreCopy                 100%
     13   1 008E   004A   000111222333 000111222333 PreCopy                  99%
     14   1 006E   009E   000111222333 000111222333 NotCreated               N/A
     15   1 0076          000111222333              N/A                     N/A
     17   3 007A   00A6   000111222333 000111222333 PreCopy                 100%
     18   3 0072   00AA   000111222333 000111222333 PreCopy                 100%
     19   3 007E   00AE   000111222333 000111222333 PreCopy                 100%
     20   3 0086   00B2   000111222333 000111222333 PreCopy                 100%
                                                                          More...

 F3=Exit F5=Refresh F9=Command Entry F10=Mapping View F11=IBM i View F12=Cancel
```

Figure 40 Disk Units screen with device paired

Creating a device file

The device file contains pairs of source and target devices. It is a record file and the maximum record length is 40.

Device file rules are as follows:

- Contains three columns, column 1 lists the array serial number, column 2 lists the source device name, and column 3 lists the target device name.
- Each array serial number and corresponding device pair must be listed on a separate line in the device file with the data delimited by a space.
- Comment lines are allowed and they must begin with the pound sign (#).
- Device file resides in the library list.

The device file can be created using the following command:

```
CRTPF FILE(QGPL/file_name) RCDLEN(40)
```

Example device file

```
# PAIRING FOR CLONE STEP
000111222333 0179 0310
000111222333 0195 0334
000111222333 0181 0314
```

The format of the device file is the same for all TimeFinder technologies, including SnapVX. The only difference is the commented header description.

Note: The data in the device file can be amended using the IBM upddta commands. For more details on this command refer to the IBM web site.

Viewing devices

This section shows how to view device information and device details.

Viewing device status and information

1. To view status and specific device information for a step, do one of the following:
   - On the Work with Associated Paths screen, enter option **5** (Work with disk units) in the **OPT** field, next to the step, and press **Enter**.
   - On the Symmetrix Configuration by Image screen, enter option **5** (Work with disk units) on the image level or ASP level.

     The Disk Units screen (Figure 41 on page 101) displays all disks for this step of the associated path. Information for non-Symmetrix physical drives displays blank.

```
CTLUNITS                          Disk Units                                HOSTNAME

 Image SYSTEM       ASP *ALL   Path SRDFMETRO     Step number  1       Step type RDFM1
 RDF mode Active               Consistency protection N/A
 Clone Emulation N/A           BCV Configuration N/A      Target Type N/A
 Snap Pool Name N/A            Clone Type N/A
 5=Show details    6=Pair    7=Unpair   8=Cancel
    Unit       Source  Target  Source       Target                Invalids/   Bias
Opt #    ASP   Dev.    Dev.    Symmetrix    Symmetrix    Status   % Copied
       1   1   0012E   00112   000196701855 000196701854 Suspended       0      T
       2   1   0012F   00113   000196701855 000196701854 Suspended       0      T
       3   1   00130   00114   000196701855 000196701854 Suspended       0      T
       4   1   00131   00115   000196701855 000196701854 Suspended       0      T
                                                                              Bottom
F3=Exit  F5=Refresh  F9=Command Entry  F10=Mapping View  F11=IBM i View  F12=Cancel
```

Figure 41 Disk Units screen — Clone type of step

Table 6 on page 51 describes the fields that display on the Disk Units screen.

For the SnapVX type of step the Disk Unit screen contains different header information (Figure 42). Table 7 on page 55 defines the fields on the Disk Units screen for SnapVX.

```
CTLUNITSVX                          Disk Units                          HOSTNAME

 Image SYSTEM      ASP *ALL  Path LSNAPVX1    Step number   1 Step type SVX1
 Snapshot Name  End_of_Quarter                   Generation 1

 5=Show details   6=Pair   7=Unpair
    Unit      Source Target Source       Target                          Invalids/
OPT #    ASP Dev.   Dev.   Symmetrix    Symmetrix    Status              % Copied
       1   1 00118  00136  000111222333 000111222333 CopyInProg              45%
       2   1 0011C  0013A  000111222333 000111222333 Copied                 100%
       3   1 0011D  0013B  000111222333 000111222333 Copied                 100%
       4   1 0011E  0013C  000111222333 000111222333 CopyInProg              30%
       5   1 0011F  0013D  000111222333 000111222333 CopyInProg              40%
       6  32 00124  0013E  000111222333 000111222333 CopyInProg              45%

                                                                          Bottom
 F3=Exit F5=Refresh F9=Command Entry F10=Mapping View F11=IBM i View F12=Cancel
```

Figure 42 Disk Units screen — SnapVX type of step

Viewing mapping information

To view mapping information (Figure 43) for all devices in the step (source and target), press **F10**=Mapping View.

```
CTLUNITMAP                  Mapping View                            HOSTNAME

 Image SYSTEM      ASP *ALL  Path LOCALNCL    Step number 1     Step type NCL1

 UNIT Source Director/                    Target Director/
 #    Dev.   Address                      Dev.   Address
    1 008A   FA-7F1/30     FA-8C1/30      0046
    8 0062   FA-7F1/20                    0092
    9 0066   FA-7F1/21                    0096
   10 006A   FA-7F1/22                    009A
   13 008E   FA-7F1/31                    004A
   14 006E   FA-7F1/32                    009E
   15 0076   FA-7F1/23                    00A2
   17 007A   FA-7F1/24                    00A6
   18 0072   FA-7F1/33     FA-8C1/33      00AA
   19 007E   FA-7F1/25     FA-8C1/25      00AE
   20 0086   FA-7F1/27                    00B2
   23 0082   FA-7F1/26                    00B6

                                                                       Bottom
 F3=Exit   F9=Command Entry   F10=Symmetrix View   F12=Cancel
```

Figure 43 Mapping View screen

Viewing Disk Units in IBM i view

To display the Disk Units screen (Figure 44) showing source devices in the IBM i view, press **F11**.

```
CTLUNITO                    Disk Units IBM i View                      HOSTNAME

 Image SYSTEM      ASP *ALL  Path N/A         Step number N/A   Step type N/A

 Unit       Serial      Resource    Disk  Disk   I/O      I/O
 #     ASP  Number      Name        Type  Model  Adapter  Bus  Ctl  Dev
    1    1  50-8A333    DD038       2107  0A85       2      0    1   10
    8    1  50-62333    DD011       2107  0A84       2      0    1    2
    9    1  50-66333    DD010       2107  0A84       2      0    1    3
   10    1  50-6A333    DD009       2107  0A84       2      0    1    4
   13    1  50-8E333    DD034       2107  0A85       2      0    1   11
   14    1  50-6E333    DD036       2107  0A84       2      0    1   12
   15    1  50-76333    DD037       2107  0A84       2      0    1    5
   17    3  50-7A333    DD013       2107  0A84       2      0    1    6
   18    3  50-72333    DD014       2107  0A84       2      0    1   13
   19    3  50-7E333    DD012       2107  0A84       2      0    1    7
   20    3  50-86333    DD032       2107  0A84       2      0    1    9
   23    3  50-82333    DD033       2107  0A84       2      0    1    8



                                                                      Bottom
 F3=Exit    F9=Command Entry    F11=Symmetrix View    F12=Cancel
```

Figure 44 Disk Units screen — IBM i View

Press **F11** to return to the Disk Units screen (Figure 41 on page 101) in the view.

Displaying device details

To display details for a device, from the Disk Units screen, enter option **5** (Show details) next to the device.

The Disk Units Details screen (Figure 45) appears. This screen displays details about the selected device for and no-devices.

```
CTLUNITD                    Disk Units Details                        HOSTNAME

 Unit 1     ASP 1     Serial number 50-03AC333

 Source device:   Symmetrix 000111222333  microcode 5875.230  unit 03AC

 Config RDF1+R-6(6+2)       RDF type RDFA1  RDF group 31    Emulation 2107_A84
 Director/Address     FA-7F0/02       FA-10F1/01

 Meta type Striped          Stripe size 1920       Bound Pool N/A
 Meta members  03AC  03AD  03AE  03AF


 Target device:   Symmetrix 000111222333  microcode 5875.230  unit 04BD
 Path LOCALSNAP   Step number 1     Step type SNP1(Save Pool Name: SNAP1)

 Config VDEV                RDF type N/A    RDF group N/A   Emulation 2107_A84
 Director/Address     N/A

 Meta type Striped          Stripe size 1920       Bound Pool N/A
 Meta members  04BD  04BE  04BF  04C0

 F3=Exit      F12=Cancel
```

Figure 45 Disk Units Details screen for step type SNAP

For SNAP type of the step, the Step Type field is expanded to display the save pool name, as shown in Figure 45.

For thin devices the thin pool that the device is bound to is displayed, as shown in Figure 46.

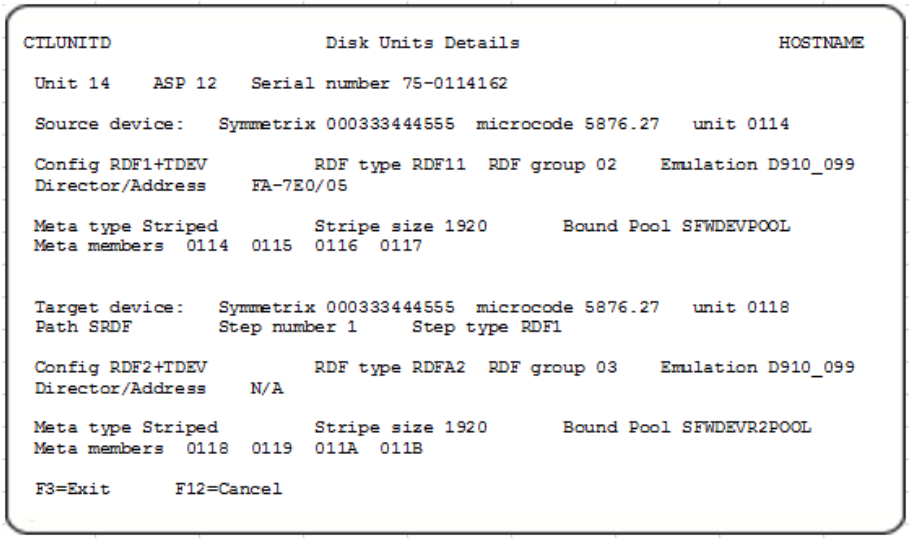


Figure 46 Disk Units Details screen for thin device

Configuring device pools

Note: For arrays running PowerMaxOS 5978 or HYPERMAX OS 5977, Snap and DSE pools are not supported, and thin pools and DATA devices are pre-configured, so pool management is not supported. Therefore, for PowerMaxOS 5978 and HYPERMAX OS 5977 arrays, creating and deleting thin pools, and management of existing pools and DATA devices (except for pool renaming and freeing devices) are not supported.

For arrays running Enginuity version 5876 and earlier, the following device pools are created and managed by SRDF/TimeFinder Manager:

- SAVE pools contain SAVE devices

  The two types of SAVE pools are:

  - SNAP pools — Used for TimeFinder/Snap operations.
  - DSE pools — Used for RDF/A DSE operations.
- Thin pools contain DATA devices. Thin pools are used for Virtual Provisioning.

Working with SAVE pools

SAVE pools are groups of SAVE devices used to store pre-update data from TimeFinder/Snap source devices or to store excess cycle data from SRDF/A Delta Set Extension (DSE) operations.

The sections that follow describe the procedures for working with SAVE pools and their associated SAVE pool devices. The procedure for disabling, enabling, or removing devices from the SAVE pool is essentially the same.

These operations can be performed through the SRDF/TimeFinder Manager main menu by selecting option **5. Working with pools** which navigates through a series of screens, or by selecting option **8. Control Pools** from the Control Menu to access the CTLPOOL command. "CTLPOOL" on page 215 describes this command in detail. The following examples use screens.

SAVE pool restrictions

The following restrictions apply to SAVE pools:

- All SAVE device pools must have the same disk type (Fibre Channel, SATA, EFD).
- All devices in a SAVE pool must have the same type. The pool itself is not defined to be a specific type. However, the first device added to the pool defines the type.

All devices in a SAVE pool must have the same protection type. If the first device added to the pool is a 2-way mirror configuration, all subsequent added devices must have 2-way mirror protection.

Creating SAVE pools

When a SAVE pool is created, it is either a DSE pool or a SNAP pool.

To create a SAVE pool:

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5.** Work with Pools.

   The List of the Pools screen(Figure 47) appears with the list of each array and the pools on each array.

```
CTLPOOLS                      List of the Pools                        HOSTNAME

 Type option, press Enter.
  5=Work with the Pool    4=Delete Pool  7=Rename Pool   8=Rebalance Thin Pool
 10=End Rebal Thin Pool  11=Change max subs  13=Compr Enable  14=Compr Disable
                     Pool           Emul     Pool  # of
 Opt  Symmetrix      Name           Type     Type  Devices    State     Compr
      000111222333  DEFAULT_POOL  N/A      SNAP  0          Enabled   N/A
      000111222333  d910          FBA      DATA  10         Enabled   Enabled
      000111222333  d910_bcv      FBA      DATA  10         Enabled   N/A
      000111222333  NON_POOLED    N/A      DATA  960        N/A       N/A
      000222333444  DEFAULT_POOL  N/A      SNAP  64         Enabled   N/A
      000222333444  mig6r6        FBA      DATA  256        Enabled   N/A
      000222333444  mig7r5        FBA      DATA  256        Enabled   N/A
      000222333444  tbcv          FBA      DATA  512        Enabled   N/A




                                                                      Bottom
 F3=Exit     F5=Refresh    F6=Create Pool    F7=Work With Unbound Thin Devices
 F9=Command Entry          F12=Cancel        F8=Work With Migration Sessions
```

Figure 47 List of the Pools screen

Note: Because the DEFAULT_POOL contains SAVE devices that have not been assigned to a pool and its technology and protections types may be mixed, in the Work with Pool screen Tech Type and Configuration display as **N/A** for the DEFAULT_POOL. Tech Type is supported for Enginuity 5876 and higher. For Enginuity versions lower than 5876, the Tech Type displays as **N/A**. Since SNAP operations are not supported in VMAX 10K systems, the Pool Type for DEFAULT_POOL is **N/A**. DEFAULT_POOL contains disabled SAVE devices that can be used for DSE pools.

2. Press **F6** (Create Pool).

The Existing Pools screen (Figure 48) appears.

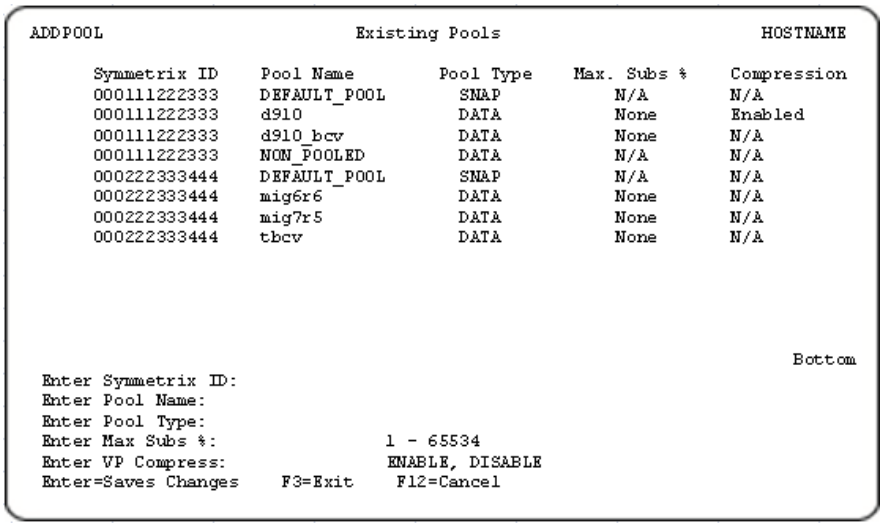


Figure 48 Existing Pools screen

3. Provide the following information:
   - **Symmetrix ID** — The name of the array on which this command is performed.
   - **Pool Name** — The name for the SAVE pool.
   - **Pool Type** — SNAP or DSE.

   Note: Max Subs% and VP Compress only apply to DATA (thin) pools.

4. Press **Enter** to save the changes.

   The List of the Pools screen updates with the new pool.

Adding devices to the SAVE pool

To add a device to a SAVE pool:

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5.** Work with Pools.

   The List of the Pools screen(Figure 47 on page 106) appears with the list of each array and the pools on each array.

2. Enter **5** in the **Opt** column of the SAVE pool where devices will be added, and press **Enter**.

   The Work with Pool screen (Figure 49 on page 108) appears. Pool Type is SNAP.

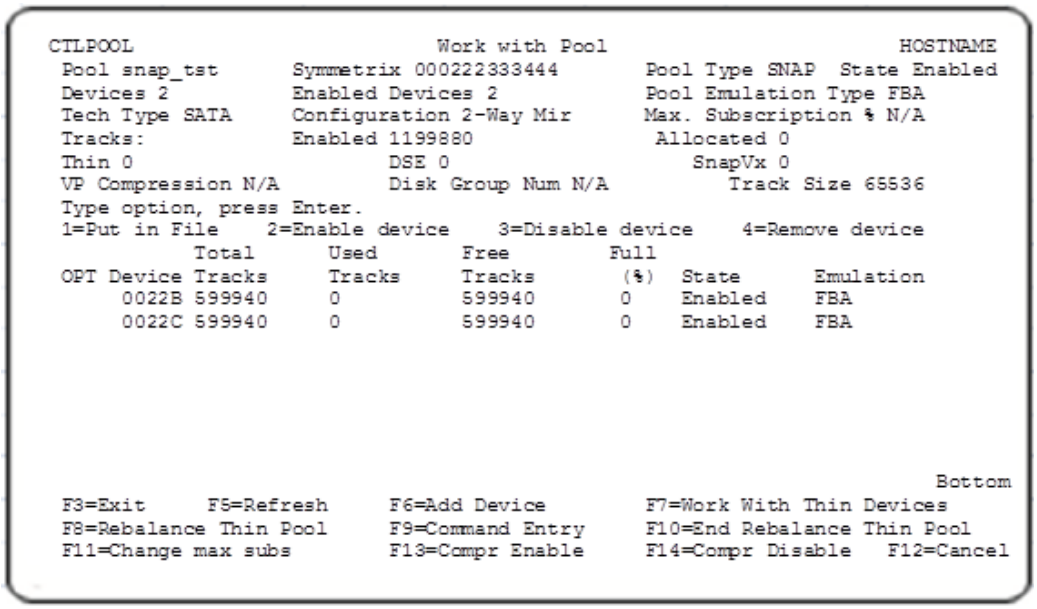


Figure 49 Work with Pool screen

3. Press **F6** (Add Device.)

   The Not Pooled Save/Data Devices screen (Figure 50 on page 109) appears.

   When the SAVE pool is not empty, only non-pooled devices with the same array type are presented. When the SAVE pool is empty, all non-pooled SAVE devices are presented.

   Note: Excluding the DEFAULT_POOL pool, SAVE pools can only have SAVE devices with the same type, disk type, and protection.

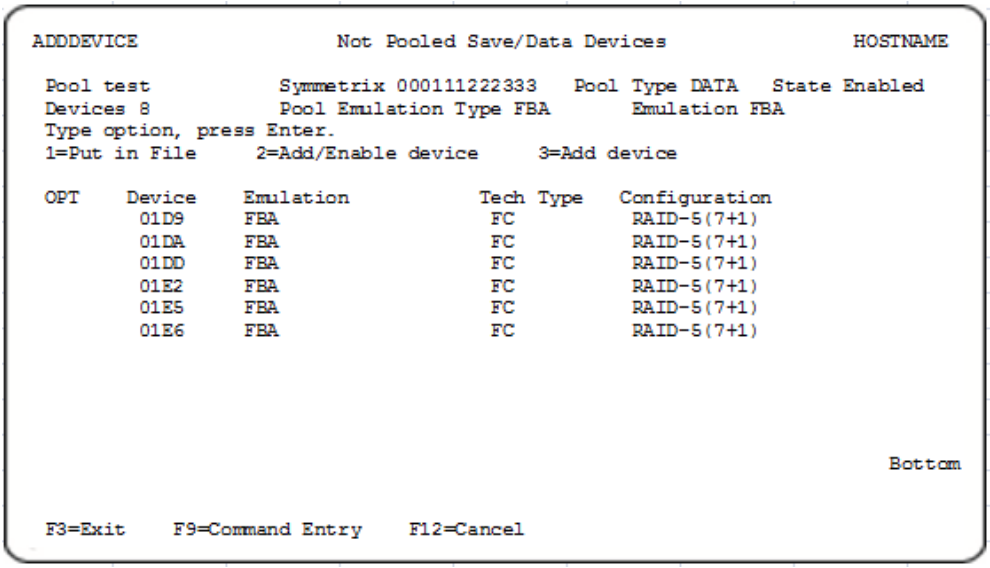


Figure 50 Not Pooled Save/Data Devices screen

4. Enter **3** in the **Opt** column of each device to be added to the SAVE pool, and press **Enter**. Added devices are disabled.

   Or

   Enter **2** in the **Opt** of each device to be added to the SAVE pool, and press **Enter**. Added devices are enabled.

   The Work with Pool screen updates with the new SAVE devices.

Creating or modifying existing device file for use with CTLPOOL command

There are four pool-related screens used for pool operations, that have **Put in File** option that can be used to create device files, or modify existing device files,

The screens that have the **Put in File** option are:

- "Work with Thin Unbound Devices screen"
- "Not Pooled Save/Data Devices screen"
- "Work with Pool screen"
- "Work with Thin Bound Devices screen"

Creating new device file

To create a new device file, from one of the screens that provides this option:

1. Enter **1** in the **Opt** column of each device to add to the file and press **Enter.**

   The Create Device File to Use with CTLPOOL screen (Figure 51 on page 110) appears.

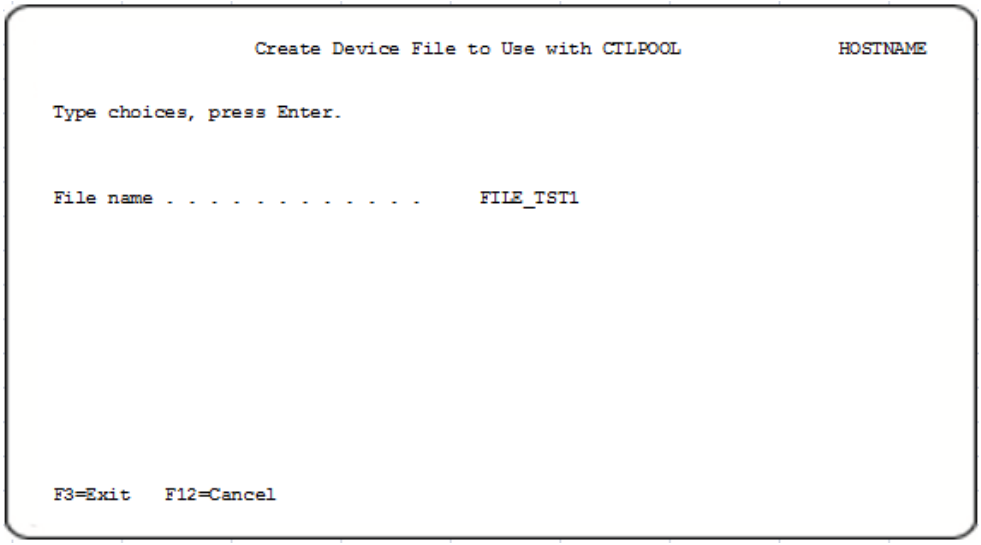


Figure 51 Create Device File to Use with CTLPOOL

- In the File Name field, enter the new device file name and press **Enter.**

  The device file is created and saved in the QGPL library.

Modifying existing device file

To modify (overwrite) an existing device file, from one of the screens that provides this option:

1. Enter **1** in the **Opt** column of each device to add to the file and press **Enter.**

   The Create Device File to Use with CTLPOOL screen (Figure 51) appears.

2. Enter the existing file name and press **Enter.**
3. A confirmation screen Figure 52 appears.

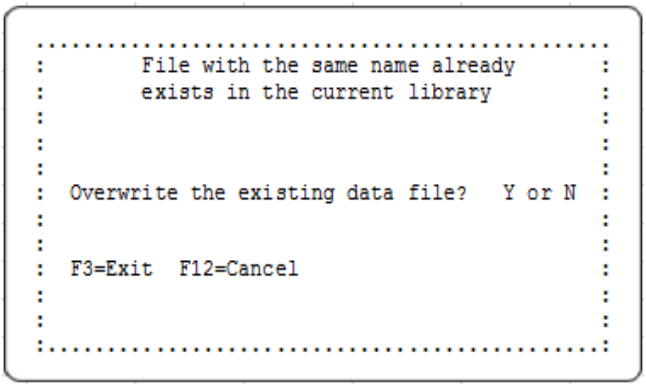


Figure 52 Confirmation screen for device file name

4. To overwrite existing file, type **Y**.

   The current file in the QGPL library is overwritten with new data.

To keep existing file, type **N** in the confirmation screen, and the Create Device File to Use with CTLPOOL screen with message that file already exists (Figure 53) appears.

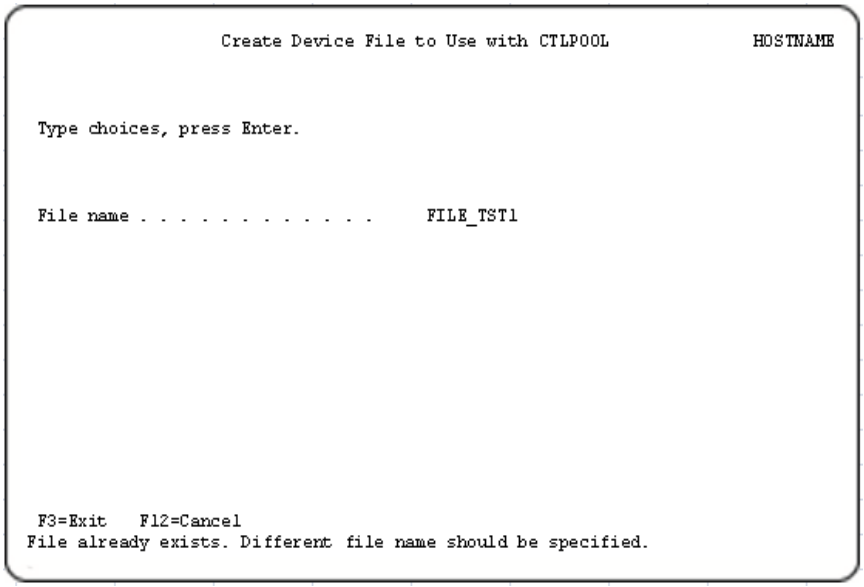


Figure 53 Create Device File to Use with CTLPOOL screen with message

Enter a different file name, press **Enter**.

The device file is created and saved in the QGPL library.

Managing pools and devices

This section describes how to:

- Enable and disable devices in a pool
- Remove devices from a pool
- Delete a device pool
- Rename a device pool

These operations are performed the same way for both SAVE and DATA pools and devices.

Enabling/disabling devices or removing devices from a pool

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5= Work with pools**.

   The List of the Pools screen (Figure 47 on page 106) appears with the list of each array and the pools on each array.

   Note: Because the DEFAULT_POOL contains SAVE devices that have not been assigned to a pool and its technology and protections types may be mixed, in the Work with Pool screen Tech Type and Configuration display as **N/A** for the DEFAULT_POOL. Tech Type is supported for Enginuity 5876 and later. For Enginuity versions earlier than 5876, the Tech Type displays as **N/A**. Since SNAP operations are not supported in VMAX 10K systems, the Pool Type for DEFAULT_POOL is **N/A**.

2. Enter **5** in the **Opt** column of the SAVE pool to perform an operation on, and press **Enter**.

   The Work with Pool screen (Figure 49 on page 108) appears.

3. In the **Opt** column of one or more devices, enter one of the following options and press **Enter**:

Note: Only one type of operation can be performed at a time.

- **2** (Enable device)
- **3** (Disable device)
- **4** (Remove device)

Note: When a SAVE device is disabled, the selected device is drained and placed in a Draining state so that it is logically removed from further Snap operations. Draining a SAVE device deactivates the device in the pool, and moves all used tracks to other devices in the same pool without affecting existing Snap sessions or DSE operations. Once the number of used tracks on the device decrements to zero, the device state changes to Disabled.

After an option is entered, the **State** field updates in the Work with Pool screen.

Note: When working with the DEFAULT_POOL pool, devices can only be disabled. For removed devices, the screen appears without confirmation of device removal from the SAVE pool. On VMAX 10K arrays, no operations are allowed with the DEFAULT_POOL.

Deleting device pools

In order to be deleted, device pools cannot have active sessions, must be disabled, and cannot contain members to be deleted.

To delete one or more pools:

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5**= Work with pools.

   The List of the Pools screen (Figure 47 on page 106) appears with the list of each array and the pools on each array.

2. Enter **4** in the **Opt** column of one or more pools to delete, and press **Enter**.

   The List of Pools screen updates.

Renaming device pools

SRDF/TimeFinder Manager allows for renaming of SNAP, DSE, and thin pools using the RENAME action. The RENAME action is also available for PowerMaxOS and HYPERMAX OS 5977 arrays.

Pool renaming restrictions

The following restrictions apply to device pool renaming:

- The array must run PowerMaxOS, HYPERMAX OS, or Enginuity 5876 or higher.
- SNAP, DSE, and thin pools can be renamed. However, the default pool name (DEFAULT_POOL) for SNAP cannot be changed.
- The new pool name must adhere to the same naming restrictions as when creating a pool.

Renaming a pool

To rename a device pool:

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5.** Work with Pools.

   The List of the Pools screen (Figure 47 on page 106) appears with the list of each array and the pools on each array.

   Enter **7** in the **Opt** column of the pool to rename, and press **Enter**.

   The Rename Pool screen (Figure 54) appears.

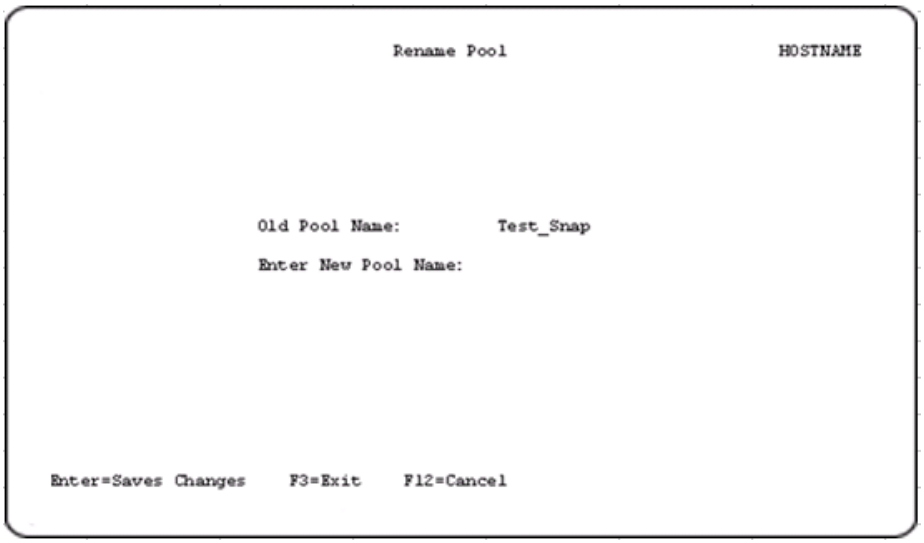


Figure 54 Rename Pool screen

   The List of the Pools screen updates with the new pool name.

Working with Virtual Provisioning™ and thin pools

Note: For arrays running PowerMaxOS 5978 or HYPERMAX OS 5977, thin pools and DATA devices are pre-configured, so pool management (see page 114) actions on thin pools) is not supported (except for pool renaming). For arrays running Enginuity 5876, the actions on thin pools described below are supported by SRDF/TimeFinder Manager, except for FREEALL which is available only on PowerMaxOS and HYPERMAX OS 5977 or later arrays.

Dell EMC SRDF/TimeFinder Manager for IBM i supports D910-099 as thin devices which also allows SRDF/TimeFinder Manager to take advantage of the Virtual Provisioning feature. Virtual Provisioning (commonly known as thin provisioning) allows for storage to be allocated and accessed on demand from a pool of storage, servicing one or many applications. Virtual Provisioning has the ability to define to the host a disk configuration that is far larger than current needs.

The Virtual Provisioning feature introduces three new concepts: *thin devices*, *DATA devices,* and *thin pools* or *DATA pools*. Thin devices can be created with an inflated capacity, because the actual storage space for the data written to thin devices is on DATA devices that are located in a thin pool. When additional storage is needed, more DATA devices can be added to the thin pool. DATA devices have FBA emulation and are not visible to the host. Thin devices bound to pools containing DATA devices are the components used for the Virtual Provisioning feature. Each write to a thin device by the host is written in the bound DATA device pool.

The same naming conventions that apply to SNAP pools and DSE pools apply to the thin pools, the only exception is there is no default thin pool.

Actions on thin pools

- BIND
- UNBIND
- REBIND
- BALANCE
- STOP_BALANCE
- CHANGE_SUBS (change subscription)
- COMPR ENABLE/DISABLE (compression enable/disable pool)
- COMP_START/END (compress devices)
- UNCOMP_START/END (uncompress devices)
- MIGRATE
- FREEALL (free up device allocations, see note)

Note: On PowerMaxOS 5978 or HYPERMAX OS 5977, all devices are under FAST control and FAST handles all device binding and allocation. However, the FREEALL action is available as a mechanism for the user to free up device allocations. This action frees up allocations regardless of whether or not data written to the device. FREEALL is only supported for arrays running PowerMaxOS 5978 or HYPERMAX OS 5977. All other actions listed above are supported on arrays running Enginuity 5876.

Thin device and pool rules and restrictions

The following restrictions apply to thin pools:

- The only IBM i thin device supported is the D910 device with 512 bytes/block geometry. Support is not provided for the legacy IBM i 2107 device with 520 bytes/block.
- D910 DATA devices are not supported. The data devices for IBM i thin pools are standard Server-type FBA devices.
- IBM i and Server-type FBA devices can share the same pool.
- All device pools must have the same disk type; SATA, Fibre Channel, or EFD.
- A thin pool can only contain DATA devices of one type. The pool itself is not defined to be a pool for a specific type. Instead, the first DATA device added to the thin pool defines the type.
- All DATA devices in a thin pool must have the same protection. For example, if a thin pool has the first DATA device defined as a 2-Way-Mir, all subsequent added DATA devices must be 2-Way-Mir devices.
- A DATA device of a specific protection can be added to an empty thin pool, or a pool that only contains DATA devices that have the same protection and type.
- A DATA device can only be removed from a thin pool if it is disabled and has no used tracks on it.
- If the total pool capacity needed to store existing thin device data becomes insufficient as a result of disabling a range of the devices, none of the devices in the requested range are disabled.
- Devices that do not belong to any pools are available to be added to a pool.
- Devices in a pool are available for use once they are enabled.

Creating thin pools

Operations for thin pools can be performed through the SRDF/TimeFinder Manager main menu using option **5. Working with pools** which navigates through a series of screens, or by using option **8. Control Pools** from the Control Menu to access the CTLPOOL command. “CTLPOOL” on page 215 describes this command in detail. The examples below use the **Working with pools** option from the main menu.

To create a thin pool:

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5. Work with Pools**.

   The List of the Pools screen (Figure 47 on page 106) appears with the list of each array and the pools on each array.

2. Press **F6** (Create Pool)**.**

   The Existing Pools screen (Figure 48 on page 107) appears.

3. Provide the following information:
   - **Symmetrix ID** — The name of the array on which this command is performed.
   - **Pool Name** — The name for the DATA pool.
   - **Pool Type** — DATA
   - **Max Subs %** — Enter a value of 1- 65534, or leave blank.
   - **VP Compress** — Enter enable or disable, or leave blank.
4. Press **Enter** to save the changes.
5. The List of the Pools screen updates with the new pool.

Adding thin devices to thin pools

To add a device to a DATA pool:

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5. Work with Pools**.

   The List of the Pools screen (Figure 47 on page 106) appears with the list of each array and the pools on each array.
2. Enter **5** in the **Opt** column of the DATA pool where devices will be added, and press **Enter**.
3. The Work with Pool screen (Figure 49 on page 108) appears and the **Pool Type** field displays as **DATA**.
4. Press **F6 (**Add Device).

   The Not Pooled Save/Data Devices screen (Figure 50 on page 109) appears and the **Pool Type** field displays as **DATA.**

   When the DATA pool is not empty, only non-pooled devices with the same type, disk type, and protection are shown. When the DATA pool is empty, all non-pooled DATA devices are shown.

   Note: DATA pools can only have DATA devices with the same type, disk type, and protection.
5. On the Not Pooled Save/Data Devices screen, enter **3** in **Opt** column of each device to add to the DATA pool and press **Enter**. Added devices are disabled.

   Or

   Enter **2** in the **Opt** column of each device to add to the DATA pool, and press **Enter**. The added devices are enabled.

   The Work with Pool screen updates with the new DATA devices.

Managing DATA pools and DATA devices

For DATA devices and pools, refer to "Managing pools and devices" on page 111 for enabling/disabling devices, removing devices from a pool, renaming pools, and deleting pools. Performing these operations is the same for both SAVE and DATA pools.

The following control operations and parameters are unique to DATA pools and thin devices:

- "Binding operations for thin devices"
- "Balancing operations for thin pools"
- "Setting subscription limit for thin pools"
- "Thin device data compression"
- "Thin device migration"

Note: With one exception, these operations are not available on arrays running PowerMaxOS 5978 or HYPERMAX OS 5977. The exception is the FREEALL command which available on PowerMaxOS 5978 and HYPERMAX OS 5977 arrays only.

Binding operations for thin devices

The following bind operations are available for thin devices:

- BIND — Binds a thin device to a thin pool of DATA devices
- UNBIND — Unbinds a thin device from a thin pool.
  A thin device cannot be unbound from a pool if it is:
  - Mapped to a front-end port or in the Ready state
  - Masked by VCM
  - Has active snap sessions
  - Held
  - Source or target of a clone session
  - Meta member
  - Part of enabled SRDF CG group
  - An RDF device
- REBIND — Changes a thin device's subsequent binding to another pool.

  Note: Rebind does not move any existing data to the new pool, but all new allocations to the thin device go to the new pool after the rebind completes.

- FREEALL — Frees up all allocations associated with a given device regardless of whether or not there is data written to the device. The following behavior applies the FREEALL command:
  - The FREEALL_START operation is blocked on any device participating in any replication operation. The device must also be not mapped or not ready.
  - The following operations are blocked if a FREEALL operation is running in the background on a device:
    - RDF createpair
    - TF/Mirror full establish and full restore
    - TF/Clone create, full establish and full restore
    - TF/SnapVX establish and link

Binding devices

There are two ways to bind thin devices to a pool.

Method 1

Use this method when the thin pool already has bound devices.

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5. Work with Pools.**

   The List of the Pools screen (Figure 47 on page 106) appears with the list of each array and the pools on each array.

2. Enter **5** in the **Opt** column of the DATA pool where devices will be bound, and press **Enter**.

   The Work with Pool screen (Figure 49 on page 108) appears.

3. Press **F7** (Work with Thin Devices).

   The Work with Thin Bound Devices screen (Figure 55) appears with the list of bound thin devices on the array.

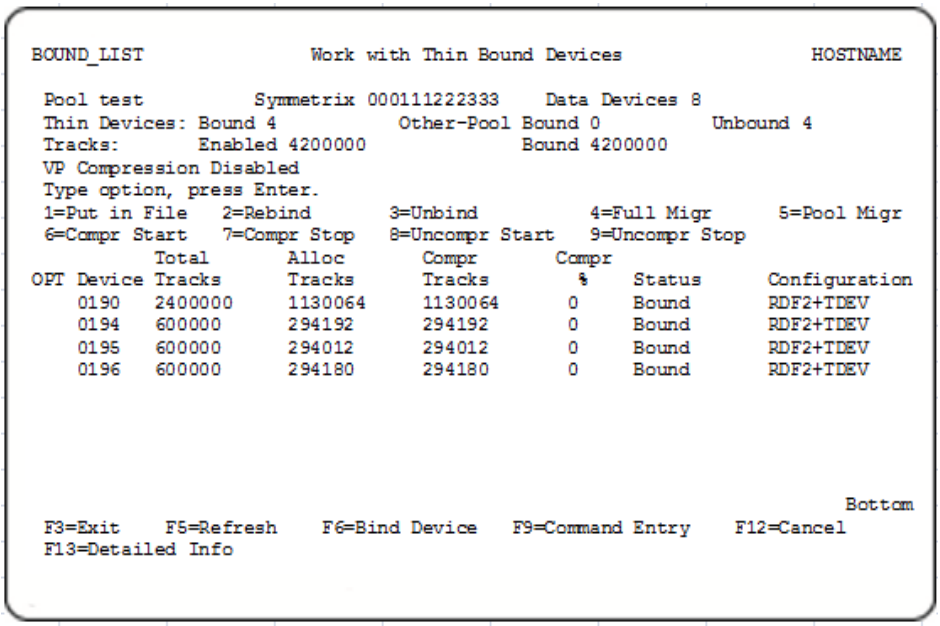


Figure 55 Work with Thin Bound Devices screen

4. Press **F6** (Bind Device). The Work with Thin Unbound Devices screen (Figure 56 on page 119) appears with the list of unbound thin devices on the array.

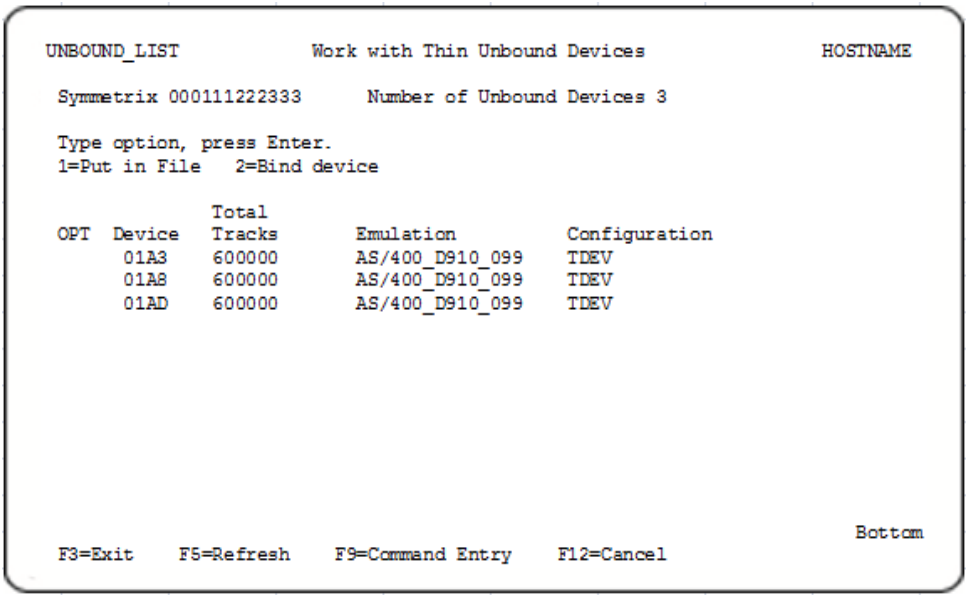


Figure 56 Work with Thin Unbound Devices screen

5. Enter **2** in the **Opt** column of each device to bind the device to the thin pool, and press **Enter.**

Method 2

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5. Work with Pools.**

   The List of the Pools screen (Figure 47 on page 106) appears with the list of each array and the pools on each array.

2. Press **F7** (Work with Unbound Thin Devices).

   Note: If more than one array is connected to the host the List of Connected Symmetrix systems appears. Otherwise, the Work with Thin Unbound Devices screen appears. See step 5.

3. The List of the Connected Symmetrix systems screen (Figure 57 on page 120) appears.

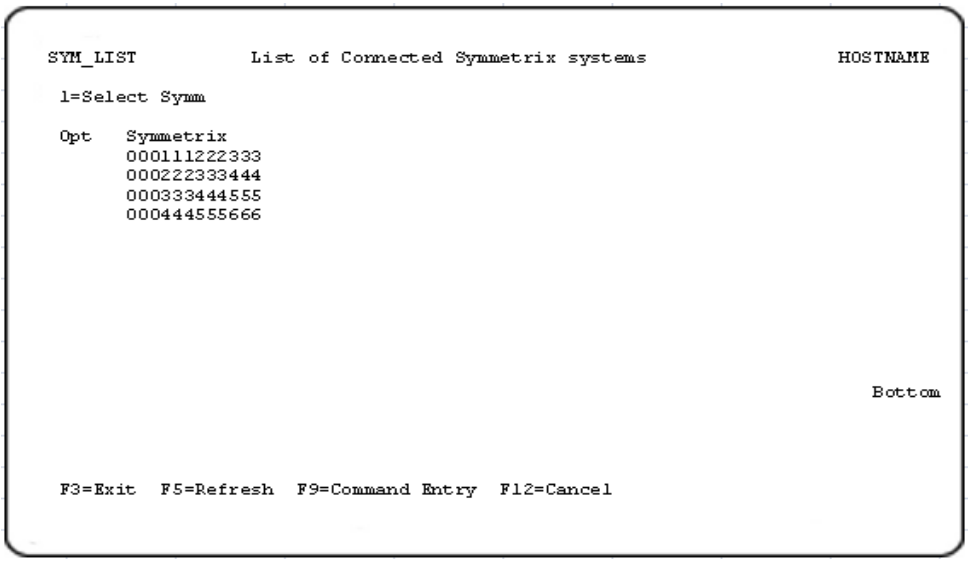


Figure 57 List of the Connected Symmetrix systems screen

4. Enter **1** in the **Opt** column of the array where devices will be bound.

   The Work with Thin Unbound Devices screen (Figure 56 on page 119) appears with the list of unbound thin devices on the specified array.

5. Enter **2** in the **OPT** column of each the thin device to bind to the pool, and press **Enter**.

   Note: If more than one thin pool exist on the array chosen in step 4, The List of Thin Pools screen (Figure 59 on page 122) appears with the list of all thin pools that exist on the array. Enter **1**=Select Pool next to the pool, press **Enter** to bind devices.

Unbinding thin devices from a thin pool

1. On the EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5. Work with pools**.

   The List of the Pools screen (Figure 47 on page 106) appears with the list of each array and the pools on each array.

2. Enter **5** in the **Opt** column of the DATA pool where devices will be unbound, and press **Enter**.

   The Work with Pool screen (Figure 49 on page 108) appears.

3. Press **F7** (Work with Thin Devices).

The Work with Thin Bound Devices screen (Figure 58) appears with only the thin devices bound to the specified pool.

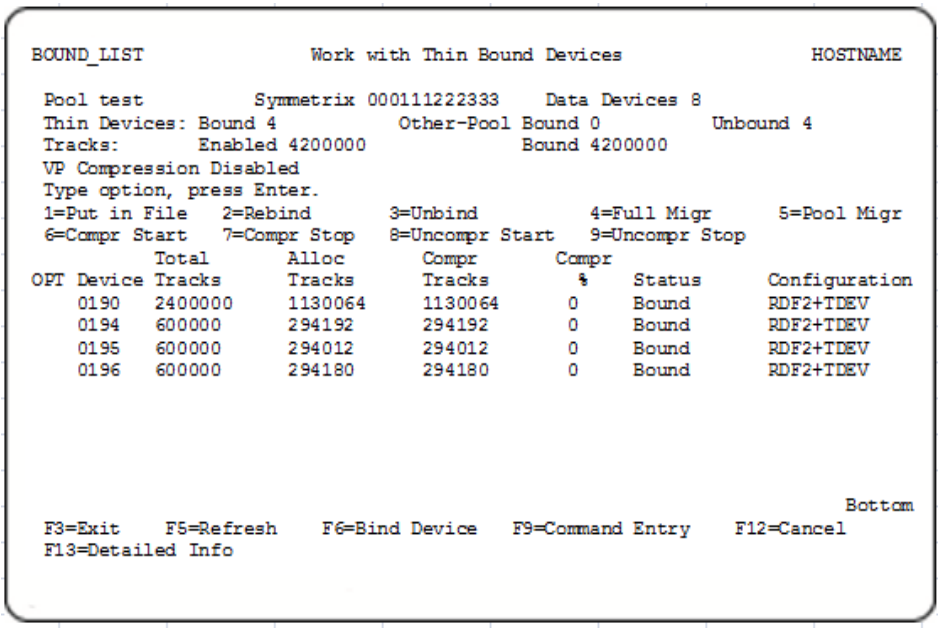


Figure 58 Work with Thin Bound Devices screen

4. Enter **3** in the **Opt** column of each thin device to unbind, and press **Enter** to unbind devices from the pool.

Rebinding thin devices

Rebinding changes current device binding to another pool. Existing data stays with the original pool and all new allocations to the thin device goes to the new pool.

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5**. Work with Pools.

   The List of the Pools screen (Figure 47 on page 106) appears.

2. Enter **5** in the **Opt** column of the DATA pool where devices will be rebound, and press **Enter**.

   The Work with Pool screen (Figure 49 on page 108) appears.

3. Press **F7** (Work with Thin Devices).

   The Work with Thin Bound Devices screen (Figure 58 on page 121) appears with the list of thin devices bound to the pool.

4. Enter **2** in the **Opt** column of each thin device to bind to another pool, and press **Enter.**

   The List of Thin Pools screen (Figure 59 on page 122) appears with the list of all thin pools that exist on the array.



Figure 59 List of Thin Pools screen

5. Enter **1** in the **Opt** column of the new pool assignment, and press **Enter** to rebind devices.

Confirming pool assignments after a rebind operation

On the Work with Thin Bound Devices screen (Figure 58 on page 121), press **F13** (Detailed Info) (F13 = Shift + F1).

The List of Bound/Allocated Devices screen (Figure 60) appears bound devices for the current pool, and allocations in any other pools for the listed devices.

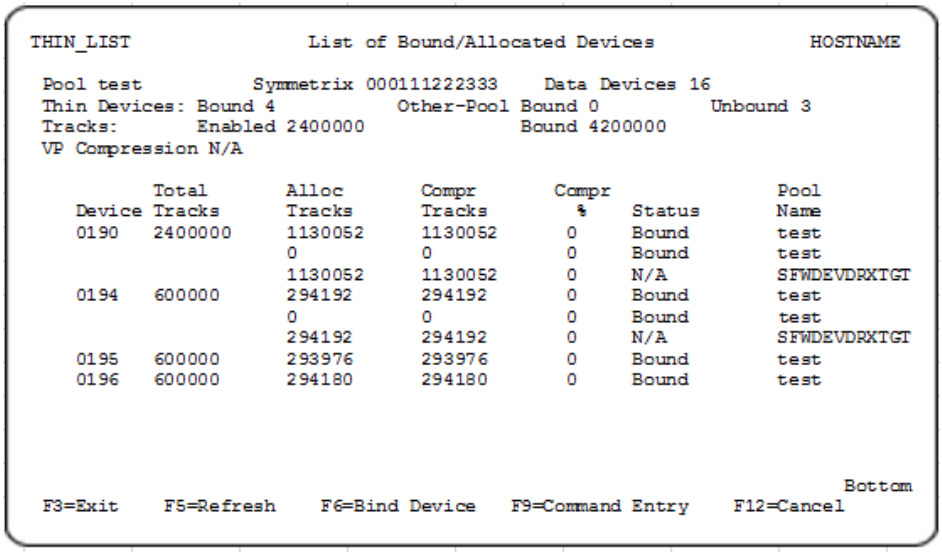


Figure 60 List of Bound/Allocated Devices screen.

Figure 60 shows device 0190 and 0194 after a rebind operation. For the current pool, allocations and status are listed. For allocations for these devices on any other pool, allocated tracks are listed, but status is not reported.

Freeing up device allocations (PowerMaxOS 5978 or HYPERMAX OS 5977)

You can start and stop the FREEALL action. In addition to the methods shown here, the FREEALL action is also available from the CTLPOOL screen.

Start FREEALL

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5. Work with pools**.

   The List of the Pools screen (Figure 47 on page 106) appears with the list of each array and the pools on each array.

2. Enter **5** in the **Opt** column of the DATA pool where device allocations will be freed up, and press **Enter**.

   The Work with Pool screen (Figure 49 on page 108) appears.

3. Press **F7** (Work with Thin Devices).

   The List of Allocated Devices screen (Figure 61) appears.

```
THIN_LIST                List of Allocated Devices                  HOSTNAME
 SRP Name DEFAULT_SRP             Pool DG4_FBA_PL       Symmetrix 000111222333
 Data Devices  40                 Number of Allocated Thin Devices 55
 Tracks:       Enabled 3754800               Allocated 2459468
 Track Size 131072                           Disk Group Num 4
 Type option, press Enter.
 1=Put in File   2=Freeall Start  3=Freeall Stop
           Total       Alloc       Compr       Compr
OPT Device Tracks      Tracks      Tracks        %    Status      Configuration
    00037  675000      795         795           0    Allocated   TDEV
    00038  675000      802         802           0    Allocated   TDEV
    0006B  45          1           1             0    Allocated   TDEV
    0006C  45          1           1             0    Allocated   TDEV
    00080  675000      598906      598906        0    Dealloc'ing BCV+TDEV
    00084  675000      455248      455248        0    Dealloc'ing BCV+TDEV
    00088  675000      75582       75582         0    Dealloc'ing BCV+TDEV




                                                                      Bottom
 F3=Exit    F5=Refresh                  F9=Command Entry    F12=Cancel
```

Figure 61 List of Allocated Devices screen

4. Enter **2** in the **Opt** column of each device where allocations are to be freed up, and press **Enter**.

Stop FREEALL

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5. Work with pools**.

   The List of the Pools screen (Figure 47 on page 106) appears with the list of each array and the pools on each array.

2. Enter **5** in the **Opt** column of the pool where allocations are to be freed up, and press **Enter**.

   The Work with Pool screen (Figure 49 on page 108) appears.

3. Press **F7** (Work with Thin Devices).

   The List of Allocated Devices screen (Figure 61) appears.

4. Enter **3** in the **Opt** column of each device where allocations will be freed up, and press **Enter**.

Balancing operations for thin pools

Whenever writes occur on a thin device, the actual data is written to the DATA devices in the thin pool to which the thin device is bound. The array distributes the I/O equally on all the DATA devices in the thin pool. Therefore, at any point in time the used capacity of each DATA device is uniform across the pool. However, when new devices are added to the pool, the allocation spread is no longer uniform. SRDF/TimeFinder Manager allows for rebalancing any unbalanced pools.

The following operations are available for starting and stopping write balancing on a pool:

- BALANCE — balances the DATA devices in the pool. On a balance operation SRDF/TimeFinder Manager checks for a balancing operation already in progress. If pool balancing is not in progress, SRDF/TimeFinder Manager initiates the operation.

  While write balancing is in progress, it is possible for data to be moved *from* deactivated data devices but never *to* deactivated data devices. The SRDF/TimeFinder Manager reports the pool state as Balancing. After the balancing operation completes, the pool state does not show as Balancing. The pool state indicates whether any balancing is in progress on a pool at any given time. All pool balancing operations are initiated by the user.

- STOP_BALANCE — stops the write balancing operation.

Note: Balancing is available on thin pools and is not applicable for SNAP or DSE pools.

Thin pool balancing and stop balancing

To balance a pool:

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5. Work with Pools.**

   The List of the Pools screen (Figure 47 on page 106) appears with the list of each array and the pools on each array.

2. Enter **8** in the **Opt** column of the DATA pool that is to be rebalanced, and press **Enter** to start the balancing operation.

   Confirm that the pool state is Balancing in the List of the Pools screen.

   Balancing can also be done from the Work with Pool screen (Figure 49 on page 108) by pressing **F8** (Rebalance Thin Pool).

To stop the balancing operation:

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5. Work with Pools.**

   The List of the Pools screen (Figure 47 on page 106) appears with the list of each array and the pools on each array.

2. Enter **10** in the **Opt** column of the DATA pool to stop balancing, and press **Enter**.

   Confirm that the balancing operation has been canceled by checking that the pool state is Enabled in the List of the Pools screen (Figure 47 on page 106).

   Stopping the balancing operation can also be done from the Work with Pool screen (Figure 49 on page 108) by pressing **F10** (End Rebalance Thin Pool).

Setting subscription limit for thin pools

The maximum subscription percent which is the limit for a thin pool can be set to any value between 1 and 65534. This number is the total host-perceived capacity of thin devices, divided by physical thin pool capacity. Maximum subscription percent is only used with thin pools.

If the value is not specified the default value is used, meaning there is no limit set and the amount of oversubscription is not restricted. Setting a value of 1 will allow only a maximum thin capacity that totals 1/100th of the actual physical capacity of the pool. Setting a value of 100 will prevent oversubscription of the pool.

The following parameters and control actions are available for controlling the maximum subscription percent:

- MAX_SUBS — sets the maximum subscription percent when creating a thin pool. This parameter is only used with thin pools.
- CHANGE_SUBS — Changes maximum subscription percent for the specified thin pool.

Changing pool maximum subscription percent

To change the maximum subscription percent for a pool:

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5.** Work with Pools.

   The List of the Pools screen (Figure 47 on page 106) appears with the list of each array and the pools on each array.

2. Enter **11** in the **Opt** column of the DATA pool where max subscription value will be changed, and press **Enter.**

The Change Maximum Subscription Percent screen (Figure 62) appears.

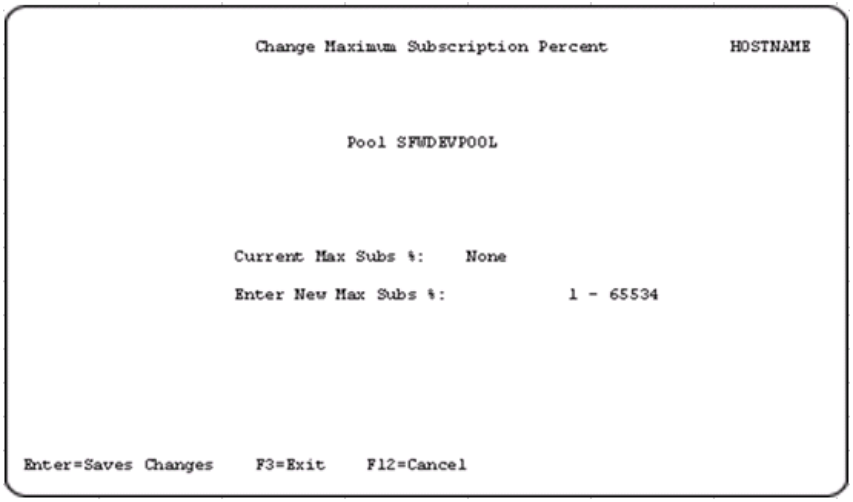


Figure 62 Change Maximum Subscription Percent screen

3. Type the new value in the **New Max Sub%** field, and press **Enter** to save the change.

   Confirm the new value of the Max Subscription% field in the Work with Pools screen (Figure 49 on page 108).

   Changing the maximum subscription percent can also be done from the Work with Pool screen (Figure 49 on page 108) by pressing **F11**.

Thin device data compression

Data within a thin device can be compressed to save space and uncompressed when needed. Data can either be compressed or uncompressed manually, or it can be compressed by FAST as part of its processing.

Enginuity allows read and write operations to compressed data. To do this, Enginuity uncompresses the data, when it is needed, into space that is reserved in the thin pool that contains the thin device compressed allocations. To reserve this space and allow allocations in a pool to be compressed, a pool must be enabled for compression. This is done by setting the compression capability on a thin pool to Enabled. When compression is no longer desired for allocations in a pool, the compression capability on a pool can be disabled.

A write operation always writes the uncompressed allocations on the thin device. Over time if the data has not been accessed, compression may occur if the device is under FAST management, and FAST is supporting compression on the associated device. The data on the device can also be manually compressed at any time.

Virtual Provisioning compression rules and restrictions

The following rules and restrictions apply to Virtual Provisioning compression:

- Once a pool is enabled for compression, a background process runs to reserve storage in the pool that is used to temporarily uncompress data. No compressions of allocations in this pool can occur until this process completes.

- When compression is disabled, the reserved storage in the pool is returned to the pool only after all compressed allocations are uncompressed. Disabling the compression capability on a pool does not cause compressed allocations to be uncompressed. The uncompressing action needs to be performed.
- When compressing a device, only allocations that have been written are compressed. Any allocations that were created during the bind or allocate processing that have not been written are reclaimed during the compression process. This means that when a compressed device is uncompressed, these allocations no longer exist.
- Once a compression request is accepted, the thin device has an associated background task that performs the compression. While this task is running, no other background task, like allocate or reclaim, can be run against the thin device.
- Requested compression action fails if all targeted devices are already in the state for the requested action. For example, if a compress start is issued for devices that are currently compressing, then SRDF/TimeFinder Manager rejects the request and displays Solutions Enabler error: SYMAPI_C_ALREADY_IN_STATE.

  Note: If some of the targeted devices are in the state for the requested action, but the remaining devices do not have a conflicting state, SRDF/TimeFinder Manager processes the request on those devices and skips the devices that are already in the desired state.

- Requested compression action fails if any of the targeted devices are in a conflicting state relative to the requested action. For example, if an uncompress start action is issued and some devices are in a compressing state, SRDF/TimeFinder Manager rejects the request and displays Solutions Enabler error: SYMAPI_C_INVALID_STATE_FOR_ACTION.
- If compression has not been enabled for a pool where a device has allocations, the SRDF/TimeFinder Manager does not fail a compress start action. However, compression does not occur for those allocations.
- Requested compression action fails if the array is not running Enginuity 5876. SRDF/TimeFinder Manager rejects the request and displays Solutions Enabler error: SYMAPI_C_FEATURE_NOT_SUPPORTED.

Compressing thin device data

To start compression:

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5. Work with Pools**.

   The List of the Pools screen (Figure 47 on page 106) appears with the list of each array and the pools on each array.

2. Enter **5** in the **Opt** column of the DATA pool where compression will be performed, press **Enter**.

   The Work with Pool screen (Figure 49 on page 108) appears.

3. If VP compression state is disabled, press **F13** (Compr Enable) to enable compression for the pool. The VP Compression state changes to Enabled.

4. Press **F7** (Work with Thin Devices).

   The Work with Thin Bound Devices screen (Figure 58 on page 121) appears with the list of thin devices bound to the pool.

5. Enter **6** in the **Opt** column of the device where compression will be performed, and press **Enter**.

6. Confirm that the device is in Compressing state in the Work with Thin Bound Devices screen (Figure 58 on page 121). This screen also shows the number of compressed tracks and the percent compression for the device.

To stop compression:

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5. Work with Pools**.

   The List of the Pools screen (Figure 47 on page 106) appears with the list of each array and the pools on each array.

2. Enter **5** in the **Opt** column of the DATA pool where compression will be performed, and press **Enter**.

   The Work with Pool screen (Figure 49 on page 108) appears.

3. Press **F7** (Work with Thin Devices). The Work with Thin Bound Devices screen (Figure 58 on page 121) appears with the list of thin devices bound to the pool.

4. Enter **7** in the **Opt** column of the device that is compressing, and press **Enter** to stop.

5. Confirm that the device is no longer in the Compressing state and shows as Bound in the Work with Thin Bound Devices screen(Figure 58 on page 121). This screen also shows the number of compressed tracks and the percent compression for the device.

Uncompressing thin device data

To start uncompression:

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5. Work with Pools**.

   The List of the Pools screen (Figure 47 on page 106) appears with the list of each array and the pools on each array.

2. Enter **5** in the **Opt** column of the DATA pool where uncompression will be performed, press **Enter**.

   The Work with Pool screen (Figure 49 on page 108) appears.

3. Press **F7** (Work with Thin Devices). The Work with Thin Bound Devices screen (Figure 58 on page 121) appears with the list of thin devices bound to the pool.

4. Enter **8** in the **Opt** column of the device to uncompress, and press **Enter**.

5. Confirm that the device is in the Uncompressing state in the Work with Thin Bound Devices screen. This screen also shows the number of compressed tracks and the percent compression for the device.

To stop uncompression:

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5. Work with Pools**.

   The List of the Pools screen (Figure 47 on page 106) appears with the list of each array and the pools on each array.

2. Enter **5** in the **Opt** column of the DATA pool where uncompression will be performed, and press **Enter**.

   The Work with Pool screen (Figure 49 on page 108) appears.

3. Press **F7** (Work with Thin Devices). The Work with Thin Bound Devices screen (Figure 58 on page 121) appears with the list of thin devices bound to the pool.

4. Enter **9** in the **Opt** column of the device that is uncompressing, and press **Enter**.

5. Confirm that the device is no longer in the Uncompressing state and shows as Bound in the Work with Thin Bound Devices screen. This screen also shows the number of compressed tracks and the percent compression for the device.

Thin device migration

Virtual LUN VP (virtual pools) migration allows thin device allocations to be moved between pools. For a given list of devices, the data tracks are moved from their current pool to a target pool.

The following rules and restrictions apply to thin device migration:

- All source devices must be thin devices.
- Allowed only on locally attached systems.
- Thin devices that are currently involved in an active migration session cannot be transitioned to other thin device states until the migration completes and is terminated.
- Thin devices must be in the bound state to be migrated.
- Thin devices that are part of any migration session cannot be migrated.

There are two types of thin device migration:

- Full migration — Full migration migrates all of the thin device's allocations. After the migration, if the thin device is not already bound to the target pool it is bound to that pool after the migration completes.
- Pool-level migration — Pool-level migration only migrates the device allocations from the designated source pool. After the migration, device binding does not change to the target pool. If rebinding the thin device to the target pool is desired, then this requires manual rebinding. Refer to "Rebinding thin devices" on page 121.

Performing full migration of thin device data

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5. Work with Pools**.

   The List of the Pools screen (Figure 47 on page 106) appears with the list of each array and the pools on each array.

2. Enter **5** in the **Opt** column of the array and pool where migration is to be performed, press **Enter**.

   The Work with Pool screen (Figure 49 on page 108) appears.

3. Press **F7** (Work with Thin Devices).

   The Work with Thin Bound Devices screen (Figure 58 on page 121) appears with the list of thin devices bound to the pool.

4. Enter **4** in the **Opt** column of each source device, and press **Enter.**

   The List of Thin Pools screen (Figure 59 on page 122) appears.

5. Enter **1** in the **Opt** column of the target pool, and press **Enter**.

6. The Name of the Migration Session to be Used screen (Figure 63 on page 130) appears.

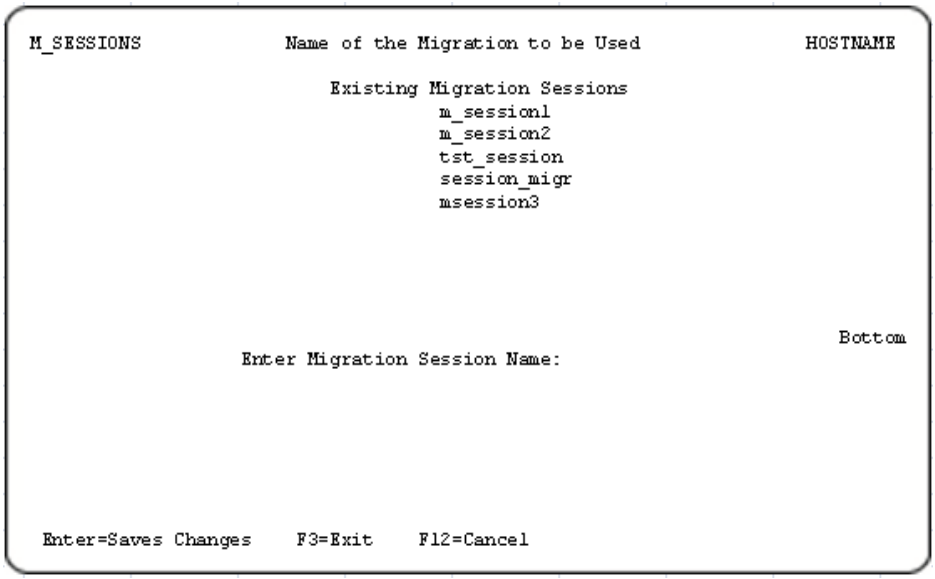


Figure 63 Name of the Migration Session to be Used screen

7. Type the new session name. If the session name entered already exists, an error message appears.

8. Press **Enter** to start the full migration session.

Performing pool-level migration of thin device data

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5. Work with Pools**.

   The List of the Pools screen (Figure 47 on page 106) appears with the list of each array and the pools on each array.

2. Enter **5** in the **Opt** column of each device where migration will be performed, press **Enter**.

   The Work with Pool screen (Figure 49 on page 108) appears.

3. Press **F7** (Work with Thin Devices).

   The Work with Thin Bound Devices screen (Figure 58 on page 121) appears with the list of thin devices bound to the pool.

4. Enter **5** in the **Opt** column of each source device, and press **Enter.**

   The List of Thin Pools screen (Figure 59 on page 122) appears.

5. Enter **1** in the **Opt** column of the target pool, and press **Enter.**

   The Name of the Migration Session to be Used screen (Figure 63) appears.

6. Type the new session name. If the session name entered already exists, an error message appears.

7. Press **Enter** to start the pool-level migration session.

Monitoring migration sessions

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5. Work with Pools**.

   The List of the Pools screen (Figure 47 on page 106) appears with the list of each array and the pools on each array.

2. Press **F8** (Work with Migration Sessions).

Note: If more than one array is connected to the host, the List of Connected Symmetrix systems screen (Figure 57 on page 120) appears, otherwise the List of Migration Sessions screen appears, see Figure 64. Only locally attached arrays display in the List of Connected Symmetrix systems screen.

---

3. The List of the Migration Sessions screen (Figure 64), which lists the migration session name, status, and percent migration complete, appears.

```
CTL_SESSIONS            List of the Migration Sessions              HOSTNAME

                            Symmetrix 000111222333

 Type option, press Enter.
 4=Terminate Session        5=Display Session Information

                Opt  Session Name  Session Status  Percent Done
                     migration       SyncInProg         87
                     migration1      SyncInProg         50

                                                                    Bottom

 F3=Exit    F5=Refresh
 F9=Command Entry        F12=Cancel
```

Figure 64 List of the Migration Sessions screen

For Further migration session details, type **5** (Display Session Information) in the **Opt** field next to the session name, and press **Enter.**

The Migration Session Information screen (Figure 65 on page 132) which lists the devices, status, migration percent complete, and migration type (full or pool-level), appears.

```
DSP_SESSION             Migration Session Information              HOSTNAME
 Symmetrix 000111222333      Session Name migration

                                                        Done
 Device   TGT Pool       SRC Pool       Status          (%)    Type
  0190    SFWDEVDRXTGT   N/A            SyncInProg      100    Full Migration
  0194    SFWDEVDRXTGT   N/A            SyncInProg      100    Full Migration
  0195    SFWDEVDRXTGT   N/A            SyncInProg       59    Full Migration
  0196    SFWDEVDRXTGT   N/A            SyncInProg       59    Full Migration

                                                                    Bottom

 F3=Exit    F5=Refresh
 F9=Command Entry        F12=Cancel
```

Figure 65 Migration Session Information screen

Terminating/Deleting a thin device migration session

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **5. Work with Pools**.

   The List of the Pools screen (Figure 47 on page 106) appears with the list of each array and the pools on each array.

2. Press **F8** (Work with Migration Sessions).

   Note: If more than one is connected to the host, the List of Connected Symmetrix systems screen (Figure 57 on page 120) appears, otherwise the List of Migration Sessions screen appears, see Figure 64 on page 132. Only locally attached arrays display in the List of Connected Symmetrix systems screen.

3. On the Work with Migration Sessions screen (Figure 64 on page 132), enter **4** in the **Opt** column of the o desired migration session, and press **Enter**.

   Note: A migration session in the “Migrated” state remains active until it is terminated. If a session in progress is terminated, the migration is stopped.

CHAPTER 5
Controlling TimeFinder

This chapter describes the procedures for controlling TimeFinder operations using the SRDF/TimeFinder Manager Control Menu.

- Overview 136
- Establish/Re-establish BCV or Clone emulation pairs 136
- Split BCV or Clone emulation pairs operation 137
- TimeFinder/Snap and TimeFinder/Clone operations 145
- Cancelling/terminating device pairs 148
- Unpairing BCV, Clone or Snap devices 148
- TimeFinder/SnapVX operations 150
- Monitoring TimeFinder operations 161
- Automating TimeFinder operations 162

Overview

SRDF/TimeFinder Manager relies on a specific sequence of operations to ensure proper execution of TimeFinder control options. SRDF/TimeFinder Manager checks the configuration and operation considerations prior to executing any command that changes the state of a device or the system.

Note: SRDF/TimeFinder Manager does not perform a discover process automatically, which can take several minutes to run. After any change to the configuration of the host or array, turn on the discover function or run the DISCOVER command (see "Discovering array and host configuration details" on page 63).

TimeFinder control operations are performed on images in which all devices are PowerMax, VMAX All Flash, VMX3, VMAX, or DMX array devices. Control operations are performed at the step level in the path for each image.

Establish/Re-establish BCV or Clone emulation pairs

The Establish operation copies the contents of the standard device to the BCV device and performs the following tasks:

- For a standard BCV, assign the BCV as the next available mirror of a standard array device.
- Copy the entire contents of the standard array device to the BCV device.
- If standard BCV devices are used in the BCV*n* step (established and split), and then used in the Clone emulation (CLN*n*) step, SRDF/TimeFinder Manager performs a full Establish. The data is replicated by TimeFinder/Mirror, and one source device can have up to four target devices.

The BCV device is unavailable to any host while the BCV device is in this established state.

After you establish BCV pairs, the array device automatically copies the data from its standard device mirrors to the newly assigned BCV device mirror. The array mirroring function also automatically writes to the BCV device all writes that occur on the standard device, while it is in the established pair state.

When you use Establish for BCV pairs that have been established previously (re-establishing a BCV pair), the array copies, to the BCV device, only those updates made to the standard device that occurred while the BCV pair was split.

Establish or re-establish BCV pairs

1. On the Control Menu, select option **1. SRDF or TimeFinder Control** to display the Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen.
2. Provide the **Image Name**, **Path Name**, and **Step Number**.
3. Select the **ESTABLISH** control action.

   The Dell Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen appears with a Protected BCV Establish prompt.
4. Type ***YES** or ***NO** for the PROT_ESTAB parameter depending upon the configuration.

The Confirm Establish BCV pairs or the Confirm Establish CLONE emulation pairs screen (Figure 66) appears.

SRDF/TimeFinder Manager attempts to perform TimeFinder operations on all ASPs in the image. Remove any ASPs that are not eligible for the operation (see "Removing ASPs from an image" on page 81).

You can monitor the synchronization process through the array configuration and status option (see "Monitoring invalid and % copied tracks" on page 161).

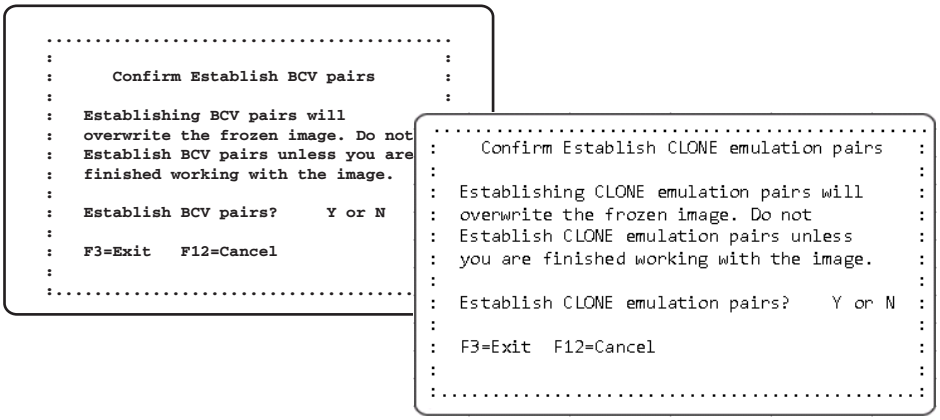


Figure 66 Confirm Establish BCV pairs and CLONE emulation pairs screens

5. Type **Y** to continue.

   SRDF/TimeFinder Manager establishes the BCV or Clone emulation pairs.

Split BCV or Clone emulation pairs operation

When a Split operation is issued on a step of the image, the Split occurs for *all* BCV devices or Clone emulation pairs in that step.

By default, SRDF/TimeFinder Manager performs an instant split, and checks to ensure that synchronization is complete between all specified standard and BCV mirrored pairs, or standard and Clone emulation pairs.

A check for complete synchronization ensures that the split pair contains an updated, consistent copy at the time of the termination. After the Split operation for the BCV or Clone emulation pairs completes, the host can resume operation with the standard array device. Figure 67 on page 138 shows the Split BCV pairs or Split Clone emulation pairs operation.

Restricted state and daemon jobs

When using split or activate you can hold all jobs, bring the system to a restricted state, use split while active, or do a maintenance split/activate. The hold jobs option holds all jobs in the system except for the daemon jobs Storapid and Storgnsd.

Additionally, choosing the restricted state option does not stop the subsystem where Storapid and Storgnsd are running. Other jobs that are running in that same subsystem are stopped using the STMENDSBS command. This command stops all subsystems.

During a split, The STMENDSBS command checks status of the subsystems for the restricted state. The split fails and you are asked to hold jobs when any of these conditions exist:

- The required daemons are not running.
- There are other jobs running in the subsystems where the daemon jobs are running.
- There are subsystems running, other than the subsystems where the daemon jobs are running.

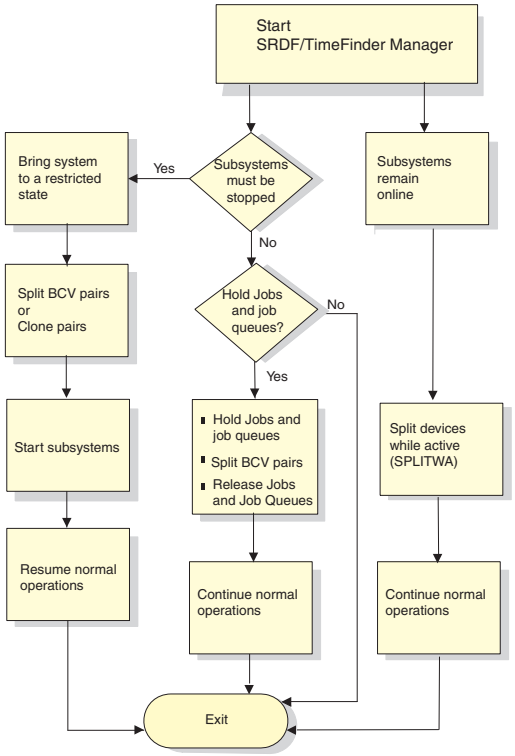


Figure 67 Split operation for BCV or Clone emulation pairs

The following sections describe the TimeFinder operations.

Split BCV pairs: Restricted state

This procedure uses the restricted state to ensure that I/O activity is halted before the BCV pair split. If you need to hold jobs and job queues rather than bring the system to a restricted state, see "Split BCV pairs: Hold jobs and job queues" on page 140.

The amount of time needed to move a host into the restricted state depends on the system configuration. By default, SRDF/TimeFinder Manager waits 1800 seconds for control to return, after issuing the command to bring the system to a restricted state. If additional time is required to bring the system to a restricted state, use the procedure "Setting the restricted state wait time" on page 60.

For IASP images, see "SRDF/TimeFinder Manager Extended Features" on page 301.

To split BCV pairs in a restricted state:

1. Bring the IBM i host to a restricted state:
   a. Prepare for bringing the system to a restricted state, using internal procedures established by your company.
   b. On the Control Menu, select option **5. Bring system to a restricted state**.

      The STM End Subsystem Confirmation screen appears.

      Note: The system is normally set to the restricted state before the TimeFinder split is performed.

   c. Press **F10**.

   The STM End All Subsystems (STMENDSBS) screen (Figure 68) appears.

```
                    STM END ALL SUBSYSTEMS (STMENDSBS)

Type choices, press Enter.

How to end . . . . . .   *CNTRLD      *CNTRLD, *IMMED
Controlled end delay time  . . .   *NOLIMIT     seconds, *NOLIMIT

                          Additional Parameters

End subsystem option . . . . . .   *DFT        *DFT, *NOJOBLOG, *CHGPTY...
               + for more values   _____

                                                                      Bottom
F3=Exit   F4=Prompt   F5=Refresh   F12=Cancel   F13=How to use this display
F24=More keys
```

Figure 68 STM End All Subsystem (STMENDSBS) screen

   d. Enter options for **How to End** and **Controlled end delay time**, and press **Enter**.

      After the End Subsystem operation completes, the Control Menu appears.

2. Split BCV pairs:
   a. Select option **1. SRDF or TimeFinder Control** to display the Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen.
   b. Provide the **Image Name**, **Path Name**, and **Step Number**.
   c. Select the **SPLIT** control action.

      The Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen reappears with a **Split/Suspend Mode** prompt.

   d. Enter **RESTRICTED** for the **SPLIT_MODE** parameter.

SRDF/TimeFinder Manager checks if the system is in a restricted state. If not, SRDF/TimeFinder Manager asks if you want to hold jobs or job queues.

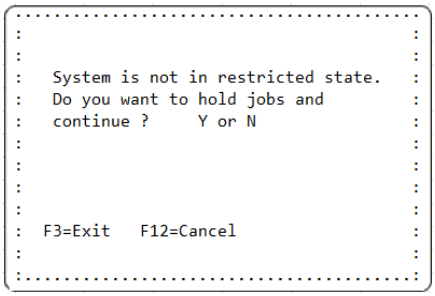


Figure 69 System not in restricted state screen

To continue operation in HOLD_JOBS mode, see "Split BCV pairs: Hold jobs and job queues" on page 140 and take the appropriate action.

A message screen appears indicating that the Split BCV pairs operation has started.

SRDF/TimeFinder Manager attempts the Split BCV pairs operation on all ASPs in the image. ASPs that are not eligible for the operation must be removed (see "Removing ASPs from an image" on page 81).

3. Start subsystems:

   a. On the Control Menu, select option **6. Start subsystems.**

   Note: Start subsystems does not power cycle the IBM i host or restart the operating system.

   Start subsystems calls the program defined in the system value QSTRUPPGM. If additional start-up procedures are used, run them separately.

   b. Resume normal operations on the source IBM i host.

Split BCV pairs: Hold jobs and job queues

This procedure holds jobs and job queues to ensure that I/O activity is halted prior to a TimeFinder Split operation.

Note: Daemon jobs are not put on hold.

CAUTION

System activity that results in a write operation can continue while jobs and job queues are on hold. For example, a restore job could have write operations continue while it is held. If you are not sure that the hold jobs and job queues option completely suspends write operations to the selected image, bring the system to a restricted state before doing a Split operation (see "Split BCV pairs: Restricted state" on page 138).

To split BCV pairs with jobs and job queues on hold:

1. On the Control Menu, select option **1. SRDF or TimeFinder Control** to display the Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen.
2. Provide the **Image Name**, **Path Name**, and **Step Number**
3. Select the **SPLIT** control action.

   The Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen reappears with a Split/Suspend mode prompt.
4. Enter **HOLD_JOBS** for the **SPLIT_MODE** parameter.

   Note: The BCV pair status for the image must be in an eligible state.

   SRDF/TimeFinder Manager repeatedly attempts to put all jobs and job queues on hold, and then executes the Split operation.

   If one or more jobs or job queues remain active, the Confirmation Window screen (Figure 70) appears.

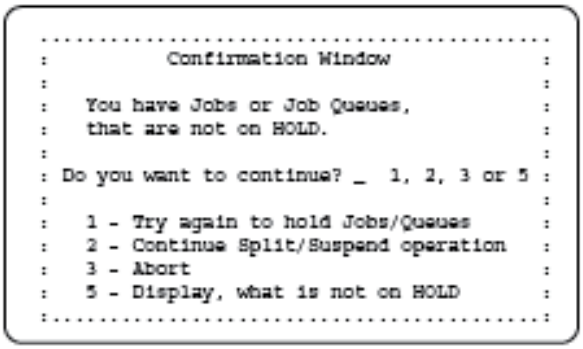


Figure 70 Confirmation Window screen

5. Select the appropriate option:
   - **1 - Try again to hold Jobs/Queues** — SRDF/TimeFinder Manager repeatedly attempts to hold all jobs and job queues.

     If jobs or job queues are not on hold, the Confirmation Window screen reappears.
   - **2 - Continue Split/Suspend operation** — Accept that some jobs remain active and continue with the Split or Suspend operation.

     ⚠CAUTION

     Selecting Continue Split/Suspend operation acknowledges that some write activity can continue during the Split operation. If any writes occur to the specified ASP during the Split operation, the resulting copy is not valid.

- **3 - Abort** — Do not perform the Split or Suspend operation and return to the Control Menu.
- **5 - Display, what is not on HOLD** — Show a list of the jobs and job queues that SRDF/TimeFinder Manager cannot hold (Figure 71 on page 142).

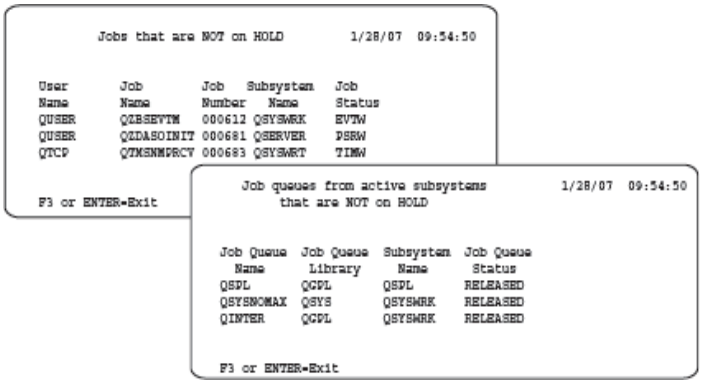


Figure 71 Jobs Not on Hold and Job Queues Not on Hold screens

6. Press **F3** or **Enter** to return to the Confirmation screen (Figure 70 on page 141).

   After all jobs and job queues are successfully on hold, or you select option **2 - Continue Split/Suspend operation**, SRDF/TimeFinder Manager performs the Split operation.

   A message screen appears indicating that the Split BCV pairs operation has started.

   After the Split operation completes, all jobs and job queues held by SRDF/TimeFinder Manager are released.
7. Optionally, use the Work with Associated Paths screen or the Disk Units screen to monitor and verify the change in BCV pair status (see "Monitoring TimeFinder operations" on page 161).
8. Resume normal operation on the source IBM i host.

Split BCV pairs: Maintenance

The Split operation in maintenance mode splits BCV pairs without interfering with the source host operations.

Note: This type of Split operation results in a BCV copy that is *not* usable as a backup. If you intend to use the BCV copy, use the RESTRICTED or HOLD_JOBS options.

Perform a Split operation in maintenance mode only when you need to IPL the secondary host with attached BCV devices. The secondary host cannot IPL while the BCVs are established with the standard devices. Therefore, perform a split before attempting an IPL.

To split BCV pairs for maintenance:

1. On the Control Menu, select option **1. SRDF or TimeFinder Control** to display the Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen.
2. Provide the **Image Name**, **Path Name**, and **Step Number**.

3. Select the **SPLIT** control action to perform.

   The Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen reappears with a **Split/Suspend mode** prompt.

4. Enter **MAINTENANCE** for the **SPLIT_MODE** parameter.

   The SRDF/TimeFinder Manager performs the Split operation without interfering with the source host operations.

TimeFinder Split: Split While Active

Split While Active enables SRDF/TimeFinder Manager to split local/remote BCVs or Clone emulations, or activate Snap or Clone images while the source system is operational. A consistent split ensures dependent write consistency for all devices in the selected step.

The SYMAPI server jobs must be running on the source and controlling hosts before you run this option.

The `netcnfg` file must be configured on the client using the CRTNETCNFG command. This ensures that the correct information points to the remote server. IBM i systems support non-secure connections only. "CRTNETCNFG" on page 205 has details of this command.

IMPORTANT

Use caution when performing this type of split as the data in main memory is not included in the image. Use SPLITWA only when the host and applications are in a quiescent state. If that is unacceptable to your business needs, use other operating system functions like journaling and database commitment control to get an image current with the last transaction.

To split BCV, Clone emulation, Snap, or Clone pairs while active:

- Run the CTLACT command from the command line (see "CTLACT" on page 206).

  Note: Since I/O is briefly being held for all devices, including the devices in the system ASP, the control action needs to be executed from a different host (see "CTLACT" on page 206).

- Access the Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen and select the appropriate control action:

  1. On the Control Menu, select option **1. SRDF or TimeFinder Control** to display the EMC SRDF or TimeFinder Control (CTLACT) screen.
  2. Provide the **Image Name**, **Path Name**, and **Step Number**
  3. Select the **SPLIT/ACTIVATE** control action to perform.

     The Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen reappears with a **Split/Suspend mode** prompt.

  4. Enter **SPLITWA** for the **SPLIT_MODE** parameter.

     The SRDF/TimeFinder Manager performs the Split while active operation without interfering with the source host operations.

Cancel BCV or Clone emulation pairs

To cancel a BCV or Clone emulation pairing relationship:

1. On the Control Menu, select option **1. SRDF or TimeFinder Control** to display the Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen.
2. Provide the **Image Name**, **Path Name**, and **Step Number**
3. Select the **CANCEL** control action.

   The Confirm Cancel BCV pairs or the Confirm Cancel CLONE emulation screen (Figure 72) appears.

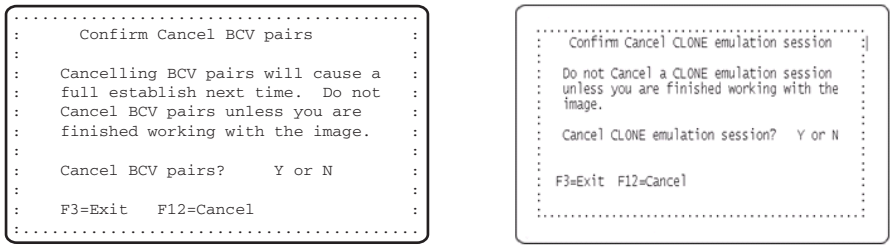


Figure 72 Confirm Cancel BCV pairs and CLONE emulation session screens

4. Type **Y** to continue.

   The incremental relationship between the standard and the BCV volume is canceled and all record of track changes is removed. If the devices are clone-enabled BCVs, the clone sessions are terminated on the array.

Note: Repairing and establishing the same devices results in a full establish.

TimeFinder/Snap and TimeFinder/Clone operations

The TimeFinder/Snap and TimeFinder/Clone specific operations are Create/Recreate, Activate, and Terminate.

Note: Recreate for virtual devices is available only for arrays running Enginuity 5876. TimeFinder/Snap is not supported on arrays running PowerMaxOS 5978 or HYPERMAX OS 5977.

Creating a Snap copy session or Clone copy session

1. On the Control Menu, select **option 1. SRDF or TimeFinder Control** to display the Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen (Figure 73).

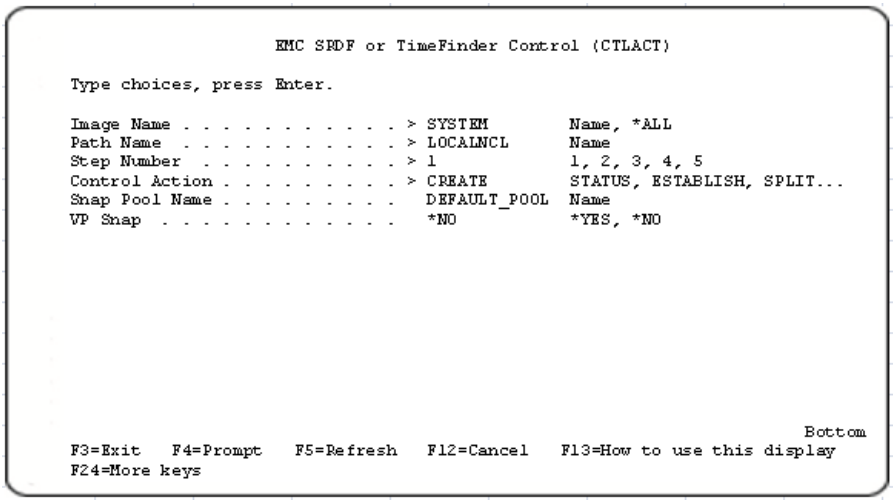


Figure 73 Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen — create Snap or Clone session

2. Provide the **Image Name**, **Path Name**, and **Step Number**.
3. Select the **CREATE** control action.

   Note: For a list of all control action parameters, press **F4** on the Control Action prompt.

4. Enter SNAP or Clone options:
   - To create a SNAP session enter the **Snap Pool Name** which is the named SAVE pool to use for the session. If pool name is not specified, the operation uses the default pool (DEFAULT_POOL).

     Note: The Snap Pool Name parameter is relevant only when creating new SNAP sessions, otherwise it is ignored.

   - To create a Clone session using VP Snap mode, set the **VP Snap** parameter to ***YES**. The VP Snap attribute can only be set when creating new clone sessions.

   For rules and restrictions for the snap pool name feature and VP Snap mode refer to "CTLACT" on page 206.

Note: If initiating Clone sessions using option **14** (Establish/Create) on the Work with Associated Paths screen, the Create VP Snap screen (Figure 74) appears only if all devices in the image are running with Enginuity 5876 or higher, and the step status is NotCreated.

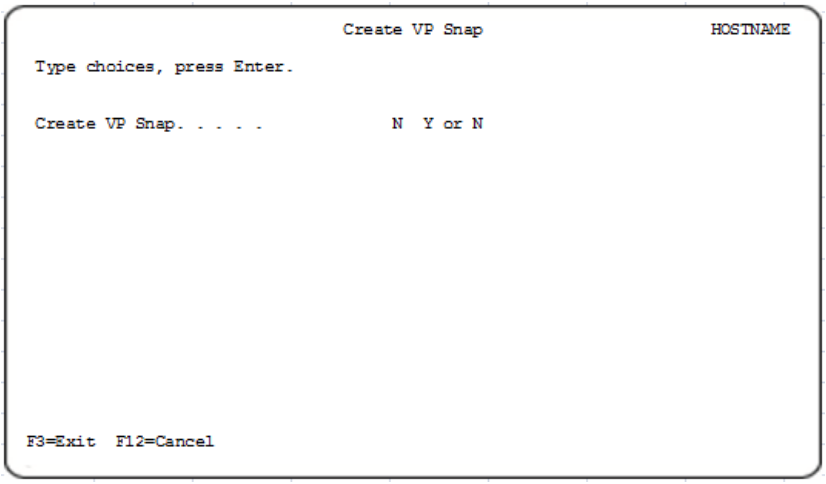


Figure 74 Create VP Snap screen

If a session previously was created and activated a confirmation screen appears (Figure 75) and recreate action is performed.

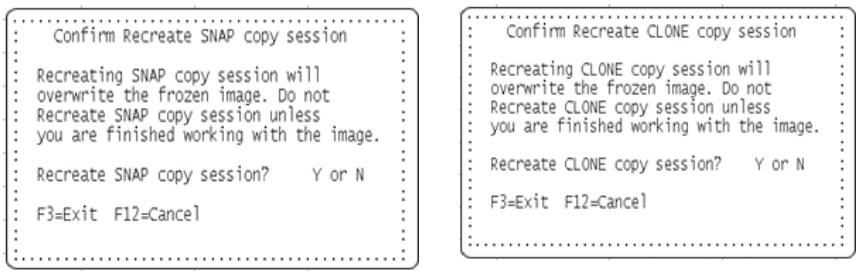


Figure 75 Confirm Recreate SNAP copy and Recreate CLONE copy session screens

SRDF/TimeFinder Manager creates the SNAP virtual copy or Clone copy session and displays a completion message.

Note: If initiating SNAP sessions using option **14** (Establish/Create), on the Work with Associated Paths screen (Figure 8 on page 48), and the status of the SNAP type of the step is NotCreated, the List of the Snap Save Pools screen(Figure 76 on page 147) appears. A pool can be selected from the list of save pools. If the image spans over several arrays, all SNAP SAVE pools are displayed, and the pool that exists on all of the arrays should be selected.

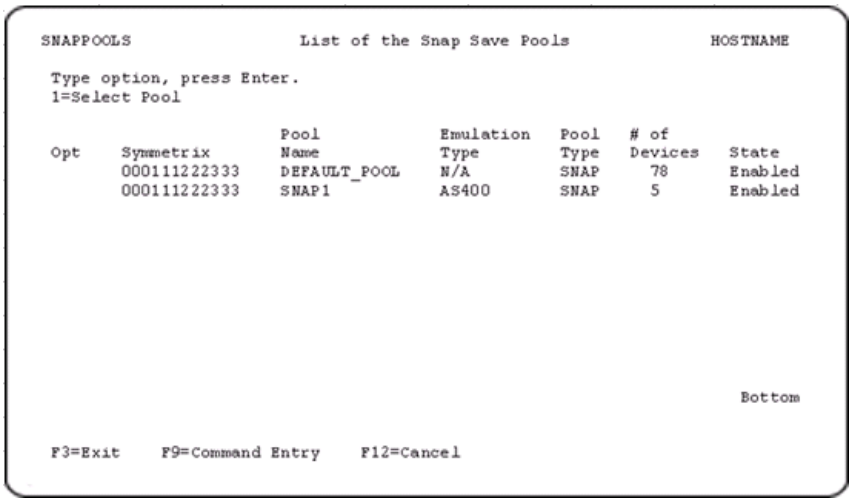


Figure 76 List of the Snap Save Pools screen

Activating a Snap virtual copy session or Clone copy session

1. On the Control Menu, select option **1. SRDF or TimeFinder Control** to display the Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen.
2. Provide the **Image Name**, **Path Name**, and **Step Number**.
3. Select the **ACTIVATE** control action.

   Note: For SPLIT_MODE, use the appropriate mode, refer to "Split BCV or Clone emulation pairs operation" on page 137.

   SRDF/TimeFinder Manager activates the copy session and displays a completion message.

Canceling/Terminating TimeFinder session

Depending upon the selected device type, use the Cancel/Terminate option to either:

- Terminate a TimeFinder/Snap virtual copy session or TimeFinder/Clone copy session.
- Cancel an incremental relationship between a standard device and a BCV device that are currently a pair in the BCV set. If the devices are clone-enabled BCVs, the clone sessions are terminated on the array.

Terminating a Snap virtual copy session or Clone copy session

1. On the Control Menu, select option **1. SRDF or TimeFinder Control** to display the Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen.
2. Provide the **Image Name**, **Path Name,** and **Step Number**.
3. Select the **TERMINATE** control action.

The Confirm Terminate SNAP Session or the Confirm Terminate CLONE session screen appears (Figure 77).

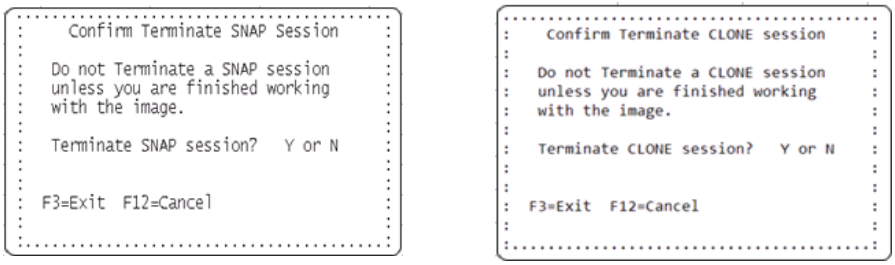


Figure 77 Confirm Terminate SNAP Session and Confirm Terminate CLONE Session

4. Type **Y** to continue.

   SRDF/TimeFinder Manager terminates the copy session and displays a completion message.

Cancelling/terminating device pairs

To cancel BCV device pairs or terminate Clone or Snap copy sessions:

- From the Disk Units screen, type **8** (Cancel) in the **OPT** field next to the selected device pair.
- To cancel multiple device pairs, type **8** (Cancel) next to each entry and the selected pairs are canceled at the same time.

After the Cancel operation, the status for BCV*n* or CLN*n* types of the step become NeverEstab, for NCL*n* , and SNP*n* types of the step become NotCreated. Any subsequent Establish/Create operation is a full Establish.

⚠CAUTION

Run the Cancel operation before doing the unpairing operation on RAID 5/RAID 6 or mixed BCVs, clone-enabled devices, clone devices, or Snap devices. Not running the Cancel operation causes hanging Clone sessions to remain on the array.

Unpairing BCV, Clone or Snap devices

Device unpairing makes BCV, Clone and Snap devices reusable.

- To unpair devices, type **7** (Unpair) on the Work with Associated Paths screen next to the step to be unpaired or from the Disk Units screen next to the unit number of the source device.
- To unpair multiple device pairs type **7** (Unpair) next to each entry and the selected pairs are unpaired at the same time.

BCV device unpairing

For a standard or R2 device paired with a BCV device, the unpair option on a device removes pairing information, for the specified image.

A BCV pair cannot be unpaired while the devices are in a Synchronized, Restored, SyncInProg, or RestInProg state. Prior to performing this operation, the pairs must be split and canceled.

If clone emulation is enabled for the BCV pair, cancel the relationship prior to unpairing. If the BCVs were established, the information about the previous standard/BCV relationship is stored in the SYMAPI database. To recover the information about the devices, the devices must be paired again.

Snap device unpairing

For a standard or BCV device paired a virtual device, the unpair option on a device removes pairing information, for the specified image.

You cannot unpair Snap devices while the devices are in a CopyInProg, Copied, Created, CopyOnWrite, CreateInProg, RestInProg, or Restored state. Run the Terminate option before unpairing Snap devices.

Clone device unpairing

For a standard or BCV device paired with a standard or BCV device, the unpair option on a device removes pairing information, for the specified image.

Clone device unpairing cannot be done unless they are in a NotCreated state. Prior to performing this operation, terminate the pairs.

Recovering from a NotConsistent state

When BCV device pairs, in a step of an image, are at different states (for example one pair is in Synchronized state and another in Split state), the BCV status for that step of the image becomes NotConsistent.

To recover from a NotConsistent state:

- For BCV or Clone emulation devices, perform the Establish operation on BCV or Clone emulation pairs.

  The Establish operation on BCV or Clone emulation pairs checks the state of each device pair that belongs to each step of the image.

  If the state is eligible for the Establish command, the operation starts for that device pair. Otherwise, the job log receives an error message providing the device numbers and their status information.

- For Snap devices or Clone devices, perform a Create operation.

  The Create operation checks the state of each device pair that belongs to each step of the image. If the state is eligible for the Create command, the operation starts for that device pair. Otherwise, the job log receives an error message providing the device numbers and their status information.

IMPORTANT

Contact the Dell EMC Customer Support Center for assistance with bringing devices back into an eligible state.

TimeFinder/SnapVX operations

TimeFinder/SnapVX specific operations are Establish, Link/Relink, Unlink, Set Copy Set NoCopy, Set TTL, Restore, Rename Snapshot, and Terminate.

Establishing SnapVX session

A SnapVX session can be established from the EMC SRDF or TimeFinder Control (CTLACT) screen, or with option **14** (Establish/Create) on the Work with Associated Paths screen. The Establish control action creates and activates a SnapVX session.

Using the Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen:

1. On the Control Menu, select option **1. SRDF or TimeFinder Control** to display the Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen (Figure 78).

```
                  EMC SRDF or TimeFinder Control (CTLACT)

Type choices, press Enter.

Image Name . . . . . . . . . . . > SYSTEM       Name, *ALL
Path Name  . . . . . . . . . . . > LSNAPVX1     Name
Step Number  . . . . . . . . . . > 1            1, 2, 3, 4, 5
Control Action . . . . . . . . . > ESTABLISH    STATUS, ESTABLISH, SPLIT...
Protected BCV Establish  . . . .   *NO          *YES, *NO
SnapVX Establish Mode  . . . . .   CONSISTENT   CONSISTENT, HOLD_JOBS...
Snapshot Name  . . . . . . . . . > END_OF_YEAR
Snapshot Time to Live  . . . . . > DATE         DAYS, DATE
 Time to Live - date . . . . . .   06/30/2014   Date

                                                                      Bottom
F3=Exit   F4=Prompt   F5=Refresh   F12=Cancel   F13=How to use this display
F24=More keys
```

Figure 78 Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen — create SnapVX session

2. Provide the **Image Name**, **Path Name**, and **Step Number**.
3. Select the **ESTABLISH** control action.
4. Enter the **SnapVX Establish Mode**: (the mode to use for the SnapVX Establish operation):
   - **RESTRICTED** — System must be put in restricted state using the STMENDSBS command.
   - **CONSISTENT** — No interference with the host operations. By default all snapshots are consistent.

IMPORTANT

Use Consistent Establish with care, as the data in main memory is not included in the image. Use this only when the host and applications are in a quiescent state. If that is unacceptable to your business needs, use other operating system functions like journaling and database commitment control to get an image current with the last transaction.

- **HOLD_JOBS —** All jobs and job queues will be put on HOLD.

Note: To display a list of all control action parameters, press F4 on the Control Action prompt.

5. Type a **Snapshot Name**.

   The name can be used for multiple devices and/or multiple snapshots. For multiple snapshots for the same device with the same name, a generation number (Gen) is used to differentiate between these point-in-time copies. Snapshots and generation numbers are time stamped and are listed from newest to oldest with Gen 0 being the most recent snapshot.

6. Optional: Enter the **Time To Live** (TTL) — Sets the time when the snapshot is automatically terminated, and is specified in days from now (**DAYS**) or a future date can be set (**DATE**). The value in DAYS can be set from 1 to 400. The DATE value can be set to a date up to 400 days in the future from the time the ESTABLISH action occurs.

   If a snapshot has TTL set and has one or more link targets, it is not terminated until links are unlinked. If the TTL value is not specified, the snapshot remains until it is manually terminated.

SRDF/TimeFinder Manager creates and activates the SnapVX session and displays the snapshot status as **EstablishInProg** while the copy is in progress. It then displays the status as **Establish** when the data is fully copied. Once established the snapshots are uniquely identified by name, Gen number, and source device for control actions.

Using the Work with Associated Paths screen

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **1. Work with Images** to display the Symmetrix Configuration by Image screen (Figure 6 on page 47).
2. Select option **2. Work with paths** to display the Work with Associated Paths screen (Figure 79 on page 152).

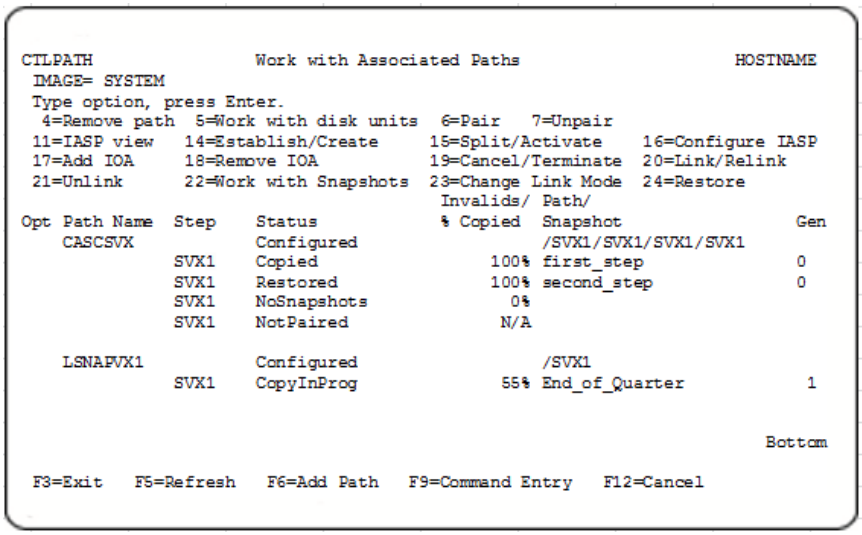


Figure 79 Work with Associated Paths

Note: If SVXn step is paired but no snapshots are established for the devices in the step, the status displays as NoSnapshots.

3. Enter **14** in the **Opt** column of a SnapVX type of step, to display the Existing Snapshots screen (Figure 80).

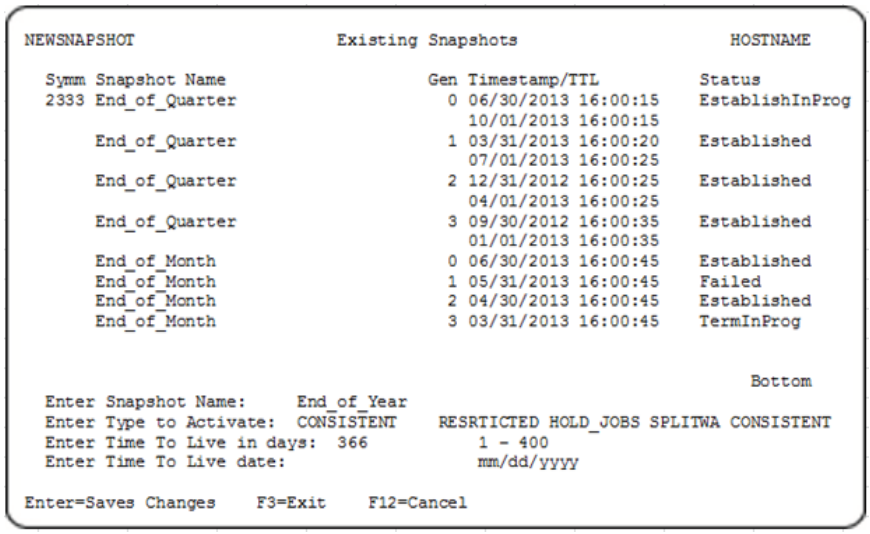


Figure 80 Existing Snapshots screen

4. Type a **Snapshot Name**.

   The name can be used for multiple devices and/or multiple snapshots. For multiple snapshots for the same device with the same name, a generation number (Gen) is used to differentiate between these point-in-time copies. Snapshots and generation numbers are time stamped and are listed from newest to oldest with Gen 0 being the most recent snapshot.

5. Enter optional **Time To Live** (TTL) — Sets the time to when the snapshot is automatically terminated, and is specified in days from now (**DAYS**) or a future date can be set (**DATE**). The value in DAYS can be set from 1 to 400. The DATE value can be set to a date up to 400 days in the future from the time the ESTABLISH action occurs.

   If a snapshot has TTL set and has one or more link targets, it is not terminated until links are unlinked. If the TTL value is not specified, the snapshot remains until it is manually terminated.

   SRDF/TimeFinder Manager creates and activates the SnapVX session and displays the snapshot status as **EstablishInProg** while the copy is in progress, and then displays the status as **Establish** when the data is fully copied.

Renaming a Snapshot

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **1. Work with Images** to display the Symmetrix Configuration by Image screen (Figure 6 on page 47).
2. Select option **2. Work with paths** to display the Work with Associated Paths screen (Figure 79 on page 152).
3. Enter **22** in the **Opt** column of a SnapVX type of step, to display the Work with Snapshots screen (Figure 81 on page 154).

Note: The operations available from the Work with Snapshot screens, such as Link/Relink, Restore, Set TTL, Rename, and Terminate, are described in "TimeFinder SnapVX" on page 24.

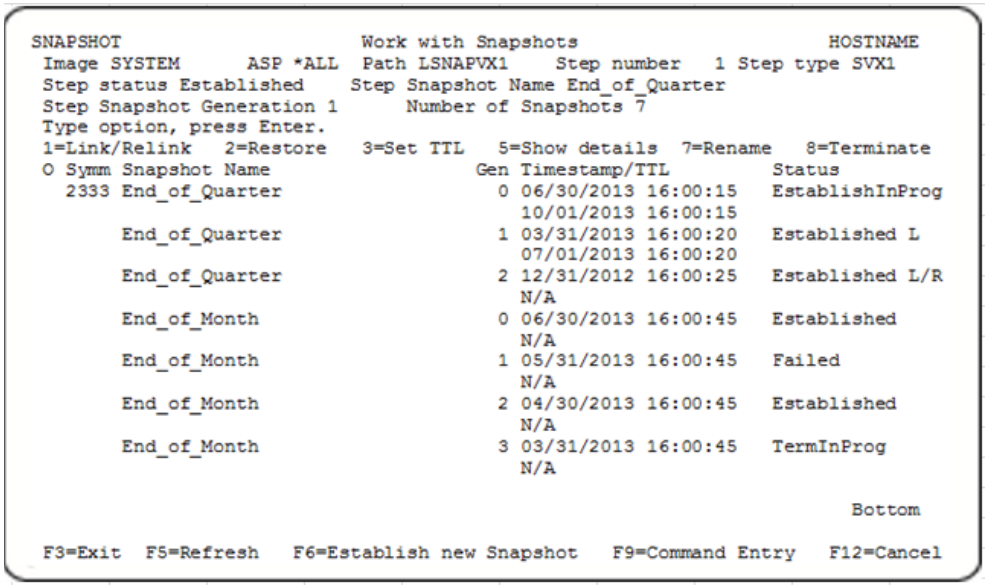


Figure 81 Work with Snapshots screen

Note: If Snapshot has links, '**L**' appears in the Status field. If Snapshot has restores, '**R**' appears in the Status field. If snapshot has links and restores '**L/R**' appears in the status field.

4. Enter **7** in the **Opt** column of a snapshot to display the Rename snapshots screen (Figure 82).

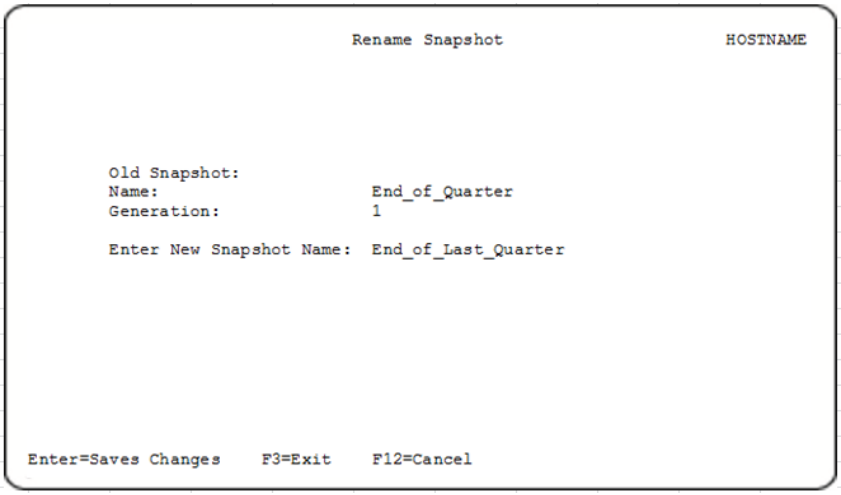


Figure 82 Rename Snapshot screen

5. Type the **New Snapshot Name**.

   Snapshot screens are updated with the new name.

Displaying Snapshot details

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **1. Work with Images** to display the Symmetrix Configuration by Image screen (Figure 6 on page 47).
2. Select option **2. Work with paths** to display the Work with Associated Paths screen (Figure 79 on page 152).
3. Enter **22** in the **Opt** field of a SnapVX type of step, to display the Work with Snapshots screen (Figure 81 on page 154).
4. Enter **5** in the **Opt** column of the snapshot to display the Snapshot Details screen.

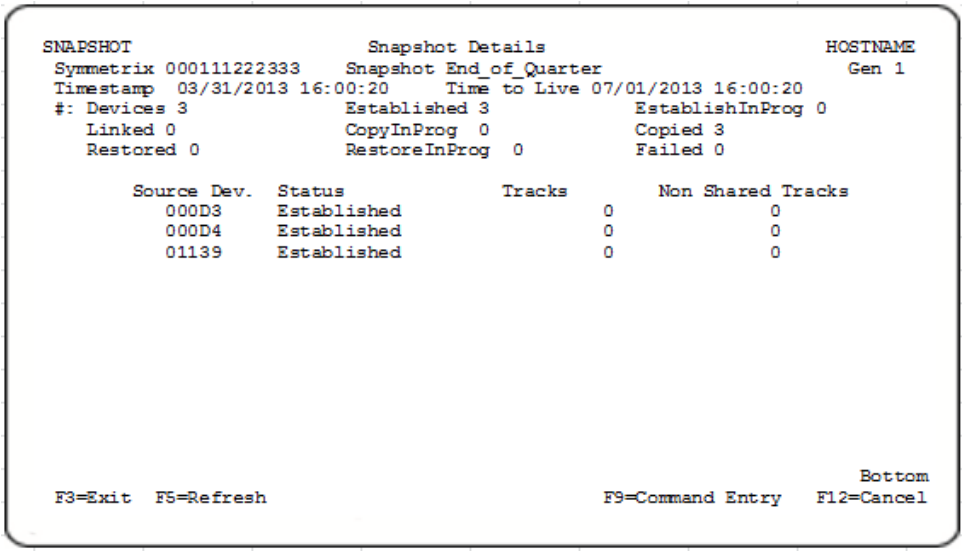


Figure 83 Snapshot Details screen

Table 12 Field descriptions For Snapshot Details screen (page 1 of 2)

Field	Description
Symmetrix	Symmetrix ID.
Snapshot	The user assigned name of the snapshot.
Gen	Generation number when there are multiple snapshots of the same name. Generation 0 is the most recent snapshot and the numbers increase for older point-in-time snapshots.
Timestamp	Time when snapshot was established.
Time to Live	The date and time when the snapshot will automatically be terminated (Time to Live feature requires auto-terminate to be the enabled state).
# Devices	Number of snapshot source devices.
Established	Number of the devices participating in the snapshot. This number will differ from the number of source devices only if of any of the devices is in a Failed or EstablishInProg state.
EstablishInProg	Number of the devices transitioning to Established state
Linked	Number of target devices linked to the snapshot in NoCopy mode.

Table 12 Field descriptions For Snapshot Details screen (page 2 of 2)

Field	Description
CopyInProg	Number of the target devices linked to the snapshot in Copy mode transitioning to the Copied state.
Copied	Number of the target devices linked to the snapshot in Copy mode when they are 100% copied.
Restored	Number of restores which are 100% restored for the snapshot.
RestoreInProg	Number of restores in transition for the snapshot.
Failed	Number of failed devices. Possible cause would be if a corruption occurs on any of the source devices.
Source Dev	List of the source devices participated in the snapshot.
Status	The state of the source device is normally Established. EstablishInProg is a transition state. If a corruption occurs on device, the status would be Failed.
Tracks	The number of source tracks that have not yet been overwritten by the host.
Non Shared Tracks	The number of tracks uniquely allocated for this snapshot's delta. This indicates the approximate number of tracks that will be returned to the SRP if this snapshot is terminated.

Note: The sum of the Established, EstablishInProg, and Failed fields should equal the number of the source devices.

Linking/Relinking Snapshots

If the host needs an access to the snapshot data, target devices can be linked or relinked to the snapshot. A link presents the point-in-time data on target devices mapped to the host. A relink removes the current link to target devices and presents a different snapshot. The relink may also be used to relink to the same snapshot. This has the effect of refreshing the point-in-time copy on the link target when it's been modified by host writes.

The links can be created in Copy mode for a permanent copy, when all the data is fully copied on the target device; or they can be created in NoCopy mode for temporary use.

Note: Previous track allocations are deallocated. Therefore, devices with persistent track allocations cannot be link targets.

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **1. Work with Images** to display the Symmetrix Configuration by Image screen (Figure 6 on page 47).
2. Select option **2. Work with paths** to display the Work with Associated Paths screen (Figure 79 on page 152).
3. Enter **20** in the **Opt** column of a SnapVX type of step, to display the List of Snapshots screen (Figure 84 on page 157).

Note: Linking and Relinking operations can also be performed from the Work with Snapshots screen.

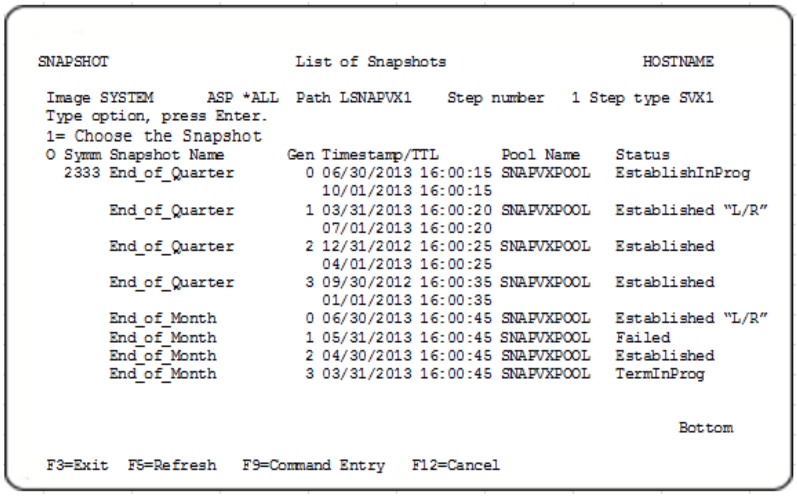


Figure 84 List of Snapshots screen

4. Enter **1** in the **Opt** column of a Snapshot Name, to display the Link Options screen (Figure 85).

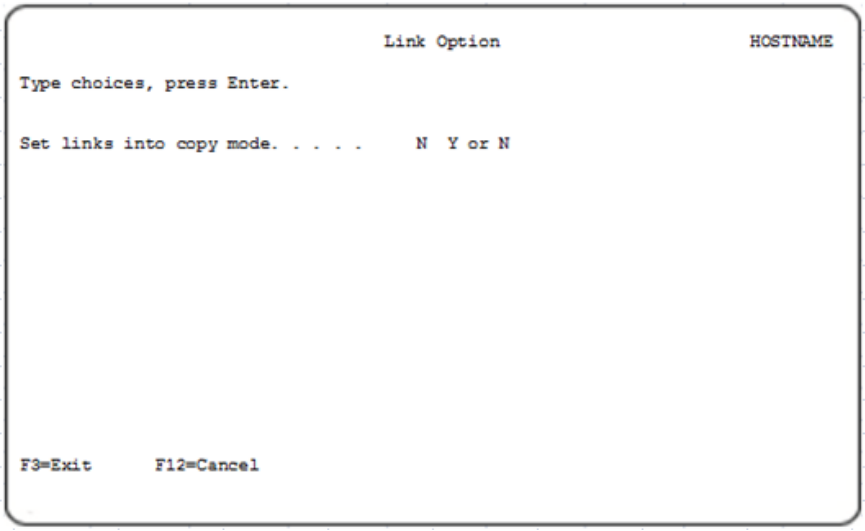


Figure 85 Link Option screen

5. Select **Link Type**:
   - Y — Creates a permanent copy when all data is fully copied to the link target.
   - N — Creates a temporary copy. Point-in-time data is removed from the target device once the link is unlinked.

Viewing status of link/relink operation

From the Disk Units screen and Work with Associated Paths screen the status displays as CopyInProg while the copy is in progress, and then as Linked when the data is fully copied. If the link is performed in copy mode, the status displays as Copied.

Snapshot linking can also be performed from the Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen using the Control Action LINK. (Figure 78 on page 150).

Option **23** on the Work with Associated Paths screen (Figure 79 on page 152) changes Link Mode, from NoCopy to Copy mode or from Copy to NoCopy, when the status of the step is Linked/CopyInProg/Copied.

Unlinking Snapshots

Unlink all links before terminating a snapshot.

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **1. Work with Images** to display the Symmetrix Configuration by Image screen (Figure 6 on page 47).
2. Select option **2. Work with paths** to display the Work with Associated Paths screen (Figure 79 on page 152).
3. Enter **21** in the **Opt** column of a SnapVX type of step.

   Note: Target devices can be unlinked if the status of the step is Linked or Copied.

   The snapshot is unlinked from the target link. From the Work with Associated Paths screen the status of the SnapVX type of step displays as Established or Restored, if source devices are restored from snapshot.

Snapshot unlinking can also be performed from the Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen using the Control Action UNLINK (Figure 78 on page 150).

Setting Time to Live

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **1. Work with Images** to display the Symmetrix Configuration by Image screen (Figure 6 on page 47).
2. Select option **2. Work with paths** to display the Work with Associated Paths screen (Figure 79 on page 152).
3. Enter **22** in the **Opt** column of a SnapVX type of step, to display the Work with Snapshots screen (Figure 81 on page 154).
4. Enter **3** in the **Opt** column of a snapshot to display the Set Time To Live (Figure 82 on page 154).

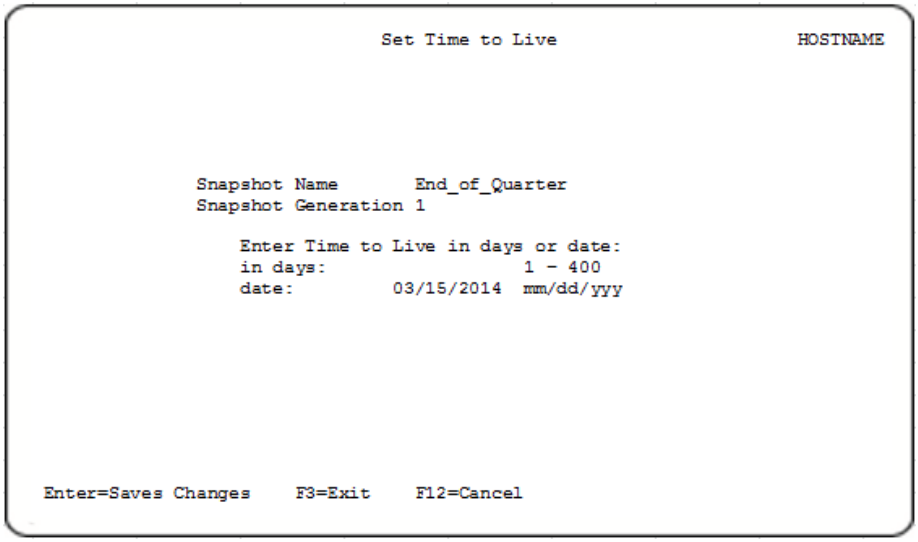


Figure 86 Set Time to Live screen

5. Enter the Time to Live in days or set a specific date. If you do not supply a value, the Time to Live value is set to unlimited and the snapshot is kept until it is manually terminated.

   Setting Time to Live can also be performed from the Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen using the Control Action SET_TTL (Figure 78 on page 150).

Restoring Snapshots

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **1. Work with Images** to display the Symmetrix Configuration by Image screen (Figure 6 on page 47).
2. Select option **2. Work with paths** to display the Work with Associated Paths screen (Figure 79 on page 152).
3. Enter **22** in trhe **Opt** column of a SnapVX type of step, to display the Work with Snapshots screen (Figure 81 on page 154).
4. Enter **2** in the **Opt** column of a snapshot to copy point-in-time data back to the original source device (Figure 82 on page 154). The snapshot must be fully established to perform a restore.

Note: For System Image, a restore is not allowed for the first step.

A restore can also be performed from the Work with Associated Paths (Figure 79 on page 152 screen, or by using the Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen using the control action RESTORE (Figure 78 on page 150).

Terminating Snapshots

Snapshots can only be terminated if there are no links (see "Unlinking Snapshots" on page 158).

1. On the Dell EMC SRDF/TimeFinder Manager for IBM i main menu, select option **1. Work with Images** to display the Symmetrix Configuration by Image screen (Figure 6 on page 47).
2. Select option **2. Work with paths** to display the Work with Associated Paths screen (Figure 79 on page 152).
3. Enter **22** in the **Opt** column of a SnapVX type of step, to display the Work with Snapshots screen (Figure 81 on page 154).
4. Enter **8** in the **Opt** column of the snapshot to be terminated.

   Note: If Snapshot has restores, only restores would be terminated. Use Option **8** again to terminate a snapshot.

Snapshot termination can also be performed from the Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen using the Control Action TERMINATE. (Figure 78 on page 150).

Monitoring TimeFinder operations

You can monitor the TimeFinder operations at the step level from the Work with Associated Paths screen or at the device level through the Disk Units screen. These procedures are in “Viewing devices” on page 101.

Monitoring invalid and % copied tracks

The Disk Unit screen and the Work with Associated Paths screen each have a column labeled **Invalids/%Copied**.

For SRDF or TimeFinder synchronization, this column lists the number of invalid tracks remaining during the synchronization operation. On the Disk Units screen invalid track counts display at the device level, and on the Work with Associated Paths screen invalid track counts display by image.

For TimeFinder/Clone, TimeFinder/Snap, and TimeFinder SnapVX operations, this column represents the percentage of copied tracks.

Note: The invalid tracks information is accurate only while synchronization is in progress. The invalid track information is not accurate while devices are Split or Activated.

For TimeFinder/SnapVX, when the COPY flag is set or when the source device is restored from the snapshot, this field displays percent of the total device size that has been copied to the target.

Automating TimeFinder operations

SRDF/TimeFinder Manager provides the option of incorporating TimeFinder control operations in customized CL programs. The CL command, CTLACT, includes the following TimeFinder control actions and parameters:

- TimeFinder/BCV control actions are as follows:
  - **ESTABLISH** — Establishes/re-establishes BCV or Clone emulation pairs for BCV control.
  - **SPLIT** — Splits BCV or Clone emulation pairs for BCV control.
  - **CANCEL** — Cancels the existing TimeFinder device relationship.
  - **STATUS** — Provides TimeFinder control status information.
- TimeFinder/Snap and TimeFinder/Clone control actions are as follows:
  - **CREATE** — Creates or recreates a virtual copy session for making Snap data copies between source devices and target virtual devices; or creates/recreates a clone copy session that defines and sets up the cloning devices selected for the clone operation.
  - **ACTIVATE** — Activates a virtual copy session or clone copy session to make data instantly accessible to target host.
  - **TERMINATE** — Terminates a virtual copy session or clone copy session to remove holds on target devices and delete device pairing information from the array.
- The following SPLIT_MODE parameter options are available for both the SPLIT command and the ACTIVATE command:
  - **SPLIT_MODE(RESTRICTED)** — Splits BCV, Clone emulation. Activates Clone, or Snap image while the system is in a restricted state.
  - **SPLIT_MODE(HOLD_JOBS)** — Holds all jobs and job queues before performing a Split or Activate control operation.
  - **SPLIT_MODE(SPLITWA)** — Splits a local or remote BCV or Clone emulation. Activates Clone, or Snap image while the source system is operational.
  - **SPLIT_MODE(MAINTENANCE)** — Splits BCV, Clone emulation. Activates Clone, or Snap image in maintenance mode (that is, no interference with host operations).
- TimeFinder/SnapVX control actions are:
  - **ESTABLISH** — Creates and activates a SnapVX snapshot. **SnapVX Establish Mode** can be set to:
    - **RESTRICTED** — Put a system in the restricted state by using option 5 from the Control menu using “Bring System to a Restricted State”, or with the STMENDSBS command. This ensures that all subsystems and jobs are ended except for the daemon jobs and their corresponding subsystems.
    - **CONSISTENT** — No interference with the host operations. By default all snapshots are consistent.

IMPORTANT

Use Consistent Establish with care as the data in main memory is not included in the image. Use this only when the host and applications are in a quiescent state. If that is unacceptable to your business needs, use other operating system functions like journaling and database commitment control to get an image current with the last transaction.

  - **HOLD_JOBS —** All jobs and job queues will be put on HOLD.

- **STATUS** — Provides control status information for the step in the associated path for the image. If status of the SVXn step is Linked, CopyInProg, Copied, Restored, RestoreInProg, RestNotPaired, or RestgNotPaired, the snapshot name and generation is provided. If Snapshot name and generation are specified, state of the specified snapshot is provided.
- **TERMINATE** — Removes an existing SnapVX snapshot.
- **LINK** — Presents the point-in-time data on target devices. If target devices are already linked, relink operation would be executed.
- **RENAME** — Changes the snapshot name.
- **RESTORE** — Copies the point-in-time data from a snapshot back to the original source device.

Note: If snapshot has restores, only restores are terminated.

- **SET_TTL** — Sets a time to live for a snapshot as either number of days from now or a future date from now.
- **SETCOPY** — Changes the copy mode of a link to “copy”.
- **SETNOCOPY** — Changes the copy mode of a link to “nocopy”.
- **UNLINK** — Removes the current link to target devices.

CHAPTER 6
Controlling SRDF

This chapter describes the procedures to control SRDF links, including how to enable and disable consistency protection for SRDF/A and set the RDF mode.

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Overview

SRDF/TimeFinder Manager relies on a specific sequence of operations to ensure proper execution of SRDF control operations. SRDF/TimeFinder Manager checks the configuration and operation considerations prior to executing a command that could change a device or system state.

SRDF Control operations are performed at the step level in the path for each image. These operations can be performed on both static and dynamic SRDF configurations, where:

- Static pairing occurs during initial array configuration
- Dynamic pairing occurs while the array is in operation.

Static RDF groups and devices are not supported on arrays running PowerMaxOS 5978 or HYPERMAX OS 5977.

Note: SRDF/TimeFinder Manager does not perform a discover process automatically, which can take several minutes to run per operation. After changing the configuration of a host or array, turn on the discover function or use the DISCOVER command. "Discovering array and host configuration details" on page 63 shows how to perform a discover operation.

RDF group/device compatibility

SRDF/TimeFinder Manager supports only the following array replication configurations when creating RDF groups between PowerMaxOS 5978 or HYPERMAX OS 5977 arrays and other Enginuity versions:

- PowerMaxOS 5978 or HYPERMAX OS 5977 array ↔ Enginuity 5876 array with Enginuity fix 67492
- PowerMaxOS 5978 or HYPERMAX OS 5977 array ↔ PowerMaxOS 5978 or HYPERMAX OS 5977 array

Device grouping/pairing rules and restrictions between Enginuity/HYPERMAX OS versions

- Static device pairing is available between arrays running Enginuity levels earlier than HYPERMAX OS 5977.
- For arrays running PowerMaxOS 5978 or HYPERMAX OS 5977, only dynamic SRDF groups and devices are supported, and, pairing of static SRDF groups/devices with dynamic SRDF groups or devices is not allowed.
- All SRDF groups created between arrays running PowerMaxOS 5978 or HYPERMAX OS 5977 and arrays runnning Enginuity 5876 can be created through the Solutions Enabler `symrdf` command or the SRDF/TimeFinder Manager "CTLRDFGRP"command, and only with dynamic RDF-capable devices.

  The `symrdf` command is supported on IBM i and is run from the IBM i command line or Qshell. See "Supported Solutions Enabler CLI commands" on page 262, and refer to *Solutions Enabler SYMCLI Reference Guide* for details of command actions and parameters, and command options.
- All devices on arrays running PowerMaxOS 5978 or HYPERMAX OS 5977 are reported as dynamic RDF-capable.

Adaptive copy write pending (ACWP) mode behavior with PowerMaxOS 5978 or HYPERMAX OS 5977

SRDF mode Adaptive Copy Write Pending (ACWP) is not available when the R1 mirror of the RDF pair is on an array running PowerMaxOS 5978 or HYPERMAX OS 5977. The CRTDLTRDFP, SETRDFMOD, and SETRDFMODC commands check the code level of the operating environment when you set the RDF mode to Adaptive Copy Write Pending.

SRDF/TimeFinder Manager sets the RDF mode to Adaptive Copy Disk mode when the R2 mirror of the SRDF pair is on an array running PowerMaxOS 5978 or HYPERMAX OS 5977 and you use:

- The CTLACTTGT command with the control action SWAP

or

- The CTLACTTGT or SRDFDRCTL commands with control action FAILOVER (with the ESTABLISH option or the RESTORE flag set)

Creating and controlling SRDF Groups

Before creating SRDF pairs, create dynamic SRDF groups. Use the CTLRDFGRP command to add, modify, or remove dynamic groups When adding dynamic SRDF groups, there are several parameters available that control the behavior of the group.

Creating dynamic SRDF groups

1. On the Control Menu, select option **10. Control Dynamic SRDF Groups** to display the Control Dynamic RDF Group (CTLRDFGRP) screen (Figure 87).

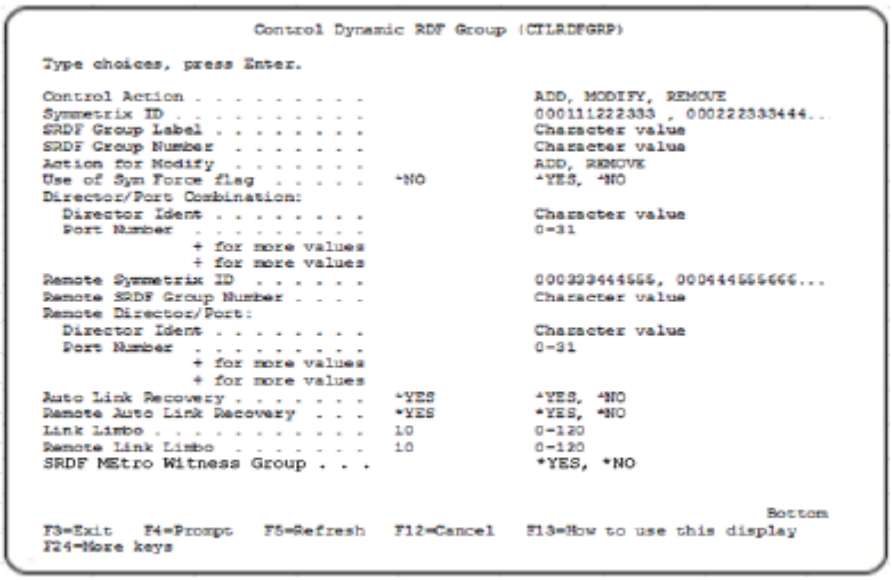


Figure 87 Control Dynamic RDF Group (CTLRDFGRP) screen

2. Provide the following information for the selected image:
   - **Control Action** — Enter **ADD.** (Create new RDF group)
   - **Symmetrix ID** — The local array ID where this command is performed.
   - **SRDF Group Label** — The RDF group label.
   - **SRDF Group Number** — The RDF group number.

     Note: For the ADD action specify both of the SRDF Group Label and SRDF Group Number parameters.

   - **SRDF Metro Witness Group** — Specify a value of ***YES** if this is a witness group for SRDF/Metro.

     There can be only one witness group between two arrays.

- **Director/Port Combination:**
  - **Director IDent** — The local array director ID, such as 1a, 1b and so on.
  - **Port Number** — The local array director port number. (0-31)
    Note: For PowerMaxOS 5978 or HYPERMAX OS 5977 RDF directors support up to 32 ports.
- **Remote Symmetrix ID** — Specifies remote array where this command is performed.
- **Remote RDF Group Number** — The remote RDF group number.
- **Remote Director/Port:**
  - **Director IDent** — The remote director ID, such as 1a, 1b and so on.
  - **Port Number** — The remote array director port number. (0-31)
    Note: For PowerMaxOS 5978 or HYPERMAX OS 5977 RDF directors support up to 32 ports.
- **Auto Link Recovery** — YES (default)
- **Remote Auto Link Recovery** — YES (default)
- **Link Limbo** — 10 (default)

  Note: Link Limbo — Only experienced SRDF users should use this option.

  Specifies the local link limbo value which is the length of time for PowerMaxOS, HYPERMAX OS, or Enginuity to wait, from the point of a link-down detection, prior to updating the link status as Down. If the link status is still sensed as Not Ready after the link limbo time expires, devices are marked Not Ready to the link. The link limbo range is 0-120 seconds. The default time is 10 seconds. This option is used only with the ADD control action.
- **Remote Link Limbo** — 10 (default). Same as link limbo except performed on a remote array.

3. Press **Enter**.

Modifying dynamic SRDF groups (associated/dissociated local and remote director/port with RDF group)

You also use the Control Dynamic RDF Group (CTLRDFGRP) screen (Figure 87 on page 168) to modify a dynamic group. When doing so, the following screen/command requirements and options are available:

- Control Action
- Symmetrix ID
- SRDF Group Label
- SRDF Group Number

  Note: You can specify either SRDF Group Label or Group Number.

- Actions for Modify — Add or Remove associated/dissociated local and remote director/port with the RDF group
- Director/Port Combination
- Remote Director/Port

Removing dynamic SRDF groups

You also use the Control Dynamic RDF Group (CTLRDFGRP) screen (Figure 87 on page 168) to remove a dynamic group. You can remove only an empty dynamic group. To remove a SRDF/Metro Witness group, there must be no SRDF/Metro groups using that witness group or there must be an alternative witness available for the SRDF/Metro groups to use. Alternatively, you can use the List SRDF Groups screen (Figure 103 on page 196) to remove a dynamic group.

When removing a dynamic group the following screen/command requirements and options are available:

- Control Action
- Symmetrix ID
- SRDF Group Label
- SRDF Group Number

  Note: You can specify either SRDF Group Label or Group Number.

- Use Sym Force Flag

Setting the SRDF group attributes

SRDF group behavior can be controlled by setting group attributes, such as whether SRDF/A DSE is automatically activated, when a SRDF/A session is activated, and how SRDF/TimeFinder Manager should handle situations where the transmission link is idling.

Note: For arrays running PowerMaxOS 5978 or HYPERMAX OS 5977, DSE autostart for SRDF groups is enabled by default.

Note: If the SRDF/A group consists of D910 devices, the FBA DSE pool should be associated with the group. If the group consists of the 2107 devices, the AS400 DSE pool should be associated with the group. If group has mixed device s (2107 and D910) both types of DSE pools (FBA and AS400) should be associated to the group. Spillover happens only to the DSE pool that has matching emulation.

Set the SRDF/A group as follows:

1. On the Control Menu, select option **11. Set RDF Group** to display the Set SRDF Group (SETRDFGRP) screen (Figure 88).

```
                        Set SRDF Group (SETRDFGRP)

Type choices, press Enter.

Symmetrix ID . . . . . .                 000111222333 , 000222333444...
SRDF Group Number  . . . . . .           Character value
Save Pool Name . . . . . .   *SAME       Character value, *SAME...
DSE auto start . . . . . .   *SAME       ENABLE, DISABLE, *SAME
DSE Threshold  . . . . . .               20-100
Transmit Idle  . . . . . .   *SAME       ENABLE, DISABLE, *SAME
Device-level Write Pacing  . .  *SAME    ENABLE, DISABLE, *SAME
Write Pace Autostart . . . . .  *SAME    ENABLE, DISABLE, *SAME
Write Pace Delay . . . . . .             1-1000000
Write Pace Threshold . . . . .           1-99
Hardware Compression . . . . .  *SAME    ENABLE, DISABLE, *SAME
Software Compression . . . . .  *SAME    ENABLE, DISABLE, *SAME
Auto Link Recovery . . . . . .  *SAME    ENABLE, DISABLE, *SAME
Link Limbo . . . . . .  *SAME            0-120, *SAME
Both Sides Flag  . . . . . .  *NO        *YES, *NO

                                                                    Bottom
F3=Exit   F4=Prompt   F5=Refresh   F12=Cancel   F13=How to use this display
F24=More keys
```

Figure 88 Set SRDF Group (SETRDFGRP) screen

2. Provide the following information:
   - **Symmetrix ID** — The name of the array on which this command is performed.

     Note: If you know the array ID, enter its name. Otherwise, press **F4** to display a list of all local and remote arrays attached to the host.

   - **SRDF Group Number** — The number of the RA group whose parameters should be modified.

- **SAVE Pool Name** — The name of the SRDF/A DSE pool to be associated with the SRDF group. Options are as follows:
  - ***SAME** — Use the pool that is currently associated.
  - **NO_POOL** — Disassociate a pool from the group. Until the pool has a guarantee that it has been drained for the group, the pool remains related to the group.

    Note: If more than one pool is associated to the group, all pools are disassociated.
  - **Pool name** — The name of the SAVE pool.

    Setting a new pool name for the group replaces the existing associated pool name. This automatically disassociates the old pool from the group. If there is data from the group spilled over in the pool, the old pool is put in the draining state.

    Note: Pools should be of the same type, if not the new pool is not associated to the group.

    Note: For arrays running PowerMaxOS 5978 or HYPERMAX OS 5977, you cannot associate a DSE pool with an RDF group. For this configuration, the SETRDFGRP command does not associate an RDF group with a DSE pool. The array is preconfigured to associate the DSE with a Storage Resource Pool (SRP), so managing pools is not required.
- **DSE auto start** — Whether SRDF/A DSE is automatically activated when the SRDF/A session is activated for the group. Options are as follows:
  - **ENABLE** — Automatically activates SRDF/A DSE when an SRDF/A session is activated for the group.
  - **DISABLE** — Does not automatically activate SRDF/A DSE when an SRDF/A session is activated for the group.
  - ***SAME** — Ignores the parameter during command processing or when ENABLE/DISABLE is specified and DSE auto start is already in the desired state.

    Note: For arrays running HYPERMAX OS 5977 or PowerMaxOS 5978, DSE autostart for SRDF groups is enabled by default.
- **DSE threshold** — Specifies the percentage of the array's write pending limit. DSE threshold is an optional parameter.

  Groups with higher threshold values keep their data in cache longer at the expense of groups with lower values. Valid values range from 20 to 100. If nothing is specified, or an SRDF group already has a desired DSE threshold, SRDF/TimeFinder Manager ignores this parameter during command processing.
- **Transmit idle** — Specifies how SRDF/TimeFinder Manager should handle situations where the transmission link is idling. Options are as follows:
  - **ENABLE** — Tolerates SRDF/A.
  - **DISABLE** — Drops SRDF/A.
  - ***SAME** — Ignores the parameter during command processing or when ENABLE or DISABLE is specified and Transmit Idle is already in the desired state.

- **Device-level Write Pacing** — Specifies whether SRDF/A device-level write pacing is automatically activated when SRDF/A session is activated for the group. Options are as follows:
  - **ENABLE** — Automatically activate SRDF/A device-level write pacing when SRDF/A session is activated for the group.
  - **DISABLE** — Do not automatically activate SRDF/A device-level write pacing when SRDF/A session is activated for the group.
  - ***SAME** — Ignores the parameter during command processing, or when ENABLE or DISABLE is specified and SRDF/A device-level write pacing is already in the desired state. This is the default.

    SRDF/A device-level write pacing feature is supported with the Enginuity 5876 or later.

    Enabling SRDF/A device-level write pacing is required when using RDF2 Async device as a Snap source.
- **Write Pace Autostart** — Specifies whether the SRDF/A group-level write pacing feature is automatically enabled or disabled when an SRDF/A session is activated for the SRDF group. Options are as follows:
  - **ENABLE** — Automatically activates SRDF/A group-level write pacing when an SRDF/A session is activated for the group.
  - **DISABLE** — Do not automatically activate SRDF/A group-level write pacing when an SRDF/A session is activated for the group.
  - ***SAME** — Ignores the parameter during command processing, or when ENABLE or DISABLE is specified, and SRDF/A group-level write pacing is already in the desired state. This is the default.
- **Write Pace Delay** — Sets the maximum host I/O delay, in microseconds, that triggers SRDF/A write pacing. Valid values are between 1 and 1000000 micro-seconds. Write pacing delay is an optional parameter. The default value is 50000 micro-seconds.

  If the write pace delay is not specified, or an SRDF group already has a desired write pacing delay, SRDF/TimeFinder Manager ignores this parameter during command processing.
- **Write Pace Threshold** — Sets the minimum percentage of the system write-pending cache at which the array begins pacing host write I/Os, for an SRDF group. Valid values are between 1 and 99. Write pacing threshold is an optional parameter. The default value is 60.

  If the write pace threshold is not specified, or an SRDF group already has a desired write pacing threshold, SRDF/TimeFinder Manager ignores this parameter during command processing.

Note: SRDF/A group-level write pacing feature is available on Enginuity 5876 or later and PowerMaxOS 5978 or HYPERMAX OS 5977. If you plan to swap the personalities of the R1 and R2 devices, configure the same SRDF/A write pacing values on both devices.

- **Hardware compression** — Specifies whether to enable or disable hardware compression for the SRDF group. Options are as follows:
  - **ENABLE** — Enables hardware compression for the SRDF group.
  - **DISABLE** — Disables hardware compression for the SRDF group.
  - ***SAME** — Ignores the parameter during command processing, or when ENABLE or DISABLE is specified, and compression for the SRDF group is already in the desired state. This is the default.
- **Software compression** — Specifies whether to enable or disable software compression for the SRDF group. Options are as follows:
  - **ENABLE** — Enables software compression for the SRDF group.
  - **DISABLE** — Disables software compression for the SRDF group.
  - ***SAME** — Ignores the parameter during command processing, or when ENABLE or DISABLE is specified, and compression for the SRDF group is already in the desired state. This is the default.

  Compression minimizes the amount of data to be transmitted over an SRDF link. Enabling hardware compression on GigE directors and software compression on Fibre and GigE directors using Solutions Enabler is available with Enginuity 5876 or later.

  Rules and restrictions for compression are as follows:
  - Enabling hardware compression on Fibre directors is available with Enginuity 5876 or later.
  - Software and hardware compression can be enabled on the R1 and R2 sides of the SRDF link, but the actual compression takes place on the side initiating the I/O, which is typically the R1 side. Since most I/O requests are initiated from the R1 side, make sure compression is enabled on this side.
- **Auto Link Recovery** — Specifies whether to use the Auto Link Recovery feature locally.
  - **ENABLE** — Enables Auto Link Recovery.
  - **DISABLE** — Disables Auto Link Recovery.
  - ***SAME** — Ignores this parameter during command processing, if *SAME is specified, or if ENABLE or DISABLE is specified and Auto Link Recovery is in the desired state. This is the default.
- **Link Limbo** — Specifies the length of time for Enginuity to wait from the point of link-down detection before actually updating the link status as down. If the link status is still sensed as Not Ready after the link limbo time expires, devices are marked Not Ready to the link. If nothing is specified, Ignores the parameter during command processing, if *SAME is specified, or if RDF group already has a desired link limbo value. The link limbo range is 0-120 seconds. The default time is 10 seconds.

  Note: Only experienced SRDF users should use this option.

- **Both Sides Flag** - Specifies whether changes should be applied only locally or both locally and remotely:
  - *__YES__ — Changes are applied locally and remotely.

    Note: If the remote side cannot be reached and the flag is set to YES, the command fails.
  - *__NO__ — Changes are applied locally. This is the default.

Dynamic SRDF pairs

Before creating SRDF device pairs:

- Determine the device pairings and create a device file that lists R1/R2 pairs. Refer to "Creating a device file" on page 176.
- Review "Dynamic pairing restrictions" on page 178.

Creating a device file

The device file is a record file and the maximum length of each record is 40.

Device file rules are as follows:

- The file contains two columns, column 1 lists the source devices and column 2 lists the target devices.
- Each device pair appears on a separate line separated by a space.
- The file can include comment lines and they begin with the pound sign (#).
- The device file resides in the library list.

Use the following command to create the device file:

```
CRTPF FILE(QGPL/file_name) RCDLEN(40)
```

Example device file

```
# PAIRING FIRST STEP OF CASCADING RDF
08C6 1167
08CA 116B
08CE 116F
```

Note: If creating concurrent SRDF pairs, refer to "Rules for dynamic concurrent SRDF pairing" on page 178.

Creating pairs

Use the Create/Delete RDF pairs screen to create SRDF pairs. There are two ways to access this screen:

- Through the Control Menu screen:

  On the Control Menu screen select option **12. Create/Delete RDF Pairs** to display the Create/Delete RDF pairs screen (Figure 89 on page 177).

- Using the CRTDLTRDFP command from the command prompt:

  ```
  CRTDLTRDFP
  ```

  The Create/Delete RDF pairs screen appears (Figure 89 on page 177).

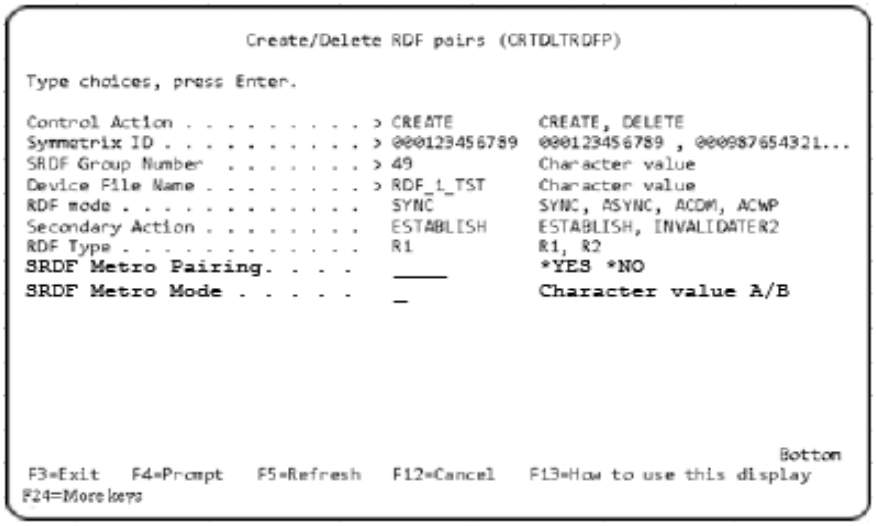


Figure 89 Create/Delete RDF pairs with Control Action —CREATE

Enter parameters (refer to "Create/Delete RDF pairs screen parameters") and press **Enter**.

Upon execution of the CRTDLTRDFP command with the **CREATE** control action the following actions occur:

- SRDF pairs become members of the specified RDF group.
- Pairing information is added to the SYMAPI database file on the host.

Create/Delete RDF pairs screen parameters

The following are some of the screen parameter details:

- **Symmetrix ID —** Specifies the array associated with the devices listed in the first column of the device file.
- **SRDF Group Number —** Specifies the SRDF group number associated with the devices listed in the first column of the device file.
- **Device File Name —** Specifies the name of the device file that defines the SRDF pairs.
- **SRDF Metro Pairing** — Specify a value of ***YES** if this pair is a SRDF/Metro device pair.
- **SRDF Metro Mode** — Specify a value of **A** if the SRDF/Metro configuration uses Witness protection or a value of **B** if the pair uses Bias protection. This value is relevant only when SRDF Metro Pairing has a value of ***YES**.
- **RDF mode —** Sets device pair SRDF mode to synchronous (**SYNC**), asynchronous (**ASYNC**), adaptive copy disk mode (**ACDM**), or adaptive copy write pending mode (**ACWP**).

Note: SRDF mode adaptive copy write pending (ACWP) is not available when the R1 mirror of the SRDF pair is on a array running PowerMaxOS 5978 or HYPERMAX OS 5977. For this configuration, the commands used to set the RDF mode return an error if the mode is set to adaptive copy write pending. In this case use another mode.

- **Secondary Action** options:
  - **ESTABLISH** (default option) – Begins copying data to the invalidated targets, synchronizing the dynamic SRDF pairs, listed in the device file, once the SRDF pairs are created.

    Note: When RDF Mode is **ASYNC** and the **ESTABLISH** option is selected, consistency protection is automatically enabled for the new device pair after executing a **CREATE** operation. Also, if the SRDF group for the new devices already contains devices, the CONS_EXEMPT (consistency exempt) flag is automatically enabled. The CONS_EXEMPT flag allows devices to be added to the SRDF group without affecting the state of the SRDF/A session, or requiring that other devices in the session be suspended. Consistency protection is automatically enabled for the new device pair if it is already enabled for other devices in the group.

  - **INVALIDATER2 —** Marks the R2 devices in the device file list to be the invalidated target for a full device copy once the SRDF pairs are created.
- **RDF Type —** Indicates whether the device in the first column of the device file is an R1 or an R2 device.

Rules for dynamic concurrent SRDF pairing

The following rules apply when creating dynamic concurrent SRDF pairs:

- Remote BCVs that have been designated as dynamic SRDF devices are not supported.
- The two SRDF mirrors of the concurrent device must be assigned to different RA groups.
- The concurrent dynamic SRDF, dynamic SRDF, and concurrent SRDF states must be enabled in the array configuration.

Two separate device files must be created when dynamically creating a second remote mirror using the CRTDLTRDFP command. One file lists the first set of R1/R2 device pairs, and the other device file lists the same R1 devices paired with different remote R2 devices. Refer to "Creating a device file" on page 176.

Dynamic pairing restrictions

The **CREATE** option fails if any of the following apply:

- The device is in one of the following BCV pair states: Synchronized, SyncInProg, Restored, RestoreInProg, or SplitInProg.
- The device is the source or target of a TimeFinder/Snap operation.
- There is a background BCV split operation in progress.

- Devices are in the back-end Not Ready state.
- The type, size of the devices, configuration are not the same, such as IBM i which has specific pairing rules.

  Note: For migration, larger R2 devices are allowed for D910 devices.

- There are existing, local invalid tracks on either the local or remote device.
- The SRDF group is in asynchronous mode and the devices being added are not the same SRDF type R1 or R2.
- The SRDF group is in asynchronous mode with the SRDF links suspended and the ESTABLISH option is selected.
- The SRDF group is enabled for SRDF consistency.
- If the operation would result in more than 64K devices being supported by any RA associated with either of the RDF groups.

Deleting SRDF pairs

1. Before deleting SRDF pairs:
   - Determine the device pairs to be deleted and create a device file that lists R1/R2 pairs to be deleted. Refer to "Creating a device file" on page 176.
   - Review "Restrictions for deleting pairs" on page 180.
2. Navigate to the Create/Delete RDF pairs screen (Figure 90).

   Steps to access this screen are the same as in creating RDF pairs. Refer to "Creating pairs" on page 176.

   Note: Before executing a DELETE operation, ensure that the SRDF pairs are in a Split, Suspended, or FailedOver state. If the **Use of Force Flag** parameter is set to **YES**, the delete operation is attempted regardless of the state of the SRDF devices.

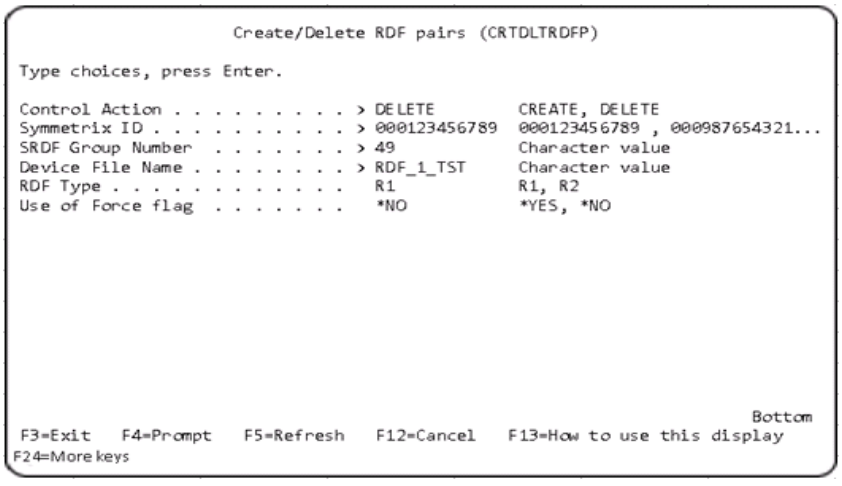


Figure 90 Create/Delete RDF pairs with Control Action — DELETE

3. Enter parameters (refer to "Create/Delete RDF pairs screen parameters" on page 177) and press **Enter**.

   Upon execution of the CRTDLTRDFP command with the **DELETE** option, the following actions occur:

   - SRDF pairs are canceled.
   - Pairing information is removed from the array and the SYMAPI database.
   - Devices are changed to non-SRDF devices (except for concurrent SRDF pairs).

Restrictions for deleting pairs

The **DELETE** operation fails if any of the following apply:

- The device is in one of the following BCV pair states: Synchronized, SyncInProg, Restored, RestoreInProg, SplitInProg, Updated, or UpdateInProg.
- The device is the source or target of a TimeFinder/Snap operation.
- There is a background BCV split operation in progress.
- Devices are in the back-end Not Ready state.
- There are existing local invalid tracks on either the local or remote device.
- There are remote invalid tracks and the FORCE flag was not specified.
- The links are not suspended.

Deleting half pairs

If a delete operation for a SRDF device pair is initiated and the remote array has been replaced, or if the status of an SRDF pair is partitioned a confirmation screen (Figure 91) appears.

Note: If running in SRDF asynchronous mode and consistency is enabled, the consistency is disabled and then delete half pair is performed.

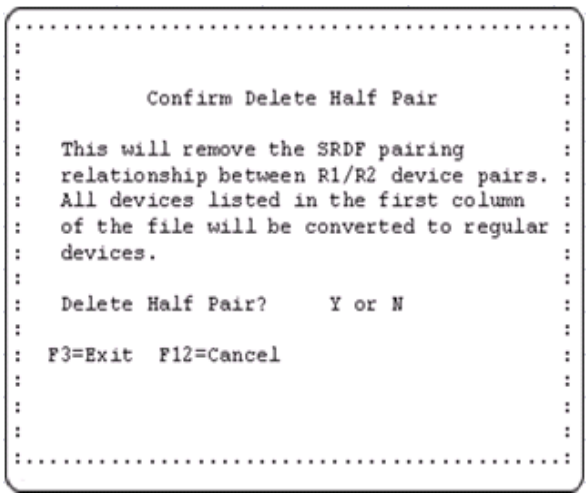


Figure 91 Confirm Delete Half Pair screen

Setting modes of SRDF operation

SRDF modes of operation are SRDF/Synchronous (SYNC), SRDF/Asynchronous (ASYNC), adaptive copy disk mode (ACDM), adaptive copy write pending (ACWP), and Metro.

Note: SRDF mode adaptive copy write pending (ACWP) is not available when the R1 mirror of the SRDF pair is on a array running PowerMaxOS 5978 or HYPERMAX OS 5977. For this configuration, the commands used to set the SRDF mode return an error if the mode is set to adaptive copy write pending. In this case use another mode.

To set the SRDF mode for a selected image, path:

1. On the Control Menu, select option **4. Set SRDF mode** to display the Set SRDF Mode (SETRDFMOD) screen (Figure 92).

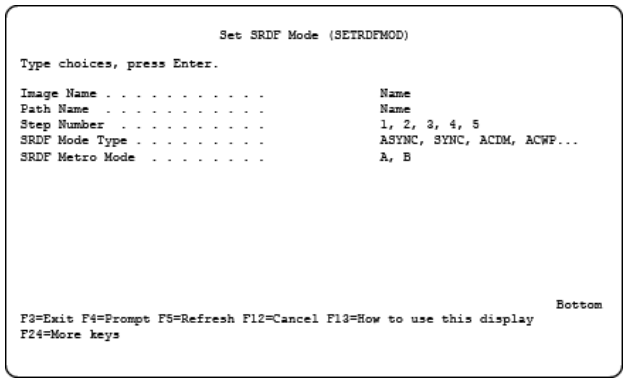


Figure 92 Set SRDF Mode (SETRDFMOD) screen

2. Provide the characteristics of the selected image:
   - **Image Name** — The name of the image to perform a control action on.
   - **Path Name** — The path name associated with the image on which to perform a control action.
   - **Step Number** — The step number for which the control action is to be initiated.
   - **SRDF Mode Type** — The mode to use for SRDF operations. The options are:
     - **SYNC** enables SRDF/Synchronous mode.
     - **ASYNC** enables SRDF/Asynchronous mode.
     - **ACDM** enables adaptive copy disk mode.
     - **ACWP** enables adaptive copy write pending mode.
     - **METRO** enables SRDF/Metro mode.

       Note: SRDF mode Adaptive Copy Write Pending (ACWP) is not available when the R1 mirror of the RDF pair is on an array running PowerMaxOS 5978 or HYPERMAX OS 5977.
   - **SRDF Metro Mode** — Specifies whether a SRDF/Metro configuration uses witness or bias protection. Enter **A** to use witness protection or **B** to use bias protection.

     Set a value for this parameter only when the value of **SRDF Mode Type** is **METRO**.
3. Press **Enter**.

If you set the SRDF mode to **METRO**, and the protection mode to **A** (witness protection) you need to:

1. Create a witness group (see "Creating dynamic SRDF groups" on page 168).
2. Suspend the SRDF link (see "Splitting/Suspending SRDF links operation" on page 184).
3. Perform a an establish control action on the link (see "Establishing SRDF links" on page 183).

IMPORTANT

For SRDF, do not use ACDM or ACWP mode while creating the RDF or remote TimeFinder images during split as this is not supported. Instead, for SRDF, use SYNC or ASYNC mode.

You can also use the SETRDFMOD command from the command line. Refer to "SETRDFMOD" on page 250 for more information.

Consistency protection in SRDF/A mode

In SRDF Asynchronous mode, consistency protection can be enabled or disabled for SRDF/A devices. When consistency protection is enabled, if data cannot be copied from the R1 to the R2, all devices in the group are made Not Ready on the link to preserve the R2 data consistency.

In Asynchronous mode, consistency protection is automatically enabled after the control action Resume or Establish is run. When the control action Split or Suspend is run, consistency protection is disabled and the Split or Suspend operation is performed. If consistency protection is enabled for the devices, consistency protection must be disabled before you attempt to change the mode from Asynchronous.

Enabling consistency protection

To enable SRDF/A consistency protection:

1. On the Control Menu, select option **1. SRDF or TimeFinder Control** to display the Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen.
2. Provide the **Image Name**, **Path Name,** and **Step Number**
3. Select the control action **ENABLE.**

   A message appears indicating that the enable consistency protection operation started successfully.

Disabling consistency protection

To disable SRDF/A consistency protection:

1. On the Control Menu, select option **1. SRDF or TimeFinder Control** to display the Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen.
2. Provide the **Image Name, Path Name**, and **Step Number**.
3. Select the control action **DISABLE**.

   A message appears indicating that the disable consistency protection operation started successfully.

Establishing SRDF links

To establish SRDF links:

1. On the Control Menu, select option **1. SRDF or TimeFinder Control** to display the Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen.
2. Provide the **Image Name**, **Path Name**, and **Step Number**.
3. Select the control action **ESTABLISH**.

   The Confirm Establish SRDF Link screen (Figure 93) appears.



Figure 93 Confirm Establish SRDF Link screen

4. Type **Y** to continue.

   SRDF/TimeFinder Manager updates SRDF device data and displays a message indicating that the Establish SRDF links processing initiated successfully.

Splitting/Suspending SRDF links operation

Both the Split and Suspend operations require I/O to be suspended on the host to create a frozen image.

To suspend I/O on SRDF links so that you can create a frozen image for backup operations, break the SRDF links between source (R1) and target (R2) volumes. Figure 94 shows this operation.

Restricted state and daemon jobs

When using split or suspend the user can select to hold all jobs or bring the system to a restricted state. When selecting the hold jobs option, all jobs in the system are held except for the daemon jobs Storapid and Storgnsd.

Additionally, when selecting the restricted state option the subsystem where Storapid and Storgnsd are running is not ended. Other jobs that are running in that same subsystem are ended using the STMENDSBS command. This command ends all subsystems.

During a split, The STMENDSBS command checks status of the subsystems for the restricted state. The split fails and the user is asked to hold jobs, if either of the following conditions exist:

- There are other jobs running in the subsystems where the daemon jobs are running.
- There are subsystems running, other than the subsystems where the daemon jobs are running.

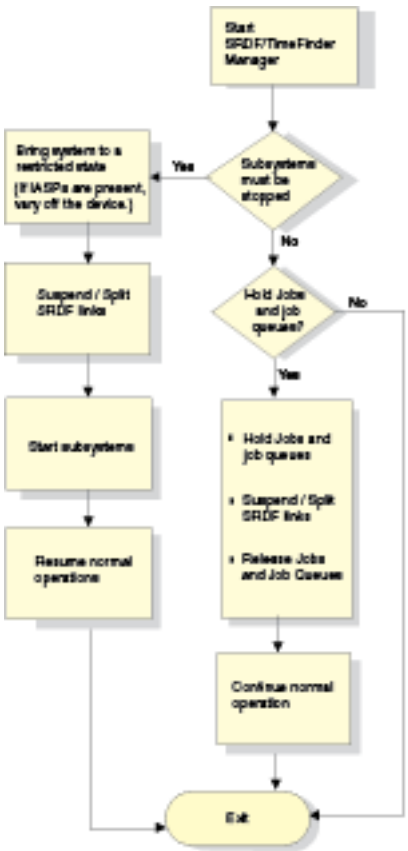


Figure 94 Split or Suspend SRDF links operation

The following sections describe the various modes for the Split/Suspend operation.

Splitting/Suspending SRDF links: Restricted state

The following procedure uses the restricted state to ensure that I/O activity is halted before the SRDF link is suspended or split.

Depending on the configuration, the time it takes for a host to go into a restricted state can vary. By default, SRDF/TimeFinder Manager waits 1800 seconds for control to return, after issuing the command to bring the system to a restricted state. If additional time (in seconds) is required to bring the system to a restricted state, use the procedure "Setting the restricted state wait time" on page 60.

For IASP images, see "SRDF/TimeFinder Manager Extended Features" on page 301.

To split/suspend SRDF links in a restricted state:

1. Bring the IBM i host to a restricted state:
   a. Prepare for bringing the system to a restricted state, using internal procedures established by your company.
   b. On the Control Menu, select option **5. Bring system to a restricted state** to display the STM End Subsystem Confirmation screen.
   c. Press **F10** to bring the IBM i host to a restricted state.

   The STM End All Subsystems (STMENDSBS) screen (Figure 95) appears.

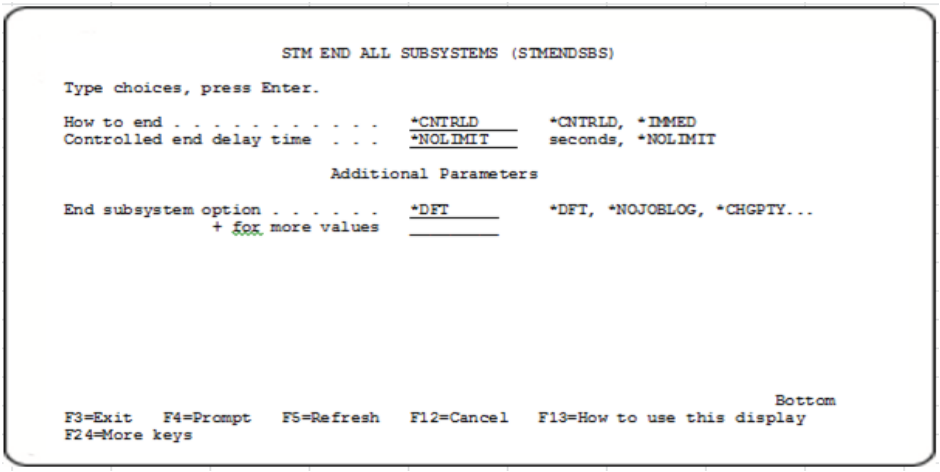


Figure 95 STM End All Subsystems (STMENDSBS) screen

   d. Select options for **How to end** and **Controlled end delay time,** and press **Enter**.

      After the End Subsystem operation completes, the Control Menu appears.

2. Split/suspend SRDF links:
   a. From the Control Menu, select option **1. SRDF or TimeFinder Control** to display the Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen.
   b. Provide the **Image Name**, **Path Name,** and **Step Number**.
   c. Select the **SPLIT** or **SUSPEND** control action.

d. The Dell EMC SRDF/TimeFinder Control (CTLACT) screen reappears with a Split/Suspend Mode prompt.

e. Enter **RESTRICTED** for the **SPLIT_MODE** parameter.

SRDF/TimeFinder Manager checks to see if the system is in a restricted state. If the system is not in a restricted state, SRDF/TimeFinder Manager displays the Confirmation Window screen (Figure 96) asking if you want to hold jobs and job queues.

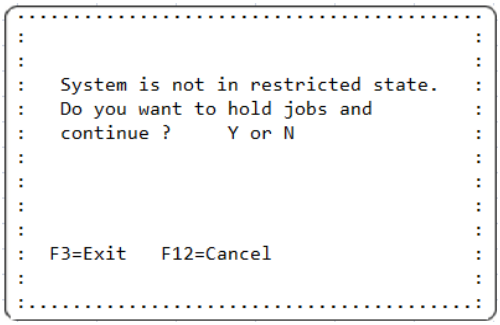


Figure 96 System not in restricted state screen

To continue the operation in **HOLD_JOBS** mode, refer to "Suspend/Split SRDF links: Hold jobs and job queues" on page 187 and take the appropriate action.

A message screen appears indicating that the Split SRDF links operation started successfully.

SRDF/TimeFinder Manager attempts to perform the SRDF links operation on all ASPs in the image. ASPs that are not eligible for the operation must be removed (see "Removing ASPs from an image" on page 81).

3. Start subsystems:

   a. On the Control Menu, select option **6. Start subsystems**.

      The Start Subsystems command resumes normal IBM i working processes from the IBM host restricted state.

      Note: The start subsystems command checks to see if the daemon jobs are the only jobs running in their dedicated subsystems. If the system is not in a restricted state, or there are other jobs running besides the daemon jobs, a message appears. This command then brings the system into a true restricted state by ending all subsystems including daemon jobs. When this is complete, the controlling subsystem is started. This should start all the required subsystems and jobs, depending on what jobs are defined as auto start jobs and what is set as a startup program defined in the system value QSTRUP. If you use additional start-up procedures, run them separately. Start Subsystems does not power cycle the IBM i host or restart the OS/400 operating system.

   b. Resume normal operations.

Suspend/Split SRDF links: Hold jobs and job queues

The following procedure holds jobs and job queues to ensure that I/O activity is halted prior to the Split or Suspend SRDF links operation.

⚠CAUTION

System activity that results in a write operation can continue while jobs and job queues are on hold. For example, a restore job could have write operations continue while it is held. If you are not sure that the hold job and job queues option completely suspend write operations to the selected image, bring the system to a restricted state before running a Split or Suspend operation. "Splitting/Suspending SRDF links: Restricted state" on page 185 contains instructions on how to do this.

To split or suspend SRDF links with jobs and job queues on hold:

1. On the Control Menu, select option **1. SRDF or TimeFinder Control** to display the Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen.
2. Provide the **Image Name, Path Name**, and **Step Number**.
3. Select the **SPLIT** or **SUSPEND** control action.

   The Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen reappears with a Split/Suspend mode prompt.
4. Enter **HOLD_JOBS** for the **SPLIT_MODE** parameter.

   Note: The SRDF link status for the image must be in an eligible state.

   SRDF/TimeFinder Manager repeatedly attempts to put all jobs and job queues on hold and then executes the Split or Suspend operation.

   When one or more jobs or job queues remain active, SRDF/TimeFinder Manager displays the Confirmation Window screen (Figure 97).

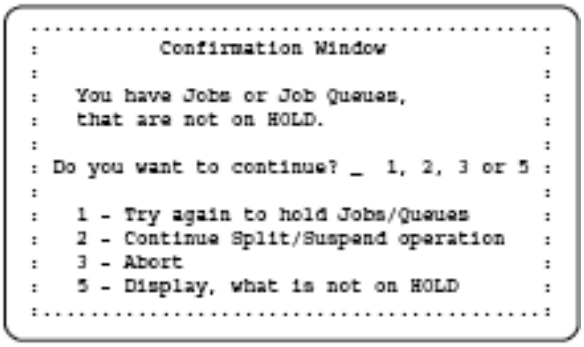


Figure 97 Confirmation Window screen

5. Select the appropriate option:
   - **1 - Try again to hold Jobs/Queues** — SRDF/TimeFinder Manager repeatedly attempts to hold all jobs and job queues.

     If any job or job queues remain not *on hold*, the Confirmation screen reappears.
   - **2 - Continue Split/Suspend operation** — Accept that some jobs remain active and continue with the Split or Suspend operation.

⚠CAUTION

Selecting the Continue Split/Suspend operation acknowledges that some write activity can continue during the Split operation. If any writes occur to the specified ASP during the Split operation, the resulting copy is not valid.

   - **3 - Abort** — Do not perform the Split or Suspend operation and return to the Control Menu.
   - **5 - Display, what is not on HOLD** — Show a list of the jobs and job queues that SRDF/TimeFinder Manager cannot hold (Figure 98).

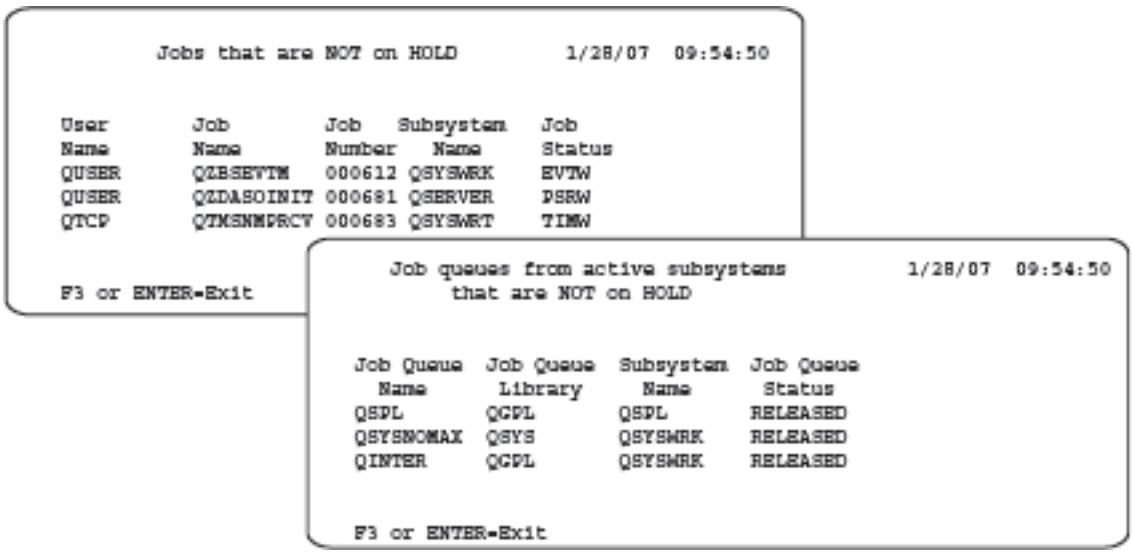


Figure 98 Jobs Not on Hold and Job Queues Not on Hold screens

6. Press **F3** or **Enter** to return to the Confirmation Window screen.

   After all jobs and job queues are successfully on hold, or if option **2 - Continue Split/Suspend operation** is selected from the Confirmation Window screen, the Suspend or Split operation is performed. A message screen appears indicating that the Split or Suspend SRDF links operation started successfully.

   After the Split or Suspend operation is complete, all jobs and job queues held by SRDF/TimeFinder Manager are released.
7. Use the Work with Associated Paths screen or the Disk Units screen to monitor and verify the change in links status.

Splitting/Suspending SRDF links: Maintenance mode

Performing a Suspend/Split operation in maintenance mode suspends or splits the SRDF links without interfering with the source host operations.

Unless you employ another method of suspending write operations, the Split or Suspend SRDF Links maintenance operation results in an R2 copy that is not usable.

Note: If you use an R2 copy in a simple SRDF R1/R2 configuration, perform a Split or Suspend with Restricted state or a Split or Suspend with HOLD_JOBS.

To suspend or split SRDF links in maintenance mode:

1. On the Control Menu, select option **1. SRDF or TimeFinder Control** to display the Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen.
2. Provide the **Image Name**, **Path Name**, and **Step Number**.
3. Select the **SPLIT** or **SUSPEND** control action.

   The Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen reappears with a Split/Suspend mode prompt.
4. Enter **MAINTENANCE** for the **SPLIT_MODE** parameter.

   The Split SRDF links or Suspend SRDF links operation performs without interfering with the source host operations.

Resuming SRDF operation

SRDF/TimeFinder Manager on the source IBM i host is used to restore SRDF operation by resuming the links between source (R1) and target (R2) volumes.

Note: The Resume SRDF links operation cannot be used when the SRDF links status is Split.

⚠CAUTION

Resuming the SRDF links before backup is complete may result in unusable backup tapes or save files. Use the Work with Associated Paths screen or the Disk Units screen to monitor and verify the change in links status.

To resume SRDF links:

1. On the Control Menu, select option **1. SRDF or TimeFinder Control** to display the Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen.
2. Provide the **Image Name, Path Name,** and **Step Number**.
3. Select the control action **RESUME.**

The Confirm Resume Link screen (Figure 99) appears.

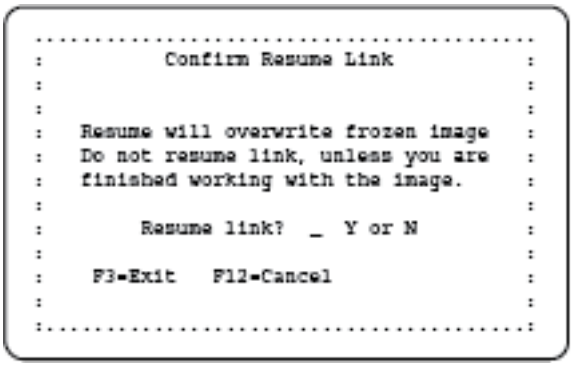


Figure 99 Confirm Resume Link screen

Note: Make sure that the R2 image is not in use on the target system before resuming the SRDF links. Backup can be performed only while the links are suspended or split (devices are only read/write accessible when they are split). Resuming links while backup is in progress has an unpredictable effect on the backup operation.

4. Type **Y**.

   A message appears indicating that the Resume operation started successfully.

Controlling SRDF/Metro

This section shows how to set up and manage SRDF/Metro configurations within SRDF/TimeFinder Manager. The *Dell EMC SRDF Introduction* has information on SRDF/Metro and its capabilities.

Setting up an SRDF/Metro configuration

To set up an SRDF/Metro configuration:

1. If the configuration is to use Array witness resilience, create a Witness group between each array in the SRDF/Metro configuration and the Witness array (see "Creating dynamic SRDF groups" on page 168).
2. If the configuration is to use vWitness resilience, create the vWitness instances on both arrays in the SRDF/Metro configuration (see "Maintain vWitness information" on page 192).
3. Create the SRDF groups on both SRDF/Metro arrays to hold the device pairs (see "Creating dynamic SRDF groups" on page 168).
4. Add the device pairs to the SRDF groups specifying them as SRDF/Metro pairs and using witness or bias, as appropriate (see "Creating pairs" on page 176).
5. Once all the devices are in a steady state (Active/Active or Active/Bias) present the R2 devices to the IBM System i host using map and mask operations.

Note: SRDF/TimeFinder Manager can use only SRDF/Metro configurations created within SRDF/TimeFinder Manager. It cannot use SRDF/Metro configurations created by Dell EMC Solutions Enabler or Unisphere.

Removing an SRDF/Metro configuration

To remove an SRDF/Metro configuration:

1. Suspend the SRDF link (see "Splitting/Suspending SRDF links operation" on page 184).
2. Delete the SRDF pair (see "Deleting SRDF pairs" on page 179).
3. (Optional) Delete the SRDF group (see "Removing dynamic SRDF groups" on page 170).

   Alternatively, you can keep the group for later reuse.
4. Make the R2 devices inaccessible to the host using unmap and unmask operations.
5. Enable the R2 devices using the ENABLEDEV command for future use (see "ENABLEDEV" on page 230).

Controlling witness and bias settings for SRDF/Metro

Maintain vWitness information

Use the vWitness Control Action screen to maintain vWitness information. There are two ways to access this screen:

- Through the Control Menu screen:

  From the Control Menu screen, select option **13. Create/Delete vWitness Info** to display the vWitness Control Action screen (Figure 100).

- The CTLVWIT command from the command prompt:

  ```
  CTLVWIT
  ```

  The vWitness Control Action screen appears (Figure 100).

```
                    vWitness Control Action (CTLVWIT)

Type choices, press Enter.

Symmetrix ID . . . . . .              000194900937 ...
Witness Name . . . . . .
Control Action . . . . . .            ADD, REMOVE, ENABLE, DISABLE
Location . . . . . .
Use of Force flag  . . . . .   *NO     *YES, *NO
Use of Sym Force flag  . . . .   *NO    *YES, *NO

                                                                  Bottom
F3=Exit  F4=Prompt  F5=Refresh  F12=Cancel  F13=How to use this display
F24=More keys
```

Figure 100 vWitness Control Action screen

Provide the following information:

- **Symmetrix ID** — Enter the identifier of the storage array.
- **Witness Name** — Enter a name for the vWitness definition:
  - The name has up to 12 characters and starts with an alphabetic character.
  - The rest of the name can contain alphanumeric characters, underscores and hyphens.
- **Control Action** — Enter the required command:
  - **ADD** — in addition, supply **Symmetrix ID**, **Witness Name**, and **Location**.
  - **REMOVE** — in addition, supply **Symmetrix ID** and **Witness Name**.
  - **ENABLE** — in addition, supply **Symmetrix ID** and **Witness Name**.
  - **DISABLE** — in addition, supply **Symmetrix ID**, **Witness Name**, **Use of Force flag** (optional), and **Use of Sym Force flag** (optional).
- **Location** — Enter the IP address or fully-qualified DNS name of the system where the vWitness instance is installed.

- **Use of Force flag** — This applies only when disabling a vWitness instance. Enter ***YES** when the instance is in use and there is another witness (virtual or physical) that can take over.
- **Use of Sym Force flag** — This applies only when disabling a vWitness instance. Enter ***YES** when the instance is in use that there is no other witness (physical or virtual) to take over.

To modify a vWitness definition, disable it, remove it, add the new definition and enable that.

Note: vWitness instances are defined outside SRDF/TimeFinder Manager. However, add a vWitness definition using the CTLVWIT screen for each of those instances that you want to use in an SRDF/TimeFinder Manager SRDF/Metro configuration. The *Dell EMC SRDF/Metro vWitness Configuration Guide* shows how to manage vWitness instances.

View vWitness information

Use the Display vWitness Information screen to view vWitness definitions on a storage array. There are two ways to access the screen:

- Through the Control menu screen:

  From the Control menu screen select option 14. Display vWitness Info (Figure 101)

- The DSPVWIT command:

  DSPVWIT

  the Display vWitness Information appears (Figure 101)

```
DSPVWIT              Display vWitness Information                  BONAIRE


Symmetrix     vWitness Name  Status  Port   vWitness Location
                             EAUI
000196701854  STMTEST        YNNN    10123  10.243.148.121
000196701855  STMTEST        YNNN    10123  10.243.148.121

                                                                   Bottom
F3=Exit   F5=Refresh   F9=Command Entry   F12=Cancel
```

Figure 101 Display vWitness Information

The screen shows the characteristics of the vWitness definitions that have been defined:

- **Symmetrix** — the identifier of the appropriate storage array.
- **vWitness Name** — the name of the vWitness definition.
- **Status** — the state of the definition.
- **Port** — the number of the IP port that the vWitness instance is using.
- **vWitness location** — the IP address of the vWitness instance.

Change the bias setting

A SRDF/Metro configuration can use bias to determine which array remains available to the host when there is a failure on the SRDF links. Use the SRDF Metro Set Bias screen (Figure 102) to define which array is on the bias side. Use the SETBIAS command to access this screen:

```
SETBIAS
```

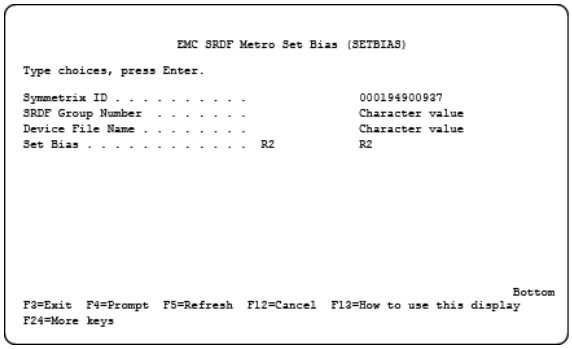


Figure 102 SRDF Metro Set Bias screen

Provide the following information:

- **Symmetrix ID** — The identity of the storage array that is to become the R2 side.
- **SRDF Group Number** — The identity of the SRDF group on the storage array whose bias setting you want to change.
- **Device File Name** — The name of the file that contains the identities of devices whose bias setting you want to change.
- **Set Bias** — which side of the pair is to become the bias side.

Monitoring SRDF operations

You can monitor the SRDF operations at the step level on the Work with Associated Paths screen or at the device level through the Disk Units screen. See "Viewing devices" on page 101.

Monitoring invalid and % copied tracks

The Disk Unit screen and the Work with Associated Paths screen have a column labeled **Invalids/%Copied**.

For SRDF or TimeFinder synchronization, this column lists the number of invalid tracks remaining during the synchronization operation. On the Disk Units screen invalid track counts display at the device level, and on the Work with Associated Paths screen invalid track counts display by image.

For TimeFinder/Clone, TimeFinder/Snap, and TimeFinder/SnapVX operations, this column represents the percentage of copied tracks.

Note: The invalid tracks information is meaningful only while synchronization is in progress. The invalid track information is not accurate while SRDF links are Suspended or Split.

Recovering from NotConsistent SRDF state

SRDF/TimeFinder Manager performs control actions only on images in which all devices are DMX, VMAX Family, VMAX All Flash, or PowerMax array devices.

When SRDF device pairs in an image are at different states, such as when one pair is in a Synchronized state and another in a Suspended state, the RDF status for the step of the image becomes NotConsistent. To recover from this situation, using SRDF/TimeFinder Manager, run the Resume SRDF links or Establish SRDF links operation.

The Resume SRDF links or Establish SRDF links control action examines all devices in the image and checks the state of each SRDF device pair that belongs to that image.

If the state is eligible for the Resume SRDF links or Establish SRDF links control action, the operation starts for that SRDF pair. Otherwise, SRDF/TimeFinder Manager issues an error message to the job log providing the device numbers and their status information.

IMPORTANT

Contact the Dell EMC Customer Support Center for assistance with bringing devices back into an eligible state.

Listing and monitoring SRDF groups

On the Control Menu, select option **9. List SRDF groups**.

The List of the SRDF Groups screen (Figure 103) appears, and displays the RDF groups for all attached arrays. For SRDF/A mode, this screen lists the number of seconds that the data on the R2 device is behind the data on the R1 device. This screen also appears if the group participates in the STAR configuration.

```
RDFGRPLST              List of the SRDF Groups                      HOSTNAME

 *N - indicates that the RA group is STAR Normal mode
 *R - indicates that the RA group is STAR Recovery mode

 4=Remove the Group   5=Display the Group

 O Group Pool         Source       Target  Target Group  Target       SRDFA R2
     #   Name         Symmetrix    Group # Pool Name     Symmetrix    Behind R1
   02    N/A          000111222333  02     N/A           000222333444
   03    N/A          000111222333  03     N/A           000222333444
   04    N/A          000111222333  04     N/A           000222333444
   05    N/A          000111222333  05     N/A           000222333444
   06    N/A          000111222333  06     N/A           000222333444
   07    N/A          000111222333  07     N/A           000222333444
   08    N/A          000111222333  08     N/A           000222333444
   09    N/A          000111222333  09     N/A           000222333444
   0A    N/A          000111222333  0A     N/A           000222333444
   0B    N/A          000111222333  0D     N/A           000222333444
   0C    N/A          000111222333  0C     N/A           000222333444
   20    N/A          000111222333  21     N/A           000222333444
                                                                  More...
 F3=Exit  F5=Refresh  F9=Command Entry  F12=Cancel
```

Figure 103 List of the SRDF Groups screen

Displaying SRDF group details

To display details for a SRDF group, on the List of SRDF Groups screen type **5** (Display the Group) next to the desired group.

The RDF Group Information screen (Figure 104) appears showing information about the local group and remote group.

```
RDFGRPD                    SRDF Group Information                    HOSTNAME
 Group Label GRP4                          Group Type Dynamic
 Group # 04      Symmetrix 000111222333    Number Devices in a Group 10
 Link Config. Fibre                        Fibre Adapter Type Switch
 Auto Link Recovery  Enabled               Link Limbo 10
 DSE:   Pool Name N/A                      Autostart Disabled    Threshold 50
 Compression:               SW Disabled       HW Disabled
 Group-Level Write Pacing:  Autostart Disabled  Max Delay 50000    Threshold 60
 Device-level Write Pacing: Autostart Disabled  State Inactive
 Transmit Idle:             State Enabled       Time 0
 Max cache usage for all SRDF/A sessions on the Sym: 75

 Remote SRDF Group Information:
 Group # 04      Symmetrix 000222333444    Number Devices in a Group 10
 Link Config. Fibre                        Fibre Adapter Type Switch
 Auto Link Recovery  Enabled               Link Limbo 10
 DSE:   Pool Name N/A                      Autostart Disabled    Threshold 50
 Compression:               SW Disabled       HW Disabled
 Group-Level Write Pacing:  Autostart Disabled  Max Delay 50000    Threshold 60
 Device-level Write Pacing: Autostart Disabled  State Inactive
 Transmit Idle:             State Enabled       Time 0
 Max cache usage for all SRDF/A sessions on the Sym: 75
 F3=Exit      F12=Cancel
```

Figure 104 SRDF Group Information screen

Performing dynamic SRDF personality swaps

SRDF/TimeFinder Manager supports Dynamic SRDF. This section explains how to use SRDF/TimeFinder Manager, after using the Solutions Enabler CLI for personality swaps.

For more information on dynamic SRDF, refer to the *Dell EMC Solutions Enabler SRDF CLI User Guide.*

Dynamic personality SRDF (R2gR1) swaps for the System image (*SYSBAS) must be performed through the Solutions Enabler CLI. After the swap, launch SRDF/TimeFinder Manager from the target host and use any remote devices (BCV, Clone, Snap, and SnapVX) as local devices. However, as Table 13 shows, depending on the step in the path, additional path association may be necessary.

Table 13 Steps in the path and its associated action after personality swap

Source	Target	Target status after swap	Associated action after swap
/BCV1 /RDF1/BCV1	/RDF1/BCV1 /BCV1	Both pairs are associated	No action required
/BCV1	/BCV1	NotConfigured	No action required
	/RDF1/BCV1	Does not exist	Need to associate
/RDF1/BCV1	/RDF1/BCV1	NotConfigured	No action required
	/BCV1	Does not exist	Need to associate

Automating SRDF operations

SRDF/TimeFinder Manager provides the option of incorporating SRDF control operations in your customized CL programs. The CL command, CTLACT, includes the following SRDF control options:

- SRDF control actions:
  - **ESTABLISH** — Establishes SRDF links for SRDF control.
  - **RESUME** — Resumes SRDF links for SRDF control.
  - **SUSPEND** — Suspends SRDF links for SRDF control.
  - **SPLIT** — Splits SRDF links for SRDF control.
  - **SPLIT_MODE(RESTRICTED)** — Splits/suspends SRDF links for SRDF control while the system is in a restricted state.
  - **SPLIT_MODE(HOLD_JOBS)** — Holds all jobs and job queues before performing a Split or Suspend control operation.
  - **SPLIT_MODE(MAINTENANCE)** — Splits/suspends SRDF links for SRDF control in maintenance mode (that is, no interference with host operations).
  - **STATUS** — Provides SRDF control status information.
- SRDF/Asynchronous control actions:
  - **ENABLE** — Enables the consistency protection for SRDF/A devices.
  - **DISABLE** — Disables the consistency protection for SRDF/A devices.

Note: Appendix A, "Standard Features Commands Reference," contains detailed command reference information.

Appendix A
Standard Features Commands Reference

This appendix provides reference information, syntax, and examples for control commands.

- Conventions 202
- CRTDLTRDFP 203
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Conventions

The descriptions of the commands in this appendix can contain the following sections:

- **Syntax** — Presents the syntax of the command. Variables that you specify are shown in *italics*. The optional parameters are indicated by brackets.
- **Description** — Describes the purpose and result of the command.
- **Required parameters** — Describes parameters that must be specified at the command line. Each description contains all possible values of the variable and states which, if any, of the values is the default. Variables that you must specify are shown in *italics*.
- **Optional parameters** — Describes parameters that can be included at the command line. Each description contains all possible values of the variable and states which, if any, of the values is the default. Variables that you can specify are shown in brackets.
- **Usage notes** — Presents specific points that you should keep in mind when using the command.
- **Examples** — Presents one or more examples of the command.

CRTDLTRDFP

Creates or deletes SRDF pairs while the array is in operation.

Syntax

```
CRTDLTRDFP ACTION(CREATE|DELETE)
      SYMID(sym_id)
      GRPNUM(group_number)
      FILENAME(device_file_name)
      METROPAIR (*YES|*NO)
      METROMODE (A|B)
      [RDF_MODE(rdf_mode)]
      [RDF_FLAG(ESTABLISH|INVALIDATER2)]
      [RDF_TYPE(R1|R2)]
      [FORCE_FLAG(*YES|*NO)]
```

Description

Creates or deletes SRDF pairs while the array is in operation. All RDF device pairs created between PowerMaxOS 5978 or HYPERMAX OS 5977 arrays and Enginuity 5876 arrays must be created with dynamic RDF-capable devices. All devices running on arrays running PowerMaxOS 5978 or HYPERMAX OS 5977 are reported as dynamic RDF-capable.

Required parameters

SYMID(*sym_id*)
: The Symmetrix ID specified in this field is associated with the devices listed in the first column of the device file.

GRPNUM(*group_number*)
: SRDF group Number. The group number specified in this field is associated with the devices listed in the first column of the device file.

FILENAME(*device_file_name*)
: Name of the file that lists the designated device pairs.

ACTION(*control_action*)
: Specifies the control operation to perform:

  - CREATE — SRDF pairs become members of the specified RDF group. Pairing information is added to the SYMAPI database file on the host.
  - DELETE — SRDF pairs are canceled and pairing information is removed from the array and the SYMAPI database. Devices are changed to non-SRDF devices (except for concurrent SRDF pairs).

METROPAIR(*YES|*NO)
: Indicates whether the pair are in a SRDF/Metro configuration.

METROMODE(A|B)
: Specifies the form of protection that a SRDF/Metro pair uses:

  - A — The pair uses witness protection.
  - B — The pair uses bias protection.

  Include this parameter only when the value of METROPAIR is *YES.

Optional parameters

`RDF_MODE(rdf_mode)`
Sets device pair SRDF mode:

- SYNC — Synchronous (default)
- ASYNC — Asynchronous
- ACDM — Adaptive copy disk mode
- ACWP — Adaptive copy write pending

Note: SRDF mode adaptive copy write pending (ACWP) is not available when the R1 mirror of the RDF pair is on an array running PowerMaxOS 5978 or HYPERMAX OS 5977. For this configuration, the CRTDLTRDFP command returns an error if the mode is set to adaptive copy write pending. In this case use another mode.

`RDF_FLAG(ESTABLISH|INVALIDATER2)`
Sets RDF copying options as follows:

- ESTABLISH (default option) — Once the SRDF pairs are created, begins copying data to the invalidated targets, synchronizing the dynamic SRDF pairs that are listed in the device file.
- INVALIDATER2 — Once the SRDF pairs are created, marks the R2 devices that are listed in the device file as the invalidated target for a full device copy.

`RDF_TYPE(R1|R2)`
Indicates whether the device in the first column of the device file is an R1 or an R2 device. R1 is the default.

`FORCE_FLAG(*YES|*NO)`
Indicates whether to use or not use the force flag during DELETE operation. *NO is the default.

CRTNETCNFG

Creates the NETCNFG file and adds network services to the file.

Syntax

```
CRTNETCNFG NODENAME(node_name)
      NETWADDR(ip_address)
      SECLVL(DEFAULT|ANY|NONSECURE|SECURE)
      PORTNBR(port_number)
```

Description

CRTNETCNFG creates the NETCNFG file in the `/var/symapi/config` directory and adds network services to the file. Specify either the node name or the IP address of the remote SYMAPI server. You cannot omit both of these parameters. In addition, the security level of the connection type is mandatory.

Required parameters

`NODENAME(node_name)`
The name of the remote SYMAPI server. NODENAME is required if the NETWADDR parameter has not been specified.

Note: When using a node name, make sure the node name is also entered into the host table of the system or is listed in the DNS tables of the attached network.

`NETWADDR(ip_address)`
The IP address of the remote SYMAPI server. NETWADDR is required if the NODENAME parameter has not been specified.

`SECLVL(DEFAULT|ANY|NONSECURE|SECURE)`
The security level for the connection type that the SYMAPI server is expecting to negotiate:

- DEFAULT — The SYMAPI client and server are initially configured to negotiate either secure or non-secure sessions. This is the default.
- ANY — Indicates that either secure or non-secure sessions are negotiated between the client and the server if the SSL (secure socket layer) capabilities do not agree and between supported and unsupported platforms.
- NONSECURE — Indicates that only non-secure sessions are negotiated between the client and the server. This level is intended as the last resort in situations where SSL cannot be used for some reason or is undesirable. NONSECURE can be useful in matters of performance and availability.
- SECURE — Indicates that only secure sessions are negotiated between the open host and the server. This level is intended in situations where SSL desirable.

Note: The SECURE option is not supported on the IBM i hosts.

`PORTNBR(port_number)`
The TCP/IP port number for the remote SYMAPI server.

The default value is **2707.**

CTLACT

Processes SRDF and TimeFinder control actions for one or all images (SYSTEM and IASP).

CAUTION

Do not use the CTLACT command with IASP images, except for initial IASP configuration. Instead, use the options on the Work with Associated Paths screen or the IASPCTL command. Using either these ensures that SRDF/TimeFinder Manager does rigorous error checking before executing the control actions against the IASPs. For more information, see "IASPCTL" on page 377.

Syntax

```
CTLACT  IMAGE_NAME(image_name|*ALL)
        PATH_NAME(path_name)
        STEPNUM(step_number)
        ACTION(control_action)
        [SPLIT_MODE(RESTRICTED|HOLD_JOBS|SPLITWA|MAINTENANCE)]
        [PROT_ESTAB(*YES|*NO)]
        [POOL_NAME(pool_name)]
        [VP_SNAP(*YES|*NO)]
        [SVX_MODE(CONSISTENT|HOLD_JOBS| RESTRICTED)]
        [SNAPVXNAME(snapshot_name)]
        [SNAPVXGEN(snapshot_generation)]
        [SNAPVXTTL(DAYS|DATE)]
        [N_SNPSHOT(new_snapshot_name)]
        [LINK_MODE(NOCOPY|COPY)]
        [TTL_DAYS(1-400)]
        [TTL_DATE(Date)]
```

Description

CTLACT processes the control action specified in the ACTION parameter for the IMAGE parameter, PATH parameter, and the STEPNUM parameter. If the specified control action is directed to any of the Suspended/Split SRDF links, Split BCV pairs, an Activate Snap virtual copy session, or Clone copy session, use the SPLIT_MODE parameter to specify which method to use to stop writes to the devices in the image.

If SnapVX Establish control action is specified, use the SVX_MODE parameter to specify which method to use when stopping writes to the devices in the image.

You can use CTLACT to control multiple images as long as all images use the same path and step number.

Required parameters

IMAGE_NAME*(image_name*|*ALL)
: The name of the image (*image_name)* or *ALL to indicated all images.

PATH_NAME*(path_name)*
: The path name associated with the image to be used in an SRDF or TimeFinder control operation.

STEPNUM(*step_number*)
: The specific step number for which the control action is initiated in an SRDF or TimeFinder control operation.

  Steps in the path are defined during configuration and are identified by their position (1 through 5) in the path.

ACTION(*control_action*)
Specifies which control operation to perform:

- STATUS — Provides SRDF, Snap, Clone, Clone emulation, BCV, or SnapVX control status information for the step in the associated path of the image. For SnapVX, if a snapshot name and generation are specified, the state of the specified snapshot is provided.

  Note: If the status of the SnapVX step is Linked, CopyInProg, Copied, Restored, RestoreInProg, RestNotPaired, or RestgNotPaired an additional message is provided with the snapshot name and generation.

- ESTABLISH — Establishes SRDF links for SRDF control or establishes/re-establishes BCV or Clone emulation pairs for BCV control. When running in SRDF/A mode, consistency protection is automatically enabled after executing an ESTABLISH operation is executed.

  For SnapVX type of step, ESTABLISH creates and activates a SnapVX snapshot session.

- SPLIT — Splits SRDF links for SRDF control or splits BCV or Clone emulation pairs for TimeFinder control. When the system is running in SRDF/A mode and a SPLIT is executed, consistency protection is disabled and then the SPLIT operation is performed.

  Note: For TimeFinder/Clone configurations, if the devices are in Restored state, use the SPLIT option to change the direction of the clone relationship (that is, the original source device becomes the source device for a future copy), which allows the use of the RECREATE command and normal processing continues. This does not apply if VP Snap is enabled.

- RESUME — Resumes SRDF links for SRDF control. When the system is running in SRDF/A mode, consistency protection is automatically enabled after a RESUME operation is executed.
- SUSPEND — Suspends SRDF links for SRDF control. When the system is running in SRDF/A mode and a SUSPEND is executed, the consistency protection is disabled and then the SUSPEND operation is performed.
- ENABLE — Enables the consistency protection for SRDF/A backed devices when running in SRDF/A mode. If data cannot be copied from the R1 to the R2, all devices in the group become Not Ready on the link to preserve the R2 data consistency.

  Note: If devices are enabled for consistency group protection, the devices must be disabled before you can change the mode to Synchronous or initiate the SPLIT/SUSPEND operation.

- DISABLE — Disables the consistency protection for SRDF/A backed devices when running in SRDF asynchronous mode.

- CREATE — Creates or recreates a TimeFinder/Snap virtual copy session or Clone copy session with devices in the image. While the operation is in progress for TimeFinder/Snap, the state of the device pair is CreateInProg. For TimeFinder/Clone the state of the device pair is PreCopy. When the Create operation completes, the state changes to Created.

  You must run CREATE before you activate the session, which has a shorter duration than an Establish operation.

  For TimeFinder Snap, Recreate is only available in environments running Enginuity 5876 or higher.

  See "TimeFinder/Snap Usage notes" on page 212 for more information about using Recreate.

- ACTIVATE — Activates a TimeFinder/Snap virtual copy session or Clone copy session with the devices in the image. For TimeFinder/Snap, while the operation is in progress the state of the device pair is CopyOnWrite. For TimeFinder/Clone, while the operation is in progress the state of the device pair is CopyInProg. Once the copy is complete, the state changes to Copied.

  Note: The point-in-time copy becomes available after an ACTIVATE operation. Create must be done before you can activate a TimeFinder/Snap copy session or Clone copy session.

- LINK — Copies the SnapVX point-in-time data to the target device.

  There are two types of links, NOCOPY (the default) or COPY. NOCOPY links do not retain point-in-time data once the link is removed and any previous target data is lost. Copy mode must be used in order for the linked target to retain data after the link is removed. When linking a snapshot, COPY or NOCOPY must be specified.

- RELINK — Removes the current SnapVX link to target device(s) and presents a different snapshot. The relink may also be used to relink to the same snapshot. This has the effect of refreshing the point-in-time copy on the link target if it has been modified by host writes.

- UNLINK - Removes the current SnapVX link to target device(s).

- RESTORE — For SnapVX , copies the point-in-time data from a snapshot back to the original source device. The snapshot must be fully established to perform a restore. Restore action is not allowed for the first step for a System Image.

- RENAME — Changes the SnapVX snapshot name. Renames on the specified generation. If generation is not specified, then generation 0 is renamed.

- SET TTL — Sets a time to live for a SnapVX snapshot, from when TTL is set, in days or a date. The value in days can be set to any value between 1 and 400. The date value is a specified date up to 400 days in the future, from which a value in days is created, based on the difference between the specified absolute time and the host time when the action is performed. When the TTL expires the snapshot automatically terminates unless there are one or more links. Automatic termination takes place when the links are removed. If TTL is not set, the snapshot remains established until it is manually terminated.

- SETCOPY - Changes the copy mode of a link to copy. Data is physically copied to the target device(s). When the link is terminated the data is retained on the target.
- SETNOCOPY - Changes the copy mode of a link to NoCopy. Data is presented on the target device(s), but not physically copied. When the link is terminated the data is not retained on the target.
- TERMINATE — Stops the existing TimeFinder/Snap virtual copy session or Clone copy session between the specified source and target devices in the image.

  For SnapVX snapshot sessions, TERMINATE removes the snapshot. All links must be unlinked before terminating a snapshot.

  Note: If a snapshot has restores, only the restored session is terminated.

- CANCEL — Cancels the existing internal session between the specified standard device and the BCV devices in the image.

  Once the session is canceled, the corresponding BCV devices are placed in a SplitNoInc state, and the BCV pair can no longer be incrementally established or restored.

  Run Cancel before you run an unpairing operation on Clone Enabled BCV devices. Not running Cancel causes hanging Clone sessions to remain on the array.

Optional parameters

`POOL_NAME(DEFAULT_POOL|`*`pool_name`*`)`
Specifies the named SAVE pool to use when creating a TimeFinder/Snap operation. If pool name is not specified, the operation uses the default pool (DEFAULT_POOL). By default, IBM i commands converts names to the upper case. If you want the name to be in lower or mixed case, put the name in quotes.

SNAP pool name rules and restrictions:

- The POOL_NAME parameter is only used when creating new SNAP sessions. Otherwise it is ignored.
- The specified pool must exist and contain at least one enabled device before creating a TimeFinder/Snap session.
- If the image spans over several arrays the specified pool must exist on all arrays.
- If at least one device in a step has established a SNAP session all other copy sessions use the pool in the established copy session.
- With mixed devices (2107 and D910) you can specify only the DEFAULT_POOL.

`VP_SNAP(*YES|*NO)`
Specifies whether to use TimeFinder VP Snap mode. The VP Snap parameter may only be set when a session is created.

VP Snap allows for concurrent sessions.

VP Snap rules and restrictions:

- Active sessions are in the CopyOnWrite state.
- Support is limited to D910 IBM i (512 byte blocksize) thin-to-thin device clone pairs. No other IBM i type devices are supported.
- All VP Snap target devices must be bound to the same thin pool. Once a VP Snap session is created for a source device, additional new sessions are rejected if the new target device is not bound to the same thin pool as the target for the existing session.

Restrictions for cascaded sessions:

- Clone sessions must be in the Copied or Split state for a VP Snap session to be created from its target device.
- The Clone session must exist before the target device is used as a VP Snap source. Creating a Clone session using a VP Snap source as its target device is not permitted.
- Snap sessions and VP Snap sessions cannot be on the same source device.
- Rebind and VLUN VP Migration operations that cause a device to be rebound are not allowed on VP Snap target devices.
- The target of a VP Snap session cannot be used as a source for any other Clone or Snap session.
- Cascading from TimeFinder Clone emulation is not supported.

`SPLIT_MODE(RESTRICTED|HOLD_JOBS|SPLITWA|MAINTENANCE)`

The mode to use for a Suspend, Split, or Activate operation:

- RESTRICTED — Executes the operation when the system is in a restricted state. This is the default. System must be put in restricted state by using option 5 from the Control menu using "Bring System to a Restricted State", or by using the STMENDSBS command. This ensures that all subsystems and jobs are ended except for the daemon jobs and their corresponding subsystems.
- HOLD_JOBS — Executes the operation after holding all jobs and job queues.
- SPLITWA — Executes the operation after holding I/Os on the array. Splits a local or remote BCV, Clone, Clone emulation, or Snap image while the source system is operational. The resulting image is consistent and holds data upon the last commit.

  With TimeFinder/CG (SPLITWA) operations, you do not have to put the IBM i host in a restricted state or perform a hold jobs operation.

  The SRDF/TimeFinder Manager client and the controlling host must have a gatekeeper configured on each locally attached array. For detailed information on array gatekeeper management, refer to the *Dell EMC Solutions Enabler Installation and Configuration Guide*.

  On the client, the netcnfg file must be configured using the CRTNETCNFG command. This ensures that correct information (node name, IP address, port, and security level) points to the remote server. Only non-secure connections are supported for IBM i.
- MAINTENANCE — Splits or Suspends SRDF links, splits BCV pairs or activates a virtual copy session or Clone copy session in maintenance mode (that is, no interference with host operations).

PROT_ESTAB(*YES|*NO)

Performs a Protected or normal BCV Establish operation:

- *YES — Establish two mirrors of a BCV device to the standard device. When the split occurs, both mirrors are split simultaneously. This operation moves all mirrors of the locally mirrored BCV devices to join the mirrors of a standard device
- *NO — Perform a normal Establish operation for the BCV device. This is the default.

SVX_MODE(CONSISTENT|HOLD_JOBS| RESTRICTED)

The mode to use for the SnapVX Establish operation:

- CONSISTENT —Executes the operation in Consistent mode (that is, no interference with the host operations). This is the default.

  IMPORTANT

  Use caution when performing this Consistent Establish mode, as the data in main memory is not included in the image. Use this mode when the host and applications are in a quiescent state. If that is unacceptable to your business needs, use other operating system functions such as journaling and database commitment control to get an image current with the last transaction.

- HOLD_JOBS — Executes the operation after holding all jobs and job queues.
- RESTRICTED — Executes the operation when the system is in a restricted state. Put the system restricted state by using option 5 from the Control menu using “Bring System to a Restricted State”, or by using the STMENDSBS command. This ensures that all subsystems and jobs are ended except for the daemon jobs and their corresponding subsystems.

SNAPVXNAME

The name of the Snapshot.

By default, IBM i commands convert names to the upper case. Put the name in quotes if you want it to be in the lower or mixed case.

Note: For ESTABLISH action SNAPVXNAME specifies the name for new snapshots. For all other control actions SNAPVXNAME specifies the name of the existing snapshots to be operated on. This parameter is required for all SnapVX control actions except STATUS where it is optional.

SNAPVXGEN

The Generation of the Snapshot. For all SnapVX control actions, except ESTABLISH, specifies the generation of the existing snapshots to be operated on. If set to zero, the most recent snapshot is used for the control action.

SNAPVXTTL

For ESTABLISH and SET_TTL SnapVX controls specifies a Time to Live:

- DAYS -Specifies the time to live in days.
- DATE - Specifies a date for the snapshot time to live.

If you do not specify a value, the Time to Live is unlimited and the Snapshot is kept until manually terminated.

N_SNPSHOT
The new name of the Snapshot. Used only for RENAME operations. By default, IBM i commands convert names to the upper case. If you want the name to be in the lower or mixed case, put the name in quotes.

LINK_MODE(NOCOPY|COPY)
The mode used for a LINK/RELINK SnapVX operation. Options are as follows:

- NOCOPY —The target does not retain point-in-time data once the link is removed, and any previous target data is lost. This is the default.
- COPY — Data is physically copied to the target devices. When the link is terminated the data is retained on the target.

TTL_DAYS
Time to Live in days. The value in days can be set to any value between 1 and 400.

TTL_DATE
Specifies a date for the snapshot time to live in the form MM/DD/YYYY (month/day/year) with reference to the host time. The specified date may be up to 400 days in the future.

TimeFinder/Snap Usage notes

In environments prior to Enginuity 5876:

- Issue TERMINATE and then CREATE before creating a new point-in-time image.
- A Snap copy session requires the following steps:
  1. Create a Snap session.
  2. Activate a Snap session.
  3. Terminate the Snap session.

In Enginuity 5876 environments:

- The Recreate action is used automatically if all devices in the image have Enginuity 5876 and sessions are already activated.
- Recreating a Snap copy session requires the following steps:
  1. Create a Snap session.
  2. Activate a Snap session.
  3. Recreate a Snap session.
  4. Activate a Snap session.
  5. Repeat the Recreate/Activate a Snap session, as necessary.
  6. Terminate the Snap session when no longer needed.

Examples

TimeFinder

```
CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(LOCALBCV) STEPNUM(1) ACTION(STATUS)
CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(LOCALBCV) STEPNUM(1) ACTION(ESTABLISH)  PROT_ESTAB(*NO)
CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(LOCALBCV) STEPNUM(1) ACTION(SPLIT) SPLIT_MODE(RESTRICTED)
CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(LOCALBCV) STEPNUM(1) ACTION(SPLIT) SPLIT_MODE(SPLITWA)
```

TimeFinder/Snap

```
CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(LOCALSNAP) STEPNUM(1) ACTION(CREATE) POOL_NAME(SNAPPOOL)
CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(LOCALSNAP) STEPNUM(1) ACTION(ACTIVATE) SPLIT_MODE(SPLITWA)
CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(LOCALSNAP) STEPNUM(1) ACTION(TERMINATE)
```

TimeFinder/SnapVX -Action Establish with TTL days and date

```
CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(LSNAPVX1) STEPNUM(1) ACTION(ESTABLISH) SNAPVXNAME(END_OF_YEAR)
SNAPVXTTL(DATE) TTL_DATE('02/15/2014')

CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(LSNAPVX1) STEPNUM(1) ACTION(ESTABLISH) SNAPVXNAME(END_OF_YEAR)
SNAPVXTTL(DAYS) TTL_DAYS(25)
```

TimeFinder/SnapVX - Action Link

```
CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(LSNAPVX1) STEPNUM(1) ACTION(LINK)
SNAPVXNAME(END_OF_YEAR) SNAPVXGEN(1))LINK_MODE(COPY)
```

TimeFinder/SnapVX - Action Set TTL days and date

```
CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(LSNAPVX1) STEPNUM(1) ACTION(SET_TTL)
SNAPVXNAME(END_OF_YEAR) SNAPVXGEN(2) SNAPVXTTL(DAYS) TTL_DAYS(25)

CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(LSNAPVX1) STEPNUM(1) ACTION(SET_TTL)
SNAPVXNAME(END_OF_YEAR) SNAPVXGEN(2) SNAPVXTTL(DATE) TTL_DATE('02/15/2014')
```

TimeFinder/SnapVX - Action Rename

```
CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(LSNAPVX1) STEPNUM(1) ACTION(RENAME)
SNAPVXNAME('End_of_Quarter') SNAPVXGEN(1) N_SNPSHOT('End_of_Last_Quarter')
```

TimeFinder/SnapVX - Action Unlink

```
CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(LSNAPVX1) STEPNUM(1) ACTION(UNLINK)
```

TimeFinder/SnapVX - Action Terminate

```
CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(LSNAPVX1) STEPNUM(1) ACTION(TERMINATE)
SNAPVXNAME('End_of_Quarter') SNAPVXGEN(2)
```

TimeFinder/SnapVX - Action Status for snapshot

```
CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(LSNAPVX1) STEPNUM(1) ACTION(STATUS)
SNAPVXNAME(END_OF_YEAR) SNAPVXGEN(2)
```

TimeFinder/SnapVX - Action Status for step

```
CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(LSNAPVX1) STEPNUM(1) ACTION(STATUS)
```

SRDF/S

```
CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(SRDF) STEPNUM(1) ACTION(STATUS)
CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(SRDF) STEPNUM(1) ACTION(RESUME)
CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(SRDF) STEPNUM(1) ACTION(SUSPEND) SPLIT_MODE(HOLD_JOBS)
CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(SRDF) STEPNUM(1) ACTION(ESTABLISH)
CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(SRDF) STEPNUM(1) ACTION(SPLIT) SPLIT_MODE(MAINTENANCE)
```

SRDF/A

```
CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(SRDFA) STEPNUM(1) ACTION(ENABLE)
CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(SRDFA) STEPNUM(1) ACTION(DISABLE)
```

CTLMASK

Creates and deletes masking views.

Syntax

```
CTLMASK SYMID(sym_id)
      VIEWNAME(mask_view_name)
      CTLACT(control_action)
      STORAGEGRP(storage_group)
      INITGRP(initiator_group)
      PORTGRP(port_group)
```

Description

Creates masking views from pre-existing storage groups, initiator groups, and port groups. Also deletes previously created masking views.

Required parameters

`SYMID(sym_id)`
The name of the array on which to perform this command.

Note: If the array ID is known, enter its name. If the array ID is not known, press F4 to display a list of all local and remote arrays attached to the host.

`VIEWNAME(mask_view_name)`
The name of the masking view to create or delete.

`CTLACT(control_action)`
Specifies the control action to perform on the view:

- CREATE — creates a new masking view.
- DELETE — deletes an existing masking view.

`STORAGEGRP (storagr_group)`
The name of the storage group to include in the new masking view.

`INITGRP(initiator_group)`
The name of the initiator group to include in the new masking view.

`PORTGRP(port_group)`
The name of the port group to include in the new masking view.

Usage notes

- Include the STORAGEGRP, INITGRP, and PORTGRP parameters only when the control action is CREATE.
- The storage group, initiator group, and port group must already exist to be able to create a new masking view.

Examples

Create

```
CTLMASK SYMID(000111222333) VIEWNAME(NEWVIEW) CTLACT(CREATE)
STORAGEGRP(STGROUPA) INITGRP(INITGROUPB) PORTGRP(PORTGROUPC)
```

Delete

```
CTLMASK SYMID(000111222333) VIEWNAME(NEWVIEW) CTLACT(DELETE)
```

CTLPOOL

Controls the device pools (SAVE and thin) and devices (SAVE and DATA) in the pools. Also allows for migration session termination.

For arrays running PowerMaxOS 5978 or HYPERMAX OS 5977, Snap and DSE pools are not supported, and thin pools and DATA devices are pre-configured, so pool management is not available. Therefore for PowerMaxOS 5978 and HYPERMAX OS 5977 arrays, creating and deleting thin pools, and management of existing pools and DATA devices (except for pool renaming) are not available. For arrays running Enginuity 5876 or earlier, the actions on thin pools described below are allowed by SRDF/TimeFinder Manager, except for FREEALL which is available on PowerMaxOS 5978 and HYPERMAX OS 5977 arrays only.

Syntax

```
CTLPOOL SYMID(sym_id)
      POOLNAME(DEFAULT_POOL|pool_name)
      DEVPLTYPE(SNAP|DSE|DATA)
      CTLACT(control_action)
      [FILENAME(file_name)]
      [STARTDEV(device_number|*ALL)]
      [ENDDEV(device_number|*ALL)]
      [OPTION(DISABLE|ENABLE)]
      [NEW_POOL(pool_name)]
      [MAX_SUBS(1-65534)]
      [COMPR(DISABLE|ENABLE)]
      [DEV_COMPR(COMP_START|COMP_END|UNCOMP_START|UNCOMP_END)]
      [FREEALL(START|STOP)]
      [SRC_POOL(pool_name)]
      [SESSION(session_name)}
```

Description

The CTLPOOL command allows the control of the device pools (SAVE and thin), and devices (SAVE and DATA) in those pools, as well as thin device association (bound/unbound) to thin pools. This command can also be used for Virtual Provisioning (migrating thin devices from one pool to another and thin device compression), and to terminate migration sessions.

CTLPOOL allows control of the device pools. SAVE pools are either SNAP (available for TimeFinder/Snap operations) or DSE pools (available for RDF/A DSE operations). Thin pools contain DATA devices (available for Virtual Provisioning).

With CTLPOOL, SAVE and DATA devices can be added to a pool, enabled or disabled in once in the pool, or removed from the pool. Thin devices can be bound to the pool (bind), unbound from the pool (unbind), bound to a different pool (rebind), migrated to a different pool, and compressed or uncompressed.

For control operations on devices, file name or range of devices (START/END) can be specified.

Required parameters

SYMMETRIX_ID(*sym_id*)
: The name of the array on which to perform this command.

Note: If the array ID is known, enter its name. If the array ID is not known, press F4 to display a list of all local and remote arrays attached to the host.

POOLNAME(DEFAULT_POOL|*Pool_name*)
The name for the device pool.:

Note: For control actions ADD, BIND, REBIND, and MIGRATE, this parameter is the name of the target pool.

- DEFAULT_POOL — The special Snap pool, which exists by default, and contains SAVE devices that have not been assigned to any named pools but are available for Snap operations. The devices in DEFAULT_POOL are either enabled or disabled but have used tracks. Devices can be disabled using a range of values or *ALL. Devices can only be enabled using *ALL.
- *Pool_name* — The name for the device pool.

  By default, IBM i commands convert names to the upper case. If you want the name to be in the lower or mixed case, put the name in quotes.

DEVPLTYPE(SNAP|DSE|DATA)
The type of the device pool:

- SNAP — Contains SAVE devices that are used for TimeFinder/Snap operations.
- DSE — RDF/A DSE contains SAVE devices that are used for SRDF/A DSE operations.
- DATA — Contains DATA devices used for Virtual Provisioning.

Note: DATA pools are supported with Enginuity 5876 and later.

Note: For unbinding of thin devices, thin device compression and termination of the migration session, pool name and pool type parameters are ignored.

CTLACT(*control_action*)
Specifies which control operation to perform on the device pool or devices in the pool:

- ADD — Adds SAVE or DATA devices to the specified pool. Devices must be disabled. SAVE devices that do not belong to any pool are available to be added to a RDF/A DSE or SNAP pool. DATA devices that do not belong to any pools are available to be added to the DATA (thin) pool.

  Disabled devices can be moved from one named pool to another pool. The target device pool must be empty, or consist of the devices with the same type as devices to be moved.
- BALANCE - Starts write balancing operation for the specified thin pool. If pool balancing is not in progress, SRDF/TimeFinder Manager initiates balancing on the specified pool.
- BIND — Binds thin devices to the DATA (thin) pool.
- CHANGE_SUBS - Changes the oversubscription ratio for the specified thin pool. The limit is set by specifying the maximum subscription value when creating thin pool. Existing thin pools can be modified to change a maximum subscription percent.
- CREATE — Creates a new device pool. When a pool is created, it can be created as a RDF/A DSE pool, a Snap pool, or a DATA (thin) pool.

- DELETE — Deletes a device pool. In order to be deleted, pools must have no active sessions, must be disabled, and cannot contain any members.
- DISABLE — Changes the state of devices in the pool to Disabled. Once the devices are disabled, they can be removed from the device pool.
- ENABLE — Changes the state of devices in the pool to Enabled. Enabled SAVE devices can be used during SNAP or SRDF/A DSE operations. Enabled DATA devices can be used for Virtual Provisioning.
- MIGRATE — Migrates allocations from one pool to another. For full migration rebinds migrated devices to the target DATA (thin) pool. If a source pool (SRC_POOL) is specified then thin device pool-level migration is performed, otherwise full migration is performed. Migration is only allowed for locally attached arrays.
- TERMINATE — Terminates thin device migration sessions during the migration process.

  Note: Termination is only allowed on locally attached arrays.

- REBIND — Rebinds thin devices to a different DATA (thin) pool. Rebind changes the thin device's current binding to a new pool. Rebind does not move any existing data to the new pool, but all new allocations to the thin device go to the new pool after the rebind action completes.
- FREEALL — Frees up all allocations associated with a given device regardless of whether or not there is data written to the device. Freeall is available only on arrays running PowerMaxOS 5978 or HYPERMAX OS 5977.
- REMOVE — Removes devices from the specified pool. In order to remove a device from a pool it must be disabled in the pool.
- RENAME — Renames the specified pool. SNAP, DATA (thin), and DSE pools can be renamed. The default pool name (DEFAULT_POOL) for Snap cannot be changed. Rename is allowed for arrays running PowerMaxOS 5978, HYPERMAX OS 5977 and earlier.
- STOP_BALANCE — Stops any write balancing operation for the specified DATA (thin) pool.
- UNBIND — Unbinds thin devices from the DATA (thin) pool.
- VP_COMPR — Allows for enabling or disabling the thin pool for compression.
- VP_DEV_COMPR — Starts or stops thin device compression or uncompression.

`FILENAME(`*`file_name`*`)`
Specifies the name of the device file used that can be used with the CTLPOOL command, and lists the designated devices for the ADD, REMOVE, ENABLE, DISABLE, BIND, UNBIND, REBIND, MIGRATE and VP_DEV_COMPR control actions.

Device files that can be used with the CTLPOOL command are created from a set of screens or created manually.

Files can be created from the following screens:

- Work with Thin Bound Devices
- Work with Thin Unbound Devices
- Work with Pool
- Not Pooled Save/Data Devices

If creating the file manually the following rules apply:

- The file is a record file and record length must not exceed 40.
- The file contains only one column that lists the source devices.
- List each device on a separate line.
- Comment lines are allowed and they must begin with the pound sign (#).
- The device file resides in the library list.

Example device file:

```
# Devices for CTLPOOL command
009A
009E
009F
```

Optional parameters

`STARTDEV(`*`device_number`*`|*ALL)`
The starting array device number or *ALL to specify device numbers, and is used with ADD, REMOVE, ENABLE, DISABLE, BIND, UNBIND, REBIND, MIGRATE and VP_DEV_COMPR control actions.

`ENDDEV(`*`device_number`*`|*ALL)`
The ending array device number or *ALL to specify device numbers, and is used with ADD, REMOVE, ENABLE, DISABLE, BIND, UNBIND, REBIND, MIGRATE and VP_DEV_COMPR control actions.

Note: Start and end devices usually specify a range of devices. To specify a pool containing a single device, specify the same device name for the start and end devices. For DEFAULT_POOL, specify *ALL for the start and end devices.

`OPTION(DISABLE|ENABLE)`
Allows for the option to also enable or disable the SAVE or DATA devices, and is used only with the ADD operation.

- DISABLE — Leaves devices disabled after adding them to the pool. DISABLE is the default.
- ENABLE — Changes the state of devices to enabled after adding them to the device pool.

`NEW_POOL(`*`pool_name`*`)`
Specifies the new device pool name, and is used only with the RENAME action.

By default, IBM i commands convert names to the upper case. Put the name in quotes if you want it to be in the lower or mixed case.

`MAX_SUBS(1-65534)`
Specifies the maximum subscription percent for a pool, and is used with the CREATE and CHANGE_SUBS control actions. This parameter is only used with (DATA) thin pools.

The maximum subscription percent is a value between 1 and 65534. If the value is not specified the default value is passed, meaning there is no limit set and the amount of oversubscription is not restricted. Setting a value of 1 allows only a maximum thin capacity that totals 1/100th of the actual physical capacity of the pool. Setting a value of 100 prevents the pool from oversubscription.

`COMPR(DISABLE|ENABLE)`
Specifies if thin pool is enabled or disabled for compression, and is used with the CREATE and VP_COMPR control actions. This parameter is only used with (DATA) thin pools.

- DISABLE — Compression is not allowed for the thin pool.
- ENABLE — Enables Thin pool for the compression.

Note: If a value is not specified, the compression parameter value assigned to the new pool is Disabled. During new pool creation, if the Enginuity version level does not support thin devices, or if the pool is a SNAP or DSE pool this parameter is ignored,

`DEV_COMPR (COMP_START|COMP_END|UNCOMP_START|UNCOMP_END)`
Specifies the compression action on the selected devices, and is used with VP_DEV_COMPR control action. This action can only be performed for DATA (thin) pools that are enabled for compression.

- COMP_START - Starts the compression on selected devices.
- COMP_END - Stops the compression on selected devices.
- UNCOMP_START - Starts the uncompression on selected devices.
- UNCOMP_END - Stops the uncompression on selected devices.

`FREEALL (START|STOP)]`
Specifies the freeall action on the selected Thin devices. This parameter is available only on arrays running PowerMaxOS 5978 or HYPERMAX OS 5977.

- START - Starts process of the freeing allocations on selected devices.
- STOP - Stops the free operation on the selected devices.

`SRC_POOL(`*`pool_name`*`)`
Specifies the source device pool name, and is used only with the MIGRATE control action. If source pool is not specified then full migration is performed. This parameter is used only with DATA (thin) pools.

By default, IBM i commands convert names to the upper case. If you want the name to be in the lower or mixed case, put the name in quotes.

`SESSION(`*`session_name`*`)`
Specifies migration session name, and is used with the MIGRATE and TERMINATE control actions. This parameter is used only with DATA (thin) pools.

By default, IBM i commands converts names to the upper case. If you want the name to be in the lower or mixed case, put the name in quotes.

Examples

Add devices to a pool type DSE

```
CTLPOOL SYMID(000123456789) POOLNAME(MYSAVPOOL) DEVPLTYPE(DSE) CTLACT(ADD) STARTDEV(0250)
ENDDEV(0252)
```

Enable devices in pool type SNAP

```
CTLPOOL SYMID(000123456789) POOLNAME(DEFAULT_POOL) DEVPLTYPE(SNAP) CTLACT(ENABLE)
STARTDEV(*ALL) ENDDEV(*ALL)
```

Add and enable devices to pool type thin

Using the optional parameter OPTION (ENABLE) available with the CTLACT(ADD) control action.

```
CTLPOOL SYMID(000123456789) POOLNAME(MYTHINPOOL) DEVPLTYPE(DATA) CTLACT(ADD) STARTDEV(0001)
ENDDEV(000A) OPTION(ENABLE)
```

Create a pool type DATA

```
CTLPOOL SYMID(000123456789) POOLNAME(MYDATAPOOL) DEVPLTYPE(DATA) CTLACT(CREATE) MAX_SUBS(100)
COMPR(ENABLE)
```

Start compression

```
CTLPOOL SYMID(000123456789) POOLNAME(POOL_NAME) DEVPLTYPE(DATA) CTLACT(VP_DEV_COMPR)
FILENAME(SRC_FILE) DEV_COMPR (COMP_START)
```

Full Migration

```
CTLPOOL SYMID(000123456789) POOLNAME(POOL_NAME) DEVPLTYPE(DATA) CTLACT(MIGRATE)
FILENAME(SRC_FILE) SESSION(MIGR_SESSION)
```

Pool Migration

```
CTLPOOL SYMID(000123456789) POOLNAME(POOL_NAME) DEVPLTYPE(DATA) CTLACT(MIGRATE)
FILENAME(SRC_FILE) SRC_POOL(SRC_POOL) SESSION(MIGR_SESSION)
```

Terminate session

```
CTLPOOL SYMID(000123456789) CTLACT(TERMINATE) SESSION(MIGR_SESSION)
```

VP Device Freeall

```
CTLPOOL SYMID(000123456789) CTLACT(FREEALL) FILENAME(TEST) FREEALL(START)
CTLPOOL SYMID(000123456789) CTLACT(FREEALL) FILENAME(TEST) FREEALL(STOP)
```

CTLRDFGRP

The CTLRDFGRP command controls the dynamic RDF group.

Syntax

```
CTLRDFGRP ACTION (control_action)
      SYMID(sym_id)
      [METROWIT(*YES|*NO)]
      [GRPLABEL (group_label)]
      [GRPNUM (group_nmber)]
      [MODIFY_ACT (modify_action)]
      [SYM_FORCE(*YES|*NO)]
      [DIRPORT((Dir1(Port1 Port2 …)[(Dir2 (Port….)]…)
      [RSYMID(sym_id)]
      [RGRPNUM (group_nmber)]
      [RDIRPORT((Dir1(Port1 Port2 …)[(Dir2 (Port….)]…)
      [AUTO_LINK(*YES|*NO)]
      [RAUTOLINK(*YES|*NO)]
      [LINK_LIMBO (0-120)]
      [RLINKLIMBO (0-120)]
```

Required Parameters

ACTION(*control action*)
Specifies the control action to perform:

- ADD — Creates a new dynamic SRDF group.
- MODIFY — Modifies an existing dynamic SRDF group

  Note: This action enables the local and remote director/port to be associated/dissociated with the RDF Group.

- REMOVE — Removes a dynamic SRDF group.

  Note: You can remove (delete) an empty RDF group only.

SYMID(*sym_id)*
The name of the array on which this command is performed. If the array ID is known it can be entered. Otherwise, press F4 key to display a list of all local and remote arrays attached to the host.

Optional Parameters

METROWIT(*YES|*NO)
Indicates whether this group is a SRDF/Metro witness group. The default value is *NO.

There can be only one witness group between two arrays.

GRPLABEL(*group_label*)
Specifies a label for the dynamic SRDF group. By default, IBM i commands converts names to the upper case. Put the label in quotes if you want it to be in the lower or mixed case.

GRPNUM(*group_number*)
Specifies the SRDF (RA) group number.

Note: If using the ADD control action, specify both the group label and the group number. Otherwise, specify only one of the parameters.

MODIFY_ACT(*modify_action*)
Specifies which control action to perform. Options are as follows:

- ADD — Adds supporting SRDF directors to a dynamic SRDF group.
- REMOVE — Removes supporting RDF directors from a dynamic SRDF group.

SYM_FORCE(*YES|*NO)
Indicates whether to use the array force operation when normally it is rejected. This option is used only with REMOVE action. Removes one side of a dynamic SRDF group if the other side is not defined or is not accessible.

- *YES — Enables Sym Force.
- *NO — Disables Sym Force. This is the default.

Note: To enable SYM_FORCE, a parameter called SYMAPI_ALLOW_RDF_SYMFORCE in the options file must be set to ENABLED.

⚠CAUTION

Use care when applying SYM_FORCE, as data could be lost or corrupted. Use of this option is not recommended, except in extreme cases.

DIRPORT((*Dir1*(*Port1 Port2* …)[(*Dir2* (*Port….*)]…)
Specifies local director/port combination. This option is used with the ADD and MODIFY control actions:

- Director ID (*Dir 1*, *Dir 2...*) — This is the list of the local array directors, such as, 1a, 1b, and so on.
- Port Number (*Port 1, Port 2...)* — This is the list of local array director ports.

Note: For PowerMaxOS 5978 and HYPERMAX OS 5977, the RDF director supports up to 32 ports.

The SYMCLI command symcfg can be used to display all available local director/port combinations. For example:

```
Addlible emcctl
CALL PGM(SYMCFG) PARM('list' '-ra' 'all' '-sid'
'000111222333')
```

RSYMID(sym_id)
The name of the remote array for adding new SRDF group. This option is used only with the ADD control action. If the array ID is known, the name could be entered. If the array ID is not known, press **F4** to display a list of all local and remote arrays attached to the host.

RGRPNUM (*group_nmber*)
Specifies the SRDF group number for the remote array. This option is used only with the ADD control action.

RDIRPORT((*Dir1*(*Port1 Port2* …)[(*Dir2* (*Port….*)]…)

Specifies remote director/port combination. This option is used with the ADD and MODIFY control actions. Options are:

- Director ID (*Dir 1*, *Dir 2...*) — This is the list of the remote array directors, such as, 1a, 1b and so on.
- Port Number (*Port 1, Port 2...)* — This is the list of remote array director ports.

Note: For PowerMaxOS 5978 and HYPERMAX 5977, the RDF director supports up to 32 ports.

You can use the SYMCLI command symcfg to display all available remote director/port combinations. For example:

```
Addlible emcctl
CALL PGM(SYMCFG) PARM('list' '-ra' 'all' '-sid'
'000123456789')
```

AUTO_LINK(*YES|*NO)

Specifies whether to use the auto link recovery feature locally. This option is used only with ADD action.

- *YES — Enables local Auto Link Recovery. This is the default.
- *NO — Disables local Auto Link Recovery.

RAUTOLINK(*YES|*NO)

Specifies whether to use the auto link recovery feature remotely. This option is used only with ADD action.

- *YES — Enables remote Auto Link Recovery. This is the default.
- *NO — Disables remote Auto Link Recovery.

LINK_LIMBO (0-120)

Note: The option should only be used by experienced SRDF users.

Specifies the local link limbo value which is the length of time for Enginuity to wait, from the point of a link-down detection, prior to updating the link status as Down. If the link status is still sensed as Not Ready after the link limbo time expires, devices are marked Not Ready to the link. The link limbo range is 0-120 seconds. The default time is 10 seconds. This option is used only with the ADD control action.

RLINK_LIMBO (0-120)

Specifies the remote link limbo value which is the length of time for Enginuity to wait, from the point of a link-down detection, prior to updating the link status as Down. If the link status is still sensed as Not Ready after the link limbo time expires, devices are marked Not Ready to the link. The link limbo range is 0-120 seconds. The default time is 10 seconds. This option is used only with the ADD control action.

Examples

Create new SRDF groups on local and remote arrays

```
CTLRDFGRP ACTION(ADD) SYMID(000111222333) GRPLABEL(GRP_20_21) GRPNUM(20)  DIRPORT((7H (10 11)) (8H (8 9 10 11))) RSYMID(000222333444) RGRPNUM(21) RDIRPORT((8H (8 9)))

CTLRDFGRP ACTION(ADD) SYMID(000111222333) GRPLABEL(GRP_22_23) GRPNUM(22)  DIRPORT((7H (10)) (8H (11))) RSYMID(000222333444) RGRPNUM(23) RDIRPORT((8H (8 9)))AUTO_LINK(*NO) R_AUTOLINK(*NO) LINK_LIMBO(20) RLINKLIMBO(20)
```

Modify SRDF groups

```
CTLRDFGRP ACTION(MODIFY) SYMID(000111222333) GRPNUM(20) MODIFY_ACT(ADD) DIRPORT((2F (10))) RDIRPORT((8F (10)))
CTLRDFGRP ACTION(MODIFY) SYMID(000111222333) GRPNUM(20) MODIFY_ACT(REMOVE) DIRPORT((8H (10 11)))
```

Remove SRDF groups from the Local array

```
CTLRDFGRP ACTION(REMOVE) SYMID(000111222333) GRPNUM(20)
```

Remove SRDF groups from the Remote array

```
CTLRDFGRP ACTION(REMOVE) SYMID(000222333444) GRPNUM(23)
```

Create a SRDF/Metro witness group

```
CTLRDFGRP ACTION(ADD) SYMID(000111222333) GRPLABEL(RKS_W) METROWIT(*YES) GRPNUM(4) DIRPORT((2F (27))) RSYMID(000222333444) RGRPNUM(18) RDIRPORT((7H (0)))
```

Create a SRDF/Metro group that uses bias

```
CTLRDFGRP ACTION(ADD) SYMID(000111222333) GRPLABEL(R1_B) GRPNUM(36) DIRPORT((2F (27))) RSYMID(000222333444) RGRPNUM(43) RDIRPORT((1G (26))) METROWIT(*NO)
```

CTLVWIT

Manages SRDF/Metro vWitness definitions on a storage array. You can create, enable, disable, and remove definitions.

Syntax

```
CTLVWIT SYMID(sym_id)
      WITNAME(witness_name)
      ACTION(control_action)
      [LOCATION(address)]
      [FORCE_FLAG(*YES|*NO)]
      [SYM_FORCE(*YES|*NO)]
```

Description

Manages the definitions of vWitness instances on a storage array. You can add a definition, enable a definition, disable a definition, and remove a definition.

Required parameters

`SYMID(sym_id)`
The name of the array on which this command is performed. If the array ID is known it can be entered. Otherwise, press F4 key to display a list of all local and remote arrays attached to the host.

`WITNAME(witness_name)`
A name for the vWitness definition. The name has up to 12 characters. The first character must be a letter; each of the remaining characters is a letter, digit, hyphen or underscore.

`ACTION(control_action)`
Specifies the control action to perform:

- ADD — Create a vWitness definition on the specified storage array.
- ENABLE — Make a previous disabled definition available for use once more.
- DISABLE — Suspend a definition from being used in a vWitness configuration.
- REMOVE — Remove a vWitness definition.

Optional parameters

`LOCATION (address)`
The address of the vWitness instance that this definition references. Specify the instances IP address or its DNS name.

`FORCE_FLAG(*YES|*NO)`
Use this option with the REMOVE control action only. Use a value of *YES if the selected vWitness instance is in use and there is another witness (physical or virtual) that can take over.

`SYM_FORCE(YES|NO)`
Use this parameter with the REMOVE control action only. Use a value of *YES when the selected vWitness instance is in use and there is no other witness (physical or virtual) to take over.

Examples

Create a vWitness definition on the storage array 000111222333

```
CTLVWIT SYMID(000111222333) WITNAME(TESTVWIT) ACTION(ADD) LOCATION('10.243.148.127')
```

Suspend (disable) the use of a vWitness definition

```
CTLVWIT SYMID(000111222333) WITNAME(TESTVWIT) ACTION(DISABLE) FORCE_FLAG(*NO) SYM_FORCE(*NO)
```

Enable the use of a previously suspended vWitness definition

```
CTLVWIT SYMID(000111222333) WITNAME(TESTVWIT) ACTION(ENABLE)
```

Remove a previously suspended vWitness definition

```
CTLVWIT SYMID(000111222333) WITNAME(TESTVWIT) ACTION(REMOVE)
```

DISCOVER

Discovers new devices, removes old device information from the SYMAPI database, and updates the database with new information. This command also updates information in the IMAGE file.

Syntax

```
DISCOVER MODE(*NO|*YES)
      CHECK(*NO|*YES)
```

Description

DISCOVER performs the discover process that is required when the configuration on the host or array changes. By default, information about all arrays visible at one time from this host remains in SYMAPI database until explicitly removed.

Required parameters

`MODE(*NO|*YES)`

- *NO — Do not remove old data from the SYMAPI database. This is the default value.
- *YES — Remove previously discovered array information from the SYMAPI database that is no longer valid (visible).

`CHECK(*NO|*YES)`

- *NO — Do not verify information stored in the IMAGE file with the current device information of the host or remove any old device information from the files. This is the default value.
- *YES — Verify the device information stored in the IMAGE file with the current device information of the host and remove any old device information from the files.

⚠CAUTION

Do not run this command with mode *YES when the RDF link is down. Doing so can remove relevant R2 information, such as remote BCV pairing for the array, from the SYMAPI database. If this information is removed, it requires additional repair or reconfiguration to restore the array data. Refer to "Running the DISCOVER command" on page 63 for more information.

DISCOVERC

Discovers new devices, removes old device information from the SYMAPI database, and updates the database with new information.

Syntax

```
DISCOVERC MODE(*NO|*YES)
```

Description

DISCOVERC performs the discover process on the target side that is required when the configuration on the host or array changes. By default, information about all arrays visible at some point from this host remains in SYMAPI database until explicitly removed.

Required parameters

`MODE(*NO|*YES)`

- *NO — Do not remove old data from the SYMAPI database. This is the default value.
- *YES — Remove previously discovered array information from the SYMAPI database that is no longer valid (visible).

⚠CAUTION

Do not run this command with MODE *YES when the RDF link is down. Doing so can remove relevant R2 information. Refer to "Running the DISCOVER command" on page 63 for more information.

DSPVWIT

Displays information on vWitness definitions.

Syntax

```
DSPVWIT
```

Description

Displays all the vWitness definitions defined in a SRDF/Metro configuration.

ENABLEDEV

Enables devices following a delete pair operation.

Note: Always make the R2 devices inaccessible to the host using unmap and unmask operations before issuing ENABLEDEV. Otherwise, data loss may ocur.

Syntax

```
ENABLE DEV SYMID(sym_id)
   FILENAME(file_name)
   RDFTYPE (rdf_type)
   SYM_FORCE(*YES|*NO)
```

Description

Following a delete pair operation, devices may be in the Not Ready state. ENABLEDEV makes those devices available once again.

Required parameters

`SYMID(sym_id)`
The name of the array on which this command is performed. Enter the array ID, if you know it. Otherwise, press F4 key to display a list of all local and remote arrays attached to the host.

`FILENAME(file_name)`
The name of the device file to use with the ENABLEDEV command. The file lists the devices to enable.

When creating the file, the following rules apply:

- The file is a record file and the maximum record length is 40 characters.
- The file contains a single column that lists the source devices.
- List each device on a separate line.
- Lines that begin with the pound sign (#) are comments.
- The device file resides in the library list.

Example device file:

```
# Devices for ENABLEDEV command
009A
009E
009F
```

`RDFTYPE(rdf_type)`
The type of SRDF identity that the devices are to assume. The available values for this parameter are:

- R1
- R2

Note: Do not present re-enabled R2 devices to the source (R1) host system as unexpected results, including data loss, can occur.

SYM_FORCE(*YES|*NO)

Indicates whether to use the array force operation when normally it is rejected.

- *YES — Enables Sym Force.
- *NO — Disables Sym Force. This is the default.

Note: To enable SYM_FORCE, set the value of the SYMAPI_ALLOW_RDF_SYMFORCE parameter in the options file to ENABLED.

⚠CAUTION

Use care when applying SYM_FORCE, as data could be lost or corrupted. Use of this option is not recommended, except in extreme cases.

Example

```
ENABLEDEV SYMID(000111222333) FILENAME(TESTFILE) RDFTYPE(R2) SYM_FORCE(*YES)
```

ENDSTMSRV

Shuts down the SRDF/TimeFinder Manager server job.

Syntax

```
ENDSTMSRV PORTNUM(port_number)
```

Description

ENDSTMSRV ends the SRDF/TimeFinder Manager server job session on the node where this command executes. If you do not include the port number, the command uses a default of 8000.

Required parameter

`PORTNUM(port_number)`
The TCP/IP port number.

If you used a port number other than the default, use that port number on the primary node, backup node, and main partition.

RELDEVLCK

Checks and releases device external locks (DEL) on an array for any image.

Syntax

```
RELDEVLCK IMAGENAME(image_name)
      PATHNAME(path_name)
      STEPNUM(step_number)
```

Description

RELDEVLCK checks for one or more SYMAPI device external locks (DEL) that were left on the array and attempts to release those locks using SYMAPI calls. Specify the image name, path name, and step number for the locks to be released.

Required parameters

`IMAGENAME(image_name)`
The name of the image for which you want to release the locks.

`PATHNAME(path_name)`
The path name associated with the image for which you want to release locks.

`STEPNUM(step_number)`
The step number associated with the path of the image for which you want to release locks.

Note: Steps in the path are defined during the configuration and are identified by their positions (1 through 5) in the path.

RELDEVLCKC

Checks and releases device external locks (DEL) on an array for any image on the target side.

Syntax

```
RELDEVLCKC IMAGENAME(image_name)
      PATHNAME(path_name)
      STEPNUM(step_number)
```

Description

RELDEVLCKC checks for one or more SYMAPI device external locks (DEL) that were left on the array and attempts to release those locks using SYMAPI calls. Specify the image name, path name, and step number for the locks to be released.

Required parameters

`IMAGENAME(image_name)`
The name of the image for which you want to release the locks.

`PATHNAME(path_name)`
The path name associated with the image for which you want to release locks.

`STEPNUM(step_number)`
The step number associated with the path of the image for which you want to release locks.

RTVSTMINF

Returns status information for the selected image, path, and step. It also returns SRDF/TimeFinder Manager, Solutions Enabler, Enginuity, and IO driver and release level information. This command can only be run within an IBM i command language program. See "Example command language program" on page 239 for an example command language program, and how to run the program.

Note: Some of the parameter return values are strings and others are decimal values. See "Return parameters" on page 236.

Syntax

```
RTVSTMINF IMAGENAME(image_name)
      PATHNAME(path_name)
      STEPNUM(step_number)
      NODENAME(node_name)
      IASPSTS(&IASPSTS)
      IOASTS(&IOASTS)
      SYMSTS(&SYMSTS)
      SYMSTSSTR(&SYMSTSSTR)
      STEPTYPE(&STEPTYPE)
      RDFTYPE(&RDFTYPE)
      RDFMODE(&RDFMODE)
      RDFACMODE(&RDFACMODE)
      INVTRACKS(&INVTRACKS)
      STMRELLVL(&STMRELLVL)
      STMEDTLVL(&STMEDTLVL)
      STMBLDLVL(&STMBLDLVL)
      SERELLVL(&SERELLVL)
      SEPCTHLVL(&SEPCTHLVL)
      LBIORELLVL(&LBIORELLVL)
      LBIOBLDLVL(&LBIOBLDLVL)
      UCODE1(&UCODE1)...
      UCODE5(&UCODE5)
      [SRCORTGT (TGTDEV|SRCDEV)]
```

Required parameters

`IMAGENAME(image_name)`
The name of the image.

`PATHNAME(path_name)`
The path name associated with the image.

`STEPNUM(step_number)`
The step number associated with the path of the image.

`NODENAME(node_name)`
The node name where the operation is performed. Node name must be a cluster node name. The node name entered must be part of the cluster that is configured on this system. If this is not the case a message appears indicating that the node cannot be found.

Note: The special node name R21NOHST is changed to R21NONOD.

Optional parameters

`SRCORTGT (TGTDEV|SRCDEV)`

This parameter is used to differentiate between source or target in the event both images are located on the same node.

This can happen, for example, when an R2 and remote BCV image are used on the same node. Only specifying the node is not enough. If the IASP file does not have an entry where the source and target node is the same, this parameter is set to the default value (TGTDEV).

Return parameters

- IASPSTS(&IASPSTS) — decimal value(5 0)
  Returns one of the following values for IASP status:
  - 0 – IASP status cannot be found.
  - 1 – IASP is varied off.
  - 2 – IASP is varied on.
  - 3 – IASP is active.
  - 4 – IASP is available

- IOASTS(&IOASTS) — decimal value(5 0)
  Returns one of the following value for IOA status:
  - 100 – IOA is active.
  - 101 – IOA is not active.
  - 102 – IOA is in a mixed state.
  - 103 – IOA status not found.

- SYMSTS(&SYMSTS)
  Returns a decimal value for the array copy status or RDF link status.

  CL var for SYMSTS (8 0)

  Specifies the name of the CL variable that receives the Symmetrix link state. The state value depends on the step type. When the setup is not in a correct state a general status is provided to indicate what is wrong. The SYMSTSSTR is set to unknown.

  Note: Values can be returned in string format using the return parameter SYMSTSSTR(&SYMSTSSTR).

  Return values for SYMSTS are dependent on the STEPTYPE:

  - STEPTYPE is BCV (1) or CLN (2) decimal value (8 0):
    - 00001000 - Never Established
    - 00001001 - Sync in progress
    - 00001002 - Synchronized
    - 00001003 - Split in progress
    - 00001004 - Split before sync
    - 00001005 - Split
    - 00001006 - Split no incremental
    - 00001007 - Restore in progress
    - 00001008 - Restored

- 00001009 - Split before restore
- 00001010 - State is invalid
- 00001011 - State not applicable

- STEPTYPE is NCL (3) or SNP (4) decimal value (8 0):
  - 00010000 - State not applicable
  - 00010001 - Copy in progress
  - 00010002 - Copied
  - 00010003 - Copy on access
  - 00010004 - Invalid
  - 00010005 - Create in progress
  - 00010006 - Created
  - 00010007 - Copy on write
  - 00010008 - Restored
  - 00010009 - Terminate in progress
  - 00010010 - Restore in progress
  - 00010011 - Failed
  - 00010012 - Recreated
  - 00010013 - Pre copy
  - 00010014 - Split

- STEPTYPE is RDF (5) decimal value (8 0):
  - 00000100 - Invalid
  - 00000101 - Sync in progress
  - 00000102 - Synchronized
  - 00000103 - Split
  - 00000104 - Suspended
  - 00000105 - Failed Over
  - 00000106 - Partitioned
  - 00000107 - R1 updated
  - 00000108 - R1 update in progress
  - 00000109 - Mixed
  - 00000110 - State not Applicable
  - 00000111 - Consistent
  - 00000112 - Transmit idle

- STEPTYPE is SVX (6) decimal value (8 0):
  - 00100000 - State not applicable
  - 00100001 - Establish in progress
  - 00100002 - Establish (Copy on write)
  - 00100003 - Restore in progress
  - 00100004 - Restored
  - 00100005 - Terminate in progress
  - 00100006 - Linked no copy
  - 00100007 - Linked copy in progress
  - 00100008 - Link copied
  - 00100009 - Snapshot failed
  - 00100010 - Link force failed

- STEPTYPE is SRDFM (7) decimal value (8 0):
  - 01000113 – Active Active
  - 01000114 – Active Bias
- STEPTYPE is BCV (1), CLN (2), NCL (3), SNP (4), RDF (5) or SVX (6) decimal variable (8 0). The following values are the same for all step types:
  - 0000061 - Configuration is not valid
  - 0000062 - Step is not configured
  - 0000063 - Step is not consistent
  - 0000064 - Config changed, image file not up to date
  - 0000065 - Not all devices are paired
  - 0000066 - Not all devices are paired but source devices are restored
  - 0000067 - All devices are paired and source devs are restored (temp status)
  - 0000068 - All devices are paired and source devs are linked (temp status)
  - 0000069 - Not all devices are paired but source devices are restore in prog.

- STEPTYPE(&STEPTYPE)—decimal value (5 0)
  Returns one of the following values for the step type that is configured for an image.
  - 0 – Step type could not be determined.
  - 1 – BCV type devices (BCV).
  - 2 – Clone type devices (CLN).
  - 3 – Native clone devices (NCL).
  - 4 – Snap type devices (SNP).
  - 5 – RDF type devices (RDF).
  - 6 – SnapVX type devices (SVX).
  - 7 – SRDF/Metro (SRDFM)

- RDFTYPE(&RDFTYPE) — decimal value (5 0)
  Returns one of the following values for the SRDF type for an image.
  - 0 – RDF type source devices is R1
  - 1 – RDF type source devices is R2
  - 2 – RDF type could not be determined.

- RDFMODE(&RDFMODE) — decimal value (5 0)
  Returns one of the following values for the SRDF link status.
  - 0 – RDF mode synchronous
  - 1 – RDF mode semi synchronous
  - 2 – RDF mode adaptive copy
  - 3 – RDF mode mixed
  - 4 – RDF mode not applicable
  - 5 – RDF mode asynchronous
  - 6 – RDF mode is SRDF/Metro

- RDFACMODE(&RDFACMODE) — decimal value (5 0)
  Returns one of the following values for the SRDF link adaptive copy status.
  - 0 – RDF AC Mode disabled
  - 1 – RDF AC Mode Write Pending Mode
  - 2 – RDF AC Mode Disk Mode

- 3 – RDF AC Mode mixed Mode
- 4 – RDF AC Mode is not applicable

- INVTRACKS(&INVTRACKS)
  Returns the value for the amount of invalid tracks or the percentage of data copied (16 characters).
  - 123456789012 - number of invalid tracks
  - 100% - percentage
  - INV_TR_NA - Invalid tracks could not be determined
- STMRELLVL(&STMRELLVL)
  Returns the value for the SRDF/TimeFinder Manager release level (5 characters).
- STMEDTLVL(&STMEDTLVL)
  Returns the value for the SRDF/TimeFinder Manager edit level (5 characters).
- STMBLDLVL(&STMBLDLVL)
  Returns the value for the SRDF/TimeFinder Manager build level (5 characters).
- SEREVLVL(&SEREVLVL)
  Returns the value for Solutions Enabler release level (15 characters).
- SEPTCLVL(&SEPTCLVL)
  Returns the value for Solutions Enabler patch level (15 characters).
- LBIORELLVL(&LBIORELLVL)
  Returns the value for I/O driver LIBIO release level (6 characters).
- LBIOBLDLVL(&LBIOBLDLVL)
  Returns the value for I/O driver LIBIO build level (5 characters).
- UCODE1(&UCOCDE1)...UCODE5(&UCOCDE5)
  Returns the value for EMC array ID, Enginuity release level, and Enginuity patch level for up to five arrays (35 characters).

Example command language program

A program must be written to run the RTVSTMINF command, and to format the display for the information that the command returns. The following is an example program named `TSTRTVINF`:

```
PGM         PARM(&IMGNAME &PATHNAME &STEPNUM &NODENAME &SRCORTGT)

              DCL         VAR(&IMGNAME)     TYPE(*CHAR) LEN(10)
              DCL         VAR(&PATHNAME)    TYPE(*CHAR) LEN(10)
              DCL         VAR(&STEPNUM)     TYPE(*CHAR) LEN(1)
              DCL         VAR(&NODENAME)    TYPE(*CHAR) LEN(8)
              DCL         VAR(&SRCORTGT)    TYPE(*CHAR) LEN(6)

              DCL         VAR(&IASPSTS)     TYPE(*DEC)  LEN(5 0)
              DCL         VAR(&IOASTS)      TYPE(*DEC)  LEN(5 0)
              DCL         VAR(&STEPTYPE)    TYPE(*DEC)  LEN(5 0)
              DCL         VAR(&SYMSTSSTR)   TYPE(*CHAR) LEN(32)
              DCL         VAR(&SYMSTS)      TYPE(*DEC)  LEN(8 0)
              DCL         VAR(&RDFTYPE)     TYPE(*DEC)  LEN(5 0)
              DCL         VAR(&RDFMODE)     TYPE(*DEC)  LEN(5 0)
              DCL         VAR(&RDFACMODE)   TYPE(*DEC)  LEN(5 0)
              DCL         VAR(&INVTRACKS)   TYPE(*CHAR) LEN(12)
              DCL         VAR(&STMRELLVL)   TYPE(*CHAR) LEN(5)
              DCL         VAR(&STMEDTLVL)   TYPE(*CHAR) LEN(5)
```

```
             DCL        VAR(&STMBLDLVL)  TYPE(*CHAR) LEN(5)
             DCL        VAR(&SERELLVL)   TYPE(*CHAR) LEN(20)
             DCL        VAR(&SEPTCHLVL)  TYPE(*CHAR) LEN(20)
             DCL        VAR(&LBIORELLVL) TYPE(*CHAR) LEN(6)
             DCL        VAR(&LBIOBLDLVL) TYPE(*CHAR) LEN(5)
             DCL        VAR(&UCODE1)     TYPE(*CHAR) LEN(35)
             DCL        VAR(&IASPSTSA)   TYPE(*CHAR) LEN(10)
             DCL        VAR(&IOASTSA)    TYPE(*CHAR) LEN(10)
             DCL        VAR(&STEPTYPEA)  TYPE(*CHAR) LEN(10)
             DCL        VAR(&SYMSTSA)    TYPE(*CHAR) LEN(10)
             DCL        VAR(&RDFTYPEA)   TYPE(*CHAR) LEN(10)
             DCL        VAR(&RDFMODEA)   TYPE(*CHAR) LEN(10)
             DCL        VAR(&RDFACMODEA) TYPE(*CHAR) LEN(10)


            RTVSTMINF   IMAGENAME(&IMGNAME) PATHNAME(&PATHNAME)
                        STEPNUM(&STEPNUM) NODENAME(&NODENAME)
                        SRCORTGT(&SRCORTGT) IASPSTS(&IASPSTS)
                        IOASTS(&IOASTS) SYMSTS(&SYMSTS)
                        STEPTYPE(&STEPTYPE) RDFTYPE(&RDFTYPE)
                        RDFMODE(&RDFMODE) RDFACMODE(&RDFACMODE)
                        INVTRACKS(&INVTRACKS) STMRELLVL(&STMRELLVL)
                        STMEDTLVL(&STMEDTLVL) STMBLDLVL(&STMBLDLVL)
                        SEREVLVL(&SEREVLVL) SEPTCLVL(&SEPTCLVL)
                        LBIORELLVL(&LBIORELLVL)LBIOBLDLVL(&LBIOBLDLVL)
                        UCODE1(&UCODE1) UCODE2(&UCODE2)UCODE3(&UCODE3)
                        UCODE4(&UCODE4) UCODE5(&UCODE5)

             CHGVAR     &IASPSTSA  &IASPSTS
             CHGVAR     &IOASTSA   &IOASTS
             CHGVAR     &STEPTYPEA &STEPTYPE
             CHGVAR     &SYMSTSA   &SYMSTS
             CHGVAR     &RDFTYPEA  &RDFTYPE
             CHGVAR     &RDFMODEA  &RDFMODE
             CHGVAR     &RDFACMODEA &RDFACMODE


SNDPGMMSG  MSG('RESULTS    STMVER:' *BCAT &STMRELLVL *BCAT +
               &STMEDTLVL *BCAT &STMBLDLVL)
SNDPGMMSG  MSG('RESULTS    LIBIOV:' *BCAT &LBIORELLVL *BCAT
               &LBIOBLDLVL)
SNDPGMMSG  MSG('RESULTS    SE_VER:' *BCAT &SERELLVL *BCAT &SEPTCHLVL)
SNDPGMMSG  MSG('RESULTS    IOASTS:' *BCAT &IOASTSA)
SNDPGMMSG  MSG('RESULTS   IASPSTS:' *BCAT &IASPSTSA)
SNDPGMMSG  MSG('RESULTS    SYMSTS:' *BCAT &SYMSTSA)
SNDPGMMSG  MSG('RESULTS SYMSTSSTR:' *BCAT &SYMSTSSTR)
SNDPGMMSG  MSG('RESULTS  STEPTYPE:' *BCAT &STEPTYPEA)
SNDPGMMSG  MSG('RESULTS   RDFTYPE:' *BCAT &RDFTYPEA)
SNDPGMMSG  MSG('RESULTS   RDFMODE:' *BCAT &RDFMODEA)
SNDPGMMSG  MSG('RESULTS RDFACMODE:' *BCAT &RDFACMODEA)
SNDPGMMSG  MSG('RESULTS INVTRACKS:' *BCAT &INVTRACKS)
SNDPGMMSG  MSG('RESULTS    UCODE1:' *BCAT &UCODE1)

ENDPGM
```

Run example program (`TSTRTVINF`) from the IBM i command line

```
CALL PGM(TSTRTVINF) PARM(PRIM1 LOCALNCL '1' NODE124)
```

A command line message returns upon completion of the RTVSTMINF command.

```
    Retrieve STM information completed..
    RESULTS    STMVER: 8.0 0 64
    RESULTS    LIBIOV: 4.4.02 10
    RESULTS    SE_VER: X8.0-1900 0.431
    RESULTS    IOASTS: 0000000100
    RESULTS   IASPSTS: 0000000001
    RESULTS    SYMSTS: 0000010002
    RESULTS SYMSTSSTR: Copied
    RESULTS  STEPTYPE: 0000000003
    RESULTS   RDFTYPE: 0000000002
    RESULTS   RDFMODE: 0000000004
    RESULTS RDFACMODE: 0000000004
    RESULTS INVTRACKS:         100%
    RESULTS    UCODE1: 000195600231 :5876      .229
                                                                    Bottom
Press Enter to continue.

F3=Exit   F5=Refresh   F12=Cancel   F17=Top   F18=Bottom
```

Figure 105 TSTRTVINF command return message

If the TSTRTVINF command fails, the following message returns:

```
Retrieve STM information failed.
```

Note: The output for this example program reports Enginuity version levels for one array. The Enginuity versions for up to five arrays can be reported.

It is not required to call for all parameters in the command line program. For example, to retrieve only the IOA status (Figure 106 on page 242), the RTVSTMINF and SNDPGMMSG commands in the CL program are as follows:

- For the RTVSTMINF command:

  ```
  RTVSTMINF IMAGENAME(&IMGNAME) PATHNAME(&PATHNAME) STEPNUM(&STEPNUM)
  NODENAME (&NODENAME) IOASTS(&IOASTS)
  ```

- For the SNDPGMMSG command:

  ```
  SNDPGMMSG  MSG('RESULTS    IOASTS:' *BCAT &IOASTSA)
  ```

```
Display All Messages
                                                              System:   XYZ
Job . . :   QPADEV0002,   User . . :   QSECOFR       Number . . . :   193487

CALL PGM(TSTRTVINF) PARM(PRIM1 LOCALNCL '1' NODE124)
    Retrieve STM information completed..
    RESULTS    IOASTS: 0000000100
                                                                      Bottom
Press Enter to continue.

F3=Exit   F5=Refresh   F12=Cancel   F17=Top   F18=Bottom
```

Figure 106 TSTRTVINF command return message when retrieving a subset of parameters

SAVEPOOL

Enables all SAVE devices in the default SAVE pool.

Syntax

```
SAVEPOOL SYMID(sym_id)
      POOLNAME(DEFAULT_POOL)
```

Description

SAVEPOOL enables all SAVE devices in the default SAVE pool.

Required parameters

`SYMID(sym_id)`
The name of the array where SAVE devices are enabled.

Note: If the array ID is known, you can enter its name. If the ID is unknown, press F4 to display a list of all local and remote arrays attached to the host.

`POOLNAME(DEFAULT_POOL)`
The name for the SAVE pool.

SETBIAS

Switches the bias between a pair of devices in a SRDF/Metro configuration.

Syntax

```
SETBIAS SYMMID(sym_id)
      GRPNUM(group_number)
      FILENAME(file_name)
      BIAS(R1|R2)
```

Description

Defines which device in a SRDF/Metro pair is the bias side when the pair use bias protection. This command is available only when the air state is Active Bias.

Required parameters

`SYMMID(sym_id)`
The identity of the storage array on which to set the bias.

`GRPNUM(group_number)`
The identifty of the SRDF group that contains the relevant device pair.

`FILENAME(file_name)`
Specifies the name of the device file to use with the SETBIAS command. The file lists the devices to change the bias setting for.

When creating the file, the following rules apply:

- The file is a record file and record length must not exceed 40.
- The file contains only one column that lists the source devices.
- List each device on a separate line.
- Comment lines are allowed and they must begin with the pound sign (#).
- The device file resides in the library list.

Example device file:

```
# Devices for SETBIAS command
009A
009E
009F
```

`BIAS(R1|R2)`
Defines the identity (R1 or R2) that each of the devices listed in the file adopts.

Example

```
SETBIAS SYMID(000111222333) GRPNUM(02) FILENAME(TESTFILE) BIAS(R2)
```

SETRDFGRP

Specifies the attribute settings for the RDF group.

Note: SETRDFGRP is currently limited to RDF/A attributes.

Syntax

```
SETRDFGRP SYMID(sym_id)
      GRPNUM(group_number)
      [POOLNAME(pool_name|NO_POOL|*SAME)]
      [DSE(ENABLE|DISABLE|*SAME)]
      [DSE_THOLD(percent)]
      [TRNIDLE(ENABLE|DISABLE|*SAME)]
      [DEVPACE(ENABLE|DISABLE|*SAME)]
      [WRP_ASTART(ENABLE|DISABLE|*SAME)]
      [WRP_DELAY(delay_time)]
      [WRP_THOLD(percent)]
      [HDW_COMPR(ENABLE|DISABLE|*SAME)]
      [SFW_COMPR(ENABLE|DISABLE|*SAME)]
      [AUTO_LINK(ENABLE|DISABLE|*SAME)]
      [LINK_LIMBO (link_limbo_time|*SAME)]
      [BOTH_SIDES(*YES|*NO)]
```

Description

SETRDFGRP manages the attribute settings for the RDF group, such as:

- Whether SRDF/A DSE is automatically activated when an SRDF/A session is activated
- How SRDF/TimeFinder Manager should handle situations where the transmission link is idling.

Required parameters

`SYMID(sym_id)`
The name of the array.

Note: If the array ID is known, enter its name. If the array ID is not known, press F4 to display a list of all local and remote arrays attached to the host.

`GRPNUM(group_number)`
The number of the RA group.

Optional parameters

`POOLNAME(pool_name|NO_POOL|*SAME)`
Specifies the name of the SRDF/A DSE pool to be associated with the SRDF group. By default, IBM i commands converts names to upper case. Put the name in quotes if you want it to be in the lower or mixed case.

Options are as follows:

- *Pool_name* — The name of the SAVE pool.
- NO_POOL — Disassociates a pool from the group. The pool is still related to the group until it is guaranteed drained for the group.

  Note: If more than one pool is associated to the group, all pools are disassociated.

- *SAME — Uses the pool currently associated. This is the default.

You can replace the existing associated pool for the group by setting a new pool name for the group. This automatically disassociates the old pool from the group. If any data from the group spilled over in the pool, the old pool is put in the Draining state.

Note: Pools should be of the same type, otherwise the new pool is associated to the group.

Note: For arrays running PowerMaxOS 5978 or HYPERMAX OS 5977, associating a DSE pool with an RDF group is not available. For this configuration, the SETRDFGRP command does not associate an RDF group with a DSE pool. The array is pre-configured to associate the DSE with a Storage Resource Pool (SRP), so managing pools is not required.

`DSE(ENABLE|DISABLE|*SAME)`

Specifies whether SRDF/A DSE is automatically activated when an SRDF/A session is activated for the group. Options are as follows:

- ENABLE — Automatically activates SRDF/A DSE when an SRDF/A session is activated for the group.
- DISABLE — Does not automatically activate SRDF/A DSE when an SRDF/A session is activated for the group.
- *SAME —Ignores this parameter during command processing, or when ENABLE or DISABLE is specified, and DSE auto start is already in the desired state. This is the default.

  Note: For arrays running PowerMaxOS 5978 or HYPERMAX OS 5977, DSE autostart for SRDF/A groups is enabled by default.

`DSE_THOLD(percent)`

Specifies the percentage of the write pending limit of the DSE threshold.

Groups with higher threshold values will keep their data in cache longer at the expense of groups with lower values. Valid values range from 20 to 100. If nothing is specified or the RDF group has a desired DSE threshold, SRDF/TimeFinder Manager ignores this parameter during command processing.

`TRNIDLE(ENABLE|DISABLE|*SAME)`

Specifies how to handle a situation where the transmission link is idling.

Transmit Idle, an SRDF/A Reserve Capacity feature, allows SRDF/A sessions to seamlessly tolerate a transient link failure. Transmit Idle provides an extra level of protection from sessions dropping when the RDF link cannot transmit data. When links associated with SRDF/A are unable to transmit data, the Transmit Idle state is activated.

Options are as follows:

- ENABLE — Tolerate SRDF/A.
- DISABLE — Drop SRDF/A.
- *SAME —Ignores this parameter during command processing, or when ENABLE or DISABLE is specified, and Transmit Idle is already in the desired state. This is the default.

DEVPACE(ENABLE|DISABLE|*SAME)

Specifies whether SRDF/A Device-level Write Pacing is automatically activated when an SRDF/A session is activated for the group:

- ENABLE — Automatically activates SRDF/A Device-level Write Pacing when an SRDF/A session is activated for the group.
- DISABLE — Do not automatically activate SRDF/A Device-level Write Pacing when an SRDF/A session is activated for the group.
- *SAME —Ignores this parameter during command processing, or when ENABLE or DISABLE is specified, and DSE device-level write pacing auto start is already in the desired state. This is the default.

  SRDF/A device-level write pacing feature is supported with the Enginuity 5876 or higher.

  Enabling SRDF/A device-level write pacing is required when using RDF2 Async device as a Snap or VP Snap source.

WRP_ASTART(ENABLE|DISABLE|*SAME)

Specifies whether the SRDF/A group-level write pacing feature is automatically enabled or disabled when an SRDF/A session is activated for the SRDF group:

- ENABLE — Automatically activates SRDF/A Group-level Write Pacing when an SRDF/A session is activated for the group.
- DISABLE — Do not automatically activate SRDF/A Group-level Write Pacing when an SRDF/A session is activated for the group.
- *SAME — Ignores this parameter during command processing, or when ENABLE or DISABLE is specified, and SRDF/A group-level write pacing is already in the desired state. This is the default.

WRP_DELAY(*delay_time*)

Sets the maximum host I/O delay, in microseconds, that triggers SRDF/A write pacing. Valid values are between 1 and 1000000 micro-seconds. The default value is 50000 micro-seconds.

If the write pace delay is not specified, or an SRDF group already has a desired write pacing delay, SRDF/TimeFinder Manager ignores this parameter during command processing.

WRP_THOLD(*percent*)

Sets the minimum percentage of the system write-pending cache at which the array begins pacing host write I/Os, for an SRDF group. Valid values are between 1 and 99. The default value is 60.

If the write pace threshold is not specified, or an SRDF group already has a desired write pacing threshold, SRDF/TimeFinder Manager ignores this parameter during command processing.

Note: SRDF/A group-level write pacing feature is supported with the Enginuity 5876 or later. If you plan on swapping the personalities of the R1 and R2 devices, configure the same SRDF/A write pacing values on both of these devices.

SRDF software and hardware compression.

Compression minimizes the amount of data to be transmitted over an SRDF link. Enabling hardware compression on GigE directors and software compression on Fibre and GigE directors using Solutions Enabler is available with Enginuity 5876 or later.

Rules and restrictions for compression are as follows:

- Enabling hardware compression on Fibre directors is available with Enginuity 5876 or later.
- Software and hardware compression can be enabled on the R1 and R2 sides of the SRDF link, but the actual compression takes place on the side initiating the I/O, which is typically the R1 side. Since most I/O requests are initiated from the R1 side, make sure compression is enabled on this side.
- Software and hardware compression can only be set at the group level.

`HDW_COMPR(ENABLE|DISABLE|*SAME)`
Specifies whether to enable or disable hardware compression for the SRDF group:

- ENABLE — Enables hardware compression for the SRDF group.
- DISABLE — Disables hardware compression for the SRDF group.
- *SAME — Ignores this parameter during command processing, or when ENABLE or DISABLE is specified, and compression for the SRDF group is already in the desired state. This is the default.

`SFW_COMPR(ENABLE|DISABLE|*SAME)`
Specifies whether to enable or disable software compression for the SRDF group:

- ENABLE — Enables software compression for the SRDF group.
- DISABLE — Disables software compression for the SRDF group.
- *SAME — Ignores this parameter during command processing, or when ENABLE or DISABLE is specified, and compression for the SRDF group is already in the desired state. This is the default.

`AUTO_LINK(ENABLE|DISABLE|*SAME)`
Specifies whether to use the Auto Link Recovery feature locally.

- ENABLE — Enables Auto Link Recovery.
- DISABLE — Disables Auto Link Recovery.
- *SAME — Ignores this parameter during command processing, if *SAME is specified, or if ENABLE or DISABLE is specified and Auto Link Recovery is in the desired state. This is the default.

`LINK_LIMBO (link_limbo_time|*SAME)`

Note: The option should only be used by experienced SRDF users.

Specifies the length of time for Enginuity to wait from the point of link-down detection before actually updating the link status as down. If the link status is still sensed as Not Ready after the link limbo time expires, devices are marked Not Ready to the link. If nothing is specified, Ignores the parameter during command processing, if *SAME is specified, or if RDF group already has a desired link limbo value. The link limbo range is 0-120 seconds. The default time is 10 seconds.

BOTH_SIDES(*YES|*NO)

Specifies whether changes should be applied only locally or both locally and remotely:

- *YES — Changes are applied locally and remotely.

  Note: If the remote side cannot be reached and the flag is set to YES, the command fails.

- *NO — Changes are applied locally. This is the default.

Examples

```
SETRDFGRP SYMID(000123456789) GRPNUM(7B) DSE_THOLD(60) TRNIDLE(ENABLE) DEVPACE(ENABLE)
WRPA_START(ENABLE) WRP_DELAY(60000) WRP_THOLD(70) HRDW_COMPR(ENABLE) SOFTW_COMPR(ENABLE)
BOTH_SIDES(*YES)


SETRDFGRP SYMID(000123456789) GRPNUM(20) TRNIDLE(ENABLE) DEVPACE(ENABLE)
WRP_ASTART(ENABLE) WRP_DELAY(60000) WRP_THOLD(70) HDW_COMPR(ENABLE) SFW_COMPR(ENABLE)
AUTO_LINK(ENABLE) LINK_LIMBO(20) BOTH_SIDES(*YES)
```

SETRDFMOD

Sets the SRDF mode for a selected image.

Syntax

```
SETRDFMOD IMAGE_NAME(image_name)
      PATH_NAME(path_name)
      STEPNUM(step_number)
      MODE_TYPE(SYNC|ASYNC|ACDM|ACWP|METRO)
      METROMODE(A|B)
```

Description

SETRFDMOD sets the SRDF mode (Synchronous, Asynchronous, adaptive copy disk mode (ACDM), and adaptive copy write pending (ACWP)) for a selected image. The name of image name, path name, and the SRDF mode associated with the image are mandatory items.

Note: SRDF mode adaptive copy write pending (ACWP) is not available when the R1 mirror of the RDF pair is on an array running PowerMaxOS 5978 or HYPERMAX OS 5977. For this configuration, the SETRDFMOD command returns an error if the mode is set to adaptive copy write pending. In this case use another node.

Required parameters

`IMAGE_NAME(image_name)`
Specifies the name of the image.

`PATH_NAME(path_name)`
Specifies the name of the path for the image.

`STEPNUM(step_number)`
Specifies the step number.

Steps in the path are defined during configuration and are identified by their positions (1 through 5) in the path.

`MODE_TYPE(SYNC|ASYNC|ACDM|ACWP|METRO)`
Specifies the SRDF mode.

IMPORTANT

Do not use ACDM or ACWP mode for SRDF while creating RDF or remote TimeFinder images (split) using SRDF/TimeFinder Manager. This is not supported. Instead, use SYNC or ASYNC mode.

`METROMODE(A|B)`
This parameter is required only when the value of MODE_TYPE is METRO. The value of this parameter defines the protection mechanism that the SRDF/Metro configuration uses:

- A — Witness protection
- B — Bias protection

Examples

```
SETRDFMOD IMAGE_NAME(SYSTEM) PATH_NAME(SRDFA) STEPNUM(1) MODE_TYPE(SYNC)
SETRDFMOD IMAGE_NAME(SYSTEM) PATH_NAME(SRDFA) STEPNUM(1) MODE_TYPE(ASYNC)
SETRDFMOD IMAGE_NAME(SYSTEM) PATH_NAME(SRDFA) STEPNUM(1) MODE_TYPE(ACDM)
SETRDFMOD IMAGE_NAME(SYSTEM) PATH_NAME(SRDFMETRO) STEPNUM(1) MODE_TYPE(METRO) METROMODE(A)
```

STMENDSBS

Stops all running subsystems except for the subsystem where the daemon jobs are running.

Syntax

```
STMENDSBS OPTION(*CNTRLD|IMMED)
      DELAY(0-99999 seconds|*NOLIMIT)
      ENDSBSOPT(*DFT|*NOJOBLOG|*CHGPTY|*CHGTSL)
```

Description

STMENDSBS stops all running subsystems prior to splitting BCV devices or Clone pairs, or SRDF links.

- You can run this command interactively or in batch mode
- To run this command, you must have the *JOBCTL privilege in your user profile or in any of the group profiles that your account is associated with.
- STMENDSBS waits for all subsystems and jobs to end. Once they have ended, STMENDSBS sends this message to QSYSOPR:

  ```
  CTL240: System ended to semi-restricted condition.
  ```

  If a subsystem or job does not complete for any reason, STMENDSBS does not continue and does not send this message to QSYSOPR.
- When STMENDSBS completes in batch mode, the QCTL is in the ACTIVE state. When the command completes interactively the QCTL state is RSTD.

Required parameters

`OPTION(*CNTRLD|*IMMED)`
Specifies whether jobs in the subsystems are ended in a controlled manner or immediately.

`DELAY(0-99999 seconds|*NOLIMIT)`
Specifies the time delay for a controlled ending for running subsystems. The default value is *NOLIMIT.

`ENDSBSOPT(*DFT|*NOJOBLOG|*CHGPTY|*CHGTSL)`
Specifies the options available when ending the active subsystems. In most cases, specifying one of these options improves the performance of the ENDSBS command. This parameter has no effect on jobs that are already in the ending status.

- *DFT — The subsystems end with special ending option. This is the default value.
- *NOJOBLOG — No job logs are created for jobs that are ended. This includes subsystem monitor jobs and all user jobs in the subsystem.
- *CHGPTY — The CPU priority of jobs that are ending changes to a higher value (lower priority) and may take longer to finish. The remaining active jobs on the system may have better performance when *CHGPTY is specified. This option is ignored if the subsystem is ending controlled. However, if the DELAY time limit expires, this option takes effect immediately.

- *CHGTSL — The time slice of jobs that are ending is changed to a lower value and may take longer to finish. The remaining active jobs on the system may have better performance when *CHGTSL is specified. This option is ignored if the subsystem is ending controlled. However, if the DELAY time limit expires, this option takes effect immediately.

Examples

```
STMENDSBS OPTION(*CNTRLD) DELAY(*NOLIMIT)
```

STMRMTCMD

Runs remote commands.

Syntax

```
STMRMTCMD CMD(stm command)
      RMTNODENAME(nodename)
      PORTNUM (8000|1-65535)
```

Description

STMRMTCMD command allows execution of commands, on a remote IBM system, through the STM server. The commands used with the STMRMTCMD command are limited to the SRDF/TimeFinder Manager commands that are available on the source or client system, and the following IBM i series commands:

- APING
- DSPASPSTS
- PING
- VRYCFG
- VFYTCPCNN
- WRKCFGSTS

Prior to running the STMRMTCMD command, the STM server must be running and configurations must be complete.

When the command completes a file is created that holds the results of the remote command. The file is located in the QGPL library and called RMTCMDOUTC at the client side and RMTCMDOUTS on the server side, and can have multiple members (SPL, SPL2, and so on) depending on the output.

See "STRSTMSRV" on page 260 for starting the STM server.

Required parameters

`STMRMTCMD CMD(stm command)`
Specifies the SRDF/TimeFinder Manager command to run from the remote IBM i system.

`RMTNODENAME(nodename)`
Specifies the node name of the remote IBM i system.

`PORTNUM(8000|1-65535)`
Specifies the port number of the remote STM server. Default port is 8000.

Examples

Run CTLACT command remotely to check the status of TimeFinder Snap.

Start STM server, see "STRSTMSRV" on page 260, then enter the following command:

```
STMRMTCMD COMMAND('CTLACT IMAGE_NAME(SYSTEM) PATH_NAME(LOCALSNAP) STEP
NUM(1) ACTION(STATUS)') RMTNODNAME(NODE100) PORTNUM(12345)
```

STMSTRSBS

Starts all subsystems.

Syntax

```
STMSTRSBS
```

Description

STMSTRSBS starts all subsystems after running the STMENDSBS command. The command checks to see if the daemon jobs are the only jobs active or if the system is in already in a restricted state. This command then brings the system into a true restricted state by ending all subsystems including the daemon jobs. When this is complete, the controlling subsystem is started. This starts all the required subsystems and jobs, depending on what jobs are defined as auto start jobs and what is set as a startup program defined in the system value QSTRUPPGM.

- You can run this command interactively or in batch mode.
- To run this command you must have the *JOBCTL privilege in your user profile or in any of the group profiles that you account is associated with.
- STMSTRSBS calls a custom startup program when running in batch mode, as long as the STARTUPPGM data area is not blank. STMSTRSBS does not call a custom startup program when running interactively.
- The custom startup program can be set up in the EMCCTL or STARTUPPGM data area in the EMCCTL library. The format of the STARTUPPGM data are is:
  - The first 10 characters contain the program name.
  - The next 10 characters contain the name of the library where the program is stored.

  If this identification is entered incorrectly, STMSTRSBS fails.

Required parameters

None

STORDAEMON

Starts, stops and queries Solutions Enabler daemons.

Syntax

```
STORDAEMON
      ACTION (START | SHUTDOWN | STARTALL | SHUTDOWNALL | STRNODEMON |
          ENDNODEMON | LIST | SHOW | SHOWLOG)
      DAEMONNAME (STORAPID | STORSRVD | STORGNSD)
      PORTNBR (port_number between 1 and 65535)
      LISTOPT (VERBOSE | RUNNING)
      SHOWLOGOPT (number between 1 and 1000)
```

Description

Starts, stops, and queries the Solutions Enabler daemons:

- Storapid (Base)
- Storsrvd (Symapi server)
- Storgnsd (GNS)

Note: See "Working with Solutions Enabler daemons" on page 72 for daemon descriptions.

Required parameters

`STORDAEMON ACTION (START | SHUTDOWN | SHUTDOWNALL | STRNODEMON | ENDNODEMON | LIST | SHOW | SHOWLOG)`
Specifies which action to perform on the specified daemon:

- START — Starts the specified daemon.
- SHUTDOWN — Stops specified daemon.
- STARTALL — Starts all daemons controlled by this command.
- STRNODEMON — Starts the node monitor.
- ENDNODEMON — Stops the node monitor.
- SHUTDOWNALL — Stops all daemons controlled by this command.
- LIST — Lists all running daemons. See examples below for list type with LISTOPT set to RUNNING and VERBOSE.
- SHOW — Shows details for a specified daemon.
- SHOWLOG — Displays the lines from a daemon log file. The number of lines to display is specified by SHOWLOGOPT, starting with the most recently written lines.

`DAEMONNAME (STORAPID | STORSRVD | STORGNSD)`
Specifies the daemon on which to perform the action.

`PORTNBR (port_number between 1 and 65535)`
Sets the port number for daemon commands. The default is 2707.

`LISTOPT (VERBOSE | RUNNING)`
When running the LIST command, specifies what data to display.

- VERBOSE — Lists the details for all running daemon.
- RUNNING — Lists all running daemons, names only.

SHOWLOGOPT(*number between 1 and 1000*)
When running the SHOWLOG commands specifies the number of lines to display from a daemon log file, starting with the most recently written lines. The default is 10.

Examples

Start and stop a daemon

```
STORDAEMON ACTION(START) DAEMONNAME(STORAPID) PORTNBR(2707)

STORDAEMON ACTION(SHUTDOWN) DAEMONNAME(STORAPID)
```

Stop all daemons

```
STORDAEMON ACTION(SHUTDOWNALL)
```

Show the details of a daemon

```
STORDAEMON ACTION(SHOW) DAEMONNAME(STORAPID)

Daemon State                                    : Running
Daemon Start Time                               : Mon Dec  2 08:55:04
   2013
Version                                         : X8.0-1900 (0.353)
Auto-Restart by Watchdog                        : Disabled

Total Number of Connections                     : 4
Number of Active Connections                    : 0
Total Number of Requests                        : 171
Gatekeeper Management                           :
Gatekeeper (GK) Management State                : Running
GK Management Total open GKs                    : 0
GK Management open GKs highwater                : 2
Allow PPath Native Devs as GKs                  : N/A
MPGK Selection                                  : conditional
Unreachable Array Time Limit                    : 7 days
Path Algorithm                                  : QOS
Alt. Path Algorithm                             : random
Allow QOS Path Failover                         : Yes
Force RDF Path Analysis                         : No
Total File Descriptors                          : 22

Feature options                                 :
  Parallel Discovery                            : Enabled
  Parallel Inquiry                              : Disabled
  Lock Information Export                       : Enabled
  Background Audit Logging                      : Disabled
  Background App Registration                   : Enabled

Config options                                  :
  Resource Recovery Interval                    : 7 second(s)
  Parallel Discovery Max Threads                : 6
  Parallel Inquiry Max Threads                  : N/A
  Device Inquiry Timeout                        : N/A
  Background App Reg Interval                   : 900 second(s)
  Persistent DEL Recovery Interval              : 60 second(s)
  SFS File Recovery Interval                    : N/A
  Maximum FDs allowed                           : N/A

QOS options                                     :
  Dev Name Inquiry Interval                     : 15 minutes
  Rec. Dev Name Inq. Interval                   : 15 minutes
```

```
  Short Term Sync interval                    : 1 minute
  Short Term Sync Limit                       : 5 minutes
  Long Term Sync Interval                     : Disabled
  Short Term Backoff Interval                 : 15 minutes
  Short Term Backoff Limit                    : 1 hour
  Long Term Backoff Interval                  : 1 hour
  Long Term Backoff Limit                     : 1 day
  Min Ping Interval                           : 5 minutes
  Discovery Interval                          : 1 day
  Max GK Utilization (%)                      : 50
```

List all running daemons

```
STORDAEMON ACTION(LIST) LISTOPT(RUNNING)
```

Output:

```
Available Daemons  ('[*]': Currently Running):

[*] storapid              EMC Solutions Enabler Base Daemon
[*] storgnsd              EMC Solutions Enabler GNS Daemon
```

List all the running daemons

LISTOPT set to VERBOSE.

Note: This example output is abbreviated. The actual output lists all the parameters as shown in the output for SHOW action.

```
STORDAEMON ACTION(LIST) LISTOPT(VERBOSE)

[*] storapid              EMC Solutions Enabler API Daemon
------------------
Daemon State                                : Running
Daemon Start Time                           : Wed Nov 27 03:49:27
   2013
Version                                     : X8.0-1900 (0.353)
Auto-Restart by Watchdog                    : Disabled

Total Number of Connections                 : 3
Number of Active Connections                : 0
Total Number of Requests                    : 4
.......

[*] storgnsd              EMC Solutions Enabler GNS Daemon
------------------
Daemon State                                : Running
Daemon Start Time                           : Wed Nov 27 03:50:27
   2013
Version                                     : X8.0-1900 (0.353)
Auto-Restart by Watchdog                    : Disabled

Total Number of Connections                 : 3
Number of Active Connections                : 0
Total Number of Requests                    : 4
......
```

Show the log file for the specified daemon

SHOWLOG option set to 10 lines.

STORDAEMON ACTION(SHOWLOG) DAEMONNAME(STORAPID)

Output:

```
The storapid daemon uses the following log files ('*' : most recently written):
  [*] /var/symapi/log/storapid.log0
      /var/symapi/log/storapid.log1

  Last 10 lines of log file:

       [19825                -sync] Dec-02 10:02:14.044 :
   [dmnInqryCreateDeviceList()] Inquiry request received from gkmgt
       [19825                -sync] Dec-02 10:02:14.861 :
   [dmnInqryCreateDeviceList()] Inquiry complete in 814.576 msec for gkmgt, 25
   devices found
       [19825                -sync] Dec-02 10:17:15.765 :
   [dmnInqryCreateDeviceList()] Inquiry request received from gkmgt
       [19825                -sync] Dec-02 10:17:16.686 :
   [dmnInqryCreateDeviceList()] Inquiry complete in 913.256 msec for gkmgt, 25
   devices found
       [19825                -sync] Dec-02 10:32:17.590 :
   [dmnInqryCreateDeviceList()] Inquiry request received from gkmgt
       [19825                -sync] Dec-02 10:32:18.409 :
   [dmnInqryCreateDeviceList()] Inquiry complete in 817.080 msec for gkmgt, 25
   devices found
       [19825                -sync] Dec-02 10:47:19.313 :
   [dmnInqryCreateDeviceList()] Inquiry request received from gkmgt
       [19825                -sync] Dec-02 10:47:20.137 :
   [dmnInqryCreateDeviceList()] Inquiry complete in 821.808 msec for gkmgt, 25
   devices found
       [19825                -sync] Dec-02 11:02:21.042 :
   [dmnInqryCreateDeviceList()] Inquiry request received from gkmgt
       [19825                -sync] Dec-02 11:02:21.858 :
   [dmnInqryCreateDeviceList()] Inquiry complete in 814.504 msec for gkmgt, 25
   devices found
```

STRSTMSRV

Starts the SRDF/TimeFinder Manager server job.

Syntax

```
STRSTMSRV
      USERID(user_id)
      PORTNUM(port_number)
      JOBQ(job_queue)
      CCSID(*CURRENT|*USRPRF|*SYSVAL|*HEX|1-65535)
```

Description

STRSTMSRV starts the job that runs the server job on the node on which the command executes. This command must be run on all nodes. If you do not supply a port number, the STRSTMSRV command uses the default of 8000.

Required parameters

`USERID(user_id)`
The user ID of the submitted job under which the server job runs. The default value is *CURRENT.

`PORTNUM(port_number)`
The TCP/IP port number. The default value is 8000. If you use a different port number, keep the port number consistent on all nodes participating in IASP operations.

`JOBQ(job_queue)`
The name of the job queue where the server job runs. The default value is *LIBL/EMCCTL.

`CCSID(*CURRENT|*USRPRF|*SYSVAL|*HEX|1-65535)`
The setting of the coded character set ID (CCSID) for the SRDF/TimeFinder Manager server job. The values are:

- *CURRENT — The CCSID specified for the job that is currently running is used.
- *USRPRF — The CCSID specified in the user profile where the submitted job initially runs is used.
- *SYSVAL — The CCSID specified in the system value QCCSID at the time the job is started is used.
- *HEX — The CCSID 65535 is used.
- 1-65535 — Specify the CCSID.

Usage notes

If STRSTMSRV fails to start, use the NETSTAT option 3 (IPV4) or option 6 (IPV6) to ensure no communication jobs are using 8000. STMSERVER runs one job constantly; others are submitted and subsequently ended during operations.

Example

```
STRSTMSRV USERID(*CURRENT) PORTNUM(8000) JOBQ(*LIBL/EMCCTL)
```

APPENDIX B
Symmetrix Integration Toolkit

This appendix lists the supported Solutions Enabler CLI commands that are packaged with the SRDF/TimeFinder Manager software in the Symmetrix Integration Toolkit, and provides some examples. These commands are run from the IBM i command line or Qshell.

- Supported Solutions Enabler CLI commands 262
- Command line examples 263
- Supporting documentation 263

Supported Solutions Enabler CLI commands

A limited set of Solutions Enabler CLI commands are available on IBM i systems and are run from the IBM i command line or Qshell.

- `symaccess` — Perform auto-provisioning group operations on storage, initiator, and port groups. Also create and manage masking views.
- `symacl -unique` — Generate a unique 24-digit access ID for the host machine or operating node to the array. The `-unique` parameter is the only one you can use with the `symacl` command.
- `symapierr` — Translate a SYMAPI return code to a string.
- `symcfg` — Discover arrays and display configuration information.
- `symcli` — Obtain a brief description of all commands in the Solutions Enabler Symmetrix Command Line Interface (SYMCLI).
- `symclone` — Perform TimeFinder/Clone operations on a device group, composite group, or devices listed in a device file.
- `symdev` — Perform operations on a specific device.
- `symfast` — Perform FAST operations on tiers, storage groups, policies and FAST controller.
- `symlist` — ???
- `symtier` — Create and manage storage tiers that can be used for FAST policies.
- `symmigrate` — Change the physical disk space associated with a device to a different data protection scheme, or relocate to disks with different performance characteristics. This command uses VLUN VP Migration technology.

  Note: For arrays running PowerMaxOS 5978 or HYPERMAX OS 5977, all devices are under internal, automated FAST control, which handles all thin device binding and allocation. Therefore, `symfast, symtier,` and `symmigrate` commands are not available.

- `syminq` — Issue a SCSI INQUIRY command, and optionally a SCSI READ CAPACITY, on one or all devices. Also, obtain a list of the local host's HBAs.
- `symlmf [list|query]` — Reports license types and usage.

  For more information on FAST technology and VLUN VP Migration, refer to *Dell EMC Solutions Enabler Array Controls and Management CLI Guide.*

- `symrdf` — Perform control operations on SRDF device pairs in an R1/R2 configuration, and concurrent and cascaded SRDF configurations.
- `symsg` — Perform operations on storage groups located in a specified array.
- `symsnapvx` — Perform TimeFinder/SnapVX control, list, and verify operations on a devie list, a list of ranges, a device group (DG), a composite group (CG), or a storage group (SG).

Command line examples

The following example displays information about all front-end directors for the specified array using the `symcfg` command in both the IBM i command line and Qshell formats.

IBM i command line

```
Addlible emcctl
CALL PGM(SYMCFG) PARM(list '-SA' 'ALL' '-sid' '710')
```

Qshell

From the IBM i command line, enter the Qshell setup commands:

```
Addlible emcctl
strqsh
cd /qsys.lib/emcctl.lib
```

From the Qshell, enter:

```
symcfg.pgm list -SA ALL -sid 710
```

Note: There is a save file available called `symapicnfg`. This file is restored during installation and is called `/var/symapi/config/alias/qshrc`. This file holds aliases for the symcli's. To enable them follow the instructions in the header of both files.

Supporting documentation

Details on command actions and parameters, options, and error codes for each supported command are in the *Dell EMC Solutions Enabler CLI Reference Guide.*

Help is also available using the `-h` parameter with the command. For example:

IBM i command line

```
CALL PGM(SYMCFG) PARM('-h')
```

Qshell

```
symcfg.pgm -h
```

Or when alias file is configured:

```
symcfg -h
```

APPENDIX C
Standard Features Messages

This appendix provides reference information about messages that SRDF/TimeFinder Manager components can generate, the reason for the message, and the recommended user action.

CTL0001

```
SYMAPI error &1 (&2).
```

Cause: SYMAPI detected an inconsistency.

Action: Refer to the Dell EMC Solutions Enabler documentation.

CTL0002

```
System level error.
```

Cause: A call to a system function such as memory allocation failed.

Action: This may be a by-product of a serious system condition that needs attention. After resolving the system problem, the software should run normally, since no internal error caused this problem.

CTL0003

```
Step number &1 was not found for the path &2.
```

Cause: A user specified the wrong step number was for the selected path.

Action: Enter the appropriate step for the path and try the operation again.

CTL0004

```
Failure opening file.
```

Cause: The file is not found, the file is locked, or the user does not have authority to open the file. Refer to the job log for details.

Action: The failure depends on what the root cause is. Wait until the file is unlocked, change the authority, or take other appropriate action.

CTL0005

```
Error within Control program. Serious program problem. Contact EMC
Customer Support.
```

Cause: Not specified, but a serious logic or memory problem occurred.

Action: Contact the Dell EMC Customer Support Center.

CTL0006

```
Path &1 is not disassociated from the image &2.
```

Cause: The path associated with the selected image has configured steps. The user opted not to disassociate the path.

Action: This is an informational message; no user action is necessary.

CTL0007

```
Cannot show logical devices for device group &1.
```

Cause: SYMAPI reports numeric error &2 (&3 - error message description).

Action: Refer to the Dell EMC Solutions Enabler documentation.

CTL0008

```
Error running SYMAPI program.
```

Cause: SYMAPI reports numeric error &1 (&2 - error message description).

Action: Refer to the Dell EMC Solutions Enabler documentation. If the documentation does not provide a solution to the problem, contact the Dell EMC Customer Support Center.

CTL0009

```
On Symmetrix &1 standard device &2 now is paired with BCV device &3.
```

Cause: The standard device was previously paired with BCV device &4, which is on the same physical disk as one of the mirrors of the standard device.

Action: Manually unpair the standard device and then pair it again.

CTL0010

```
No available unpaired BCV/SNAP devices.
```

Cause: There are no available BCV or Snap devices of the right model and capacity.

Action: Either free up the BCV or Snap devices by unpairing them from another use, or have the Dell Technologies Customer Support Engineer configure additional BCV or Snap devices within the array.

CTL0011

```
Cannot unpair BCV devices which are synchronized or in progress or
   unpaired.
```

Cause: An attempt was made to unpair devices for the step where devices are established or in progress.

Action: Use control options to split BCV pairs before trying this option. To change device pairing, you must first split the established pairs. If the devices are already unpaired, do nothing. If the devices are in a Synchronized or SyncInProg state, split the devices before making another attempt to unpair the devices.

CTL0012

```
On Symmetrix &1 standard device &2 is not paired.
```

Cause: There are no available BCV/Snap devices of the matching configuration.

Action: Make the appropriate device changes for the configuration.

CTL0036

```
User cancelled SRDF Links processing.
```

Cause: A user opted to cancel the SRDF links control action processing.

Action: This is an informational message; no user action is necessary.

CTL0037

```
Processing failed with SYMAPI error &1(&2).
```

Cause: An error occurred while processing a control action.

Action: Error number and description are provided. Either use this information to correct the problem, or contact Dell Technologies Customer Support.

CTL0041

```
System is not in restricted state. Split control action cannot be run.
```

Cause: A Split control action can be run only when the system is in a restricted state.

Action: Refer to "Split BCV pairs: Restricted state" on page 138 for information to bring a system to a Restricted state.

CTL0042

```
System is not in restricted state. Suspend SRDF links and action cannot
   be run.
```

Cause: A Suspend control action can be run only when the system is in a Restricted state.

Action: Refer to "Splitting/Suspending SRDF links: Restricted state" on page 185 for information to bring a system to a restricted state.

CTL0048

```
BCV device &1 attached to STD device &2 is in process of being Split.
```

Cause: The BCV device pair status is SplitInProg. An Establish control action cannot be performed on this pair.

Action: Wait until the Split operation completes and try the operation again.

CTL0049

```
Control action failed for device pair &1(S) and &2(B).
```

Cause: The control action &3 processing failed.

Action: Check the job log for errors.

CTL0050

```
Processing complete for device pair &1(S) and &2(B).
```

Cause: Control action &3 initiated successfully.

Action: This is an informational message; no user action is necessary.

CTL0051

```
Error with devices %s(S) and %s(T).
```

Cause: This device pair is in invalid state.

Action: Check the devices and their statuses.

CTL0052

```
Source device &1 has a Target device &2.
```

Cause: Synchronizing RDF pair is in progress or completed.

Action: Resume SRDF links can be run only when the RDF pair state is in the Suspended state. Establish SRDF links can be run only when the RDF pair state is in a Suspended or Split state.

CTL0053

```
Source device is &1 and Target device is &2.
```

Cause: This RDF pair state is in &3 state.

Action: Resume SRDF links can only be run when the links are in a Suspended state. Establish SRDF links can only be run when the links are in a Suspended or Split state.

CTL0055

```
Control action is initiated for the Image &1.
```

Cause: &2 control action initiated successfully.

Action: This is an informational message; no user action is necessary.

CTL0056

```
Standard device &1 (S) has a BCV device &2 (B).
```

Cause: Synchronizing this BCV pair is in progress or completed.

Action: This is an informational message; no user action is necessary.

CTL0057

```
User cancelled Establish/Re-establish BCV pairs processing.
```

Cause: A user opted to cancel the Establish/Re-establish BCV pairs control action processing at this time.

Action: This is an informational message; no user action is necessary.

CTL0060

```
SRDF status for the &1 image is not in Synchronized state so the remote
BCV operation may not present a current image of data.
```

Cause: The SRDF links are not synchronized at this time, so the remote BCV operation may not present a current copy of data on the remote BCV devices.

Action: Make sure that the SRDF links are synchronized to get a current image of data from R1 devices to R2 devices and remote BCV devices.

CTL0064

```
Error message &1 was received during the IBM i &2 API Call.
```

Cause: An error occurred while &2 API call was made on the IBM i host.

Action: Refer to the job log for errors. Either use this information to correct the problem, or contact Dell Technologies Customer Support

CTL0076

```
User cancelled Establishing SRDF Links processing.
```

Cause: A user opted to cancel the Establish SRDF Links control action processing.

Action: This is an informational message; no user action is necessary.

CTL0077

```
Job not on Hold (User Name: &1, Job Name: &2, Job Number: &3).
```

Cause: The job could not be put on hold.

Action: This is an informational message; no user action is necessary.

CTL0078

```
Job Queue not on Hold (Job Queue Name: &1, Job Queue Library Name: &2).
```

Cause: The job could not be put on hold.

Action: This is an informational message; no user action is necessary.

CTL0081

```
Vary off failed for the Independent ASP &1.
```

Cause: Check the job log for errors.

Action: Make sure that no job is using the independent ASP and then retry the operation.

CTL0083

```
On Symmetrix &1 paired BCV &2 is on the same physical disk as one of
the mirrors of the standard device &3.
```

Cause: No alternate matching BCV device was found on the separate disk.

Action: This informational message indicates that no user action is required, unless performance of the BCV synchronization needs improvement. Sometimes unpairing of other devices of similar emulation makes other candidates available for pairing. Contact Dell Technologies Customer Support if an array reconfiguration is required to reassign volumes or add new BCV devices.

CTL0084

```
Inconsistencies were detected in the configuration. The Discover
command had not been run.
```

Cause: There is a disk with serial number &1 currently in the ASP &2 with unit number &3, but the configuration database contains ASP &4 with unit number &5.

Action: Run the DISCOVER command to refresh the configuration database.

CTL0085

```
Inconsistencies were detected in the configuration.
```

Cause: There is a disk with serial number &1 currently in the ASP &2 with unit number &3, but the configuration database contains ASP &4 with unit number &5.

Action: Refer to the job log for errors. Use this information to correct the problem or contact Dell Technologies Customer Support.

CTL0086

```
SymSync failed with error code 34.
```

Cause: SRDF/TimeFinder Manager cannot communicate with the array with serial number &1. The array could be offline (temporarily or permanently) or the gatekeeper device has been removed or changed.

Action: In case of permanently removed arrays or problems with gatekeeper devices, the configuration database should be refreshed. Run the DISCOVER command with option *YES.

CTL0087

```
Path &1 was not found for the image &2.
```

Cause: The wrong path name was specified.

Action: Enter the appropriate path and try the operation again.

CTL0088

```
Cannot remove device &1 from the device group &2.
```

Cause: SYMAPI reports numeric error &3 (&4 - error message description).

Action: Refer to the Dell EMC Solutions Enabler documentation.

CTL0089

`Cannot delete device group &1.`

Cause: SYMAPI reports numeric error &3 (&4 - error message description).

Action: Refer to the Dell EMC Solutions Enabler documentation.

CTL0090

`File &1 does not exist.`

Cause: The file does not exist in the system.

Action: Make sure that the installation process completed successfully.

CTL0091

`Cannot connect to server on &1 with port number &2.`

Cause: There are many possible causes for this error, such as the server name may be wrong, the port number may be wrong, and so on.

Action: Check the TCP/IP configuration.

CTL0092

`Target IOA is still active.`

Cause: Target IOA is still active, but should not be active for this control action.

Action: Remove the target IOA from the node and try the control action again.

CTL0094

`Target devices cannot be presented to the host.`

Cause: The target devices are already presented to the host.

Action: Remove the target devices from the host.

CTL0095

`Target devices cannot be added because the link is synchronized or Synchronize in progress.`

Cause: Target devices cannot be added to a node if the devices in a Synchronized or SyncInProg state.

Action: Split the link before adding target devices.

CTL0096

`IOA file &1 is empty, please rebuild it first.`

Cause: The IOA file &1 contains no record(s).

Action: Rebuild the IOA file &1 before proceeding with the operation.

CTL0097

```
Error messages were issued during EMC control action &1 processing.
Check the job log for details about errors.
```

Cause: Error messages were issued while processing the control action &1.

Action: Refer to the job log for errors. Either use this information to correct the problem, or contact Dell Technologies Customer Support Center.

CTL0098

```
EMC control action &1 processing initiated successfully. Check the job
log for details.
```

Cause: Control action &1 processing initiated successfully.

Action: This is an informational message; no user action is necessary.

CTL0099

```
Error messages were issued during EMC control action &1 processing.
Check the job log for details about errors.
```

Cause: Error messages were issued while processing the control action &1.

Action: Refer to the job log for errors. Either use this information to correct the problem, or contact the Dell Technologies Customer Support Center.

CTL0100

```
Socket operation failed, the failed operation is &1.
```

Cause: The socket operation failed.

Action: Check the TCP/IP configuration and job log.

CTL0104

```
Operation failed, please check job log for more detail.
```

Cause: The operation failed.

Action: Check the job log to view a detailed message and then take the appropriate action.

CTL0105

```
Hold jobs option can be used only when the program is running from the
console in an interactive session.
```

Cause: The application is not running from the console.

Action: Try the operation again from the console.

CTL0106

```
Path &1 is not deleted from the path file.
```

Cause: A user opted not to delete the path at this time.

Action: This is an informational message; no user action is necessary.

CTL0107

`Image &1 was not found.`

Cause: A user specified the wrong image name.

Action: Enter the appropriate image name and try the operation again.

CTL0108

`ASP &1 was not found in the list of asps for the image &2.`

Cause: Some errors occurred.

Action: Try the operation again.

CTL0109

`Path &1 was not found in the list of path types.`

Cause: A user specified a wrong path name.

Action: Enter the appropriate path and try the operation again.

CTL0110

`Specified step number &1 is greater, than number of steps for path &2.`

Cause: A user specified the wrong step number.

Action: Enter the appropriate step number and try the operation again.

CTL0111

`Cannot add device &1 to the device group &2.`

Cause: SYMAPI reports numeric error &3 (&4 - error message description).

Action: Refer to the Dell EMC Solutions Enabler documentation.

CTL0112

`Cannot create device group &1.`

Cause: SYMAPI reports numeric error &2 (&3 - error message description).

Action: Refer to the Dell EMC Solutions Enabler documentation.

CTL0121

`Path &1 is invalid for image &2.`

Cause: Not all source devices in the step &3 are RDF capable.

Action: Move ASPs containing one or more non-RDF capable devices to the NON_CONFIG image.

CTL0123

`The &1 is &5 for the devices in the &2 image, &3 path, step number &4.`

Cause: Control action processing initiated for the image &2.

Action: This is an informational message; no user action is necessary.

CTL0124

```
&1 processing failed for the &2 image, &3 path and step number &4.
```

Cause: Control action processing failed for the image &2.

Action: Check the errors in the job log. Contact the Dell Technologies Customer Support Center if needed.

CTL0125

```
&1 processing initiated for the &2 image, &3 path and step number &4.
```

Cause: Control action processing initiated successfully for the image &2.

Action: This is an informational message; no user action is necessary.

CTL0126

```
BCV devices are in Sync_in_progress/Restore_in_progress state or
Synchronized/Restored for the &1 image, &2 path and step number &3.
```

Cause: A user tried to Establish the BCV devices, but the BCV state is Synchronized or SyncInProg for the &1 image.

Action: This is an informational message; no user action is necessary.

CTL0127

```
SPLIT control action cannot be run. Synchronizing/Restoring BCV pairs
is in progress for the &1 image, &2 path and step number &3.
```

Cause: Control action Establish is in progress for the &1 image. The Split action cannot be run at this time.

Action: Wait until the BCV state becomes Synchronized and then try the Split control action.

CTL0128

```
BCV devices are already in the Split state for the &1 image, &2 path
and step number &3.
```

Cause: A user tried to Split the BCV, but the devices are already in the Split state for the &1 image.

Action: BCV devices state must be Synchronized to run the Split BCV pairs option.

CTL0129

```
SRDF links are already Synchronized or Sync_in_progress state for the
&1 image, &2 path and step number &3.
```

Cause: A user tried to Resume, Establish, or Restore the SRDF links, but the links are already in a Synchronized or SyncInProg state.

Action: This is an informational message; no user action is necessary.

CTL0130

```
SRDF links are already Suspended for the &1 image, &2 path and step
number &3.
```

Cause: A user tried to Suspend the SRDF links, but the links are already in a Suspended state.

Action: SRDF links must be in a Synchronized state to run the Suspend SRDF links operation.

CTL0131

```
Split or Suspend SRDF Links cannot be run. SRDF links are in
Sync_in_progress state for the &1 image, &2 path and step number &3.
```

Cause: The SRDF links are in SyncInProg state. Split or Suspend SRDF links action cannot be run at this time.

Action: Wait until the SRDF link state becomes Synchronized and then try the Split or Suspend control action.

CTL0132

```
SRDF links are already Split for the &1 image, &2 path and step number
&3.
```

Cause: A user tried to Split the SRDF links, but the links are already in a Split state.

Action: The SRDF links must be in a Synchronized state to run the Split SRDF links option.

CTL0133

```
Resume SRDF Links cannot be run. SRDF links are in Split state for the
&1 image, &2 path and step number &3.
```

Cause: The SRDF links are in a Split state. The Resume SRDF links control action cannot be run at this time.

Action: The SRDF links must be in a Suspended state to run the Resume SRDF links option.

CTL0134

```
The devices for the &1 image, &2 path and step number &3 are in
ineligible state to perform the control action specified.
```

Cause: Check the status of the devices on the Configuration screen.

Action: Change any configuration details needed. Make sure that the devices are in eligible state for the control action requested.

CTL0135

```
SNAP devices are not in eligible state to be created, status must be
NotCreated.
```

Cause: It is most likely that the devices have not been paired with the Snap target devices.

Action: Pair the devices with the Snap target devices.

CTL0136

```
The &1 devices cannot be activated because they are being created.
```

Cause: The `&1` devices cannot be activated because the devices are in the process of being created.

Action: Wait until the devices are created before activating them.

CTL0137

```
The &1 devices have already been activated.
```

Cause: The `&1` devices have already been activated.

Action: This is an informational message; no user action is necessary.

CTL0138

```
The System is not in restricted state. The ACTIVATE control action
cannot be run.
```

Cause: The system is not in a Restricted state.

Action: Put the system in a Restricted state and try the operation again.

CTL0139

```
Path &1 is not associated with the image &2.
```

Cause: The wrong path name was specified.

Action: Associate a path with the image first and then proceed with the operation.

CTL0140

```
For &1 image path &2 step &3 unit &4 source device is not known.
```

Cause: The target device for the previous step is unknown. The previous step is not configured.

Action: Check the configuration.

CTL0141

```
For &1 image path &2 and step &3 target device &4 is in ineligible
state to be unpaired from standard device &5.
```

Cause: Target device is in &6 state.

Action: This is an informational message; no user action is necessary.

CTL0142

```
Not all devices in the &1 image are Symmetrix devices. Control actions
can be performed only for images having Symmetrix devices.
```

Cause: &1 image has &2 type of devices.

Action: Move ASPs with non-Symmetrix devices to the NON_CONFIG image.

CTL0143

`Additional time required to ensure SRDF/A consistent split.`

Cause: Insufficient time was specified in the SRDFATIME data area to perform a consistent Split.

Action: Increase the value in the SRDFATIME data area and try the Split operation again.

CTL0144

`&1.`

Cause: The load source mirror is in an ineligible state.

Action: From the Display Disk Configuration Status screen in SST, check the status of the disks and make appropriate changes.

CTL0145

`&1 action failed for the &2 image, &3 path and step number &4.`

Cause: A user tried the operation, but the load source is in the &5 state.

Action: From the Display Disk Configuration Status screen in SST, check the status of the disks and make appropriate changes.

CTL0146

`Non-Symmetrix devices exist in the configuration.`

Cause: The configuration contains devices that are not recognized by the array.

Action: Run the DISCOVER command to refresh the configuration database.

CTL0147

`SNAP devices for step &1 and &2 path could not be paired.`

Cause: Some or all source devices for the step are RAID 5 BCVs or RAID 6 BCVs and are clone enabled.

Action: This is an informational message; no user action is necessary.

CTL0148

`User cancelled the Cancel BCV pairs processing.`

Cause: A user opted to cancel the Cancel BCV pairs control action processing at this time.

Action: This is an informational message; no user action is necessary.

CTL0149

`Devices for step &1 and &2 path could not be unpaired.`

Cause: Some or all BCV devices for the step are RAID 5 BCVs or RAID 6 BCVs and are Clone enabled.

Action: The Cancel BCV pairs control action should be run first.

CTL0150

```
Devices in step &1 and &2 path for &3 image have a mixed BCV
configuration.
```

Cause: Some BCV devices in the step are standard devices and some are RAID 5 or RAID 6 BCVs. Refer to Table 6 on page 51 for greater detail.

Action: This is an informational message; no user action is necessary.

CTL0151

```
System is not ended to restricted condition.
```

Cause: The system is not ended to the restricted condition in the allocated time.

Action: Use the WRKSBS command to check the status of the controlling subsystem and other subsystems.

CTL0152

```
Establish control action cannot be run.
```

Cause: All BCV devices are standard BCVs and there are one or more Clone enabled BCV devices.

Action: Use the Cancel option to remove the incremental relationship between standard devices and the split BCV devices.

CTL0153

```
On Symmetrix &1 for the step &2 target device &3 is in ineligible state
to be unpaired from standard device &4.
```

Cause: Target device is in &5 state.

Action: This is an informational message; no user action is necessary.

CTL0154

Cannot associate &1 device &2 with Composite Group &3.

Cause: The SYMAPI reports error &2(&3).

Action: Refer to the Dell EMC Solutions Enabler documentation.

CTL0156

```
User cancelled the RDF Clear Consistency state processing.
```

Cause: A user opted to cancel the RDF Clear Consistency state control action processing.

Action: This is an informational message; no user action is necessary.

CTL0157

User cancelled the Disable SRDF ECA/MSC processing.

Cause: A user cancelled the Disable SRDF ECA/MSC control action processing.

Action: This is an informational message only. No user action is required.

CTL0158

```
Cannot create composite group &1.
```

Cause: The SYMAPI reports error &2(&3).

Action: Refer to the EMC Solutions Enabler documentation.

CTL0159

```
Cannot add device &1 to the composite group &2.
```

Cause: The SYMAPI reports error &3(&4).

Action: Refer to the Dell EMC Solutions Enabler documentation.

CTL0160

```
Cannot delete Composite Group &1.
```

Cause: The SYMAPI reports error &3(&4).

Action: Refer to the Dell EMC Solutions Enabler documentation.

CTL0162

```
Split while active not completed for step &1 , &2 path and &3 image.
SRDF Consistency state not enabled.
```

Cause: The SRDF Consistency state is not enabled while the SPLIT with SPLITWA option is selected.

Action: Either enable the RDF Consistency State with the CTLACT command (RDFCSENABLE) or use another option with the SPLIT control action (RESTRICTED, MAINTENANCE or HOLDJOBS).

CTL0163

```
Cannot use SPLITWA option for step that is greater than 2, unless if it
is step 3 after cascading RDF.
```

Cause: Splitting with the SPLITWA option is unnecessary on steps greater than 2, unless it is a step 3 after cascading RDF. For example, a multihop configuration uses a remote BCV/R1 that must be split first to create a valid copy. A BCV/R1 does not have a step number greater than 2.

CTL0164

```
Network address (NETWADDR) and node name (NODENAME) cannot be both
empty. Specify either network address or node name.
```

Cause: User did not specify NETWADDR and/or NODENAME.

Action: Specify a network address or node name. When specifying both parameters, make sure they are unique.

CTL0165

```
SRDF ECA/MSC protection is not disabled.
```

Cause: SRDF ECA/MSC protection cannot be disabled because the consistency group does not exist.

Action: Enable the SRDF ECA/MSC protection by executing the CTLACT command with ECAENABLE option.

CTL0166

```
SRDF SPLITWA cannot be run.
```

Cause: The user selected the SPLITWA control action for an unsupported feature.

Action: Select a different split control action.

CTL0167

```
User cancelled the SN-ID Read/Clear action.
```

Cause: A user opted to cancel the SN-ID Read/Clear action processing.

Action: This is an Informational message. No action is necessary.

CTL0168

```
Path &1 is invalid for image &2.
```

Cause: There are RAID 6 devices in the step &3.

Action: Check the configuration.

CTL0169

```
Path &1 is invalid for image &2.
```

Cause: There are devices in the step &3 with the RDF group number greater than 64.

Action: Check the configuration.

CTL0170

```
User cancelled the Terminate SNAP pairs processing.
```

Cause: A user opted to cancel the Terminate the Snap pairs control action.

Action: This is an informational message; no action is required.

CTL0171

```
User cancelled Failover processing.
```

Cause: A user opted to cancel the Failover control action.

Action: This is an informational message; no action is required.

CTL0172

```
User cancelled Failback processing.
```

Cause: A user opted to cancel the Failback control action.

Action: This is an informational message; no action is required.

CTL0173

```
On the Symmetrix &1 changes are not done for the RDF group
```

Cause: SRDF/TimeFinder Manager did not detect any changes from the current configuration.

Action: This is an informational message; no action is required.

CTL0174

```
SRDF mode for the &1 image is Adaptive Copy, so the remote BCV
operation will not present a current image of data.
```

Cause: The SRDF links are in Adaptive Copy mode. Therefore, the remote BCV operation may not present a current copy of data on the remote BCV devices.

Action: Make sure that the SRDF links are in Synchronous or Asynchronous mode and the links are in Synchronized or Consistent state to get a current image of data from R1 devices to R2 devices and remote BCV devices.

CTL0176

```
SymSync failed for the Symmetrix with serial number &1.
```

Cause: SymSync failed with SYMAPI error &2(&3). Array could be removed temporally or permanently, or the gatekeeper device removed or changed.

Action: In case of permanently removed array or a problem with gatekeeper device, configuration database should be refreshed. Command Discover should be run with option *YES.

CTL0177

```
Option is not allowed for current version of the operation system.
```

Cause: This option is only allowed for the V6R1M0 operating system or higher.

Action: Only MAINTENANCE SPLIT/SUSPEND/ACTIVATE is allowed for an IASP image.

CTL0178

```
Devices are not Varied OFF. Option is not allowed for current version
of the operating system.
```

Cause: This option is only allowed for the V6R1M0 operating system or higher.

Action: Vary off the devices.

CTL0179

```
Control action &1 will be processed.
```

Cause: The devices are neither unavailable nor online.

Action: This is an informational message; no user action is necessary.

CTL0180

```
Devices are not Varied OFF. Control action &1 failed.
```

Cause: This option is not allowed when devices are varied on.

Action: Vary off the devices.

CTL0181

```
Data area &1 not found. Default values used.
```

Cause: Default values have been used: Suspend timeout: &2 Timeout action: &3 Auto-resume timeout: &4.

Action: Contact Dell Technologies support to obtain the &1 data area.

CTL0182

```
Access to ASP &1 is suspended.
```

Cause: Access to ASP &1 is suspended. New transactions cannot start until access is resumed. Transaction quiescing used &2 seconds. Operations quiescing used &3 seconds. Writes to disk used &4 seconds, for a total of &5 seconds.

Action: This is an informational message; no action is necessary.

CTL0183

```
&1 Processing timed out after &2 seconds for ASP &3.
```

Cause: &1 Processing for ASP &3 did not complete within the Suspend Timeout value of &2 seconds.

Action: Change the Suspend Timeout value using the command
&P CHGDTAARA (&5/&4 (40 6)) 'nnnnnn'
where nnnnnn is the desired Suspend Timeout value in seconds, which must be six digits with leading zeros.

CTL0184

```
The command processing will continue for ASP &1.
```

Cause: The Timeout Action has been specified as Continue.

Action: Change the Timeout Action to end the operation on a timeout using the following command:
CHGDTAARA (&3/&2 (90 1)) 'E'

CTL0185

```
The command processing has been ended for ASP &1
```

Cause: Refer to the messages previously listed. The Timeout Action has been specified as End.

Action: Change the Timeout Action to continue command processing on a timeout with the following command:
CHGDTAARA (&3/&2 (90 1)) 'C'

CTL0186

```
Access to ASP &1 successfully resumed.
```

Cause: Access to ASP &1 is successfully resumed. Users may access data within the ASP.

Action: This is an informational message; no action is necessary.

CTL0187

```
Source IOA is still active.
```

Cause: The target IOA has not been removed from that node.

Action: Remove the target IOA from the node.

CTL0188

```
For &1 image path &2 step &3 not all source devices are reachable.
```

Cause: The Problems occur in the previous step.

Action: Check status of the previous step.

CTL0189

```
SRDF Failover operation failed.
```

Cause: SRDF Failover with Establish option is specified but these devices are not Dynamic SRDF devices.

Action: Use SRDF Failover with option *NO.

CTL0190

```
User cancelled the Recreate CLONE pairs processing.
```

Cause: User opted to cancel the Recreate CLONE pairs control action processing at this time.

Action: This is an informational message; no action is necessary.

CTL0191

```
User cancelled the Recreate SNAP pairs processing.
```

Cause: User opted to cancel the Recreate SNAP pairs control action processing at this time.

Action: Informational message. No action is necessary.

CTL0192

```
User cancelled the Terminate CLONE pairs processing.
```

Cause: User opted to cancel the Terminate CLONE pairs control action processing at this time.

Action: Informational message. No action is necessary.

CTL0193

```
Path &1 is invalid for image &2.
```

Cause: Not all source devices in the step &3 are standard devices.

Action: You can change pairing for previous step.

CTL0194

```
SRDF Create/Delete pair operation failed.
```

Cause: Control action &1 operation is specified but devices listed in the Device File are not Dynamic SRDF devices.

Action: Use Dynamic RDF devices.

CTL0195

```
Path &1 is invalid for image &2 for current code level.
```

Cause: Path is valid for Enginuity 5876 or later.

Action: Informational message. No action is necessary.

CTL0196

```
Path &1 is invalid for image &2 for current code level.
```

Cause: Duplicate Snap is valid for Enginuity 5876 or later.

Action: Informational message. No action is necessary.

CTL0197

```
Device &1 on the Symmetrix &2 is a target device for TF/Clone Emulation
session and cannot be used for pairing.
```

Cause: Target device is already in a Clone emulation session.

Action: Informational message. No action is necessary.

CTL0198

```
Device &1 on the Symmetrix &2 is a target device for TF/Clone session
and cannot be used for pairing.
```

Cause: Target device is already in a Clone session.

Action: Informational message. No action is necessary.

CTL0199

```
Cannot use SPLITWA option for second step if first step is not RDF.
```

Cause: There is no use splitting with SPLITWA option on the second step, unless if it is step 2 after RDF.

Action: Use other split option with the selected step.

CTL0200

```
Symmetrix &1 does not have &2 license.
```

Cause: Array does not have requested license.

Action: Informational message. No action is necessary.

CTL0201

```
Device &1 is not found on the Symmetrix &2 and cannot be used for
pairing.
```

Cause: Target device is not found on the array.

Action: Informational message. Check the device file.

CTL0202

```
Device &1 on the Symmetrix &2 is &3 and cannot be used for pairing for
this type of the step.
```

Cause: Target device has not valid configuration type.

Action: Informational message. Check the device file.

CTL0203

```
Device &1 on the Symmetrix &2 is in use, i.e. paired with another
device and cannot be used for pairing.
```

Cause: Target device is already in use.

Action: Informational message. Check the device file.

CTL0204

```
SRDF type for source devices in this step is &1. The SRDF mode for the
step is &2.
```

Cause: Additional status for RDF type of the step is provided.

Action: This is an informational message; no user action is necessary.

CTL0205

```
SRDF mode for the &1 image, &2 path and step number &3 is Adaptive
Copy, so the Failover operation is not allowed.
```

Cause: Informs the user that the SRDF links are in Adaptive Copy mode.

Action: Make sure that the SRDF links are in Synchronous or Asynchronous mode.

CTL0206

```
Save pool &1, specified for create virtual copy sessions is not valid
for image &2, path &3 and step number &4.
```

Cause: Not all arrays in the configuration have a specified save pool.

Action: You can call program wrk_pool (option 5 from the main screen) and check available save pools.

CTL0207

```
Operation to create virtual copy sessions cannot be executed for image
&1, path &2 and step number &3.
```

Cause: Virtual copy session(s) already exists with &4 and &5 save pools.

Action: You can cancel existing sessions and create new ones with desired pool name.

CTL0208

```
Save pool &1, specified for create virtual copy sessions for image &2,
path &3 and step number &4 will be ignored.
```

Cause: Virtual copy session(s) exists with &5 save pool name.

Action: You can specify correct pool name or cancel existing sessions and create new ones with desired pool name.

CTL0209

```
Path &1 is invalid for image &2.
```

Cause: VP Snap should be the last step in the path.

Action: This is an informational message; no user action is necessary.

CTL0210

```
Create VP Snap control action cannot be run.
```

Cause: Some or all devices in previous step are clone enabled.

Action: This is an informational message; no user action is necessary.

CTL0211

```
Configuration change &1 for pool &2 is failed.
```

Cause: Configuration change processing failed for the specified pool.

Action: Check the errors in the symapi log. Contact the Dell Technologies Customer Support if needed.

CTL0212

```
Configuration change &1 for pool &2 is successful.
```

Cause: Configuration change processing was successful for the specified pool.

Action: This is an informational message; no user action is necessary. Check symapi log for more details.

CTL0213

```
Configuration change &1 for pool &2 is successful. Termination of the
configuration session failed.
```

Cause: Configuration change processing was successful for the specified pool.

Action: Check the errors in the symapi log. Contact the Dell Technologies Customer Support if needed.

CTL0214

```
RTVSTMINF Error: &1
```

Cause: RTVSTMINF failed with an error

Action: Check joblog for earlier messages.

CTL0215

```
&1 cannot be run for the &2 image, &3 path and step number &4.
```

Cause: Multisession Consistency (MSC) for SRDF/A is enabled. This configuration is not supported.

Action: Change the configuration if needed. Contact the Dell Technologies Customer Support if needed.

CTL0216

```
&1 cannot be run for the &2 image, &3 path and step number &4.
```

Cause: SRDF Enginuity Consistency Assist (RDF-ECA) for SRDF/S is enabled. This configuration is not supported.

Action: Change the configuration if needed. Contact the Dell Technologies Customer Support if needed.

CTL0217

```
The devices for the &1 image, &2 path and step number &3 are in
ineligible state to perform the control action specified.
```

Cause: Snap restore from clone/clone emulation target and VP snap restore from clone target is not supported for this Enginuity version.

Action: Change any configuration details needed. Make sure that the devices are in eligible state for the control action requested.

CTL0218

```
The devices for the &1 image, &2 path and step number &3 are in
ineligible state to perform the control action specified.
```

Cause: Source devices for snap restore from clone/clone emulation target or VP snap restore from clone target are in invalid state.

Action: Change any configuration details needed. Make sure that the devices are in eligible state for the control action requested.

CTL0219

```
The devices for the &1 image, &2 path and step number &3 are in
ineligible state to perform the control action specified.
```

Cause: Clone operation with Snap/VP Snap from target is not available on this Enginuity version.

Action: Change any configuration details needed. Make sure that the devices are in eligible state for the control action requested.

CTL0220

```
The devices for the &1 image, &2 path and step number &3 are in
ineligible state to perform the control action specified.
```

Cause: State for Snap/VP Snap from clone target is invalid for clone operation.

Action: Change any configuration details needed. Make sure that the devices are in eligible state for the control action requested.

CTL0221

```
The devices for the &1 image, &2 path and step number &3 are in
ineligible state to perform the control action specified.
```

Cause: Clone incremental restore to Snap or VP Snap Source is not supported for this Enginuity version.

Action: Change any configuration details needed. Make sure that the devices are in eligible state for the control action requested.

CTL0222

```
The devices for the &1 image, &2 path and step number &3 are in
ineligible state to perform the control action specified.
```

Cause: State for Snap or VP Snap is invalid for clone incremental Restore to Snap or VP Snap Source.

Action: Change any configuration details needed. Make sure that the devices are in eligible state for the control action requested.

CTL0223

```
Not all devices in a range are &1 devices.
```

Cause: All devices in a range should be &1 devices.

Action: Check devices in a range.

CTL0224

```
Not all devices in a file are &1 devices.
```

Cause: All devices in a file should be &1 devices.

Action: Check devices in a file.

CTL0225

```
Device &1(&2), specified in the range of devices is invalid for the
control operation.
```

Cause: All devices in a range should be &3 devices.

Action: Check devices in a range.

CTL0226

```
Device &1(&2), specified in the device file is invalid for the control
operation.
```

Cause: All devices in a file should be &3 devices.

Action: Check devices in a file.

CTL0227

```
There are no SNAP pools on the Symmetrix &1.
```

Cause: Array does not have SNAP pools.

Action: Check the configuration.

CTL0230

```
The System is not in restricted state. New Snapshots could not be
ESTABLISHED.
```

Cause: The system is not in restricted state.

Action: Please put the system in restricted state and retry the operation.

CTL0231

```
The status of &1 (generation &2) snapshot from source devices of the
step &3, path &4 and &5 image is &6. &7
```

Cause: Control action processing initiated for the image &5.

Action: Informational message. No action is necessary.

CTL0232

```
Snapshot is established from the &1 devices of the step. There are &2
devices in the step.
```

Cause: Additional status for the snapshot is provided.

Action: Informational message. No action is necessary.

CTL0233

```
&1 snapshot &2 generation &3.
```

Cause: Additional information for the step is provided.

Action: Informational message. No action is necessary.

CTL0234

```
Operation is not supported for current code level.
```

Cause: Operation is supported for arrays running PowerMaxOS 5978 or HYPERMAX 5977.

Action: Informational message. No action is necessary.

CTL0235

```
Snapshot has links. Terminate snapshot is not
allowed.
```

Cause: Operation is not allowed.

Action: Informational message. No action is necessary.

CTL0236

```
There are no snapshots for the devices in the &1 image &2 path and step
   number &3.
```

Cause: No control operations except Establish are allowed.

Action: Informational message. No action is necessary.

CTL0237

```
There are no snapshots for the devices in the &1 image &2 path and step
number &3 or devices for this step are not paired
```

Cause: Link target devices is not allowed.

Action: Informational message. No action is necessary.

CTL0238

```
There are no snapshots for the devices in the &1 image &2 path and step
number &3 or devices for this step are already restored.
```

Cause: Restore operation is not allowed.

Action: Informational message. No action is necessary.

CTL0239

```
Status of the snapshot for the devices in the &1 image, &2 path and
step number &3 is Restore in Progress. Terminate action cannot be run
at this time.
```

Cause: Terminate snapshot is not allowed.

Action: Wait until the snapshot state becomes Restored and then try the Terminate control action.

CTL0240

```
System ended to a semi-restricted condition.
```

Cause: The STMSENDSBS command has run and completed successfully. The system is in a semi-restricted state and is available for maintenance activities.

CTL0241

```
Error found on CALL command Check STARTUPPGM data area for the
   configured startup program. Corret the error and pressR to retry.
```

Cause: The command STMSTRSBS has ended in error. The program configured in the STARTUPPGM data area is in the wrong format or is not found. The STARTUPPGM must be configured as:

- From the 1st position, the program name (up to 10 characters)
- From the 11th position, the library name (up to 10 characters long)

CTL0242

```
Incorrect IP address hostname config found in NETCFNG file.
```

Cause: The SPLITWA action has ended in error. The NETCNFG file contains incorrect IP address/hostname configuration. Check the SPLITWA_SERVICE configuration in `/var/symapi/config/netcnfg`.

CTL0244

```
Set Bias operation failed.
```

Cause: Control action &1 operation is specified bu the devices listed in the device file are not SRDF devices.

Recovery: Specify SRDF devices.

APPENDIX D
Miscellaneous Device Information

This appendix lists miscellaneous device information such as device s and device pairing compatibility.

- D910 device support (thin devices) 294
- Variable-sized LUNs 296

D910 device support (thin devices)

Enginuity and Solutions Enabler provide support for 512 byte block FBA D910-099 as thin devices. This allows users of Dell EMC SRDF/TimeFinder Manager for IBM i to take advantage of VMAX features FAST VP, VLUN VP Migration, and Virtual Provisioning which is available for arrays running Enginuity 5876; and automated FAST which is available for arrays running PowerMaxOS 5978 or HYPERMAX OS 5977.

Requirements and rules for D910 support

The rules and requirements for D910 device support are:

- Arrays running Enginuity 5876.
- For VMAX 10K arrays, and arrays running PowerMaxOS 5978 or HYPERMAX OS 5977:
  - D910 device only. Emulation for 2107 devices is not supported.
  - Thin devices only. Thick or meta devices are not supported.
  - Thin (bound) gatekeeper devices only.
- Server hardware must be POWER6 or higher.
- POWER6 model systems firmware must be FW3.5.0 or later.

  Note: There is no minimum firmware level for POWER7 systems.

- Fibre Channel adapters must be Smart IOA Fibre Channel (IOPless) adapters FC5749, FC5774, FC5735, or future IBM i Fibre Channel adapters. Use of IOP-based adapters is not allowed.
- IBM i operating system must be version 6.1.1 or later.
- In a VIOS configuration that uses D910 devices with NPIV, the IBM i operating system must be version 7.1 or higher.
- Allows for pairing between like devices (same emulation type) for SRDF and TimeFinder replication.
- Allows for SRDF/TimeFinder Manager support for thin gatekeepers.
- When configuring D910 devices, the IBM i operating system must be version 6.1.1 or later, and the following minimum PTF levels must be applied to IBM i operating system:
  - 6.1.1 requires a minimum CUM C2122610 (2122) MAY 14, 2012 (NPIV support is excluded)
  - 7.1 requires a minimum of TECH REFRESH 4 (TR4) or CUM C2115710 (2115) May 17, 2012.
- 512-byte and 520-byte LUNs can reside on the same IBM i partition, and on the same fiber channel adapter.

- For disk units formatted with 520 byte sectors, the 520 bytes in each sector consists of 512 bytes of data and 8 bytes of header information. The 8 bytes of header are used by the IBM i licensed internal code to self-identify the data in the page, and during recovery operations to rebuild the directories of permanent extents. A page of data consists of 8 sectors.
- For disk units formatted with 512 byte sectors, the IBM i licensed internal code uses nine sectors for each page, since the physical sectors on a disk unit formatted with 512-byte sectors do not have room for an 8 byte header. The 9 sectors of the page consist of eight data sectors and one sector containing the eight headers for those data sectors (along with other control information). Since one of every nine sectors contains header and control information, the capacity usable for the client is approximately 11% smaller than the actual capacity of the logical unit.
- Only LUN Model Number 099 (protected) is used. Other LUN models such as A01, A81, and 050 are not allowed.

  Note: Since the LUNs report to the IBM i partition as a protected unit, the operating system will not allow system level mirroring to be used.

- D910 device capacities (raw) can range from 3GB up to, but not including, 2TB.

  Note: Once assigned, LUN capacities must remain constant, which is the case with any other storage attachment to IBM i (both 520 and 512). To change the capacity of a LUN, it must be removed from the configuration, destroyed, and recreated before adding it back into the partition's configuration.

- D910 devices can be used in the System ASP, User ASPs and IASPs.
- D910 devices can be used as an IBM i load source device. The minimum raw capacity for a D910 load source device is 20GB.
- The SCSI FORMAT UNIT command is supported as the **Add to ASP** action requires that the LUN is zeroed.
- D910 devices do not support SCSI Skip Read and Skip Write operations. Instead, IBM i sends aggregated data as several individual SCSI Read or Write commands. The 2107 device does support these operations.

Application impact

Only the low level IBM i licensed internal code Storage Management layer of the operating system is directly impacted by the way the disk sectors are formatted. Applications, databases, and programs do not need any modifications to use disks that are formatted with 512-byte sectors.

Performance impact

There is some additional data traffic through the I/O Adapter, because each “page” of data transfers 4608 bytes (9 * 512) instead of 4160 bytes (8 * 520). There is a very small amount of increased usage of the processors when the IBM i licensed internal code builds the header information into the 9th sector of the page.

Variable-sized LUNs

SRDF/TimeFinder Manager for IBM i recognizes and tolerates volumes of non-fixed capacities (variable-sized LUNs) introduced in IBM i V6.1.1. These include devices 2107-099 and 2107-050. SRDF and TimeFinder replications are supported between like devices of the same size and characteristics.

Devices are:

- 2107-099 — Where 099 indicates the volume is treated as "Protected" DPY Parity devices (equivalent to the current IBM A0x models).
- 2107-050 — Where 050 indicates the volume is treated as "Unprotected" by the OS, and allows it to be mirrored by the OS (equivalent to current IBM A8x models).

Note: Enginuity supports a minimum size of 165 cylinders and a maximum of 2236962 cylinders for these device types.

Minimum Enginuity versions supported are 5876.

APPENDIX E
Troubleshooting

This appendix provides some troubleshooting information for SRDF/TimeFinder Manager. For information on backup, recovery, journaling, and journal management, refer to IBM documentation on the IBM website

- Handling failed or non-reporting hardware resources 298
- Error generated when running Discover command within SST 298
- Multipath reset 298
- Error when running CRTSYMAPI or CRTSYMAPIC commands 299

Handling failed or non-reporting hardware resources

In some cases, non-reporting or failed hardware resources for disk units can interfere with operation of SRDF/TimeFinder Manager. For example, you may be unable to group or pair some devices.

Non-reporting or failed hardware resources can appear on a system after a hardware reconfiguration. You should remove non-reporting or failed hardware resources for disk units before using SRDF/TimeFinder Manager. This can be accomplished after a grouping or pairing error has occurred.

Remove failed and non-reporting hardware resources for disk units as follows:

1. Sign on as the authorized user and issue the IBM i `STRSST` command to display the System Service Tools (SST) Main menu.
2. From **SST**, select option **1** to start a service tool.
3. Select option **7** to start the hardware service manager.
4. Select option **4** to list failed and non-reporting hardware resources.
5. Select option **4** next to each failed and non-reporting hardware resource to remove it.

Refer to the IBM documentation for more information.

Error generated when running Discover command within SST

On a host running IBM i 7.1 operating system, an error may occur when running the STM Discover command after removing a disk unit from an ASP using SST (rather than DST). To avoid this issue, it is recommended to completely exit the SST environment before running the Discover command.

Multipath reset

Multipath Resetting can be used to stop error messaging due to a multipath missing connection. For example, a missing connection occurs when the image is loaded on a different hardware set.

Multipath resetting is done from the IBM SST (Service Tool).

From the IBM console:

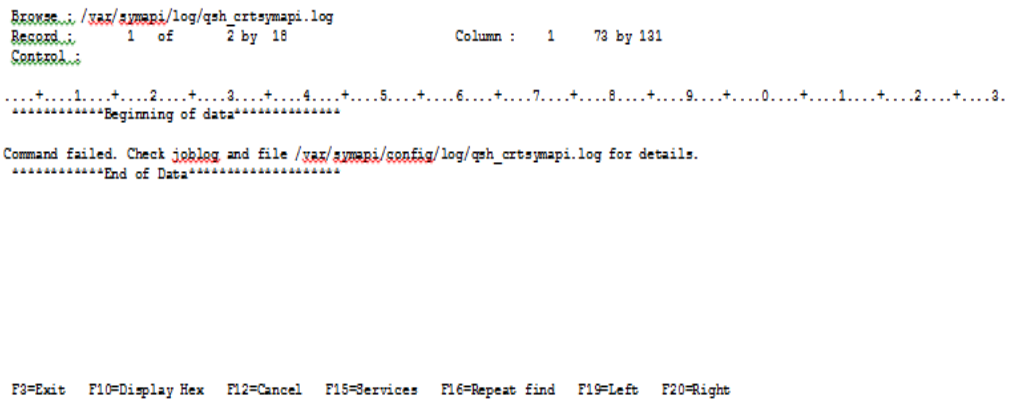
1. Sign on to **SST**.
2. Select option **1. Start a Service Tool**.
3. Select option **7**, **Hardware service manager**.
4. Select option **1**, **Packaging hardware resources**.
5. Select option **9** for the required Disk Unit System. Select option **7**, **Paths to multiple path disk**, to check the status of the paths.

6. Press **F14** to reset all the paths.
7. When prompted for confirmation, verify the information and press **F10** to reset the paths.

   When the reset is completed, a confirmation message is displayed at the end of the screen.

Error when running CRTSYMAPI or CRTSYMAPIC commands

The error message “Hostname not found” may be generated when running the CRTSYMAPI or CRTSYMAPIC command.



To fix this issue, add the host name to the TCP domain information using the CHGTCPDMN command.

PART 2
SRDF/TimeFinder Manager Extended Features

Part 2 of this product guide describes how to prepare and configure SRDF/TimeFinder Manager to support IASP use in a clustered environment. It contains the following chapters and appendixes:

Chapter 7, "Extended Features Introduction"

Introduces the extended features, that EMC SRDF/TimeFinder Manager for IBM i provides, to support IBM Independent Auxiliary Storage Pools (IASPs) functionality.

Chapter 8, "Configuring for IASP Environment"

Explains how to prepare and configure SRDF/TimeFinder Manager to support IASP use in a clustered environment.

Chapter 9, "IASP-Supported Operations"

Describes how to use the SRDF/TimeFinder Manager interface to perform IASP-supported operations when working with SRDF and TimeFinder (BCV, Clones, or Snap) devices configured for multi-node backup and disaster recovery (DR) solutions.

Appendix F, "Non-Dell EMC Procedures"

Provides guidelines when using IBM-related procedures.

Appendix G, "Extended Features Commands Reference"

Provides reference information, syntax, and examples for control commands.

Appendix H, "Extended Features Messages"

Provides reference information about messages that may display for the extended features components.

CHAPTER 7
Extended Features Introduction

This chapter introduces the extended features that Dell EMC SRDF/TimeFinder Manager for IBM i provides to support IBM independent ASPs (IASPs) functionality.

- Extended features description 304
- Independent ASP functionality 305

Extended features description

Dell EMC SRDF/TimeFinder Manager for IBM i Extended Features provide support for the IBM independent ASP (IASP) functionality. An IASP is a set of switchable or private auxiliary disk pools (up to 223) that can be brought online or offline on an IBM i host without affecting the rest of the system.

This support, when combined with SRDF/TimeFinder Manager for IBM i, enables you to control SRDF or TimeFinder operations on PowerMax, VMAX All Flash, VMAX3, VMAX Family, and DMX arrays attached to IBM i hosts as described in "Standard features description" on page 22. In addition, the extended features can use IASPs to present one or more target devices containing an IASP image to another host for business continuance (BC) processes.

Extended features control operations can be performed two ways:

- From the SRDF/TimeFinder Manager menu-driven interface.
- From the command line using SRDF/TimeFinder Manager commands and associated IBM i commands.

Independent ASP functionality

SRDF/TimeFinder Manager supports two-node, three-node and four-node solutions for performing disaster recovery (DR) and normal operations on a backup node using IASP images in a combined IBM i and PowerMax, VMAX All Flash, VMAX3, VMAX, and DMX array environment for business continuance (BC) operations.

SRDF/TimeFinder Manager supports Multipath operations for IASP.

Depending on the disaster recover (DR) solution, IASP operations can be performed on several different IBM i systems. IASP operations are similar for a two and three-node solution.

Two-node solution

A two-node solution (Figure 107) can be used for disaster recovery or backup services.

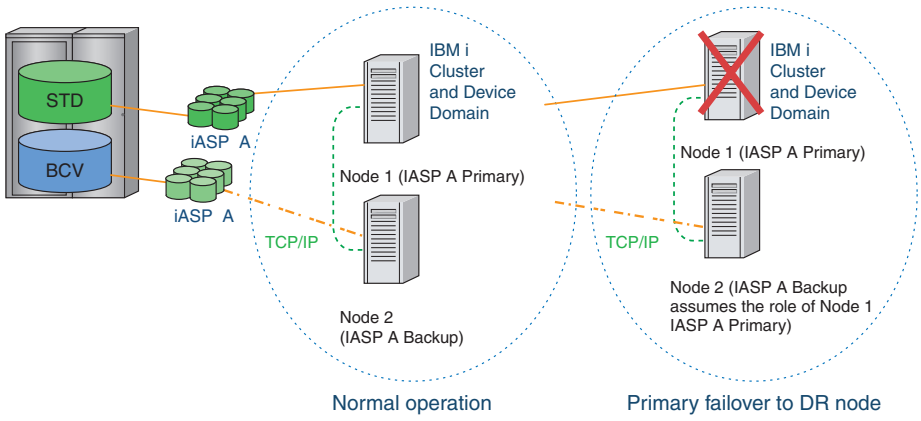


Figure 107 Two-node cluster and two-node DR configuration

Three-node solution

In a three-node solution (Figure 108), disaster recovery and backup services are provided simultaneously as follows:

- One node serves as primary node.
- One node serves as a DR node.
- One backup node is configured with remote BCVs, clones, or snap devices for backing up an IASP image.

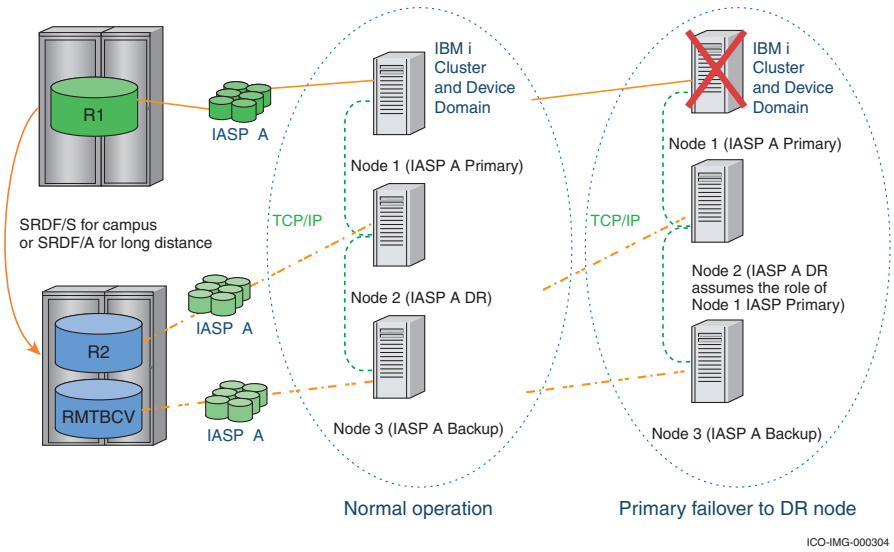


Figure 108 Three-node DR environment

Four-node solution

A four-node solution (Figure 109) can be used for disaster recovery, high availability (HA), and backup services as follows:

- One node serves as a primary node.
- One node serves as an HA node.
- One node serves as a DR node.
- One node serves as a backup node.

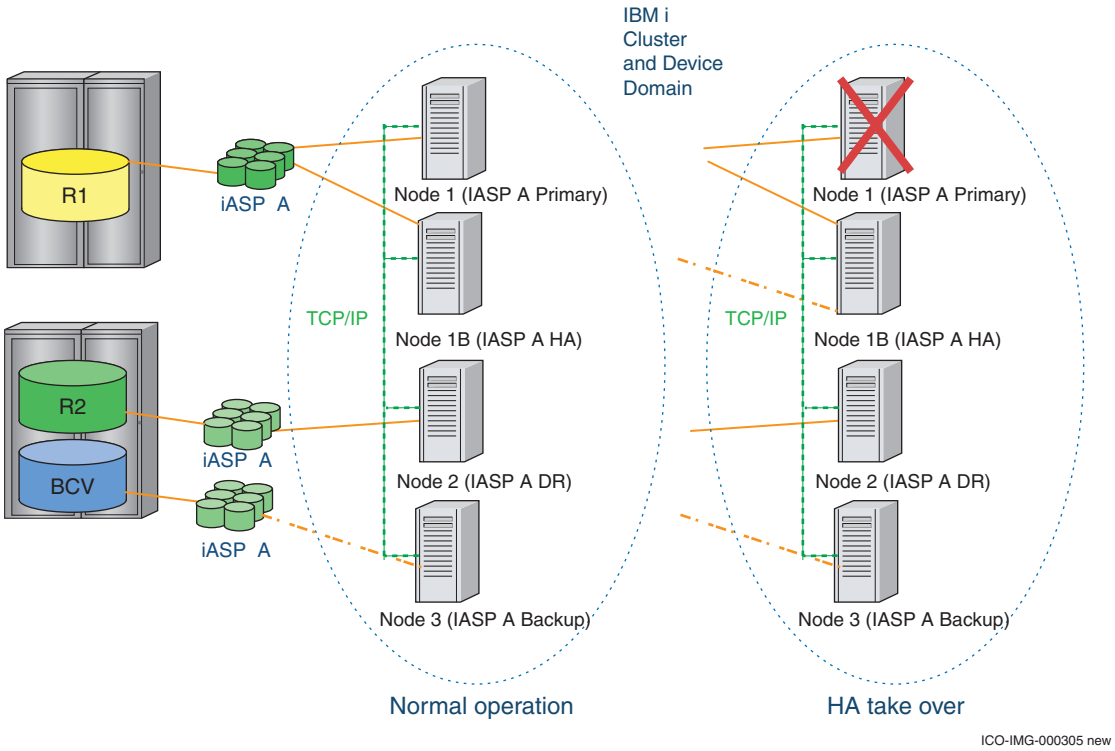


Figure 109 Four-node high availability setup

The function of the HA node is to take over for the primary node in case of a failure of the primary node.

Note: Throughout this guide, the word backup refers to IASP operations (backup, application testing, and load balancing) that are performed on a backup node.

Some of the operations that you can perform on an IASP image are:

- Establish/create — Establish SRDF links, establish BCV or clone emulation devices, create a TimeFinder/Snap virtual copy session or TimeFinder/Clone copy session.
- Split/activate — Split SRDF links, split BCV or clone emulation devices, or activate a TimeFinder/Snap virtual copy session or TimeFinder/Clone copy session.
- Cancel/terminate — Terminate a TimeFinder/Snap virtual copy session, TimeFinder/Clone copy session, or cancel a BCV or clone emulation pairs.

- Create logical point-in-time snapshots of images for arrays running PowerMaxOS 5978 or HYPERMAX OS 5977, using TimeFinder SnapVX.

  Note: TimeFinder SnapVX is supported only on arrays running PowerMaxOS 5978 or HYPERMAX OS 5977.

- Add one or more IOAs to all nodes, or remove one or more IOAs from all nodes participating in the replication solution.
- Rebuild the IOA information file for source and target IOAs.

Note: IASP Vary on/Vary off operations are now performed through IBM commands. Chapter 9, "IASP-Supported Operations," contains greater detail.

CHAPTER 8
Configuring for IASP Environment

This chapter shows how to prepare and configure SRDF/TimeFinder Manager to support IASP use in a clustered environment.

- Requirements 310
- Configuration procedures 311
- Configuring the environment for IASP 318

Requirements

The prerequisites for using an IASP and its image with SRDF/TimeFinder Manager are:

- All IBM software is installed and operational. Refer to the *Dell EMC SRDF/TimeFinder Manager for IBM i Release Notes* for a complete list of requirements.
- The latest IBM IASP and cluster PTFs, CUMs, or any other required PTFs have been applied. Refer to the IBM product documentation.
- SRDF/TimeFinder Manager is installed on the primary and HA nodes, and SRDF/TimeFinder Manager Client is installed on the Disaster Recovery and backup nodes. The *Dell EMC SRDF/TimeFinder Manager for IBM i Release Notes* show how to install SRDF/TimeFinder Manager.
- SRDF/TimeFinder Manager operations from the DR node require a gatekeeper device configured on the target array and the DR node.

Cluster requirements

The following are requirements for clusters:

- The cluster must be set up for IASP use according to IBM specifications. Create one or more secondary IASP pools, as necessary.
- The backup node must be the last node in the backup order.
- If clustering is configured through System i Navigator or IBM systems Director Navigator, the Service Tools Server must be configured and the port name “as-sts” is unblocked and enabled. Refer to the IBM product documentation for more information.

IOA requirements

The following are requirements for IOAs:

- When using IASPs, check that no two primary ASP groups share the same IOA. Primary and secondary IASPs are operated on simultaneously.
- All devices in each IASP group (or its image) must be on dedicated IOAs.
- No devices on the IOA can be part of a User ASP or System ASP (ASPs 1-32).
- An IASP image cannot share an IOA with another IASP image.
- A gatekeeper device cannot be presented to an IOA that contains an IASP image.

Configuration procedures

The following sections provide detailed instructions on how to configure the SRDF/TimeFinder Manager environment to support IASPs in a multi-node environment:

- "Starting and stopping the server job"
- "Building/rebuilding the IOA file"
- "Making target devices visible to the backup node"
- "Performing initial TimeFinder/SRDF operations"

These sections are intended to be used as a reference for the section "Configuring the environment for IASP" on page 318.

Starting and stopping the server job

The SRDF/TimeFinder Manager server job, which runs in the EMCCTL subsystem (the default), must be running on all participating nodes. Depending on the request type received, the server job can perform tasks such as:

- Adding and removing IOAs
- Running a system command on the node
- Generating IOA files on all nodes
- Updating the IASP configuration files on the nodes

Prior to starting the SRDF/TimeFinder Manager server job, the following conditions are required:

- TCP/IP must be active on all nodes.
- The appropriate EMCCTL or EMCCTLC library must be in the library list.

  Refer to the *Dell EMC SRDF/TimeFinder Manager for IBM i Release Notes* for more information about adding libraries.
- The subsystem job EMCCTL must be started as follows:

  ```
  STRSBS SBSD(EMCCTL)
  ```

  Note: If another subsystem other than the default EMCCTL is used, start that subsystem.

Starting the server job

To start the server job on each node, enter the following STRSTMSRV command:

```
STRSTMSRV USERID(*CURRENT) PORTNUM(8000) JOBQ(*LIBL/EMCCTL)
```

- USERID — The parameter that submits the job under the user ID that issues the command. The default is *CURRENT.

  This is the user ID that was created and is used to run SRDF/TimeFinder Manager.

  All other nodes can use the user profile that has *IOSYSCFG, *JOBCTL, and *SERVICE special authority and *USE access to the following *SRVPGM objects in QSYS:

  – QCSTCTL

  – QCSTCRG1

  – QCSTCRG4

  For the server job STMSERVER, if the EMCCTL subsystem is not used, check that more than one job is allowed to run in that subsystem.

- PORTNUM — The port used for TCP/IP communications.

  The default port number is 8000. If you use a port number other than 8000, make sure the port number is the same on the primary node, HA node, backup node, and DR node.

- JOBQ — The subsystem where you want the STRSTMSRV server job to run. The default is *LIBL/EMCCTL.

  If you use a different subsystem, enter that job queue name associated with the subsystem, for example:

  jobq(*libl/<*jobqname*>)

Stopping the server job

To stop the SRDF/TimeFinder server job, from the command line, enter the ENDSTMSRV command. Supply the port number that was used when the STRSTMSRV command was issued on the primary, backup, and DR nodes. For example:

```
ENDSTMSRV PORTNUM(8000)
```

The default port number is 8000.

Performing initial TimeFinder/SRDF operations

When configuring SRDF/TimeFinder Manager, perform initial SRDF or TimeFinder operations, such as Establish/Split, Create/Activate, or Link on the devices when IASP images are present in the configuration.

To perform SRDF or TimeFinder operations on an IASP image:

1. From the command line, enter the CTLACT command and press **F4** to prompt.

   The Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen (Figure 110) appears.

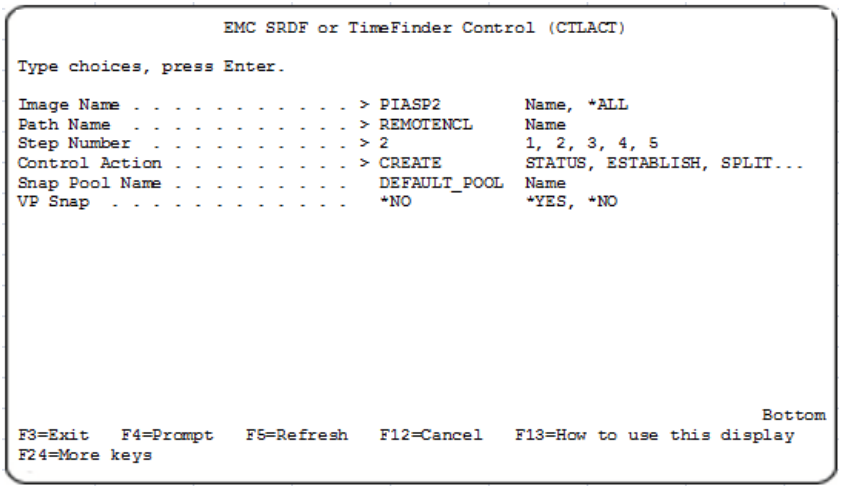


Figure 110 Dell EMC SRDF/TimeFinder Ctrl (CTLACT) screen with CREATE action

   At this point in the configuration process, the IOA file for the target devices has not been built, and SRDF/TimeFinder Manager cannot perform its validation check. Therefore, make sure target devices are not visible to the DR node or backup node.

2. Enter the following information:
   - **Image Name** — The name of the image on which to perform a control action.
   - **Path Name** — The path name associated with the image on which to perform a control action.
   - **Step Number** — The specific step number for which the control action is to be initiated.
   - **Control Action** — The SRDF or TimeFinder control action to be performed.
3. If you are creating a SNAP session enter the **Snap Pool Name** which is the named SAVE pool to use for the session.

   Note: You can supply the Snap Pool Name parameter only when creating new SNAP sessions. In all other cases, the name is ignored.

4. If you are creating a Clone session using VP Snap mode, set the **VP Snap** parameter to ***YES**. The VP Snap attribute can only be set when creating new clone sessions.

   For rules and restrictions and additional information for the snap pool name feature and VP Snap mode refer to "CTLACT" on page 206.

   Set the control action to **ESTABLISH** for SRDF, TimeFinder Clone Emulation, SnapVX, or BCV devices. Use the control action **CREATE** for TimeFinder/Snap or TimeFinder/Clone devices.

   If creating snapshots, first establish a session and then linking can be performed.

5. After the IASP image is synchronized, created, or for snapshots, established:

   a. Vary off the source devices.

   b. Re-enter the **CTLACT** command with the same IASP name and image name.

   c. Set **Control Action** to **SPLIT** or **ACTIVATE,** or for SnapVX, **LINK**.

   d. Set **Split/Suspend Mode** to **MAINTENANCE.**

   Refer to "CTLACT" on page 362 for more information on the CTLACT command. Refer to "Controlling TimeFinder" on page 135 and "" on page 165 for more information on Establish/Create, Split/Activate/Link and other SRDF and TimeFinder control operations.

   Once SRDF/TimeFinder Manager is configured for IASP use, you can use the Work with Associated Paths screen (Figure 111 on page 315) to perform ESTABLISH/CREATE and SPLIT/ACTIVATE and SnapVX operations.

Configuring an IASP image

Before using IASPs in an SRDF/TimeFinder Manager environment, you must configure one or more IASP images as needed for the replication solution.

IMPORTANT

To configure an IASP image, a valid copy of the IASP image must first be presented to the selected target node. Use the CTLACT command to do this. For SRDF, BCV, CLONE or SNAP type of step, carry out a Split/Activate action for this image to be visible at the target side. For SnapVX type of step, carry out a Link action on the image. Refer to "Performing initial TimeFinder/SRDF operations" on page 313 for using the CTLACT command. Once this is done a valid image is present at the target side and the IOA's can be presented to this node using the ADDIOA command described in "Making target devices visible to the backup node" on page 317.

To configure an IASP image:

1. Select option **1. Work with Images** from the Dell EMC SRDF/TimeFinder Manager for IBM i main menu.

   The Symmetrix Configuration by Image screen appears.

2. Select option **2. Work with paths** from the Symmetrix Configuration by Image screen.

3. The Work with Associated Paths screen (Figure 111 on page 315) appears.

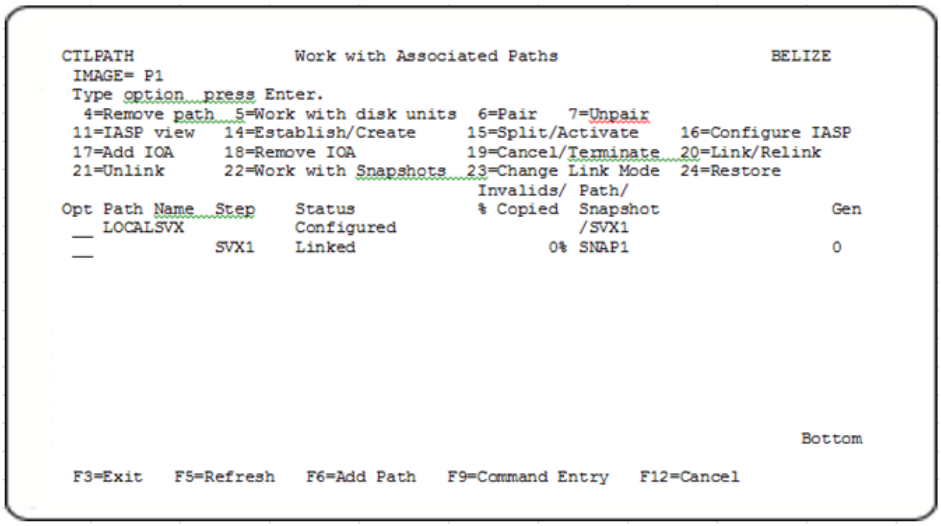


Figure 111 Work with Associated Paths screen

4. Enter **16** in the **Opt** column of the step associated with the path.

   The Configure IASP Session screen (Figure 112) appears.

5. Enter the following information:

   - **Target Node** — The name of the target node for this step. The node name entered must be part of the cluster that is configured on this system. If this is not the case, a message appears indicating that the node cannot be found.

     Note: The cluster node name can be found by running DSPCLUINF command. This command also displays the node IP address.

   - **Port Number** — The TCP/IP port that the SRDF/TimeFinder Manager server job uses for communications. The default is 8000.

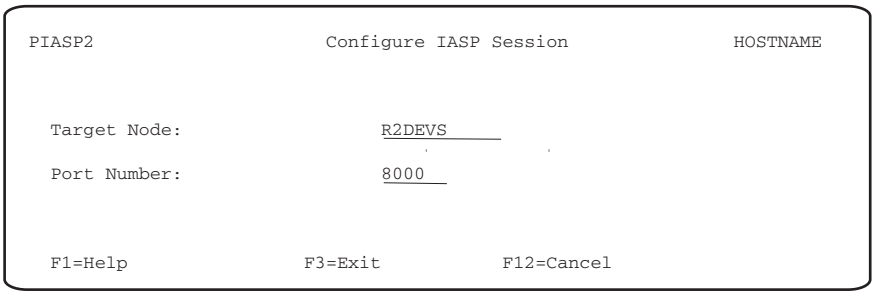


Figure 112 Configure IASP Session screen with configuration detail

Note: SRDF/TimeFinder Manager checks the validity of the information supplied and then sends the information to all nodes currently configured. If the node name is not part of the configured cluster, a message appears at the bottom of the screen indicating that the node name is not part of the configured cluster on this system.

6. Repeat steps 2 - 5 for each IASP image that you plan to use for SRDF/TimeFinder Manager.

IMPORTANT

If the server job on any node is unavailable at the time of the request, the IASP configuration process fails. If this occurs, the IASPS file on the primary node and the IASPS file on other nodes are not updated.

Allowing multiple TimeFinder IASP copies to share the same IOAs on a single host

To allow for this feature, each IASP copy needs to be presented to the target node making it possible to control each copy individually.

From the Work with Associated Path screen (Figure 111 on page 315) use Option 16, or use the "IASPCTL" command with CONFIG control action to create an entry that is written to the configuration file for each copy. A CONFIG action must be run for each copy on the host, and during the configuration process the other copies that share the IOA must be offline.

Note: When the IOA is shared between multiple TimeFinder copies, or when R2 and backup devices are sharing IOA's, be aware that if the IOA is enabled for one copy it is active at the same time for the other copy. For example, when trying to terminate a SNAP session for one copy the IOA needs to be inactive. If needed, the CTLACT command can then be used to terminate one copy while keeping the IOA active. Also one copy can be presented at any time, so make sure that access to the devices is only to one set of devices. This can be accomplished with the CLI command SYMACCESS.

Building/rebuilding the IOA file

When the IASP configuration is run, a rebuild action is automatically performed. As each IASP session is configured, the IOA files are also copied to the appropriate nodes.

However, IOA files must be current. SRDF/TimeFinder Manager depends on information stored in the files such as error checking or IOA status, prior to performing any operations against an IASP image. As a result, if the IOA files need to be reconfigured, the REBUILDIOA operation may need to be run as a separate action.

⚠CAUTION

Failure to rebuild and maintain current IOA files may result in critical data loss and/or system crash.

To rebuild the IOA file, use the IASPCTL command. In the following example, the IASPCTL command adds an entry into the ISRCIOA file, and rebuilds the IOA file for the IASP image (PIASP2) associated with a path name of REMOTEBCV.

```
IASPCTL IMAGENAME(PIASP2) PATHNAME(REMOTEBCV) STEPNUM(1) ACTION(REBUILDIOA) NODENAME(NODE1)
IASPCTL IMAGENAME(PIASP2) PATHNAME(REMOTEBCV) STEPNUM(2) ACTION(REBUILDIOA) NODENAME(NODE2)
```

Note: The REBUILDIOA control action can also be run from the EMC Independent ASP Control (IASPCTL) screen. Refer to "Executing IASP control actions on primary node" on page 343 for more information on executing actions from this screen.

Making target devices visible to the backup node

There are many ways to make target devices visible to the backup node, for example:

1. With the storage devices online, after the Split operation is complete, connect the cables to the backup node.
2. Use the ADDRMVOIA command to add IO resources. Refer to “ADDRMVIOA” on page 359.

Do not present target devices until the first Split operation completes. After making the target devices visible to the backup node, rebuild the IOA file for the target devices. “Building/rebuilding the IOA file” on page 316 describes this procedure.

Configuring the environment for IASP

Configuring the SRDF/TimeFinder Manager environment to support IASP use in a multi- node environment, requires using a combination of IBM commands, SRDF/TimeFinder Manager commands, and screen options.

The following sections describe how to configure the IASP environment (SRDF/S or SRDF/A with remote BCVs):

- "Prepare cluster and IASP nodes" on page 318
- "Start the server" on page 318
- "Configure paths" on page 319
- "Configure an IASP session" on page 320
- "Populate the IOAs for each node in the cluster" on page 321
- "Post-configuration considerations" on page 322

Note: In the IASP View by ASP screen, IOAs that are removed from an IASP configuration display a Not Active status, and IOAs that are added to an IASP configuration display an Active status. When there are secondary IASPs included in the configuration and the ADDRMVIOA command is being used, the primary and all secondary IASPs must be specified with the corresponding IOAs in the command. Refer to "ADDRMVIOA" on page 359 for more details.

Prepare cluster and IASP nodes

After creating all cluster nodes, start the cluster node on each node as follows:

```
STRCLUNOD CLUSTER(CLUSTERNAME) NODE(NODENAME)
```

1. Create IASP device domains.
2. On the DR node (R2) and backup node (RMTBCV):
   a. Using the ADDRMVIOA command, make sure the IASP IOAs for the DR and backup nodes are removed. Refer to "ADDRMVIOA" on page 359 for more information on the ADDRMVIOA command.
   b. Make sure the IOAs are active on the primary node.
   c. Vary on the IASP for the primary node.

Start the server

1. On the primary node (R1) and HA node, add EMCCTL to the library list.
2. On the DR and backup (RMTBCV) nodes, add EMCCTLC to the library list.
3. Make sure TCP/IP is active on all nodes.
4. Start the subsystem job EMCCTL, or start another available subsystem. If you use a subsystem other than the default EMCCTL, start that subsystem first:

   ```
   STRSBS SBSD(EMCCTL)
   ```

5. Start the SRDF/TimeFinder Manager server job (STRSTMSRV) on all nodes in the cluster:

   ```
   EMCCTL/STRSTMSRV USERID(*CURRENT) PORTNUM(8000) JOBQ(*LIBL/EMCCTL)
   EMCCTLC/STRSTMSRV USERID(*CURRENT) PORTNUM(8000) JOBQ(*LIBL/EMCCTL)
   ```

   If you started a different subsystem job in step 4, enter the jobq name associated with the subsystem, for example:

   jobq(*libl/<*jobq name*>)

"Starting and stopping the server job" on page 311 shows how to start the SRDF/TimeFinder Manager server job using STRSTMSRV.

Configure paths

1. Call the following program:

   ```
   CALL PGM(EMCCTL/CRTSYMAPI)
   ```

2. From the EMCCTL main menu, start SRDF/TimeFinder Manager:

   ```
   GO MAINCTL
   ```

   "Dell EMC SRDF/TimeFinder Manager for IBM i main menu" on page 43 shows how to access the SRDF/TimeFinder Manager Main menu.

3. Select option **1=Work with images.**

   The Symmetrix Configuration by Image screen appears.

4. Scroll down to the IASP image name and enter **8** in the **Opt** column for that image**.**

   The Paths not Associated with the Image screen appears.

5. For a three-node configuration, enter **1** in the **Opt** column of the **REMOTEBCV /RDF1/BCV1** path, and press **Enter**. For all other configurations, enter **1** in he **Opt** column of the appropriate path, and press **Enter.**

   The Work with Associated Paths screen appears.

   Refer to "Work with Associated Paths screen" on page 48 for more information on accessing the Work with Associated Paths screen.

6. Enter **2** in the **Opt** column of the IASP image.

   The Work with Associated Paths screen appears.

   Note: SRDF devices are automatically associated.

7. Enter **6** in the **Opt** column of the BCV1 step.

   The remote BCVs are associated with the R2 devices.

8. From the command line, use **CTLACT** to establish the BCVs and synchronize the RDF devices as needed. "Performing initial TimeFinder/SRDF operations" on page 313 contains greater detail.

9. Make sure that IOA is active and primary node is varied on.

10. Split RDF devices and remote BCVs using CTLACT. "Split BCV pairs: Maintenance" on page 142 shows how to carry out a maintenance split.

Configure an IASP session

To configure the IASP session:

1. For the RDF1 step, select option **1. Work with Images** from the Dell EMC SRDF/TimeFinder Manager for IBM i main menu to display the Symmetrix Configuration by Image.
2. Select option **2. Work with paths** from the Symmetrix Configuration by Image screen to display the Work with Associated Paths screen.
3. From the Work with Associated Paths screen, enter **16** in the **Opt** column of the step associated with the path.

   The Configure IASP Session screen (Figure 113) appears.

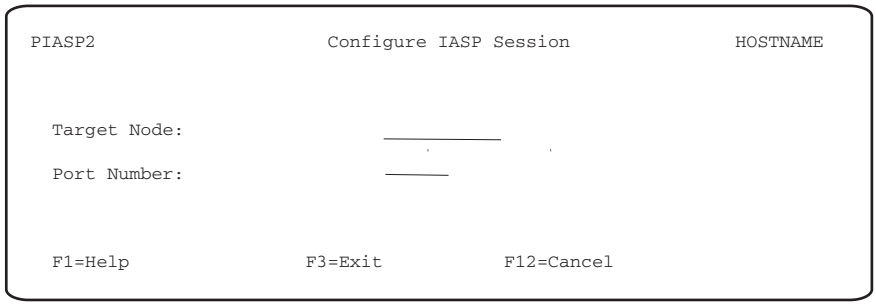


Figure 113 Configure IASP Session screen

4. Enter the appropriate information for the DR node and the server port number.

   Note: The node name entered must be part of the cluster that is configured on this system. If this is not the case, a message appears indicating that the node cannot be found.

5. For the BCV1 step, repeat step 1–4, and enter the appropriate information for the backup node in the Configure IASP Session screen.

   The above steps populate the EMCCTLC/IASPs physical file on the backup node. "Configuring an IASP image" on page 314 describes this procedure.

6. On the primary node, enter:

   ```
   DSPPFM EMCCTL/IASPS
   ```

7. On the DR and backup nodes, enter:

   ```
   DSPPFM EMCCTLC/IASPS
   ```

   The Display Physical File Member screen appears. Steps 6 and 7 verify that the IASP configuration file has been successfully propagated to the appropriate nodes.

   Refer to "Displaying contents of a physical file on each node" on page 355 for more information.

The CONFIG action automatically performs a REBUILDIOA operation. As each IASP session is configured, the IOA files are also copied to the appropriate nodes.

However, IOA files must be current. SRDF/TimeFinder Manager depends on information stored in the files such as error checking or IOA status, prior to performing any operations against an IASP image. As a result, if the IOA files need to be reconfigured, the REBUILDIOA operation may need to be run as a separate action.

Populate the IOAs for each node in the cluster

These configuration steps are used only when issuing a REBUILD as a separate operation from a CONFIG operation.

The IOAs for each node must be presented to the system. The primary node IOAs are presented first followed by the backup and DR nodes.

Perform the following steps on the primary node:

1. Issue the following command for SRDF setup:

   ```
   EMCCTL/IASPCTL IMAGENAME(IASPNAME)PATHNAME(REMOTEBCV) STEPNUM(1)
   ACTION(REBUILDIOA)NODENAME(PRIMARYNODENAME)
   ```

   Note: STEPNUM(1) denotes the SRDF step.

   Refer to "Building/rebuilding the IOA file" on page 316 for an example command to rebuild the IOA file.

   Note: To rebuild primary node IOA file, you can also issue the IASPCTL which displays the EMC Independent ASP Control (IASPCTL) screen. Refer to "Executing IASP control actions on primary node" on page 343 for more information on using this screen.

2. Ensure that all pairs in the path are in a split state.
3. Vary off the IASP.
4. Remove the IOA from the primary node using the SRDF/TimeFinder Manager IASPCTL or ADDRMVIOA command. Refer to "ADDRMVIOA" on page 359.
5. Perform the following steps on the DR node:
   a. Add the IOA to the DR node using the ADDRMVIOA command.
   b. For SRDF setup, issue the following commands:

      ```
      EMCCTLC/IASPCTLC IMAGENAME(IASPNAME) PATHNAME(REMOTEBCV)
      STEPNUM(1) ACTION(REBUILDIOA)NODENAME(DRNODENAME)
      ```

6. Remove IOA from the DR node using the ADDRMVIOA command.
7. Perform the following steps on the backup node:
   a. Add the IOA to the backup node using the ADDRMVIOA command.
   b. For BCV setup, issue the following commands:

      ```
      EMCCTLC/IASPCTLC IMAGENAME(IASPNAME) PATHNAME(REMOTEBCV)
      STEPNUM(2)ACTION(REBUILDIOA)NODENAME(BACKUPNODENAME)
      ```

      Note: STEPNUM(2) denotes the BCV step.

Note: To rebuild the backup or DR node IOA file, you can also issue the IASPCTLC which displays the Dell EMC Independent ASP Control (IASPCTLC) screen. Refer to "Executing IASP control actions on DR or backup node" on page 345 for more information on using this screen.

8. Remove the IOA from the backup node using the ADDRMVIOA command. Refer to "ADDRMVIOA" on page 359.
9. Verify the IOA location information on all nodes by viewing the physical file (ISRCIOA file) in the EMCCTL and EMCCTLC libraries.
10. On the primary node, verify the IASP setup from SRDF/TimeFinder Manager as follows:
    a. From the Work with Associated Paths screen, run the IASP view option against the step of the IASP.
    b. View and verify the IASP information from the IASP view by ASP screen.
11. Add the IOA back to the primary node using the IASPCTL command, and then vary on the IASP using the IBM commands.

Post-configuration considerations

After SRDF/TimeFinder Manager is configured for IASP support, make a list of the hardware configurations, such as partitions and IOA bus and slot location information, for all nodes. If a disaster occurs, having this information readily accessible makes the DR go more smoothly.

CHAPTER 9
IASP-Supported Operations

This chapter shows how to use the SRDF/TimeFinder Manager interface to perform IASP-supported operations when working with SRDF and TimeFinder (BCV, Clones, Snap, or SnapVX) devices configured for multi-node backup and disaster recovery (DR) solutions.

- IASP-supported operations 324
- SRDF/TimeFinder Manager IASP-specific menu options 336
- Executing IASP control actions 343
- Automated operations 351

Note: This chapter covers the use of IASP in the SRDF/TimeFinder Manager environment only. It does not provide details on how to perform IBM-specific operations, except where necessary. For information on IASPs, refer to the IBM documentation.

IASP-supported operations

The following sections show how to perform operations, such as disaster recovery (planned downtime and unplanned DR) and backup, in SRDF/TimeFinder Manager configured for IASP support:

- Three-node planned failover (downtime) operations on the primary node. See "Three-node planned failover (DR) and recovery solution" on page 324.
- Unplanned disaster recovery. See "Three-node failover and recovery (unplanned) solution" on page 326.
- Routine backup operation. See "Backup operations" on page 327.
- Switch to high availability node. See "Managing a four-node solution with a high availability node" on page 331.

See "Configuring the environment for IASP" on page 318 to ensure all cluster requirements are met. Many operations, such as failover and recovery processes, use a combination of commands (SRDF/TimeFinder Manager, Operations Navigator, or IBM i) and System Service Tools (SST) functions.

The DR and recovery operations detailed below require IASP images to be varied on or off. See "Varying on and varying off IASP images locally" on page 340 for more information.

Three-node planned failover (DR) and recovery solution

This section shows how to recover from a planned primary node downtime in a three-node environment. The following rules apply to this recovery solution:

- The primary node is where the IASP source devices (R1 devices) reside.
- The DR node is the node that takes over when the primary node fails or is taken offline.
- The backup node acts as the secondary owner for an IASP image. This node is configured with remote BCV, Clones Snap, or Snap VX for backing up an IASP image.

⚠CAUTION

All tasks for a planned downtime must be completed as defined by the chosen failover process. Any deviation from this process may result in a system crash or possible data loss.

Perform the following steps for a three-node planned recovery solution:

1. End IASP operations using the internal procedures established at your site.
2. Vary off the source IASP image, using the IBM VRYCFG command issued from a source host.
3. Remove the source IOA from the primary node using the IASPCTL command. See "IASPCTL" on page 377 for more information
4. Do one of the following:
   - If the primary node is available, from the primary node, perform a failover using the SRDFDRCTL command with the Establish option.

- If the primary node is not available, run the CTLACTTGT command on the DR node and perform a failover with the ESTABLISH option to the target.

Note: Use the ESTABLISH/RESTORE option only with dynamic SRDF. This performs a dynamic directional swap of the replication and establish the R1 and R2 devices.

See "SRDFDRCTL" on page 406 and "CTLACTTGT" on page 363 for more information.

5. Add IOAs for target devices to the DR node, using SRDF/TimeFinder Manager.

   See "IASPCTL" on page 377 (if the primary node is available) or "IASPCTLC" on page 384 (if the primary node is down).

6. Vary on the IASP image on the DR node, using the IBM VRYCFG command from the DR node.
7. Verify the data, using IBM commands.

   Refer to the IBM documentation for more information.

When the primary node is ready to re-assume normal operations:

1. Vary off the IASP image on the DR node, using the IBM VRYCFG command from the DR node.
2. Remove IOAs from the DR node.

   See "Adding or removing IOAs" on page 338.

3. Do one of the following:

   - If the primary node is available, and it is a dynamic SRDF configuration, on the primary node, run the SRDFDRCTL command using the FAILOVER control action with the Establish option.

     See "SRDFDRCTL" on page 406.

   - If the primary node is available, and it is *not* a dynamic SRDF configuration, perform a FAILBACK.
   - If the primary node is not available, and it is a dynamic SRDF configuration, on the DR node, run the CTLACTTGT command with the FAILOVER control action and the ESTABLISH option. Otherwise, perform a FAILBACK.

     The CTLACTTGT command with the FAILOVER/FAILBACK control action performs all the steps listed as action items when the primary node is down.

     See "CTLACTTGT" on page 363.

4. Check status of the cluster. See the IBM documentation for more information.
5. Add IOAs for the source devices on the primary node.

   See "IASPCTL" on page 377, if primary node is available, and "IASPCTLC" on page 384, if the primary node is down.

6. Vary on the source IASP image on the primary node, using the IBM VRYCFG command.

Three-node failover and recovery (unplanned) solution

This section shows how to recover from a primary node failure in a three-node environment. The following rules apply to this recovery solution:

- The primary node is where the IASP source devices (R1s) reside.
- The DR node is the node that takes over when the primary node fails or is taken offline.
- The backup node acts as the secondary owner for an IASP image. This node is configured with remote BCV, Clones, Snap, or SnapVX for backing up an IASP image.

To recover from a primary node failure in a three-node environment:

1. If the primary node is down:
   a. Run the CTLACTTGT command on the DR node using the FAILOVER control action with the Establish option.
   b. If the SRDF links are not available, run the CTLACTTGT command on the DR node with the FAILOVER *NO option.

   Note: Use the Establish/Restore option only with dynamic SRDF. Establish performs a dynamic directional swap of the replication and establishes the R1 and R2 devices. For static SRDF configurations, use FAILOVER with *NO.

   See "CTLACTTGT" on page 363 for more information.
2. Add IOA for target devices on the DR node, using the IASPCTLC command. See "IASPCTLC" on page 384 for more information.
3. Vary on the IASP image on the DR node using the IBM VRYCFG command.
4. Verify the data, using IBM commands.

   Refer to the IBM documentation for more information.

To resume normal operations, perform the following steps:

1. Vary off the target IASP image on the DR node using the IBM VRYCFG command.
2. Remove the target IOA from the DR node using the IASPCTLC command. See "IASPCTLC" on page 384 for more information.
3. On the DR node, with dynamic SRDF, run the CTLACTTGT command with the FAILOVER control action and the ESTABLISH option.

   On the DR node, without dynamic SRDF, perform a FAILBACK.

   If a FAILOVER with option *NO was performed in step 1 , run the CTLACTTGT command with FAILBACK. See "CTLACTTGT" on page 363 for more information.
4. Verify the primary node is available to resume normal operations. Refer to the IBM documentation for more information.
5. If not already done, add EMCCTL to the library list, and run the STRSTMSRV command to start the SRDF/TimeFinder Manager Server. See "STRSTMSRV" on page 410 for more information.

6. Make sure the source IASP devices are available to the host. If not, add IOAs for the source devices to the primary node. See "IASPCTL" on page 377, if primary node is available and "IASPCTLC" on page 384, if primary node is down.
7. Check status of the cluster. See the IBM documentation for more information.
8. Vary on the source IASP on the primary node, using the IBM VRYCFG command.
9. Verify the data.

   Refer to the IBM documentation for more information.

Backup operations

This section shows how to do routine backup operations in a three-node IASP environment.

The following rules apply to backup operations:

- The primary node is where the IASP source devices (R1s) reside.
- The DR node is the node that takes over when the primary node fails or is taken offline.
- The backup node acts as the secondary owner for an IASP image. This node is configured with remote BCV, Clones, Snap, or SnapVX for backing up an IASP image.

If necessary, perform an IPL of the host. You do not need to restore any IASP data updates from this copy.

Managing remote TimeFinder devices from the DR or primary node for backup operations

When planned or unplanned DR events occur, and the primary node is unavailable, use the CTLACTTGT command on the DR node to control remote TimeFinder devices on the backup node.

Do the following (starting at either step 1 or 2), for backup operations, depending upon the availability of the primary node:

1. On the primary node, split the remote BCV devices to make them available for backup:
   - If the IASP is varied off, use the IASPCTL command to perform a maintenance split.
   - If the IASP is varied on, use the IASPCTL command for the split. SRDF/TimeFinder Manager uses the Change ASP Activity CL command (CHGASPACT) to quiesce IASP data for the split. See "IASPCTL" on page 377 for more information.
2. On the DR node, split the remote BCV devices to make them available for backup:
   - If the IASP is varied off, use the CTLACTTGT command to perform a maintenance split.
   - If IASP is varied on use the CTLACTTGT command with the CHG_ASP_ACT quiesce option for the split. This option uses the Change ASP Activity CL command (CHGASPACT) to quiesce IASP data for the split. See "CTLACTTGT" on page 363 for more information.
3. Vary on the source IASP image if it is still varied off.

4. Add IOA for target (remote BCV) devices to the backup node. See "Adding or removing IOAs" on page 338 for more information.
5. Vary on the IASP copy for the backup node.
6. Run the backup on the IASP data using the procedure established by your company.
7. Vary off the IASP image for the backup node using the IBM VRYCFG command issued from the backup node.
8. Remove IOA for target (remote BCV) devices from the backup node. See "Adding or removing IOAs" on page 338 for more information.
9. Establish the remote BCV devices.
10. IPL the backup node in preparation for the next backup operation.

    This step is needed only if an error occurs when varying on the IASP copy of the backup node.

Managing remote SnapVX devices from DR node or primary node for backup operations

When planned or unplanned DR events occur, and the primary node is unavailable, use the CTLACTTGT command on the DR node to control Remote SnapVX devices on the backup node.

Do the following (starting at either step 1 or 2), for backup operations, depending upon the availability of the primary node:

1. On the primary node, establish a snapshot:
   - If the IASP is varied off use the IASPCTL command to perform a consistent establish of a snapshot.
   - If the IASP is varied on, use the IASPCTL command for the establish snapshot operation.

   SRDF/TimeFinder Manager uses the Change ASP Activity CL command (CHGASPACT) to quiesce IASP data for the establish snap shot operation. See "IASPCTL" on page 377 for more information.
2. If the primary node is not available, establish a snapshot on the DR node for backup availability:
   - If the IASP is varied off use the CTLACTTGT command to perform a consistent establish of a snapshot.
   - If IASP is varied on, use the CTLACTTGT command with the CHG_ASP_ACT quiesce option for the establish. This option uses the Change ASP activity CL command (CHGASPACT) to quiesce IASP data for the establish. See "CTLACTTGT" on page 363 for more information.
3. When the snapshot is established use the IASPCTL command (or CTLACTTGT on the DR node) for the LINK snapshot operation. This makes the data available on the devices attached to the target node for the selected snapshot. To link a different snapshot to the target devices, use the LINK/RELINK action.

   Note: If creating multiple snapshots over a period of time, use the link command to present each snapshot to the target node and create backups of all snapshots.

4. Vary on the source IASP image.
5. Add IOA for target (remote SnapVX) devices to the backup node. See “Adding or removing IOAs” on page 338 on page 305 for more information.
6. Vary on the IASP copy for the backup node.
7. Run the backup on the IASP data using the procedure established by your company.
8. Vary off the IASP image for the backup node using the IBM VRYCFG command issued from the backup node.
9. If an error occurs when varying on the IASP copy on the backup node, remove IOA for target devices from the backup node.
10. To link a different snapshot to the target devices, use the LINK/RELINK action.

Restore operations

This section shows how to do routine restore operations in a three-node IASP environment. The following rules apply to restore operations:

- The primary node is where the IASP source devices (R1s) reside.
- The DR node is the node that takes over when the primary node fails or is taken offline.
- The backup node acts as the secondary owner for an IASP image. This node is configured with remote BCV, Clones, Snap, or SnapVX for backing up an IASP image. If necessary, perform an IPL of the host.

Managing remote SnapVX devices from the backup node for restore operations

1. On the primary node use the IASPCTL command or menu option to perform a consistent Establish to create a snapshot for step 2 (SVX1).

   SRDF/TimeFinder Manager uses the Change ASP Activity CL command (CHGASPACT) to quiesce IASP data for the establish snapshot operation if the IASP is active, otherwise it uses maintenance. See "IASPCTL" on page 377 for more information.
2. When the snapshot is established, use the IASPCTL command or menu option to link the snapshot using Copy Mode. This makes the data available on the target devices for the selected snapshot (SVX1 step 2).
3. Add IOA for target devices (SVX1 step 2) to the backup node. See "Adding or removing IOAs" on page 338.
4. Vary on the IASP copy on the backup node (SVX1 step 2).
5. Create some data on the IASP copy, this data is restored and available later on the R1 devices of the IASP copy.
6. Vary off the IASP copy for the backup node using the IBM VRYCFG command.
7. Remove IOA for the target devices (SVX1 step 2) from the backup node. See "Adding or removing IOAs" on page 338.
8. Use the IASPCTL command or menu option to unlink the snapshot (SVX1 step 2). This makes the data unavailable to the target devices.
9. Create a dummy path DUMMY.
   - For a local snapshot, the path should be /SVX1/SVX2.
   - For a remote snapshot, the path should be /RDF1/SVX1/SVX2 for recovery purpose only.
10. Pair step 3 (SVX2) with R2 devices (step 1 - SVX1 for a local snapshot and RDF1 for a remote snapshot) using a file. See "SnapVX device pairing" on page 96.
11. On the primary node use the CTLACT command or menu option to perform a consistent Establish to create a snapshot for step 3 (SVX2).
12. On the primary node use the IASPCTL command or menu option to split step 1 (SVX1 is step 1 for a local snapshot and RDF1 is for a remote snapshot).

13. When the snapshot (SVX2) is established, use the CTLACT command to link the snapshot with Copy Mode to the target devices.

    Note: SVX2 is step 2 for a local snapshot and step 3 for a remote snapshot.

14. When the snapshot (SVX2) is 100% copied, use the CTLACT command to restore step 2.
15. When the restore is 100% copied, use the CTLACT command to unlink the snapshot from the target devices.

    Note: SVX2 is step 2 for a local snapshot and step 3 for a remote snapshot.

    Note: Steps 15 -17 are only needed when the user decides to verify that the data was restored to the R2 devices, in preparation for restoring this data to the primary node devices.

16. Add IOA for target R2 devices (RDF1 step 1) to the DR node. See “Adding or removing IOAs” on page 338.
17. Vary on the IASP copy on the DR node for step 1 (RDF1). When satisfied the data is correct for restore, vary off the IASP copy on the DR node.
18. Remove IOA for target R2 devices (RDF1 step 1) from the DR node. See “Adding or removing IOAs” on page 338.
19. On the primary node use the SRDFDRCTL command to Execute a failback for step 1 (RDF1). See “SRDFDRCTL” on page 406.
20. Add IOA for R1 devices (RDF1 step 1) to the primary node. See “Adding or removing IOAs” on page 338.
21. Vary on the IASP copy on the primary node for step1 (RDF1) and verify data is restored.

Managing a four-node solution with a high availability node

This section shows how to switch to the high availability node.

Steps 1 through 5 need to be done only at initial setup or after a configuration change.

To switch to the high availability node:

1. On the primary node, vary off the IASP using the IBM VRYCFG command.

   If the primary node has failed, go to step 5. Otherwise, proceed to step 2.

2. Remove the source IOA from the primary node using the IASPCTL command.

   Refer to “IASPCTL” on page 377 for more information.

3. Start the STM servers on the primary and HA nodes in the configuration.

   Refer to “STRSTMSRV” on page 410 for more information.

4. On the primary node, run the IASPCTLHA command using the control action CFGHAIASP.

   Refer to “IASPCTLHA” on page 386 for more information.

5. End the STM server on the primary node using the ENDSTMSRV command.

   Refer to "ENDSTMSRV" on page 376 for more information.

6. Using DST/SST (service tools), make sure that the IOAs are added to the HA machine and they are reporting in. If they are not added or reporting, use the ADDRMVIOA command for initial setup to add the IOAs to the HA machine. After the initial setup use the IASPCTL command to add IOAs.

7. On the HA node, run the IASPCTLHA command with the control action SWHAIASP.

   Make sure the STM server is up and running on the HA, DR, and backup nodes.

8. On the HA node, vary on the IASP using the IBM VRYCFG command.

   Continue normal operations on the HA node.

To switch back to the primary node from the HA node:

1. On the HA node, vary off the IASP using the IBM VRYCFG command.

2. Remove the source IOA from the HA node using the IASPCTL command.

   Refer to "IASPCTL" on page 377 for more information.

3. Start the STM server on the primary and HA node.

   Refer to "STRSTMSRV" on page 410 for more information.

4. Add the source IOA to the primary node using the IASPCTL or ADDRMVIOA command. Refer to "ADDRMVIOA" on page 359 or "IASPCTL" on page 377 for more information.

5. On the HA node, run the IASPCTLHA command using the control action CFGHAIASP. Use the primary node as the target node.

6. End the STM server on the HA node using the ENDSTMSRV command.

   Refer to "ENDSTMSRV" on page 376 for more information.

7. On the primary node, run the IASPCTLHA command with the control action SWHAIASP.

   Refer to "IASPCTLHA" on page 386 for more information.

   Make sure the STM server is up and running on the primary, DR, and backup nodes.

8. On the primary node, vary on the IASP using the IBM VRYCFG command.

9. Continue normal operations on the primary node.

Source to source processing for IASPs

Dell EMC SRDF/TimeFinder Manager for IBM i V9.2 and later contains a facility for updating an IASP from a SnapVX snapshot without using a RESTORE operation. The architecture of such a configuration is like this:

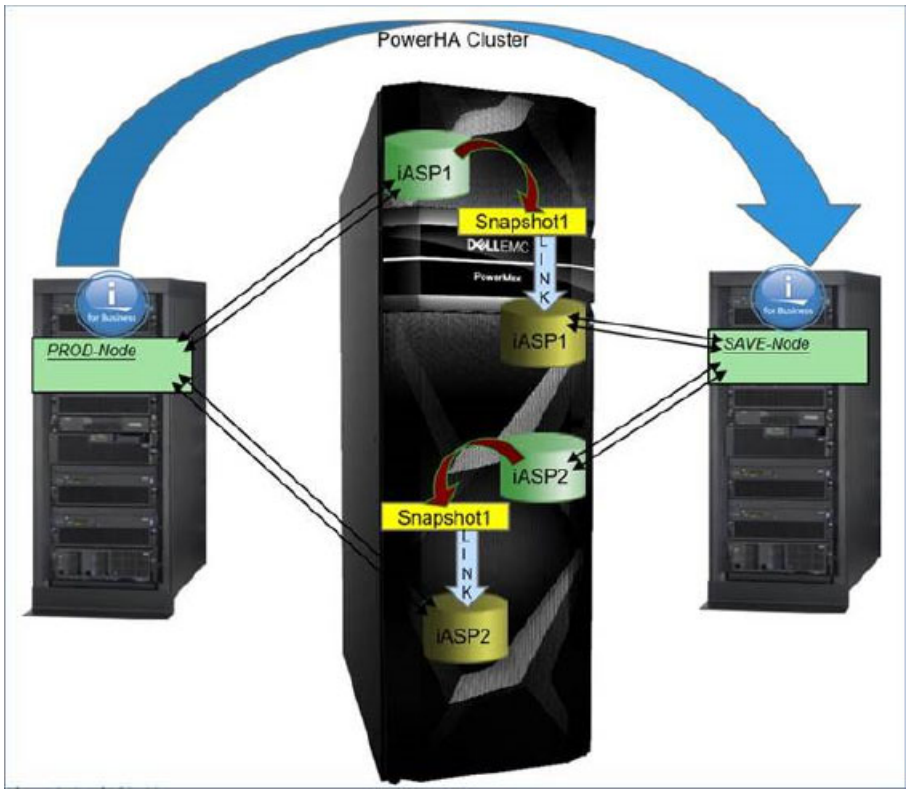


Figure 114 Source to source processing for IASPs

Operational overview

The overall operation of the source to source configuration is:

1. Applications on the PROD-Node create and process data on IASP1.
2. A point-in-time snapshot of IASP1 is taken using the path LOCALSVX.
3. The snapshot is linked and varied online to the SAVE-Node.
4. Applications on the SAVE-Node copies process the data on this IASP1. The applications create modified data that needs porting back to PROD-Node.
5. The modified data on the IASP1 snapshot is copied to IASP2.
6. A point-in-time snapshot of IASP2 is taken using the path LOCALSVX.
7. The snapshot is linked and varied online to the PROD-Node.
8. The PROD-Node can now access the modified data on IASP2 snapshot and integrate that with the data on IASP1.

Configuring IASP1 on PROD-Node

This section summarizes the steps to configure IASP1 on the PROD-Node ("Configuring an IASP image" on page 314 contains details of how to configure an IASP image).

1. On the PROD-Node:
    1. Use ADDRMVIOA to ADDIOA for IASP1 to PROD-Node.
    2. Establish the snapshot for IASP1 using path LOCALSVX.
    3. Link the IASP1 snapshot using path LOCALSVX.
2. On the SAVE-Node:
    1. Use ADDRMVIOA to ADDIOA for IASP1 to SAVE-Node.
3. On the PROD-Node:
    1. On the Work with Associated Path screen, use Option 16 to point IASP1 with path LOCALSVX at SAVE-NODE.
    2. On the Work with Associated Path screen, use Option 17 with path LOCALSVX to remove the IOA for IASP1 from the SAVE-Node.
    3. With the IASPCTL command, use REMOVEIOA to remove the IOA associated with the LOCALSVX path for IASP1 on the PROD-Node.

Configuring IASP2 on SAVE-Node

This section summarizes the steps to configure IASP2 on the SAVE-Node.

1. On the SAVE-Node:
    1. Use ADDRMVIOA to ADDIOA for IASP2 to SAVE-Node.
    2. Establish the snapshot for IASP2 using path LOCALSVX.
    3. Link the IASP2 snapshot using path LOCALSVX.
2. On the PROD-Node:
    1. Use ADDRMVIOA to ADDIOA for IASP2 to PROD-Node.
3. On the SAVE-Node:
    1. On the Work with Associated Path screen, use Option 16 to point IASP2 with path LOCALSVX at PROD-NODE.
    2. On the Work with Associated Path screen, use Option 17 with path LOCALSVX to remove the IOA for IASP2 from the PROD-Node.
    3. With the IASPCTL command, use REMOVEIOA to remove the IOA associated with the LOCALSVX path for IASP2 on the SAVE-Node.

Processing cycle

Once the two IASPs are configured the daily processing cycle is:

1. On the PROD-Node, enter:

   ```
   IASPCTL IMAGENAME(IASP1) PATHNAME(LOCALSVX) STEPNUM(1)
   ACTION(ADDIOA) NODENAME(PROD-Node)
   ```

2. Vary IASP1 online on the PROD-Node.

   Leave it online for the entire production cycle. The applications on PROD-Node can now create and process data on IASP1.

3. Establish a snapshot of IASP1 with path LOCALSVX using SnapVX.
4. Link the snapshot of IASP1 with path LOCALSVX.
5. Use Option 17 on the Work with Associated Paths screen to add an IOA to the SAVE-Node.
6. On the SAVE-Node:
   1. Vary IASP1 online.
   2. Enter:

      ```
      IASPCTL IMAGENAME(IASP2) PATHNAME(LOCALSVX) STEPNUM(1)
      ACTION(ADDIOA) NODENAME(SAVE-Node)
      ```

   3. Vary IASP2 online.

      Leave it onlne for the entire production cycle. The application on SAVE-Node can now process the data on IASP2.

   4. Establish a snapshot of IASP2 with path LOCALSVX using SnapVX.
   5. Use Option 17 on the Work with Associate Paths screen to add an IOA to the PROD-Node.
   6. Vary IASP1 offline.
7. On the PROD-Node:
   1. Use Option 18 on the Work with Associated Paths screen to remove the IOA for IASP1 with path LOCALSVX.
   2. Use Option 21 on the Work with Associated Paths screen to unlink the snapshot for IASP1.
   3. Vary IASP2 online and process its data.
   4. Once data processing is complete, vary IASP2 offline.
8. On the SAVE_Node:
   1. Use Option 18 on the Work with Associated Paths screen to remove the IOA for IASP2 with path LOCALSVX from the PROD-Node.
   2. Use Option 21 on the Work with Associated Paths screen to unlink the snapshot for IASP2.

SRDF/TimeFinder Manager IASP-specific menu options

The following sections describe the menu options you use when working with IASP images and SRDF and TimeFinder devices.

- "Establish/create, split/activate/link/relink, and cancel/terminate operations" on page 336.
- "Adding or removing IOAs" on page 338
- "Varying on and varying off IASP images locally" on page 340
- "Varying on and varying off IASP images remotely" on page 340
- "Displaying an IASP configuration by ASP" on page 341
- "Removing image information" on page 342

Establish/create, split/activate/link/relink, and cancel/terminate operations

With SRDF/TimeFinder Manager, you can perform Establish/Create, Split/Activate, Link/Relink, work with snapshots, change Link Mode, Restore, and Cancel/Terminate operations on devices the following ways:

- Executing the IASPCTL command from the command line or through the Control Menu. Refer to "Executing IASP control actions" on page 343 and "IASPCTL" on page 377.
- From the Work with Associated Paths screen, as described in the following sections.

  "Work with Associated Paths screen" on page 48 contains details of the Work with Associated Paths screen.

An IASP pair always consists of a source device and its target device. After an IASP image is configured and initialized, the target device contains no data.

After an Establish/Create operation is performed on an IASP pair, the IASP target device is synchronized with the IASP source device. The target device contains an identical copy of the data from the source device, which is valid until a Split or Activate process begins. The data on the target device can be accessed after the Split or Activate operation completes successfully.

Note: With SnapVX a target device can only be used after Link command is performed. A Link command with the Copy option copies the data to the target devices. Links can be created in Copy mode for a permanent copy on the target device, or in NoCopy mode for temporary use. With NoCopy mode, an Unlink command removes the data from the target device.

Establish/create

To perform an Establish/Create operation from a screen, from the Work with Associated Paths screen, enter **14** in the **Opt** column of the step for the associated IASP image.

The Establish Operations screen appears for you to complete the operation.

The establish operation for SnapVX creates and activates a snapshot of the selected image, path and step. When the source IASP is online a CHG_ASP_ACT is done to flush the memory to the disks.

⚠CAUTION

In normal operations, make sure the server job is running on all nodes before you issue an Establish or Create command. Failure to do so may cause unpredictable results.

Split/activate

To perform a Split/Activate operation, from the Work with Associated Paths screen, type **15** (Split/Activate) next to the step of the associated IASP image.

Link/Relink

From the Work with Associated Paths screen, enter **20** in the **Opt** column of the step for the associated IASP image.

The List of Snapshots screen appears for you to complete the operation. There you can select the snapshot that you want to select for the Link/Relink operation.

Note: Link/relink operations can also be performed from the Work with Shapshots screen. Enter **1** in the **Opt** column of the desired snapshot.

Unlink

From the Work with Associated Paths screen, enter **21** in the **Opt** column of the step for the associated IASP image.

This removes a link that exists to the target devices. To complete the operation the target IOA's connected to these devices need to removed first.

Work with snapshots

From the Work with Associated Paths screen, enter **22** in the **Opt** column of the step for the associated IASP image.

The work with snapshots screen appears for selecting the following available options for the snapshots listed: **1** (Link/Relink), **2** (Restore), **3** (Set TTL), **5** (Show details), **7** (Rename), and **8** (Terminate).

Change Link Mode

The change link mode changes the link mode of the linked snapshot to Copy or NoCopy. In Copy mode, the data is copied to the target devices, and when the snapshot is unlinked the data remains on the target devices. In NoCopy mode, when the snapshot is unlinked the data is removed from the target devices.

To perform a Change Link Mode operation, from the Work with Associated Paths screen, enter **23** in the **Opt** column of the step for the associated IASP image.

Restore

From the Work with Associated Paths screen, enter **24** in the **Opt** column of the step for the associated IASP image.

The List of Snapshots screen appears for selecting the snapshot for the Restore operation.

Note: Note: Restore operations can also be performed from the Work with Shapshots screen. Enter **2** in the **Opt** column of the desired snapshot.

Cancel/terminate

Depending upon the selected device type, you use the **Cancel/Terminate** option to perform either of the following:

- Cancel an incremental relationship between a standard device and a BCV device that are currently a pair in the BCV set. If the devices are clone-enabled BCVs, the clone sessions are terminated on the array.
- Terminate a TimeFinder/Snap virtual copy session or TimeFinder/Clone copy session.

To cancel or terminate a BCV, Snap, or Clone relationship, from the Work with Associated Paths screen, enter **19** in the **Opt** column of the step of the configured path.

The Work with Associated Paths screen reappears with the status for the device changed to NeverEstab or NotCreated. SRDF/TimeFinder Manager displays a confirmation message.

Note: Terminate SnapVX snapshots can only be done from the Work with Snapshots screen.

Adding or removing IOAs

All disk units within a primary IASP and secondary IASP are operated on as one automatic operation.

Note: The ADDRMVIOA command does not physically add or remove the IOA from the system (LPAR), it enables or disables the IOA. Removal or addition of an IOA from the system or LPAR is performed from the Hardware Management Console.

Rules and restrictions for adding or removing IOAs

The following rules apply to adding and removing IOAs in normal operations:

- When adding IOAs:
  - The IOA cannot be active on any node when the SRDF link is in sync or a BCV is established.
  - Target IOAs cannot be added if the devices are Established.
  - After an IOA is added, all disk units belonging to that IOA are presented to the node, and the selected image can be varied on for user access.
  - The target and source devices cannot be simultaneously presented to the same node after the addition. Otherwise, the operation fails when an attempt to add another copy of the same IASP image is presented.
- When removing IOAs:
  - The respective image must be varied off.
  - If using the ADDRMVIOA command to remove IOAs, the primary and secondary IASPs must be specified in the command. Specifying only the primary IASP removes only the resources for the primary IASP. See "ADDRMVIOA" on page 359 for more details.
  - After an IOA is removed, all disk units are removed from the configuration, and the data on those disks cannot be accessed.

Adding or removing IOAs on the primary node

To add or remove IOAs on the primary node use one of the following:

- IASPCTL on the primary node. See "IASPCTL" on page 377.
- IASPCTLC on the target DR or backup node. See "IASPCTLC" on page 384.

See "ADDRMVIOA" on page 359 for more information on adding or removing IOAs.

Note: For hosts running V6R1 or higher, when varying on an IASP on a different system with a different amount of fiber connections to the IASP devices, the vary on goes through DASD checks to detect if all paths can be reestablished. To avoid a longer vary on time, a multi-path reset executes after the ADDIOA is completed and the devices have reported back in.

Adding or removing IOAs on the DR or backup nodes

To add or remove IOAs on a DR or backup node, from the Work with Associated Paths screen, enter **17** (to add IOA) or **18** (to remove IOA) in the **OPT** column of the step associated with the path.

The Work with Associated Paths screen reappears.

Note: In a three-node remote BCV configuration, when options 17 or 18 are entered on the RDF1 step, the IOAs are added to or removed from the DR node. When options 17 or 18 are entered on the BCV1 step, the IOAs are added to or removed from the backup node. There is no interim screen that appears. The operation automatically occurs.

Varying on and varying off IASP images locally

To vary on and vary off IASP images locally, use the following VRYCFG IBM i command with STATUS set to *ON or *OFF:

```
VRYCFG CFGOBJ(IASP832) CFGTYPE(*DEV) STATUS(*ON)
VRYCFG CFGOBJ(IASP832) CFGTYPE(*DEV) STATUS(*OFF)
```

Varying on and varying off IASP images remotely

Using the REXEC server

To vary on an IASP image:

1. Start the *REXEC server on the target (STRTCPSRV *REXEC).
2. Enter the following:

```
RUNRMTCMD CMD('VRYCFG CFGOBJ(IASP832) CFGTYPE(*DEV) STATUS(*ON)')
RMTLOCNAME('10.xxx.xxx.xxx' *IP) RMTUSER(QSECOFR) RMTPWD(XXXX)
```

To vary off an IASP image:

1. Start the *REXEC server on the target (STRTCPSRV *REXEC).
2. Enter the following:

```
RUNRMTCMD CMD('VRYCFG CFGOBJ(IASP832) CFGTYPE(*DEV) STATUS(*OFF)')
RMTLOCNAME('10.xxx.xxx.xx' *IP) RMTUSER(QSECOFR) RMTPWD(XXXX)
```

Using the STM server

To vary on an IASP image:

1. Start the STM server (See "STRSTMSRV" on page 410.)
2. Enter the following:

```
STMRMTCMD CMD('VRYCFG CFGOBJ(IASP832) CFGTYPE(*DEV) STATUS(*ON)')
RMTNODNAME(nodexyz) PORTNUM(8000)
```

To vary off an IASP image:

1. Start the STM server (See "STRSTMSRV" on page 410.)
2. Enter the following:

```
STMRMTCMD CMD('VRYCFG CFGOBJ(IASP832) CFGTYPE(*DEV) STATUS(*OFF)')
RMTNODNAME(nodexyz) PORTNUM(8000)
```

Displaying an IASP configuration by ASP

To display configuration details of an IASP image by ASP:

1. On the Work with Associated Paths screen, enter **11** in the **Opt** column of the step of the IASP image.

   The IASP View by ASP screen (Figure 115) appears displaying the IASP view for the selected image and its associated path.

   Note that only the step you selected (source and target) displays.

```
CTLDASPVW                IASP View by ASP                     HOSTNAME

Image PIASP1       Path REMOTEBCV    Step number 1    Step type RDF1

           # of  SRCIOA        TGTIOA        SRCDEV          TGTDEV
 Opt  ASP  Units Status        Status        Status          Status
  ___   52     1  Not active    Active        Available       Varied Off



                                                                  Bottom
F3=Exit   F5=Refresh   F12=Cancel
```

Figure 115 IASP View by ASP screen

Table 14 describes the fields on the IASP View by ASP screen.

Table 14 IASP View by ASP screen

Field	Description
Image	The name of the IASP image.
Path	The name of the path used to move data on devices of the source image to devices of the target image.
Step number	The description of how data is copied (the number of steps in the path identified by its position 1:5) from source devices to target devices, such as 1, 2, 3, 4, and 5.
Step type	The replication step type for the device, such as *BCVn, RDFn, CLN*n, *NCL*n, *SNPn, or SVXn*.
Opt	Entry field for configuration action options. This field is reserved for future use.
ASP	ASP number.
# of Units	The total number of devices in the ASP.
SRCIOA Status	Active — The IOA is present in the configuration for the selected IASP. Not active — The IOA is not present in the configuration for the selected IASP.
TGTIOA Status	Active — The IOA is present in the configuration for the selected IASP. Not active — The IOA is not present in the configuration for the selected IASP.
SRCDEV Status	Any status that IBM i presents. The IBM WRKCFGSTS command contains greater detail.
TGTDEV Status	Any status that IBM i presents. The IBM WRKCFGSTS command contains greater detail.

Removing image information

Image information can be removed from the IASPs and ISRCIOA files using the IASPFILE command.

To remove image information:

1. On the command line enter IASPFILE.

   The EMC IASP File Utility (IASPFILE) screen appears (Figure 116).

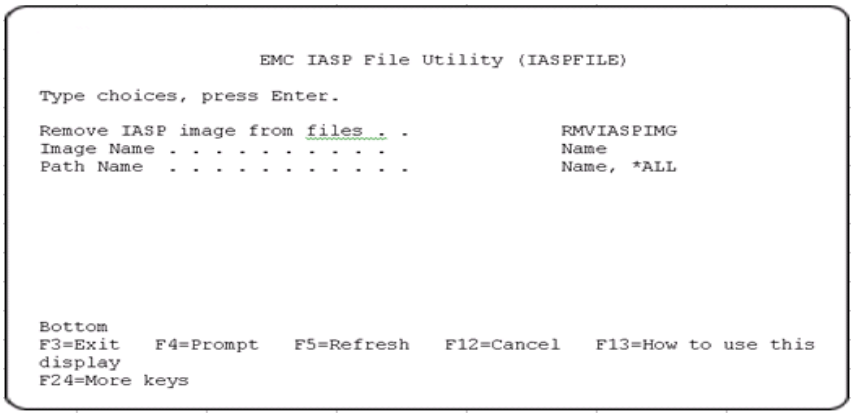


Figure 116 IASP File Utility (IASPFILE) screen

2. Enter the control action **RMVIASPIMG**.
3. Enter the image name.
4. Enter the path name for the image, or enter ***ALL** for all paths of the image entered in step 3 .

   Note: To remove image information, run the command from each node in the DR setup.

Executing IASP control actions

Control actions are performed against devices in an IASP environment for SRDF, TimeFinder, TimeFinder/Clone, TimeFinder/Snap, and TimeFinder SnapVX operations:

- On the primary node, use the IASPCTL command from either the SRDF/TimeFinder Manager interface or the command line.
- On the backup or DR node, use the IASPCTLC, and CTLACTTGT command from the command line.

Executing IASP control actions on primary node

To execute IASP control actions on the primary node:

1. On the Dell EMC SRDF/TimeFinder Manager main menu, select option **3. Control menu**.
2. Select option **2. Independent IASP Control**.

   You can also run the IASPCTL command from the command line.

   The Dell EMC Independent ASP Ctrl (IASPCTL) screen (Figure 117) appears.

```
                    EMC Independent ASP Control (IASPCTL)

Type choices, press Enter.

Image Name . . . . . .   P1          Name
Path Name  . . . . . .   LOCALNCL    Name
Step Number  . . . . .   1           1  2  3  4  5
Control Action . . . .   ADDIOA      CONFIG  ADDIOA  FORCEADD...

                                                                  Bottom
F3=Exit   F4=Prompt   F5=Refresh   F12=Cancel   F13=How to use this display
F24=More keys
```

Figure 117 Dell EMC Independent ASP Ctrl (IASPCTL) screen with ADDIOA action

3. Provide the following information:
   - **Image Name** — The name of the image on which to perform a control action.
   - **Path Name** — The path name associated with the image on which to perform a control action.
   - **Step Number** — The step number on which the control action operates.
   - **Control Action** — The IASP control action to be performed.

     To display a list of all control action parameters, press **F4** on the **Control Action** prompt (Figure 117) and the Specify Value for Parameter Action screen (Figure 118 on page 344) appears. See "IASPCTL" on page 377 for more information on the control action parameters.

4. If creating a SNAP session enter the **Snap Pool Name** which is the named SAVE pool to use for the session. If pool name is not specified, the operation uses the default pool (DEFAULT_POOL).

   Note: The Snap Pool Name parameter is relevant for creating new SNAP sessions only, otherwise it is ignored.

5. If creating a Clone session using VP Snap mode, set the **VP Snap** parameter to ***YES**. The VP Snap attribute can only be set when creating new clone sessions.

   For rules and restrictions for the snap pool name feature and VP Snap mode, see "IASPCTL" on page 377.

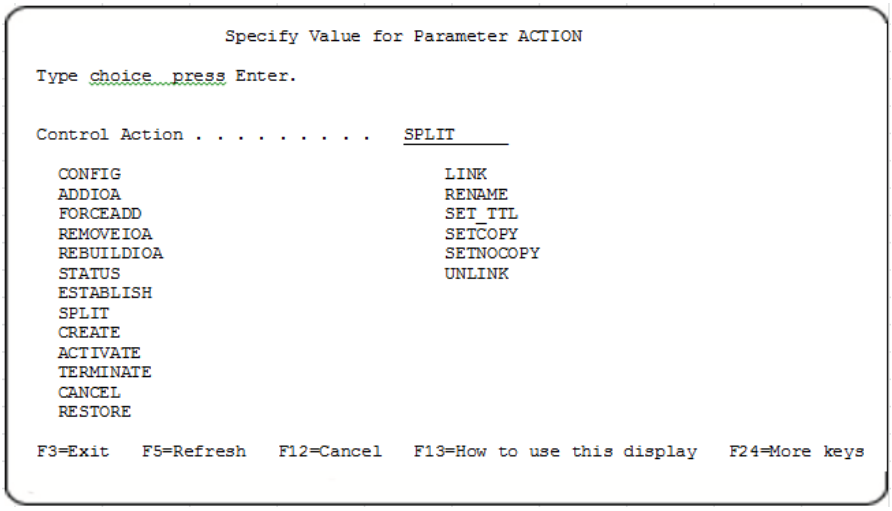


Figure 118 Specify Value for Parameter Action screen

6. Enter the desired control action, and press **Enter**.

   The Dell EMC Independent ASP Cntrl (IASPCTL) screen appears with the selected IASP control parameters.

   Depending on the control action chosen, the IASPCTL screen may appear with other required parameters.

7. Enter additional parameters as needed.

   See "IASPCTL" on page 377 for more information on additional control parameters.

Executing IASP control actions on DR or backup node

1. On the command line, run the IASPCTLC command.

   The Dell EMC Independent ASP Ctrl (IASPCTLC) screen (Figure 119) appears.

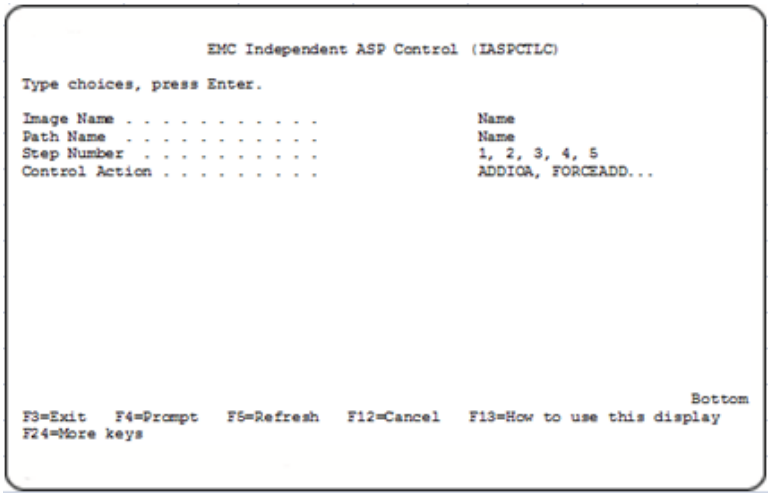


Figure 119 Dell EMC Independent ASP Ctrl (IASPCTLC) screen

2. Provide the following information:
   - **Image Name** — The name of the image on which to perform a control action.
   - **Path Name** — The path name associated with the image on which to perform a control action.
   - **Step Number** — The step number on which the control action operates.

     To display a list of the node names in cluster, refer to "Executing IASP control actions on primary node" on page 343.
   - **Control Action** — The IASP control action you want to perform.

     To display a list of all control action parameters, press **F4** on the **Control Action** prompt (Figure 119) and the Specify Value for Parameter Action screen (Figure 120) appears.

     See "IASPCTLC" on page 384 for more information on the control action parameters.

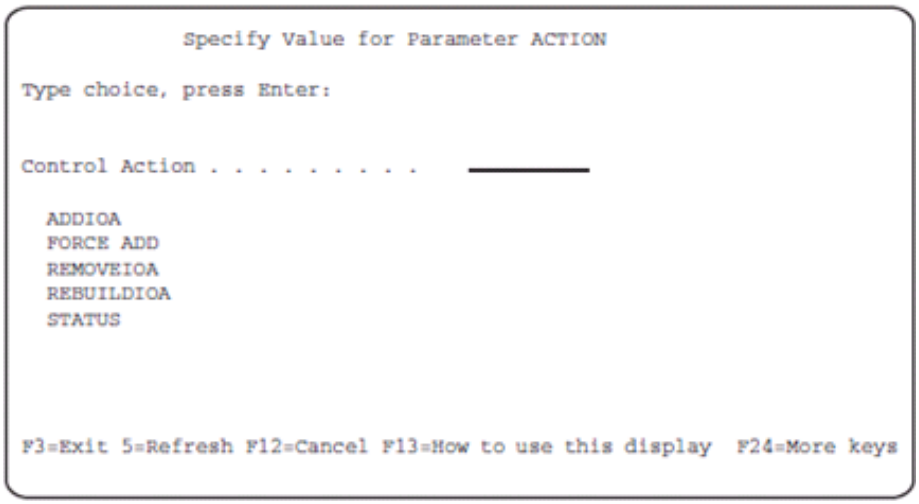


Figure 120 Specify Value for Parameter Action screen

3. Enter the desired control action (Figure 120 on page 346), and press **Enter.**

   The Dell EMC Independent ASP Ctrl (IASPCTLC) screen (Figure 121) appears with the IASP control parameters.

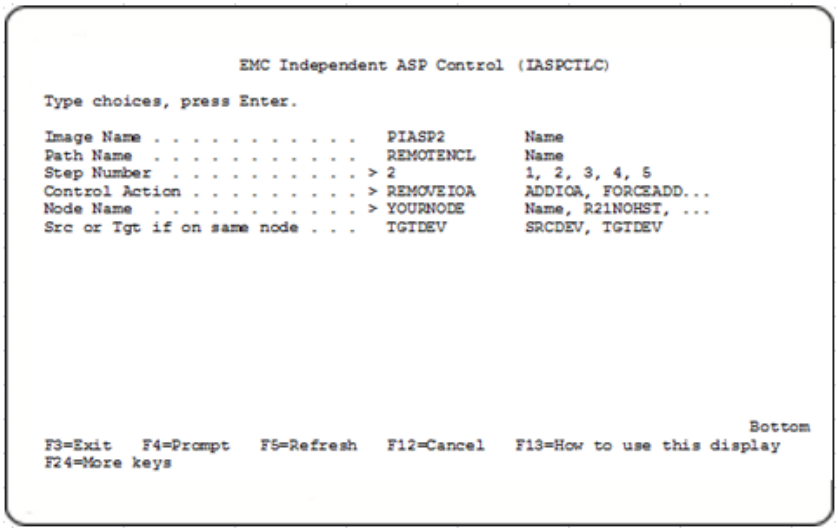


Figure 121 Control Action REMOVEIOA

- **Node Name** — Parameter used in conjunction with the control action REBUILDIOA when the configuration file needs to be rebuilt for a node and to specify where the operation must be performed. This parameter is also used for STATUS, ADDIOA, REMOVEIOA, and FORCEADD.

  The node name you provide must be part of the cluster that is configured on this system. If this is not the case a message appears indicating that the node cannot be found.

  Note: The cluster node name can be found by running DSPCLUINF command. This command also displays the node IP address.

See "IASPCTLC" on page 384 for more information on additional control action parameters.

Executing SRDF or TimeFinder control actions on DR or backup node

1. On the command line, run the CTLACTTGT command.

   The Dell EMC SRDF/TimeFinder Ctrl (CTLACTTGT) screen (Figure 122 on page 347) appears.

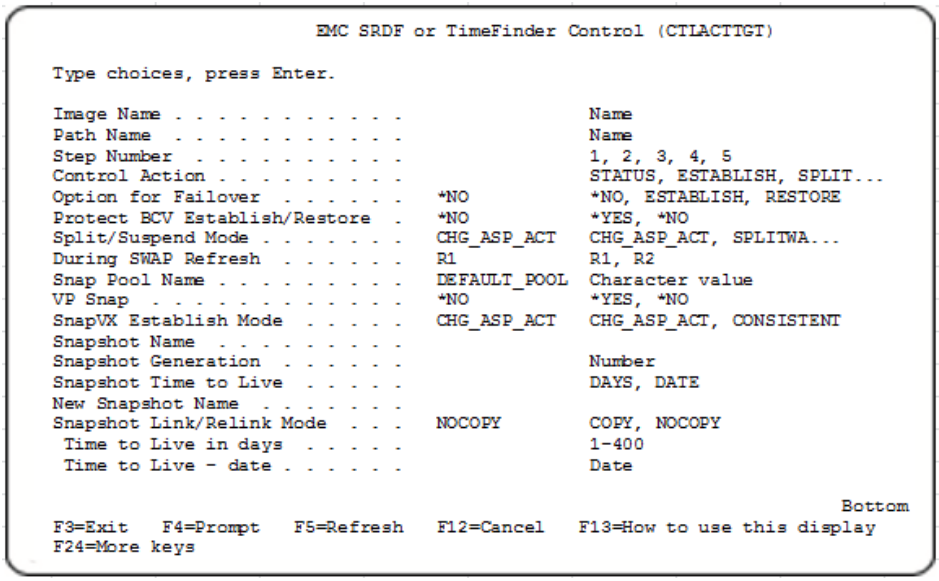


Figure 122 Dell EMC SRDF/TimeFinder Control (CTLACTTGT) screen with CREATE action

2. Enter the following information:
   - **Image Name** — The name of the image to be used in an SRDF or TimeFinder control operation. Use this command for IASPs images only.
   - **Path Name** — The name of the path for the image to be used in an SRDF or TimeFinder operation.
   - **Step Number** — The step number for which the control action is initiated in an SRDF or TimeFinder operation.
   - **Control Action** — The SRDF or TimeFinder control action to perform on the IASP image. See "CTLACTTGT" on page 363 for a list of control actions.
3. If creating a SNAP session, enter the **Snap Pool Name** which is the named SAVE pool to use for the session. If pool name is not specified, the operation uses the default pool (DEFAULT_POOL).

   Note: The Snap Pool Name parameter is relevant only when creating SNAP sessions.

4. If creating a Clone session using VP Snap mode, set the **VP Snap** parameter to ***YES**. The VP Snap attribute can only be set when creating new clone sessions.

   For rules and restrictions for the snap pool name feature and VP Snap mode, see "CTLACT" on page 362.

Note: If executing SnapVX operations additional parameters are required. See "CTLACTTGT" on page 363.

Preparing target HA node

The IASPCTLHA command pushes configuration files to the specified target HA node and prepares the node to assume the role of the primary node.

To push configuration files and prepare the target HA node for a primary take over:

1. Vary off the IASP that is connected to the production node and the HA node. Only one path can be active at any time. When switching from the primary to the HA node, make sure that the path to the primary node is removed and the path to HA node is active.

   You can perform this through the fiber switch or using the FA ports that have a path to both the production and HA node devices. See the example setup in Figure 109 on page 307.

2. Ensure that EMCCTL is installed on the primary and the HA node and that the SRDF/TimeFinder Manager Server is started on all nodes.

3. On the primary node, execute the IASPCTLHA command.

   The Dell EMC IASP HA Control (IASPCTLHA) screen (Figure 123) appears.

```
                        EMC IASP HA Control (IASPCTLHA)

Type choices, press Enter.,

Control HA Action  . . . . . .   CFGHAIASP     CFGHAIASP, SWHAIASP...
Image name . . . . . . . . . .   *ALL          Name, *ALL
             + for more values   _____
Target HA Node . . . . . . . .   _____          Character value
Port Number  . . . . . . . . .   8000           Character value

                                                                   Bottom,
F3=Exit   F4=Prompt   F5=Refresh   F12=Cancel   F13=How to use this display
F24=More keys
```

Figure 123 Dell EMC IASP HA Control (IASPCTLHA) screen

4. Provide the following information:

   - **Control HA Action** — Enter CFGHAIASP.
   - **Image Name** — Specifies which IASP image configuration files to push to the HA node. Use this command for IASPs images only.
     - ***ALL** — Specifies configuration files for all IASP images to be pushed to the HA node. This is the default.

  - **Name** — Name of the IASP image to be pushed to the HA node.
  - **+ for more values** — Allows for multiple IASP image configuration files to be pushed to the HA node.

- **Target HA Node**— The name of the target HA node.

  The node name entered must be part of the cluster that is configured on this system. If this is not the case a message appears indicating that the node cannot be found.

  Note: The cluster node name can be found by running DSPCLUINF command. This command also displays the node IP address.

- **Port Number** — Port number for HA node. Default is 8000. If using a different port number, keep the port number consistent on all nodes participating in IASP operations.

5. With the server ended on the Primary node and the path added to the HA node, run IASPCTLHA with the SWHAIASP control action on the HA node. This performs a REBUILDIOA operation on the HA node as well.

   The HA system is now ready to perform normal STM operations.

6. To get to the original configuration, run IASPCTLHA with action CFGHAIASP on the HA machine, and then select the production host as the target node. On the production node, run IASPCTLHA with control action SWHAIASP.

If a rebuild of the primary node is necessary, follow the normal configuration steps.

When returning to the primary node from the HA node, or when going to the HA node from the Primary node (if the IASPCTLHA command with CFGHAIASP control action was run at least once on both nodes during initial configuration) and no configuration changes occurred, you do not need to run IASPCTLHA (CFGHAIASP) or IASPCTLHA (SWHAIASP) on the required nodes. You only need to run IASPCTLHA with RSTHAIASP on the node taking control of the configuration (primary or HA node).

Note: When using RSTHAIASP on a node, you do not need to issue the STRSTMSRV command on that node. One action of the RSTHAIASP command is to start the server.

Executing SRDF control actions during DR on primary node

1. On the command line, run the SRDFDRCTL command.

   The Dell EMC SRDF DR Control (SRDFDRCTL) screen (Figure 124) appears.

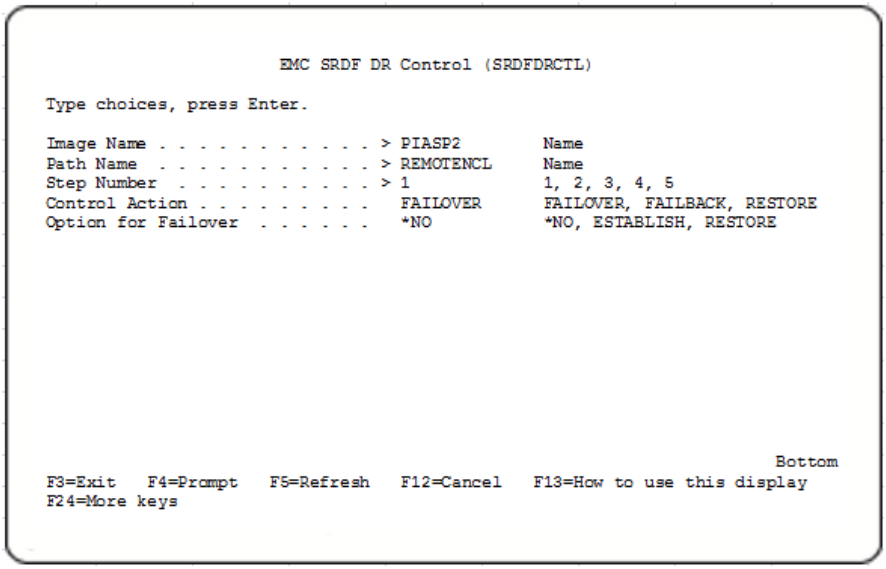


Figure 124 Dell EMC SRDF DR Control (SRDFDRCTL) screen

2. Provide the following information:
   - **Image Name** — The name of the image to be used in an SRDF control operation.
   - **Path Name** — The name of the path for the image to be used in an SRDF operation.
   - **Step Number** — The step number that the control action affects in an SRDF operation.
   - **Control Action** — The SRDF control action to perform on the IASP image:
     - **FAILOVER** — Switches data processing from the source (R1) side to the target (R2) side. When running in SRDF asynchronous mode and a FAILOVER is executed, the consistency protection is disabled and then the FAILOVER operation is performed. With dynamic RDF, if ESTABLISH/RESTORE is specified, devices are failed over, swapped and then re-established. In SRDF asynchronous mode, consistency protection is automatically enabled after executing an ESTABLISH/RESTORE.
     - **FAILBACK** — Performs a source (R1) device takeover when you are ready to resume normal SRDF operations by initiating read/write operations on the source (R1) devices, and stopping read/write operations on the target (R2) devices. The target (R2) devices become read-only to their local hosts while the source (R1) devices are read/write enabled to their local hosts.
     - **RESTORE** — Restores data from SRDF R2 data devices back to source devices.
3. Enter the optional control action parameter, if appropriate:

   Option for Failover — *NO, ESTABLISH, RESTORE. *NO is the default.

   "SRDFDRCTL" on page 406 completely describes this optional parameter and its actions.

Automated operations

Use the following commands to perform select IASP operations that are typically run using multiple manual steps. These commands have combined the manual steps into a single process to improve ease-of-use of SRDF/TimeFinder Manager.

Other batch process type commands are available besides the commands described below, and source code is provided in the install library file called QCLSRC.

Failing over/failing back

In addition to the multi-step recovery procedures described in this chapter, there are two commands that can run in batch mode and perform all the steps for a failover or failback of IASP images. The commands are as follows and are run from the DR node:

- IASPFAILOV — Performs the failover to the DR node.
- IASPFAILBK — Performs the failback to the primary node.

Note: When the initial command IASPFAILOV is used with the option parameter set to ESTABLISH or RESTORE the command IASPFAILOV needs to be run again on the DR node to get back to the original state. When the option parameter *NO is used on the initial IASPFAILOV command the IASPFAILBK command needs to be run to get back to the original state.

When these commands are run, a screen appears. After the required parameters are entered, the system then performs all the necessary steps to complete the IASP failover or failback. These commands can also be run from the command line. When running from the command line, a confirmation screen appears and asks if you want to continue with the operation. For more details on these commands, see "IASPFAILOV" on page 391 and "IASPFAILBK" on page 388.

The IASPFAILOV and IASPFAILBK commands are typically used if no additional user intervention is required for the failover or failback operations. If there are situations where user-supplied setup scripts need to be run or the DR machine requires changes before the IASP is varied on, then the multi-step procedures that follow should be used. However, the SRDF/TimeFinder Manager scripts used to run these commands can be modified to add user-specific actions. See "Modifying the script for the IASPFAILOV command" on page 393 and "Modifying the script for the IASPFAILBK command" on page 390.

Note: IASPFAILOV and IASPFAILBK commands are not available in Cascaded SRDF configurations.

Preparing source/target nodes for backup and restore

There are two commands used to prepare the source and target nodes for TimeFinder backup, as well as restore operations of IASP images. The commands are as follows and are run from the target node:

- IASPTFBCK — Prepares the source and target nodes for a TimeFinder backup operations.
- IASPTFEST — Prepares source and target to go back to original state prior to issuing the IASPTFBCK command.

RTVSTMINF is used within the two scripts to return status information for the selected image, path and step. It also returns SRDF/TimeFinder Manager, Solutions Enabler, Enginuity, and IO driver release level information.

For more details on these commands, see "IASPTFBCK" on page 395, "IASPTFEST" on page 399, and "RTVSTMINF" on page 403.

APPENDIX F
Non-Dell EMC Procedures

This appendix provides guidelines when using IBM-related procedures.

- Overview 354
- Displaying contents of a physical file on each node 355
- Verifying IOA information 356

Overview

During setup, configuration, or recovering from a disaster, there are occasions when you must use IBM commands, perform DST or SST procedures, or use other third-party software. The sections that follow describe those procedures.

Refer to the IBM documentation for more information about IBM i and IASPs

Displaying contents of a physical file on each node

To display the contents of the physical file IASPs on each node, on any node, type:

```
DSPPFM EMCCTLC/IASPS
```

The Display Physical File Member screen (Figure 125) appears.

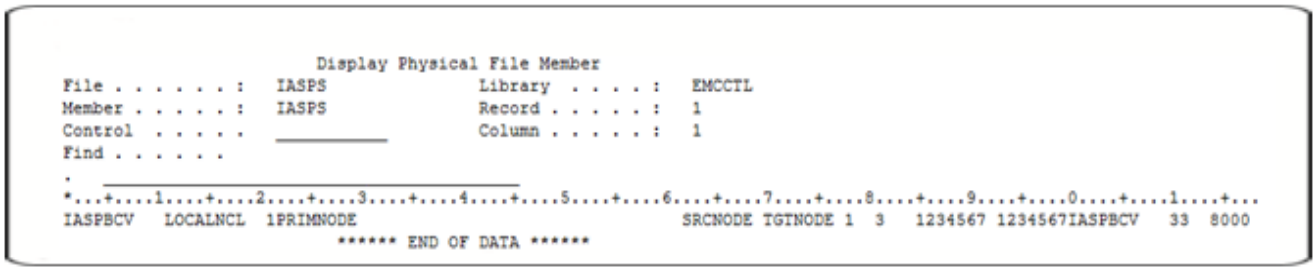


Figure 125 Displaying the contents of a physical file

The screen displays the following information:

IASPBCV	IASP device
LOCALNCL	Configured path for the SRDF image
1	Step number
PRIMNODE	Primary node name
SRCNODE	Source node for this step
TGTNODE	Target node for this step
1	Source node LPAR ID
3	Target node LPAR ID
1234567	System serial number for the source node
1234567	System serial number for the target node
IASPBCV	IASP name
33	ASP number
8000	Port number of the SRDF/TimeFinder Manager server

Verifying IOA information

To verify IOA information stored in the physical file ISRCIOA on any node, use the Display Physical File Member screen (Figure 126).

To verify one or more IOA serial numbers:

1. On the command line, type the following command:

   DSPPFM EMCCTL/ISRCIOA

   The Display Physical File Member screen or DSPPFM EMCCTLC/ISRCIOA on a DR or backup node appears.

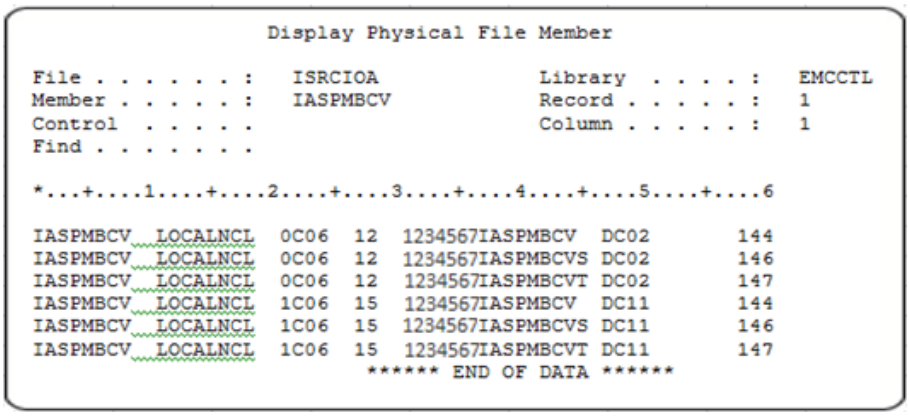


Figure 126 Verifying IOA information

   The screen shows the slot number, bus number, resource name, ASP number, and system serial number for each IOA.

2. If necessary, verify the slot number and bus number:

   DSPHDWRSC TYPE(*STG) OUTPUT(*PRINT)

APPENDIX G
Extended Features Commands Reference

This appendix provides reference information about the control commands.

- Conventions 358
- ADDRMVIOA 359
- CRTNETCNFG 361
- CTLACT 362
- CTLACTTGT 363
- DISCOVER 374
- DISCOVERC 375
- ENDSTMSRV 376
- IASPCTL 377
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- RTVSTMINF 403
- SETRDFMOD 404
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- SRDFDRCTL 406
- STMRMTCMD 409
- STRSTMSRV 410

Conventions

The descriptions of the commands in this appendix can contain the following sections:

- **Syntax** — Presents the syntax of the command. Variables that you specify are shown in *italics*. Optional parameters are indicated by brackets.
- **Description** — Describes the purpose and result of the command.
- **Required parameters** — Describes one or more parameters that must be specified at the command line. Each description contains all possible values of the variable and states which, if any, of the values is the default. Variables that you must specify are shown in *italics*.
- **Optional parameters** — Describes one or more parameters that can be included at the command line. Each description contains all possible values of the variable and states which, if any, of the values is the default. Variables that you can specify are shown in brackets [ ].
- **Usage notes** — Presents specific points that you should keep in mind when using the command.
- **Examples** — Presents examples of how to invoke the command.

ADDRMVIOA

Adds or removes the IOAs for a selected IASP.

Note: The ADDRMVIOA command does not physically add or remove the IOA from the system (LPAR), it enables or disables the IOA. Removal or addition of an IOA from the system or LPAR is performed from the Hardware Management Console.

Syntax

```
ADDRMVIOA ACTION(ADDIOA|REMOVEIOA)
IASPRESC((iasp_name_1(ioa_resource_name_1, ioa_resource_name_2))
(iasp_name_2(ioa_resource_name_1, ioa_resource_name_2)))
```

Description

Adds or removes the IOAs for a selected IASP. This command replaces the CTLIOA command. Run this command on the node where the IOAs are added or removed.

Required parameters

`ACTION(ADDIOA|REMOVEIOA)`
Add or remove command

`IASPRESC((iasp_name_1(ioa_resource_name_1 ioa_resource_name_2))`
`(iasp_name_2(ioa_resource_name_1 ioa_resource_name_2)))`

IASP name and IOA resource name. Up to 256 ASPs can be added, and up to 128 IOAs can be added for each ASP.

Note: The IOA resource name can be found in the ISRCIOA file or by using the System Service Tools (SST).

`IASPRESC((primary_iasp_name(ioa_res_name_Prim_1 ioa_res_name__Prim_2))`
`(secondary_iasp_name_1(ioa_res_name_Sec1_1 ioa_res_name_Sec1_2))`
`(secondary_iasp_name_2(ioa_res_name_Sec2_1 ioa_res_name_Sec2_2)))`

Primary and secondary IASP names, and primary and secondary IOA resource names.

Note: When the add or remove command is run on a primary IASP that has secondary IASPs, the secondary IASPs must to be entered into the same command.

Usage notes

The ADDRMVIOA command can be used to add an IOA even if SRDF/TimeFinder Manager is not yet configured.

This command can be executed from the command line with the required parameters, as shown in the example below, or the command ADDRMVIOA can be entered without parameters and the Dell EMC Add/Remove IOA screen appears (Figure 127 on page 360).

Examples

Add single and multiple IOAs to IASP images

```
ADDRMVIOA ACTION(ADDIOA) IASPRESC((IASP1(IAO1)) (IASP2(IOA1 IOA2)))
```

Add primary and secondary IOAs to IASP images

```
ADDRMVIOA ACTION(ADDIOA) IASPRESC((primIASP(primIOA1 primIOA2)) (sec1IASP(sec1IOA1 sec1IOA2))
(sec2ASP(sec2IOA1 sec2IOA2)))
```

.

```
                              EMC Add/Remove IOA (ADDRMVIOA)

Type choices, press Enter.

Add or Remove IOA  . . . . . > ADDIOA    ADDIOA, REMOVEIOA
IASP resources:
  IASP Name  . . . . . > IASP1     Character value
  IOA Resource Name  . . . . . > IOA1      Character value
             + for more values

  IASP Name . . . . . > IASP2     Character value
  IOA Resource Name  . . . . . > IOA1      Character value
             + for more values  > IOA2
             + for more values,



                                                                        Bottom
F3=Exit   F4=Prompt   F5=Refresh   F12=Cancel   F13=How to use this display
F24=More keys
```

Figure 127 Dell EMC Add/Remove IOA screen

CRTNETCNFG

Creates the NETCNFG file and adds network services to the file.

See “CRTNETCNFG” on page 205.

CTLACT

Processes SRDF and TimeFinder control actions for one or all images (SYSTEM and IASP).

See “CTLACT” on page 206.

CTLACTTGT

Processes SRDF or TimeFinder control actions for IASP images on the target side.

Syntax

```
CTLACTTGT IMAGE_NAME(image_name|*ALL)
PATH_NAME(path_name)
STEPNUM(step_number)
ACTION(control_action)
[SPLIT_MODE(CHG_ASP_ACT|SPLITWA|MAINTENANCE)]
[PROT_ESTAB(*YES|*NO)]
[POOL_NAME(pool_name)]
[VP_SNAP(*YES|*NO)]
[SVX_MODE(CHG_ASP_ACT|CONSISTENT)]
[SNAPVXNAME(snapshot_name)]
[SNAPVXGEN(snapshot_generation)]
[SNAPVXTTL(DAYS|DATE)]
[N_SNPSHOT(new_snapshot_name)]
[LINK_MODE(NOCOPY|COPY)]
[TTL_DAYS(1-400)]
[TTL_DATE(Date)]
```

Description

CTLACTTGT processes SRDF or TimeFinder control actions for IASP images on the target side.

Required parameters

IMAGE_NAME*(image_name|*ALL)*
: The name of the image (*image_name)* or *ALL to indicated all images.

PATH_NAME*(path_name)*
: The path name associated with the image to be used in an SRDF or TimeFinder control operation.

STEPNUM(*step_number*)
: The specific step number for which the control action operates on in an SRDF or TimeFinder control operation.

  Steps in the path are defined during configuration and are identified by their position (1 through 5) in the path.

ACTION(*control_action*)
: Specifies the control operation to perform:

  - STATUS — Provides SRDF, Snap, Clone, Clone emulation, or BCV control status information for the step in the associated path of the image. For SnapVX, if snapshot name and generation are specified, the state of the specified snapshot is provided.

    Note: If the status of the SnapVX step is Linked, CopyInProg, Copied, Restored, RestoreInProg, RestNotPaired, or RestgNotPaired an additional message is provided with the snapshot name and generation.

- ESTABLISH — Establishes SRDF links for SRDF control or establishes/re-establishes BCV or Clone emulation pairs for BCV control. When running in SRDF/A mode, consistency protection is automatically enabled after executing an ESTABLISH operation is executed.

  For SnapVX type of step, ESTABLISH creates and activates a SnapVX snapshot session.

- SPLIT — Splits SRDF links for SRDF control or splits BCV or Clone emulation pairs for TimeFinder control. When the system is running in SRDF/A mode and a SPLIT is executed, consistency protection is disabled and then the SPLIT operation is performed.

  Note: For TimeFinder/Clone configurations, if the devices are in Restored state, use the SPLIT option to change the direction of the clone relationship (that is, the original source device becomes the source device for a future copy), which allows the use of the RECREATE command and normal processing continues. This does not apply if VP Snap is enabled.

- FAILOVER — Switches data processing from the source (R1) side to the target (R2) side. If the source (R1) is operational, FAILOVER suspends I/O traffic on the RDF links, and write disables the devices on the source (R1) side to their hosts, and then write enables the devices on the target side to their local hosts. When running in SRDF asynchronous mode and a FAILOVER is executed, the consistency protection is disabled and then the FAILOVER operation occurs.

  With dynamic RDF, when you specify the ESTABLISH option, devices failover, swap, and then re-establish. In SRDF/A mode, consistency protection is automatically enabled after an ESTABLISH operation executes. For arrays or dynamic volumes, EMC recommends FAILOVER with the ESTABLISH option which swaps personality (the R1 devices become R2 devices and the R2 devices become R1 devices) and then starts the synchronization.

  For dynamic RDF, if the RESTORE option is specified, devices failover, swap and then re-establish. Invalid tracks on the new R2 side (originally the R1 side) are restored to the new R1 side (formerly the R2 side).

  When running in SRDF asynchronous mode, consistency protection is automatically enabled after executing a RESTORE operation.

- FAILBACK — Performs a fail back, or source (R1) device takeover, when you are ready to resume normal SRDF operations by initiating read/write operations on the source (R1) devices, and stopping read/write operations on the target (R2) devices. The target (R2) devices become read-only to their local hosts while the source (R1) devices are read/write enabled to their local hosts. In SRDF/A mode, consistency protection is enabled after a FAILBACK operation executes.

- R2RW — Enables the R2 read/write, enables R2 mirror write, and enables the target (R2) devices to their local hosts.

  If the source (R1) is operational, R2RW suspends I/O traffic on the RDF links. Then, R2RW write enables the devices on the target side to their local hosts.

  In SRDF asynchronous mode, when an R2RW executes, the consistency protection is disabled, and then the read/write enable R2 mirror operation is performed.

- SUSPEND — Suspends SRDF links for SRDF control. When the system is running in SRDF/A mode and a SUSPEND is executed, the consistency protection is disabled and then the SUSPEND operation is performed.
- ENABLE — Enables the consistency protection for SRDF/A backed devices when running in SRDF/A mode. If data cannot be copied from the R1 to the R2, all devices in the group are made Not Ready on the link to preserve the R2 data consistency.

  Note: If devices are enabled for consistency group protection, the devices must be disabled before you change the mode to Synchronous or initiate the SPLIT/SUSPEND operation.

- DISABLE — Disables the consistency protection for SRDF/A backed devices when running in SRDF asynchronous mode.
- CREATE — Creates or recreates a TimeFinder/Snap virtual copy session or Clone copy session with devices in the image. While the operation is in progress for TimeFinder/Snap, the state of the device pair is CreateInProg. For TimeFinder/Clone the state of the device pair is PreCopy. When the Create operation completes, the state changes to Created.

  Run CREATE before you activate the session, which has a shorter duration than an Establish operation.

  For TimeFinder Snap, Recreate is only available in systems running Enginuity 5876 or higher.

  See “TimeFinder/Snap Usage notes” on page 212 for more information about using Recreate.

- ACTIVATE — Activates a TimeFinder/Snap virtual copy session or Clone copy session with the devices in the image. For TimeFinder/Snap, while the operation is in progress the state of the device pair is CopyOnWrite. For TimeFinder/Clone, while the operation is in progress the state of the device pair is CopyInProg. Once the copy is complete, the state changes to Copied.

  Note: The point-in-time copy becomes available after an ACTIVATE operation. Create must be done before you can activate a TimeFinder/Snap copy session or Clone copy session.

- LINK — Copies the SnapVX point-in-time data to the target device.

  There are two types of links, NOCOPY (the default) or COPY. NOCOPY links do not retain point-in-time data once the link is removed and any previous target data is lost. Copy mode must be used in order for the linked target to retain data after the link is removed. When linking a snapshot, COPY or NOCOPY must be specified.

- UNLINK - Removes the current SnapVX link to target device(s).
- RESTORE — Restores data from target TimeFinder replication devices or from SRDF R2 data devices back to source devices or other devices. This can be a full or incremental restore operation.

  The following behavior and rules apply to the restore operation for the different TimeFinder replication devices:

TimeFinder/Mirror and TimeFinder /Clone Emulation

A full restore operation copies the entire contents of the BCV device to the standard device. For an incremental restore process, the BCV device copies only the new data to the standard device that was written to the BCV device while the BCV pair was split. Any changed tracks on the standard device are also overwritten with the data on the corresponding tracks of the BCV device. This process is useful for porting an application that was tested on the BCV device, to the standard device. The restore process is complete when the standard device and the BCV device contain identical data.

The BCV device is not available to the host while it is assigned as a BCV mirror on a standard device. However, any new data written to the standard device is copied to the BCV device while the BCV pair exists.

When using native TimeFinder/Mirror and a BCV device is used to restore a point-in-time copy to the standard device, the point-in-time copy on the BCV device is modified if the host writes to the standard device during the restore operation. The `-protected` option used on the restore operation preserves the point-in-time stored on the BCV. When using mode, a protected restore operation is always in effect.

To continue normal operations, the BCV pairs should be split.

TimeFinder/Clone

The restore operation copies target data to another device (full restore), or back to the original source device (incremental restore). For a full restore, the original session terminates and a copy session to the target of the restore operation begins. For an incremental restore operation, the original session copy direction reverses, and changed data is copied from the target device to the source device. To support the restore operation, the session must be in a Copied state.

Use the SPLIT command to split a clone device pair that is in the Restored state. This command changes the direction of the clone relationship (that is, the original source device becomes the source device for a future copy), which allows the use of the recreate command. TimeFinder/Clone supports the SPLIT command. This does not apply if VP Snap is enabled.

TimeFinder/Snap

The restore operation creates a new restore copy session between the source device and the target virtual device. You can carry out a restore operation when an additional copy session is available for use. Any existing copy sessions persist until manually terminated. After the virtual device has been restored to another device, the restore copy session must be terminated before another restore operation is allowed from that virtual device. All commands (CTLACTTGT, CTLACT, IASPCTL and IASPCTLC) and screens on the source side would show RestInProg or Restored state. The original snap copy session remains in the CopyOnWrite state but is not shown. For an incremental restore, any changes made to the virtual device tracks during the active copy session are restored back to the original source device. Any changes written to the original source device during the copy session are overwritten by the virtual device tracks when the source device is restored.

The restore target device (source device) and virtual device are automatically set to the Not Ready state while the track protection bitmaps prepare to copy any changed tracks. The source device automatically becomes available for use (Ready state) once the track protection bitmap completes. The changed tracks then begin copying and continue to copy in the background until all the protected tracks are restored.

To continue using the original copy session, the virtual device must be manually set to the Ready state after the restore operation has completed.

Once the restore operation completes, both the original and restore copy sessions are maintained, and must be manually terminated if not needed for future use. When an original session is terminated, the device pointers are deleted from the virtual device and the SAVE device space becomes available for future use. The restored copy session must be terminated first, before the original copy session is allowed to be terminated.

TimeFinder/VP Snap

VP Snap supports incremental restore operations back to the original source device. Unlike TimeFinder/Clone restore operations, the source device does not have to be fully copied and the original session between the source device and the target device is maintained. The following restrictions apply:

- Only one restore session is allowed for the source device.
- Because the original session is maintained, an additional session is charged to the source device.
- If you have concurrent sessions off of a source device, all of the concurrent sessions must be in the Copied or CopyOnWrite state.
- Splitting a clone device pair is not supported for VP Snap restored sessions.
- Starting from the time of the initiation of the incremental restore command until the restore copy is complete, existing sessions cannot be changed from one mode to another, nor can they perform a differential recreate operation. New sessions cannot be created with another create command.

TimeFinder/SnapVX

The restore operation copies the point-in-time data from a snapshot back to the original source device. The snapshot must be fully established to perform a restore.

SRDF restore operation rules and restrictions

The new data, that was updated on the target (R2) device while the SRDF pair was split, is copied to the source (R1) device. Any changed tracks on the source (R1) device are refreshed from the corresponding tracks on the target (R2) device. After the restore operation has successfully completed, the SRDF pairs synchronize. You can use the RTVSMINF command to confirm that the SRDF pairs are in a Synchronized state.

Note: The CTLACTTGT and IASPCTLC commands using control action STATUS can be used to confirm the status of SRDF pairs.

When an incremental restore is initiated for each specified SRDF pair in a device file:

- The source (R1) device is write disabled for the local hosts.
- The target (R2) device is write disabled for the local hosts.
- Data traffic is suspended on the SRDF links.
- The invalid tracks on the source (R1) device are refreshed with the changed tracks on the target (R2) side. The R1 and R2 track tables are merged.
- Traffic is resumed on the SRDF links.
- The source (R1) device is read/write enabled for the local hosts.
- The SRDF pair is in the Synchronized state when the source (R1) device and the target (R2) device contain identical data.
- The incremental restore operation is rejected if the target has invalid local (R2) tracks.

Note: When running in SRDF asynchronous mode, consistency protection is automatically enabled after executing a RESTORE operation.

- RENAME — Changes the SnapVX snapshot name of a specified generation. If you do not specify a generation, the control operation renames generation.
- SET TTL — Sets a time to live (TTL) for a SnapVX snapshot, from when TTL is set, in days or a date. The value in days can be set to a value between 1 and 400. The date value is a specified date up to 400 days in the future, from which a value in days is created, based on the difference between the specified absolute time and the host time when the action is performed. When the TTL expires the snapshot automatically terminates unless there are one or more links. Automatic termination takes place when the links are removed. If TTL is not set, the snapshot remains established until it is manually terminated.
- SETCOPY - Changes the copy mode of a link to copy. Data is physically copied to the target device(s). When the link is terminated the data is retained on the target.
- SETNOCOPY - Changes the copy mode of a link to NoCopy. Data is presented on the target device(s), but not physically copied. When the link is terminated the data is not retained on the target.
- TERMINATE — Stops the existing TimeFinder/Snap virtual copy session or Clone copy session between the specified source and target devices in the image.

  For SnapVX snapshot sessions, TERMINATE removes the snapshot. All links must be unlinked before terminating a snapshot.

  Note: If a snapshot has restores, only the restored session is terminated.

- CANCEL — Cancels the existing internal session between the specified standard device and the BCV devices in the image.

  Once the session is cancelled, the corresponding BCV devices are placed in a SplitNoInc state, and the BCV pair can no longer be incrementally established or restored.

Use Cancel before performing an unpairing operation on Clone Enabled BCV devices. Failure to run Cancel causes hanging Clone sessions to remain on the array.

Optional parameters

`POOL_NAME(DEFAULT_POOL|`*`pool_name`*`)`
Specifies the named SAVE pool to use when creating a TimeFinder/Snap operation. If pool name is not specified, the operation uses the default pool (DEFAULT_POOL).

SNAP pool name rules and restrictions:

- The POOL_NAME parameter is relevant only for creating new SNAP sessions, otherwise it isignored.
- The specified pool must exist and contain at least one enabled device before creating a TimeFinder/Snap session.
- If the image spans over several arrays the specified pool must exist on all arrays.
- If at least one device in a step has established a SNAP session all other copy sessions will use the pool in the established copy session.
- With mixed device s (2107 and D910) only the DEFAULT_POOL can be used.

`VP_SNAP(*YES|*NO)`
Specifies whether to use TimeFinder VP Snap mode. The VP Snap parameter may only be set when a session is created.

VP Snap allows for concurrent sessions.

VP Snap rules and restrictions:

- Active sessions are in the CopyOnWrite state.
- Support is limited to D910 IBM i (512-byte block size) thin-to-thin device clone pairs. All other IBM i type devices are not supported.
- All VP Snap target devices must be bound to the same thin pool. Once a VP Snap session is created for a source device, additional new sessions are rejected if the new target device is not bound to the same thin pool as the target for the existing session.

Restrictions for cascaded sessions:

- Clone sessions must be in the Copied or Split state for a VP Snap session to be created from its target device.
- The Clone session must exist before the target device is used as a VP Snap source. Creating a Clone session using a VP Snap source as its target device is not permitted.
- Snap sessions and VP Snap sessions cannot be on the same source device.
- Rebind and VLUN VP Migration operations that cause a device to be rebound are not allowed on VP Snap target devices.
- The target of a VP Snap session cannot be used as a source for any other Clone or Snap session.
- Cascading from TimeFinder Clone emulation is not supported.

Restrictions for SRDF/S and SRDF/A:

- SRDF devices used as VP Snap targets are not supported.
- The use of R2 devices as VP Snap source devices requires that SRDF/A Device-level Write Pacing (Device-level Pacing) is configured with autostart on the R1 side, or that the Device-level Pacing is activated and supported on the SRDF/A session.
- If the VP Snap source is an R2 device and the VP Snap session is in the RestoreInProgress state, SRDF restore to the R1 device is blocked.

Restrictions for incremental restore of VP Snap sessions:

- The VP Snap restore is similar to the snap persistent restore and results in an additional session charged to the source device. The original CopyOnWrite session is preserved.
- All clone sessions associated with the selected source device must be fully copied or split and any VP Snap sessions must be in the CopyOnWrite or Copied state. If any concurrent session is in any other state the incremental restore of the VP Snap session is blocked.
- The VP Snap source device can only have one restored session at a time.
- Once restored, the only operation allowed is termination of the restored session. The CopyOnWrite session is preserved.
- Clone Split is available for VP Snap restored sessions.

Restrictions for incremental restore operation until the restore data copy is complete

- One of the existing sessions may be the target of a differential Recreate or incremental Establish operation.
- No new session can be created using another Create or full Establish command.

`SPLIT_MODE(CHG_ASP_ACT|SPLITWA|MAINTENANCE)`

Specifies method for stopping writes to the devices in the image. The options are:

- CHG_ASP_ACT — Executes the operation after suspending initiation of new transactions and operations.

  Note: The Change ASP Activity CL command (CHGASPACT) is used for hosts running V6R1M0 or higher. The selected IASP image that uses the command is quiesced at the start of new commitment control transactions, new database change operations, and new Integrated File System (IFS) change operations. The command suspends initiation of new transactions and operations before the split, writes changes to auxiliary storage, and resumes initiations of new transactions and operations after the split.

- SPLITWA — Executes the operation after holding I/Os on the array. Splits a local or remote BCV, Clone, Clone emulation, or Snap image while the source system is operational. The resulting image is consistent and holds data upon the last commit.

  With TimeFinder/CG (SPLITWA) operations, you do not have to set the IBM i host in a restricted state or perform a hold jobs operation.

The SRDF/TimeFinder Manager client and the controlling host must have a gatekeeper configured on each locally attached array. For detailed information on array gatekeeper management, refer to the *Dell EMC Solutions Enabler Installation and Configuration Guide*.

On the client, the netcnfg file must be configured using the CRTNETCNFG command. This ensures that correct information (node name, IP address, port, and security level) points to the remote server. Only non-secure connections are supported for IBM i.

- MAINTENANCE — Splits or Suspends SRDF links, splits BCV pairs or activates a virtual copy session or Clone copy session in maintenance mode (that is, no interference with host operations).

`PROT_ESTAB(*YES|*NO)`
Performs a Protected or normal BCV Establish operation:

- *YES — Establish two mirrors of a BCV device to the standard device. When the split occurs, both mirrors are split simultaneously. This operation moves all mirrors of the locally mirrored BCV devices to join the mirrors of a standard device
- *NO — Perform a normal Establish operation for the BCV device. This is the default.

`SVX_MODE(CHG_ASP_ACT|CONSISTENT)`
The mode to use for the SnapVX Establish operation:

- • CHG_ASP_ACT — Executes the operation after suspending initiation of new transactions and operations.
- CONSISTENT —Executes the operation in Consistent mode (that is, no interference with the host operations). This is the default.

IMPORTANT

Use caution when using this Consistent Establish mode, as the data in main memory is not included in the image. Use this only when the host and applications are in a quiescent state. If that is unacceptable to your business needs, use other operating system functions such as journaling and database commitment control to obtain an image current with the last transaction.

`SNAPVXNAME`
The name of the Snapshot.

Note: For ESTABLISH action SNAPVXNAME specifies the name for new snapshots. For all other control actions SNAPVXNAME specifies the name of the existing snapshots to be operated on. This parameter is required for all SnapVX control actions except STATUS where it is optional.

`SNAPVXGEN`
The Generation of the Snapshot. For all SnapVX control actions, except ESTABLISH, specifies the generation of the existing snapshots to be operated on. If set to zero, the most recent snapshot is used for the control action.

`SNAPVXTTL`
For ESTABLISH and SET_TTL SnapVX controls specifies a Time to Live. The possible values are:

- DAYS -Specifies the time to live in days.
- DATE - Specifies a date for the snapshot time to live.

If you do not specify a value, the Time to Live is set to unlimited and the Snapshot is kept until manually terminated.

`N_SNPSHOT`
The new name of the Snapshot. Used only for RENAME operations.

`LINK_MODE(NOCOPY|COPY)`
The mode used for a LINK SnapVX operation. Options are as follows:

- NOCOPY —The target does not retain point-in-time data once the link is removed, and any previous target data is lost. This is the default.
- COPY — Data is physically copied to the target devices. When the link is terminated the data is retained on the target.

`TTL_DAYS`
Time to Live in days. The value in days can be set to any value between 1 and 400.

`TTL_DATE`
Specifies a date for the snapshot time to live in the form MM/DD/YYYY (month/day/year) with reference to the host time. The specified date may be up to 400 days in the future.

Usage notes

All control commands executed on the DR or backup node use the image file built on the primary node and sent to the corresponding node. The PATH_NAME and STEPNUM parameters should use the path name and step used on the primary node for corresponding devices. For example, if the path REMOTEBCV /RDF1/BCV1 exists in the image file, use the path name REMOTEBCV and step number 2 to execute TimeFinder control operations on the DR node.

In Enginuity 5876:

- The Recreate action occurs automatically if all devices in the image have Enginuity 5876 and sessions are already activated.
- Recreating a Snap copy session requires the following steps:
  1. Create a Snap session.
  2. Activate a Snap session.
  3. Recreate a Snap session.

     Note: Each Create issued after the first Activate recreates the Snap session.

  4. Activate a Snap session.
  5. Repeat steps 3 and 4, as necessary.
  6. Terminate the Snap session when it is no longer needed.

Examples

```
CTLACTTGT IMAGE_NAME (DISKPOOL) PATH_NAME(REMOTEBCV) STEPNUM(1)ACTION(FAILOVER)
OPTION(ESTABLISH)

CTLACTTGT IMAGE_NAME(IASP1) PATH_NAME(SRDF) STEPNUM(1) ACTION(SWAP)REFRESH(R1)

CTLACTTGT IMAGE_NAME(IASP1) PATH_NAME(REMOTEBCV) STEPNUM(2)ACTION(SPLIT) SPLIT_MODE(SPLITWA)
```

For control action command examples for TimeFinder/SnapVX, see the examples section in “CTLACT” on page 206. Replace “CTLACT” with “CTLACTTGT”.

DISCOVER

Discovers new devices, removes old device information from the SYMAPI database, and updates the database with new information. This command also updates information in the IMAGE file and controls checking and clearing of serial number IDs for all devices.

See "DISCOVER" on page 227.

DISCOVERC

Discovers new devices, removes old device information from the SYMAPI database, and updates the database with new information. DISCOVERC performs the discover process on the target side that is required when the configuration on the host or array changes.

See “DISCOVERC” on page 228.

ENDSTMSRV

Shuts down the SRDF/TimeFinder Manager server job.

"ENDSTMSRV" on page 232.

IASPCTL

Performs IASP control functions on the primary node or HA node.

Syntax

```
IASPCTL IMAGENAME(Image_name)
PATHNAME(Path_name)
STEPNUM(Step_number)
ACTION(Control_action)
NODENAME(Node)
SRCORTGT (TGTDEV|SRCDEV)
[POOL_NAME(pool_name)]
[VP_SNAP(*YES|*NO)]
[PROT_ESTAB(*YES|*NO)]
[TGTNAME(Node_name)]
[PORTNUM(port_number)]
[SNAPVXNAME(snapshot_name)]
[SNAPVXGEN(snapshot_generation)]
[SNAPVXTTL(DAYS|DATE)]
[N_SNPSHOT(new_snapshot_name)]
[LINK_MODE(NOCOPY|COPY)]
[TTL_DAYS(1-400)]
```

Description

IASPCTL performs the control action specified on the devices associated with the path of the selected IASP image.

Required parameters

`IMAGENAME(Image_name)`
The name of the image on which to perform a control action.

`PATHNAME(Path_name)`
The path name associated with the image to be used in an SRDF or TimeFinder control operation.

`STEPNUM(Step_number)`
The specific step number for which the control action is initiated in an SRDF or TimeFinder control operation.

`ACTION(control_action)`
The control action to perform:

- ESTABLISH — Establishes SRDF links for SRDF control or Establishes/Re-establishes BCV pairs for BCV control. When the system is running in SRDF asynchronous mode, consistency protection is automatically enabled after executing an ESTABLISH operation.

  For SnapVX type of step, ESTABLISH creates and activates a SnapVX snapshot session.

- SPLIT — Splits SRDF links for SRDF control or splits BCV pairs for TimeFinder control. When the system is running in SRDF asynchronous mode and a SPLIT executes, the consistency protection is disabled and then the Split operation is performed. For TimeFinder/Clone configurations, if the devices are in Restored state, use the SPLIT option to change the direction of the clone relationship

(that is, the original source device becomes the source device for a future copy), which allows the use of the RECREATE command and normal processing continues. This does not apply if VP Snap is enabled.

On the primary node, when performing a split operation follow the steps described below.

Status	Step
Varied off	Use the IASPCTL command to perform a maintenance split.
Varied on	Use the IASPCTL command for the split. SRDF/TimeFinder Manager will use the Change ASP Activity CL command (CHGASPACT) to quiesce IASP data for the split.

- ADDIOA — Adds one or more IOAs with target TimeFinder replication devices to the backup node.

  IMPORTANT

  When varying on an IASP on a different system with different number of fiber connections to the IASP devices, the vary on goes through DASD checks to detect if all paths can be reestablished. To avoid a longer vary on time, a multi-path reset executes automatically after the ADDIOA completes and the devices have reported back in.

- FORCEADD — Adds the IOA to the target host when the SRDF link is in a partitioned or transmit idle state when the regular ADDIOA control action fails, or when the source STM Server is not available. The message that appears when the regular add fails indicates this as well.
- REMOVEIOA — Removes one or more IOAs with target devices from the backup node.
- REBUILDIOA — Rebuilds the IOA file with IOA configuration information (slot number, bus number, and system serial number for the IOA). The rebuild action automatically runs with a CONFIG action.
- CREATE — Creates or recreates a TimeFinder/Snap virtual copy session or Clone copy session with devices in the image. For TimeFinder/Snap, while the operation is in progress, the state of the device pair is CreateInProg. For TimeFinder/Clone the state of the device pair is PreCopy. When the Create operation completes, the state changes to Created.

  Note: CREATE must be executed before you activate the session, which has a shorter duration than an ESTABLISH operation.

  See "Usage notes" on page 383 for more information.

- ACTIVATE — Activates a TimeFinder/Snap virtual copy session or Clone copy session with the devices in the image. For TimeFinder/Snap, while the operation is in progress, the state of the device pair is CopyOnWrite. For TimeFinder/Clone, while the operation is in progress the state of the device pair is CopyInProg. Once the copy is complete, the state changes to Copied.

  Note: CREATE must be executed before you activate a TimeFinder/Snap copy session.

  When performing an ACTIVATE operation on the primary node, use the steps described in the next table.

Status	Step
Varied off	Use the IASPCTL command to perform the ACTIVATE operation.
Varied on	Use the IASPCTL command to perform the ACTIVATE operation. SRDF/TimeFinder Manager uses the Change ASP Activity CL command (CHGASPACT) to quiesce IASP data for the split.

- RESTORE — Restores data from target devices back to source devices or other devices. This can be a full or incremental restore operation. See "CTLACTTGT" on page 363 for details on the behavior and rules that apply to the restore operation for TimeFinder replication devices.
- TERMINATE — Stops the existing TimeFinder/Snap virtual copy session or Clone copy session between the specified source and target devices in the image.

  For SnapVX snapshot sessions, Terminate removes the snapshot. All links must be unlinked before terminating a snapshot.

  Note: Note: If a snapshot has restores, only the restored session is terminated.

- CANCEL — Cancels the existing internal session between the specified standard device and the BCV devices in the image.

  Once the session is cancelled, the corresponding BCV devices are placed in the SplitNoInc state, and the BCV pair can no longer be incrementally established or restored. CANCEL must be executed before you perform an unpairing operation on RAID 5/RAID 6 or mixed BCV devices. Failure to run CANCEL causes hanging Clone sessions to remain on the array.
- LINK — Copies the SnapVX point-in-time data to a target device.

  There are two types of links, NOCOPY (the default) or COPY. NOCOPY links do not retain point-in-time data once the link is removed and any previous target data is lost. Use copy mode so that the linked target retains data after the link is removed. When linking a snapshot, COPY or NOCOPY must be specified.
- RELINK — Removes the current SnapVX link to target device(s) and presents a different snapshot. The relink may also be used to relink to the same snapshot. This has the effect of refreshing the point-in-time copy on the link target if it has been modified by host writes.

- UNLINK - Removes the current SnapVX link to target device(s).
- RENAME — Changes the SnapVX snapshot name. Renames on the specified generation. If generation is not specified, then generation 0 is renamed.
- SET TTL — Sets a time to live (TTL) for a SnapVX snapshot, from when TTL is set, in days or a date. The value in days can be set to a value between 1 and 400. The date value is a specified date up to 400 days in the future, from which a value in days is created, based on the difference between the specified absolute time and the host time when the action is performed. When the TTL expires the snapshot terminates unless there are one or more links. Automatic termination occurs when the links are removed. If TTL is set to 0, the snapshot remains established until it is manually terminated.
- STATUS — Retrieves link status, IOA status, and device status for the specified device. For SnapVX, if snapshot name and generation are specified then the state of the generation is provided.
- CONFIG — Configure IASP operation and writes data to the IASP file. Also rebuilds the IOA file.

`NODENAME(`*`Node`*`)`
The parameter used in conjunction with the control action REBUILDIOA when the configuration file needs to be rebuilt for a node and to specify where the operation must be performed. This parameter is also used for STATUS, ADDIOA, REMOVEIOA, and FORCEADD. The node name entered must be part of the cluster that is configured on this system. If this is not the case a message appears indicating that the node cannot be found.

`SRCORTGT (TGTDEV|SRCDEV)`
This parameter is used in conjunction with the control actions ADDIOA, REMOVEIOA, STATUS, and FORCEADD to differentiate between source or target operations when both images are located on the same node. This parameter is only shown when both images are on the same node.

For example, if an R2 and remote BCV image are being used on the same node, only specifying the node is not enough. If the IASP file does not have an entry where the source and target node is the same, this parameter is set to the default value (TGTDEV).

It is required when the source and target node are the same in the IASPs file. This can be the case when using remote BCV's where the R2's and the BCV's are presented to the same host.

This parameter is available only when this is the case, in all other cases this parameter is available and has the default value TGTDEV.

For any other control actions, this parameter is set to default and does not have an effect.

Optional Parameters

`POOL_NAME(DEFAULT_POOL|`*`pool_name`*`)`
Specifies the named SAVE pool to use when creating a TimeFinder/Snap operation. If pool name is not specified, the operation uses the default pool (DEFAULT_POOL).

SNAP pool name rules and restrictions:

- The POOL_NAME parameter is only allowed for creating new SNAP sessions, otherwise it is ignored.
- The specified pool must exist and contain at least one enabled device before creating a TimeFinder/Snap session.
- If the image spans over several arrays the specified pool should exist on all arrays.
- If at least one device in a step has established a SNAP session all other copy sessions use the pool in the established copy session.
- With mixed device s (2107 and D910) only the DEFAULT_POOL can be used.

`VP_SNAP(*YES|*NO)`
Specifies whether to use TimeFinder VP Snap mode. The VP Snap parameter may only be set when a session is created.

VP Snap allows for concurrent sessions.

VP Snap rules and restrictions:

- Active sessions are in the CopyOnWrite state.
- Support is limited to D910 IBM i (512-byte block size) thin-to-thin device clone pairs. No other IBM i-type devices are not supported.
- All VP Snap target devices must be bound to the same thin pool. Once a VP Snap session is created for a source device, additional new sessions are rejected if the new target device is not bound to the same thin pool as the target for the existing session.

Restrictions for cascaded sessions:

- Clone sessions must be in the Copied or Split state for a VP Snap session to be created from its target device.
- The Clone session must exist before the target device is used as a VP Snap source. Creating a Clone session using a VP Snap source as its target device is not permitted.
- Snap sessions and VP Snap sessions cannot be on the same source device.
- Rebind and VLUN VP Migration operations that cause a device to be rebound are not allowed on VP Snap target devices.
- The target of a VP Snap session cannot be used as a source for any other Clone or Snap session.
- Cascading from TimeFinder Clone emulation is not supported.

Restrictions for SRDF/S and SRDF/A:

- SRDF devices used as VP Snap targets are not supported.
- The use of R2 devices as VP Snap source devices requires that SRDF/A Device-level Write Pacing (Device-level Pacing) is configured with autostart on the R1 side, or that the Device-level Pacing is activated and supported on the SRDF/A session.
- If the VP Snap source is an R2 device and the VP Snap session is in the RestoreInProgress state, SRDF restore to the R1 device is blocked.

Restrictions for incremental restore of VP Snap sessions:

- The VP Snap restore is similar to the snap persistent restore and results in an additional session charged to the source device. The original CopyOnWrite session is preserved.
- All clone sessions associated with the selected source device must be fully copied or split and any VP Snap sessions must be in the CopyOnWrite or Copied state. If any concurrent session is in any other state the incremental restore of the VP Snap session is blocked.
- The VP Snap source device can only have one restored session at a time.
- Once restored, the only operation allowed is termination of the restored session. The CopyOnWrite session is preserved.
- Clone Split is not supported for VP Snap restored sessions.

Restrictions for incremental restore operation until the restore data copy is complete

- One of the existing sessions may be the target of a differential Recreate or incremental Establish operation.
- No new session can be created using another Create or full Establish command.

`PROT_ESTAB(*YES | *NO)`
Performs protected BCV ESTABLISH/RESTORE operation:

- *YES — Establishes two mirrors of a BCV device to the standard device. When the Split operation occurs, both mirrors are split simultaneously. This operation moves all mirrors of the locally mirrored BCV devices to join the mirrors of a standard device.

  Note: Use the *YES parameter only when protected BCVs are in the configuration.

- *NO — Performs a normal Establish operation for the BCV device. This is the default.

  Note: When using native TimeFinder/Mirror and a BCV device is used to restore a point-in-time copy to the standard device, the point-in-time copy on the BCV device is modified if the host writes to the standard device during the restore operation. The protected option used on the restore operation preserves the point-in-time stored on the BCV. When using mode, a protected restore operation is always in effect.

`TGTNAME(Node_name)`
The target name location to where the configuration of the selected step for this IASP is being set. The node name entered must be part of the cluster that is configured on this system. If this is not the case a message appears indicating that the node cannot be found.

`PORTNUM(port_number)`
The TCP/IP port number.

The default value is 8000. If you used a port number other than the default, use that port number on the primary node, backup node, and main partition.

Usage notes

In systems running Enginuity 5876 or higher:

- The Recreate action is used automatically if all devices in the image have Enginuity 5876 or higher and sessions are already activated.
- Recreating a Snap copy session requires the following steps:
  1. Create a Snap session.
  2. Activate a Snap session.
  3. Recreate a Snap session.

     Note: Each Create issued after the first Activate recreates the Snap session.
  4. Activate a Snap session.
  5. Repeat steps 3 and 4 as necessary.
  6. Terminate the Snap session when no longer needed.

Examples

Split the IASP image (PIASP2)

Associated path name is (REMOTEBCV) on the primary node.

```
IASPCTL IMAGENAME(PIASP2) PATHNAME(REMOTEBCV) STEPNUM(1) ACTION(SPLIT)
```

Establish the IASP image (PIASP2)

Associated path name is (REMOTEBCV) on the primary node.

```
IASPCTL IMAGENAME(PIASP2) PATHNAME(REMOTEBCV) STEPNUM(1) ACTION(ESTABLISH) PROT_ESTAB(*NO)
```

Rebuild IOA files on target devices for the IASP image (PIASP2) on the backup node

```
IASPCTL IMAGENAME(PIASP2) PATHNAME(REMOTEBCV) STEPNUM(2) ACTION(REBUILDIOA) NODENAME(NODE)
```

For control action command examples for TimeFinder/SnapVX, see the examples section for “CTLACT” on page 206. Replace “CTLACT” with “IASPCTL”.

IASPCTLC

Performs IASP control functions on the backup node or DR node.

Syntax

```
IASPCTLC IMAGENAME(Image_name)
PATHNAME(Path_name)
STEPNUM(Step_number)
ACTION(Control_action)
NODENAME(Node)
SRCORTGT (TGTDEV|SRCDEV)
```

Description

IASPCTLC performs the control action specified on the devices associated with the path of the selected IASP image.

Required parameters

`IMAGENAME(Image_name)`
The name of the image on which to perform a control action.

`PATHNAME(Path_name)`
The path name associated with the image to be used in an SRDF or TimeFinder control operation.

`STEPNUM(Step_number)`
The specific step number for which the control action is initiated in an SRDF or TimeFinder control operation.

`ACTION(Control_action)`
Specifies the control action to perform:

- STATUS — Retrieves link status, IOA status, and device status for the specified device.
- ADDIOA — Adds IOA(s) with source or target devices to the primary or backup node.

  IMPORTANT

  For hosts running V6R1 and higher, when you vary on an IASP on a different system with a different number of fiber connections to the IASP devices, the vary on goes through DASD checks to detect if all paths can be reestablished. To avoid a longer vary on time, a multi-path reset executes automatically after the ADDIOA is completed and the devices have reported back in.

- FORCEADD — Adds IOAs to the target host when the SRDF link is in a partitioned or transmit idle state when the regular ADDIOA control action fails, or when the source STM Server is not available. The message that appears when the regular add fails indicates this as well.
- REMOVEIOA — Removes IOAs with source or target devices from the primary or backup node.
- REBUILDIOA — Rebuilds the IOA file with IOA configuration information (slot number, bus number, and system serial number for the IOA). The rebuild action automatically runs with a CONFIG action.

NODENAME(*Node*)
: The parameter used in conjunction with the control action REBUILDIOA when the configuration file needs to be rebuilt for a node and the node where the operation must be performed must be varied on or off. This parameter is also used for STATUS, ADDIOA, REMOVEIOA, and FORCEADD. The node name entered must be part of the cluster that is configured on this system. If this is not the case a message appears indicating that the node cannot be found.

SRCORTGT (TGTDEV|SRCDEV)
: This parameter is used in conjunction with control actions ADDIOA, REMOVEIOA, STATUS and FORCEADD to be able to differentiate between source or target operations in case both images are located on the same node. This parameter is available only when both images are on the same node.

  This can be the case if, for example, an R2 and remote BCV image are used on the same node. Only specifying the node is not enough in this case. If the IASP file does not have an entry where the source and target node is the same, this parameter is set to the default value (TGTDEV).

  For any other control actions this parameter is set to default and has no effect.

Usage notes

All control commands executed on the DR or backup node use the image file built on the primary node and sent to the corresponding node. The path and step parameters should use the path name and step used on the primary node for corresponding devices.

When adding and removing I/O in an unplanned disaster recovery situation, you cannot verify whether:

- The target IASP image is varied on or varied off on the backup node before the IOA removal.
- The target device IOA has not been presented to any node before adding the IOA.

Examples

Add IOAs to the IASP image (PIASP2) with target devices to the backup node

```
IASPCTLC IMAGENAME(PIASP2) PATHNAME(REMOTEBCV) STEPNUM(2) ACTION(ADDIOA) NODENAME(NODE)
```

Rebuild the IOA file for the IASP image (PIASP2)

```
IASPCTLC IMAGENAME(PIASP2) PATHNAME(REMOTEBCV) STEPNUM(1) ACTION(REBUILDIOA) NODENAME(NODE)
```

IASPCTLHA

Pushes configuration files to the selected target HA host and prepares this host to act as the primary host.

Syntax

```
IASPCTLHA
ACTION(Control_action)
IMAGENAME(image_name1 image_name2...image_name(n)|*ALL)
TGTNAME(node)
PORTNUM(port_number)
```

Description

IASPCTLHA pushes configuration files to the selected target HA host and prepares this host to act as the primary host.

Required parameters

`ACTION(Control_action)`
Specifies the control action to perform:

- CFGHAIASP — Makes copies of the original configuration files, then pushes configuration files to the selected target HA node.
- SWHAIASP — Switches configuration files on the HA node so that this node acts as the primary node. It pushes the updated configuration files to target nodes in the cluster, as defined in the IASP file, and performs a rebuild of the IOAs for all steps 1 in the configuration.
- RSTHAIASP — Restores the original configuration files to the Primary node that were saved during the CFGHAIASP action.

`IMAGENAME(Image_name1... imagename(n))|*ALL))`
Specifies which IASP image configuration files to push to the HA node. The command should be used only for IASP images. Use *ALL for all images. This is the default.

`TGTNAME(node)`
The HA node where the configuration files must be sent. The node name entered must be part of the cluster that is configured on this system. If this is not the case a message appears indicating that the node cannot be found.

`PORTNUM(Port_number)`
The TCP/IP port number.

The default value is 8000. If you use a different port number, keep the port number consistent on all nodes participating in IASP operations.

Usage notes

Using IASPCTLHA with the control action SWHAIASP can return the configuration files to their original state if the command is run on the primary node.

Control action CFGHAIASP is typically used on the primary node, while the control action SWHAIASP is typically used on the HA node.

RSTHAIASP is run on the Primary node to restore original configuration files, and pushes configuration files to target nodes in the cluster, as defined in the IASP file.

Note: When RSTHAIASP is used on a node, you do not need to run the STRSTMSRV command. RSTHAIASP automatically restarts the server.

Examples

```
IASPCTLHA IMAGENAME(IASP1 IASP2 IASP3) TGTNAME(HANODE) PORTNUM(8000) ACTION(CFGHAIASP)
IASPCTLHA IMAGENAME(*ALL) TGTNAME(HANODE) PORTNUM(8000) ACTION(CFGHAIASP)
```

IASPFAILBK

Performs a failback of IASP images from the DR node to the primary node in one step, and is used when IASPFAILOV command is used with OPTION *NO.

The IASPFAILBK command is typically used if no additional user intervention is required for the failback operation. If there are situations where user setup scripts need to be run or the DR machine requires changes before the IASP is varied off, you can use the multi-step failback procedure (see "IASP-supported operations" on page 324). However, you can modify the script used to run this command to add user-specific actions. See "Modifying the script for the IASPFAILBK command" on page 390 for instructions on modifying the script.

Syntax

```
IASPFAILBK
IMAGE_NAME(Image_name)
PATH_NAME(Path_name)
STEPNUM(Step_number)
```

Description

`IASPFAILBK` performs a failback on IASP images from the DR node to the primary node. After the required parameters are entered, the system then performs all the necessary steps to complete the IASP failback. See "Command execution steps" on page 392 for a description of the command steps.

Required parameters

`IMAGENAME(Image_name)`
: The name of the IASP image to be used in a failback operation.

`PATHNAME(Path_name)`
: The path name associated with an IASP image to be used in a failback operation.

`STEPNUM(Step_number)`
: The specific step number for which the control action is initiated in a failback operation.

Usage notes

IASPFAILBK command must be executed from the DR node.

You can enter the command, with the required parameters, on the command line as shown in the example below. If you enter the command without parameters, the IASP fail Back Procedure screen appears (Figure 128 on page 389).

Examples

```
IASPFAILBK IMAGENAME(P1IASP2) PATHNAME(REMOTEBCV) STEPNUM(1)
```

Note: When running IASPFAILBK from the command line, a confirmation screen appears and asks if you want to continue with the operation.

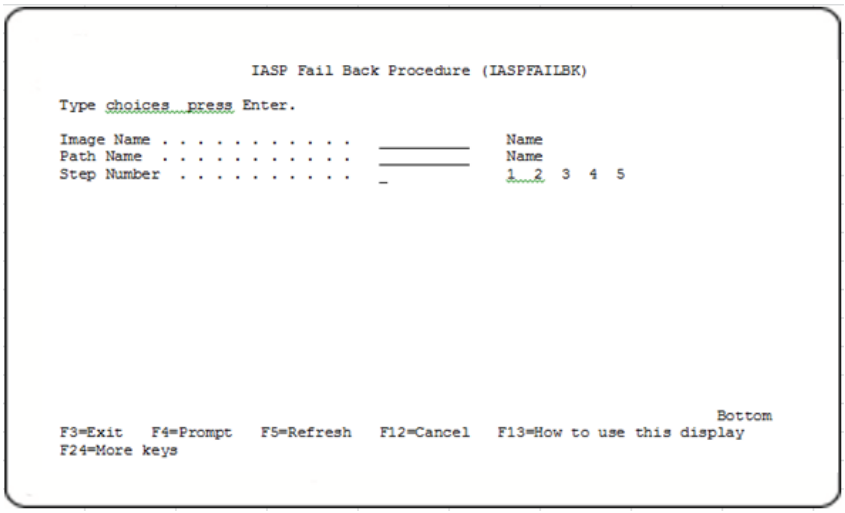


Figure 128 IASP Fail Back Procedure screen

Command execution steps

The IASPFAILBK command does the following:

1. Checks the IASP files for valid input and configuration.
2. Checks the local IASP status to ensure it is offline.
3. Checks the SRDF status to ensure it is in failed over state.
4. Checks that the remote system is active.
5. Checks the remote IASP status to ensure it is offline.
6. Removes the IOAs for both the source and target IASP.
7. Carries out the failback operation.
8. Once failback is complete, adds an IOA for the user-selected IASP and step number on the primary node.
9. Varies on the user-selected IASP image on the primary node.

Modifying the script for the IASPFAILBK command

User-specific actions can be included in the IASPFAILBK command by editing the script that generates the failback command, as follows:

Note: In the EMCCTLC client library there is a Source Physical File called QCLSRC that contains the scripts needed for the failback operation, COMPILE.CLP and IASPFAILBK.CLLE.

1. Locate the IASPFAILBK.CLLE script and make desired changes to the script.
2. Compile the COMPILE.CLP script as follows:

   ```
   CRTCLPGM PGM(EMCCTLC/COMPILE) SRCFILE(EMCCTLC/QCLSRC)
   ```

3. Execute the COMPILE program as follows:

   ```
   CALL COMPILE PARM(<SRCLIB> <TGTLIB> <RELEASE>)
   ```

   Where:

   - *SRCLIB* is the library where the IASPUTIL module and the QCLSRC file are located. Both files are located in the EMCCTLC library.
   - *TGTLIB* is the library where the compiled programs will be located.
   - *RELEASE* is the earliest operating system release level the program is allowed to run on.

   Running the COMPILE program generates the user-modified IASPFAILOV, IASPFAILBK, IASPTFBCK, and IASPTFEST commands and programs.

   Note: The COMPILE.CLP can be also be edited and compiled (as shown in step 2).

IASPFAILOV

Performs a failover of IASP images from the primary node to the DR node in one step. To go back to the original state, run IASPFAILOV again on the DR node. When OPTION *NO is used the first time to do a failover, run the IASPFAILBK command to go back to the original state.

The IASPFAILOV command is typically used if no additional user intervention is required for the failover operation. If there are situations where user setup scripts need to be run, or the DR machine requires changes before the IASP is varied on, use the multi-step failover procedure (see "IASP-supported operations" on page 324). However, you can modify the script used to run this command to add user-specific actions. See "Modifying the script for the IASPFAILOV command" on page 393 for instructions on modifying the script.

Syntax

```
IASPFAILOV
IMAGE_NAME(Image_name)
PATH_NAME(Path_name)
STEPNUM(Step_number)
[OPTION(*NO|ESTABLISH|RESTORE)]
```

Description

`IASPFAILOV` performs a failover of IASP images from the primary node to the DR node. After the required parameters are entered, the system then performs all the necessary steps to complete the IASP failover. See "Command execution steps" on page 392 for a description of the command steps.

Required parameters

`IMAGE_NAME(Image_name)`
The name of the IASP image to be used in a failover operation.

`PATH_NAME(Path_name)`
The path name associated with an IASP image to be used in a failover operation.

`STEPNUM(Step_number)`
The specific step number for which the control action is initiated in a failover operation.

Optional parameters

`OPTION(*NO|ESTABLISH|RESTORE)`
Controls the failover operation:

- *NO — Devices are left in the failed over state. This is the default.
- ESTABLISH — With dynamic SRDF, the specified devices are failed over, swapped and then re-established. In SRDF asynchronous mode, consistency protection is automatically enabled after an ESTABLISH operation executes.
- RESTORE — same as the ESTABLISH option except that after the swap, invalid tracks on the new R2 side (the original R1 side) are restored on the new R1 side (the original R2 side).

Usage notes

Run the IASPFAILOV command from the DR node.

You can enter the command from the command line with the required parameters, as shown in the example below. If you enter the command without parameters, the IASP fail over Procedure screen appears (Figure 129).

Examples

```
IASPFAILOV IMAGENAME(P1IASP2) PATHNAME(REMOTEBCV) STEPNUM(1) OPTION(*NO)
```

Note: When running IASPFAILOV from the command line, a confirmation screen appears that asks if you want to continue with the operation.

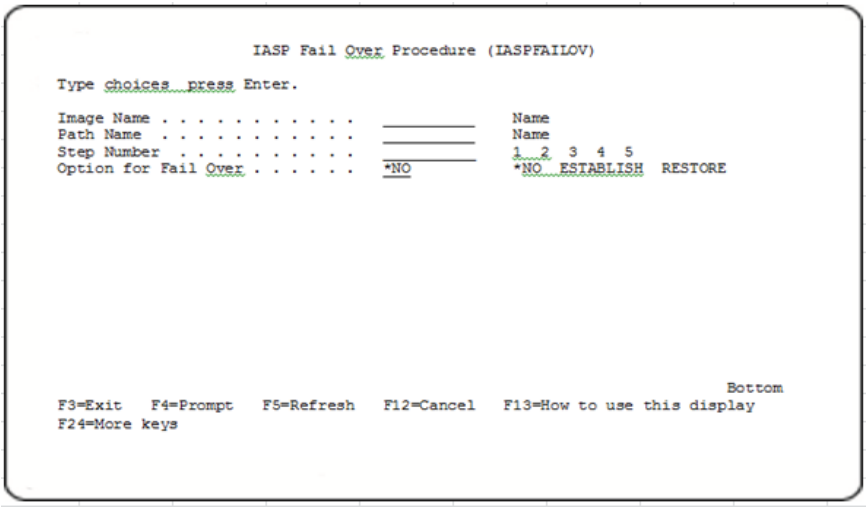


Figure 129 IASP Fail Over Procedure screen

Command execution steps

The IASPFAILOV command does the following:

1. Checks the IASP files for valid input and configuration.
2. Checks the local IASP status to ensure it is offline.
3. Checks the SRDF status to ensure it is synchronized.
4. Checks that the remote system is active.
5. Checks the remote IASP status to ensure it is offline.
6. Removes the IOAs for both the source and target IASP.
7. Carries out the failover operation.
8. Once failover is complete, adds an IOA for the user-selected IASP and step number.
9. Varies on the user-selected IASP image on the DR node.

Modifying the script for the IASPFAILOV command

You can add user-specific actions to the IASPFAILOV command by editing the script that generates the command, as follows:

Note: In the EMCCTLC client library there is a Source Physical File called QCLSRC that contains the scripts needed for the failback operation, COMPILE.CLP and IASPFAILOV.CLLE.

1. Locate the IASPFAILOV.CLLE script and make desired changes to the script.
2. Compile the COMPILE.CLP script as follows:

   ```
   CRTCLPGM PGM(EMCCTLC/COMPILE) SRCFILE(EMCCTLC/QCLSRC)
   ```

3. Execute the COMPILE program as follows:

   ```
   CALL COMPILE PARM(<SRCLIB> <TGTLIB> <RELEASE>)
   ```

   Where:

   - *SRCLIB* is the library where the IASPUTIL module and the QCLSRC file are located. Both files are located in the EMCCTLC library.
   - *TGTLIB* is the library where the compiled programs are located.
   - *RELEASE* is the earliest operating system release level the program can run on.

   Running the COMPILE program generates the user-modified IASPFAILOV, IASPFAILBK, IASPTFBCK, and IASPTFEST commands and programs.

Note: The COMPILE.CLP can be also be edited and compiled (as shown in step 2).

IASPFILE

Removes image information.

Syntax

```
IASPFILE
ACTION(Control_action)
IMAGENAME(image_name)
[PATHNAME(pathname|*ALL)]
```

Description

IASPFILE removes image information from the IASPs and ISRCIOA files for a specified image.

Required parameters

`ACTION(Control_action)`
Specifies the control action to perform:

- RMVIASPIMG — removes image information.

`IMAGENAME(image_name)`
The name of the image.

Optional parameters

`PATHNAME(path_name|*ALL)`
The name of the path associated with the image specified in the `IMAGENAME` parameter. Use *ALL to specify all paths of the image specified in the `IMAGENAME` parameter.

Usage notes

To remove image information run the command from each node in the DR setup.

Examples

```
IASPFILE ACTION(RMVIASPIMG) IMAGENAME(IASPIMAGE) PATHNAME(*ALL)
```

IASPTFBCK

Prepares the source and target node for TimeFinder backup operations on IASPs.

```
IASPTFBCK
IMAGE_NAME(Image_name)
PATH_NAME(Path_name)
STEPNUM(Step_number)
[SNAPVX(*YES|*NO)]
SPLIT_MODE(CHG_ASP_ACT|SPLITWA|MAINTENANCE)
[SNAPVXNAME(snapshot_name)]
[SNAPVXGEN(snapshot_generation)]
[SNAPVXTTL(DAYS|DATE)]
[LINK_MODE(NOCOPY|COPY)]
[TTL_DAYS(1-400)]
[TTL_DATE(Date)]
```

Description

Prepares the source and target node for TimeFinder backup operations on IASPs. After the required parameters are entered, the system performs all the necessary steps to prepare the source and target nodes for backup. See "Command execution steps" on page 397 for a description of the command steps.

Required parameters

`IMAGE_NAME(Image_name)`
The name of the IASP image to be used in a backup operation.

`PATH_NAME(Path_name)`
The path name associated with an IASP image to be used in a backup operation.

`STEPNUM(Step_number)`
The specific step number for which the control action is initiated in a backup operation.

`SPLIT_MODE(CHG_ASP_ACT|SPLITWA|MAINTENANCE)`
Specifies the method for stopping writes to the devices in the image:

- CHG_ASP_ACT — Executes the operation after suspending initiation of new transactions and operations.

  Note: The selected IASP image that uses the Change ASP Activity CL command (CHGASPACT) is quiesced at the start of new commitment control transactions, new database change operations, and new Integrated File System (IFS) change operations. The command suspends initiation of new transactions and operations before the split, writes changes to auxiliary storage, and resumes initiations of new transactions and operations after the split.

- SPLITWA - Executes the operation after holding I/Os on the array. It splits a local or remote BCV, Clone, or Snap image while the source system is operational. The resulting image is consistent and holds data upon the last commit.

  With TimeFinder/CG (SPLITWA) operations, you do not have to put the selected IASP in the quiesced state.

SRDF/TimeFinder Manager client (the DR node) and the controlling host must have a gatekeeper configured on each locally attached array. When using the same controlling host for Primary and DR nodes, the controlling host must have additional gatekeeper device configured on each remotely attached array (local to the DR node). For detailed information on array gatekeeper management, refer to the *Dell EMC Solutions Enabler Installation and Configuration Guide*.

On the client, the netcnfg file must be configured using the "CRTNETCNFG" command. This ensures that correct information (node name, IP address, port, and security level) points to the remote server. Only non-secure connections are supported for IBM i.

Note: When using this split type, the data in main memory is not included in the image. Only use SPLITWA when the host and applications are in a quiescent state. If this is not suitable for your business needs, use other operating system functions such as journaling and database commitment control to create an image that includes the last transaction.

- MAINTENANCE — Splits or suspends SRDF links or split BCV pairs or activates a virtual copy session or Clone copy session in maintenance mode (that is, no interference with host operations).

Optional parameters

`SNAPVX`
Controls whether the system displays values of the parameters relating to SnapVX to the user. Set the value to `*YES` to display the values. The default value is `*NO`.

`SNAPVXNAME`
The name of the Snapshot.

`SNAPVXGEN`
The Generation of the Snapshot. For all SnapVX control actions, except ESTABLISH, specifies the generation of the existing snapshots to be operated on. If set to zero, the most recent snapshot is used for the control action.

`SNAPVXTTL`
The Time to Live for the Snapshot. The possible values are:

- DAYS -Specifies the time to live in days.
- DATE - Specifies a date for the snapshot time to live.

If you do not specify a value, the Time to Live is set to unlimited and the Snapshot is kept until manually terminated.

`LINK_MODE(NOCOPY|COPY)`
The mode used for a LINK SnapVX operation. Options are as follows:

- NOCOPY —The target does not retain point-in-time data once the link is removed, and any previous target data is lost. This is the default.
- COPY — Data is physically copied to the target devices. When the link is terminated the data is retained on the target.

`TTL_DAYS`
Time to Live in days. The value in days can be set to any value between 1 and 400.

TTL_DATE
Specifies a date for the snapshot time to live in the form MM/DD/YYYY (month/day/year) with reference to the host time. The specified date may be up to 400 days in the future.

Usage notes

The IASPTFBCK command must be executed from the target node, and the STM server must be running.

Examples

You can enter the command from the command line with the required parameters, as shown in the example below. If you enter the command without parameters, the IASP TimeFinder Backup Procedure screen appears (Figure 130).

```
IASPTFBCK IMAGE_NAME(P1IASP2) PATH_NAME(REMOTEBCV) STEPNUM(2)SPLIT_MODE(SPLITWA)
```

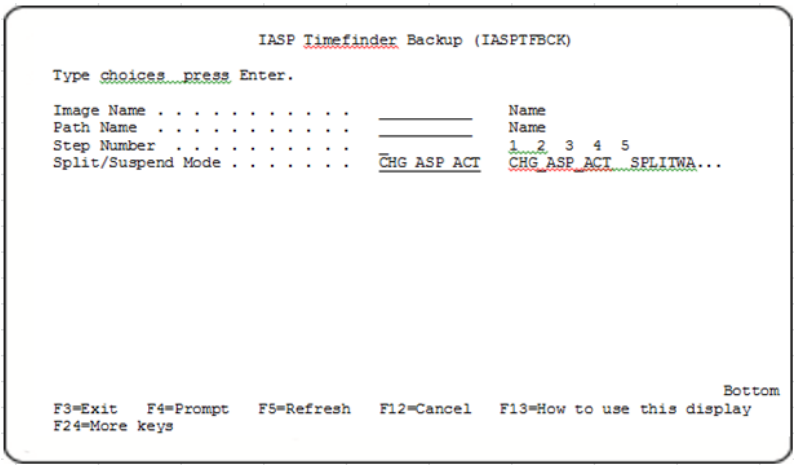


Figure 130 IASP TimeFinder Backup procedure screen

Command execution steps

1. Checks the IASP files for valid input and configuration.
2. Checks the local IASP status to ensure it is offline.
3. Checks that the source system is up.
4. Checks that the link status is Synchonized, Precopy or Recreated.
5. Checks for SnapVX.
6. Carries out the Split/Activate control action.
7. Adds the target IOAs and varies on the target node.
8. Checks that the source IASP status is offline.
9. Removes the IOAs for the IASP from the source and target system.

Modifying the script for the IASPTFBCK command

You can add user-specific actions to the IASPTFBCK command by editing the script that generates the command, as follows:

Note: In the EMCCTLC client library, there is a Source Physical File called QCLSRC that contains the scripts needed for the failback operation, COMPILE.CLP and IASPTFBCK.CLLE.

1. Locate the IASPTFBCK.CLLE script and make desired changes to the script.
2. Compile the COMPILE.CLP script:

   ```
   CRTCLPGM PGM(EMCCTLC/COMPILE) SRCFILE(EMCCTLC/QCLSRC)
   ```

3. Execute the COMPILE program:

   ```
   CALL COMPILE PARM(<SRCLIB> <TGTLIB> <RELEASE>)
   ```

   Where:

   - *SRCLIB* is the library where the IASPUTIL module and the QCLSRC file are located. Both files are located in the EMCCTLC library.
   - *TGTLIB* is the library where the compiled programs are located.
   - *RELEASE* is the earliest operating system release level the program can run on.

   Running the COMPILE program generates the user-modified IASPFAILBK, IASPTFBCK, and IASPTFEST commands and programs.
4. The COMPILE.CLP can be also be edited and compiled (as shown in step 2).

IASPTFEST

Prepares the source and target to go back to original state prior to running the IASPTFBCK operation.

```
IASPTFEST
IMAGE_NAME(Image_name)
PATH_NAME(Path_name)
STEPNUM(Step_number)
```

Description

Prepares the source and target to go back to original state prior to running the IASPTFBCK operation. After the required parameters are entered, the system then performs all the necessary steps to prepare the source and target nodes for restore. See "Command execution steps" on page 400 for a description of the command steps.

Required parameters

`IMAGE_NAME(Image_name)`
The name of the IASP image to be used in a restore operation.

`PATH_NAME(Path_name)`
The path name associated with an IASP image to be used in a restore operation.

`STEPNUM(Step_number)`
The specific step number for which the control action is initiated in a restore operation.

Usage notes

Run the IASPTFEST command from the target node.

Examples

You can enter the command can be executed from the command line with the required parameters, as shown in the example below. If you enter the command without parameters, the IASP TimeFinder restore Procedure screen appears (Figure 128 on page 389).

```
IASPTFEST IMAGE_NAME(P1IASP2) PATH_NAME(REMOTEBCV) STEPNUM(2)
```

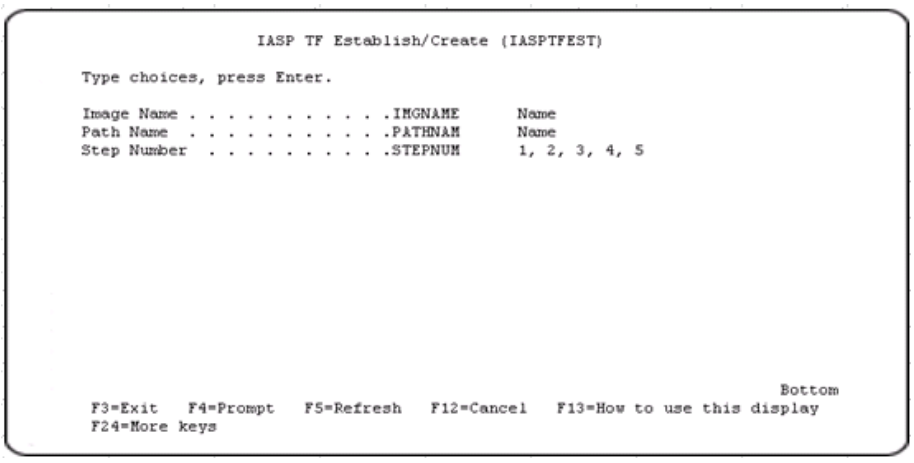


Figure 131 IASP TimeFinder Restore procedure screen

Command execution steps

1. Checks the IASP files for valid input and configuration.
2. Checks that the local IASP status is offline.
3. Checks that the source system is up.
4. Removes the IOAs for the IASP from the target system.
5. Carries out an ESTABLISH/CREATE control action.
6. Carries out ADDIOA on the source node.
7. Varies on the source IASP.

Modifying the script for the IASPTFEST command

You can add user-specific actions to the IASPTFEST command by editing the script that generates the command, as follows:

Note: In the EMCCTLC client library there is a Source Physical File called QCLSRC that contains the scripts needed for the failback operation, COMPILE.CLP and IASPTFEST.CLLE.

1. Locate the IASPTFEST.CLLE script and make desired changes to the script.
2. Compile the COMPILE.CLP script:

   ```
   CRTCLPGM PGM(EMCCTLC/COMPILE) SRCFILE(EMCCTLC/QCLSRC)
   ```

3. Execute the COMPILE program:

   CALL COMPILE PARM(<*SRCLIB*> <*TGTLIB*> <*RELEASE*>)
   Where:

   - *SRCLIB* is the library where the IASPUTIL module and the QCLSRC file are located. Both files are located in the EMCCTLC library.
   - *TGTLIB* is the library where the compiled programs will be located.
   - *RELEASE* is the earliest operating system release level the program is allowed to run on.

   Running the COMPILE program generates the user-modified IASPFAILBK, IASPTFBCK, and IASPTFEST commands and programs.

4. The COMPILE.CLP can be also be edited and compiled (as shown in step 2).

RELDEVLCK

Checks and releases device external locks (DEL) on an array for any image.

See “RELDEVLCK” on page 233.

RELDEVLCKC

Checks and releases device external locks (DEL) on an array for any image on the target side.

See “RELDEVLCKC” on page 234.

RTVSTMINF

Returns status information for the selected image, path, and step. It also returns SRDF/TimeFinder Manager, Solutions Enabler, Enginuity, and IO driver and release level information.

See “RTVSTMINF” on page 235.

SETRDFMOD

Sets the SRDF mode for a selected image.

See “SETRDFMOD” on page 250.

SETRDFMODC

Sets the SRDF mode for a selected image on the target side.

Syntax

```
SETRDFMODC IMAGE_NAME(image_name)
PATH_NAME(path_name)
STEPNUM(step_number)
MODE_TYPE(SYNC|ASYNC|ACDM|ACWP)
```

Description

SETRFDMODC sets the SRDF mode (synchronous, asynchronous, adaptive copy disk mode (ACDM), and adaptive copy write pending (ACWP)) for a selected image on the target side only. You must specify the image name, path name, and the SRDF mode associated with the image.

Note: SRDF mode adaptive copy write pending (ACWP) is not supported when the R1 mirror of the RDF pair is on an array running PowerMaxOS 5978 or HYPERMAX OS 5977. For this configuration, the SETRDFMODC command returns an error if the mode is set to adaptive copy write pending. In this case, set a different mode.

Required parameters

IMAGE_NAME(*image_name*)
: The name of the image to be used in an SRDF control operation.

PATH_NAME(path_name)
: The name of the path for the image to be used in an SRDF operation.

STEPNUM(*step_number*)
: The specific step number for which the control action is initiated in an SRDF control operation.

MODE_TYPE(SYNC|ASYNC|ACDM|ACWP)
: The parameter that sets the SRDF mode for a selected image to synchronous, asynchronous, adaptive copy disk mode, or adaptive copy write pending.

IMPORTANT

Do not use ACWP or ACDM mode for SRDF while creating RDF or remote TimeFinder images (split) using SRDF/TimeFinder Manager. This is not supported. Instead, use Sync or Async mode.

Usage notes

All control commands executed on the DR or backup node use the image file built on the primary node and sent to the corresponding node. The path and step parameters should use the path name and step used on the primary node for corresponding devices.

Examples

```
SETRDFMODC IMAGE_NAME(PIASP2) PATH_NAME(SRDFA) STEPNUM(1) MODE_TYPE(SYNC)
SETRDFMODC IMAGE_NAME(PIASP2) PATH_NAME(SRDFA) STEPNUM(1) MODE_TYPE(ASYNC)
SETRDFMODC IMAGE_NAME(PIASP2) PATH_NAME(SRDFA) STEPNUM(1) MODE_TYPE(ACDM)
```

SRDFDRCTL

Processes SRDF control actions for IASP images on the source host during disaster recovery situations.

Syntax

```
SRDFDRCTL IMAGE_NAME(Image_name)
PATH_NAME(Path_name)
STEPNUM(Step_number)
ACTION(Control_action)
[option(*NO|ESTABLISH|RESTORE)]
```

Description

SRDFDRCTL processes SRDF control actions for IASP images on the target (R2) side.

Required parameters

`IMAGE_NAME(Image_name)`
The name of the IASP image to be used in an SRDF control operation.

`PATH_NAME(Path_name)`
The path name associated with an IASP image to be used in an SRDF operation.

`STEPNUM(Step_number)`
The specific step number for which the control action is initiated in an SRDF operation.

`ACTION(Control_action)`
Specifies the control action to perform:

- FAILOVER — Switches data processing from the source (R1) side to the target (R2) side. If the source (R1) is operational, FAILOVER suspends I/O traffic on the RDF links and write disables devices on the source (R1) side to their hosts. FAILOVER then write enables the devices on the target side to their local hosts. When running in SRDF asynchronous mode and a FAILOVER is executed, the consistency protection is disabled and then the FAILOVER operation is performed.

  With dynamic RDF, when you specify the ESTABLISH option, devices failover, swap, and then re-establish.

  Note: With control action FAILOVER with the ESTABLISH option or the RESTORE flag is set, and the R2 mirror of the SRDF pair are on an array running PowerMaxOS 5978 or HYPERMAX OS 5977, the RDF mode is automatically set to Adaptive Copy Disk

  In SRDF/A mode, consistency protection is automatically enabled after an ESTABLISH operation executes.

  For dynamic RDF, if option RESTORE is specified, devices failover, swap and then re-establish. Invalid tracks on the new R2 side (originally the R1 side) are restored to the new R1 side (formerly the R2 side).

  When running in SRDF asynchronous mode, consistency protection is automatically enabled after executing an RESTORE operation.

- FAILBACK — Performs a source (R1) device takeover when you are ready to resume normal SRDF operations by initiating read/write operations on the source (R1) devices, and stopping read/write operations on the target (R2) devices. The target (R2) devices become read-only to their local hosts while the source (R1) devices are read/write enabled to their local hosts.

  Note: In SRDF/A mode, consistency protection is automatically enabled after a FAILBACK operation executes.

- RESTORE — Restores data from SRDF R2 data devices back to source devices.

  The new data, that was updated on the target (R2) device while the SRDF pair was split, is copied to the source (R1) device. Any changed tracks on the source (R1) device are refreshed from the corresponding tracks on the target (R2) device. After the restore operation has successfully completed, the SRDF pairs synchronize. The RTVSMINF command can be used to confirm that the SRDF pairs are in a Synchronized state.

  Note: The CLACT and IASPCTL commands using control action STATUS or the Work with Associated Paths screen can be used to confirm the status of SRDF pairs.

  When an incremental restore is initiated for each specified SRDF pair in a device file:

  – The source (R1) device is write disabled for the local hosts.
  – The target (R2) device is write disabled for the local hosts.
  – Data traffic is suspended on the SRDF links.
  – The invalid tracks on the source (R1) device are refreshed with the changed tracks on the target (R2) side. The R1 and R2 track tables are merged.
  – Traffic is resumed on the SRDF links.
  – The source (R1) device is read/write enabled for the local hosts.
  – The SRDF pair is in the Synchronized state when the source (R1) device and the target (R2) device contain identical data.
  – The incremental restore operation will be rejected if the target has invalid local (R2) tracks.

  Note: In SRDF/A mode, consistency protection is automatically enabled after a RESTORE operation executes.

`Option(*NO|ESTABLISH|RESTORE)`
Controls the failover operation:

- *NO — Devices are left in the failed over state. This is the default.
- ESTABLISH — With dynamic SRDF, the specified devices are failed over, swapped and then re-established. When running in SRDF asynchronous mode, consistency protection is automatically enabled after executing an ESTABLISH operation.
- RESTORE — Same as the ESTABLISH option except that after the swap, invalid tracks on the new R2 side (the original R1 side) are restored on the new R1 side (the original R2 side).

Examples

```
SRDFDRCTL IMAGE_NAME(DISKPOOL) PATH_NAME(REMOTEBCV) STEPNUM(1) ACTION(FAILOVER)
OPTION(ESTABLISH)
```

STMRMTCMD

Runs remote commands.

See “STMRMTCMD” on page 254.

STRSTMSRV

Starts the SRDF/TimeFinder Manager server job.

See “STRSTMSRV” on page 260.

APPENDIX H
Extended Features Messages

This appendix provides reference information about messages that may display for the extended features components, the reason for the message, and the recommended user action. See "Standard Features Messages" on page 265 for messages that may display for standard feature components.

IAP0001

```
Application is waiting for synchronization to finish.
```

Cause: SRDF/TimeFinder Manager is waiting for device synchronization to complete.

Action: This is an informational message; no user action is necessary.

IAP0002

```
Application is waiting for split to finish.
```

Cause: SRDF/TimeFinder Manager is waiting for a Split operation to complete.

Action: This is an informational message; no user action is necessary.

IAP0003

```
Operation is not supported, please check product guide.
```

Cause: The attempted operation is not supported with IASP images.

Action: Refer to the appropriate section in this guide.

IAP0004

```
Operation failed, please use option #16 to configure the primary IASP.
```

Cause: The attempted operation on the primary IASP failed.

Action: Reconfigure the primary IASP and try the operation again.

IAP0005

```
ASP number in IASP file does not match ASP number generated by
   operating system, there might be an IASP system setup error.
```

Cause: The check to verify the IASP number returned an IASP number mismatch.

Action: Refer to the MATRMD section in the IBM documentation for details and take the following steps to verify the status of the IASP. Use SST to verify that all IOAs have been successfully added to the system and the devices are active.

If all devices in the IASP are active, use WRKCFGSTS CFGTYPE(*DEV) CFGD(*IASP name*) to verify the IASP status. If this was an ADDIOA operation, and the IOAs are on the system and the devices are active, continue with the VARYON of the IASP. If the IASP status is 'AVAILABLE', normal IASP operations can continue.

IAP0006

```
Failure opening ISRCIOA file with member &1 on node &2.
```

Cause: ISRCIOA file with member &1 was not found or was not able to be opened.

Action: Make sure that correct image name is used or that the product installation is completed successfully.

IAP0007

```
No IOAs of the target devices are found in file ISRCIOA.
```

Cause: The IOA serial numbers of the target devices are not present in the ISRCIOA file.

Action: Rebuild the target device IOA file as documented in "Building/rebuilding the IOA file" on page 316.

IAP0008

```
The device SYMAPI status must be Synchronized/Created before a
Split/Activate can occur.
```

Cause: An attempt was made to Split or Activate a device before the device has been synchronized or created.

Action: Correct the mistake and then attempt the Split or Activate operation again.

IAP0009

```
ISRCIOA file with member &1 on node &2 has no records.
```

Cause: ISRCIOA file with member &1 has no records. Make sure that the configuration process completed correctly.

Action: Rebuild the IOA file for both source devices and target devices as documented in "Building/rebuilding the IOA file" on page 316.

IAP0010

```
Application is locating disk units.
```

Cause: Missing disk units appear in the selected IASP.

Action: Wait until all missing disk units report for the selected IASP.

IAP0011

```
Missing disk units are detected in IASP prior to varying on the IASP.
```

Cause: Not all missing disk units report in the IASP image.

Action: Reset one or more IOAs to recover them. If the problem persists, IPL the host.

IAP0012

```
Devices need to be varied off before removal.
```

Cause: One or more selected devices must be varied off before removal.

Action: Vary off the selected devices.

IAP0013

`Missing disk units are detected in IASP prior to varying on IASP.`

Cause: Not all missing disk units report in the IASP image.

Action: Complete adding one or more IOAs before varying on the IOAs. If ADDIOA has been completed, try resetting the IOAs to recover the IASP devices. If the problem persists, IPL the host.

IAP0014

`Devices must be varied on before rebuilding ISRCIOA file.`

Cause: The selected devices are varied off.

Action: Vary on the devices to make them available.

IAP0015

`File IASPS does not exist.`

Cause: The file IASPs does not exist for the configuration.

Action: Make sure the installation completed successfully.

IAP0016

`Cannot locate IASPS record to be sent.`

Cause: The record for the selected IASP does not exist.

Action: Reconfigure the selected IASP.

IAP0017

`Socket communication failed, the operation is SEND.`

Cause: The socket communication failed for the SEND operation.

Action: Check the TCP/IP configuration and hardware.

IAP0018

`Socket communication failed, the operation is RECEIVE.`

Cause: The socket communication failed for the RECEIVE operation.

Action: Check the TCP/IP configuration and hardware.

IAP0019

`&1 finished successfully, please check job log for detail.`

Cause: The operation finished successfully.

Action: This is an informational message; no user action is necessary.

IAP0020

```
&1.
```

Cause: This is the result of a specific action that is determined when the operation occurs.

Action: The corrective action to take, if any, depends on the operation performed. Most often this is an informational message; no user action is necessary.

IAP0021

```
&1.
```

Cause: This is the result of a specific action that is determined when the operation occurs.

Action: The corrective action to take, if any, depends on the operation performed. Most often this is an informational message; no user action is necessary.

IAP0022

```
Range of port number does not include value entered, or parameter
entered for the port number must be numeric
```

Cause: The value specified for the port number must be a numeric value that is greater than or equal to 1 or less than or equal to 65535.

Action: Enter a numeric value that is within the specified range.

IAP0023

```
RESTORE action not allowed for path with R21 configured.
```

Cause: Path has R21 configured and cannot be used to do a restore.

Action: Select correct path and try again.

IAP0024

```
Parameter NODENAME required.
```

Cause: A required parameter was not specified.

Action: Enter a value for parameter NODENAME.

IAP0025

```
Parameter TGTNAME required.
```

Cause: A required parameter was not specified.

Action: Enter a value for parameter TGTNAME.

IAP0026

```
Node with name &2 not found in cluster.
```

Cause: Node name entered cannot be found in list of nodes that are configured in the cluster on this system.

Action: Enter a valid node name or add the node to the cluster and try again.

IAP0027

```
Cluster information could not be retrieved.
```

Cause: Cluster information could not be retrieved.

Action: Check if a cluster is configured correctly on this system. Cluster information is collected into a user space before processing. Check if user space CLUINFO is created in library SYMAPI.

IAP0028

```
Split operation is not allowed for VP Snap copy sessions.
```

Action: Use a valid control action for the selected path and try again.

IAP0029

```
Node name &1 not found in IASP file.
```

Cause: Node name &1 cannot be found in IASP file for selected image &2 path &3 and step

Action: Make sure that configuration is completed correctly and the IASP file contains the correct node name for the selected image path and step that is used.

IAP0030

```
&1 Node name &2 not valid for this control action.
```

Cause: &1 Node name &2 cannot be used or is not valid for this control action.

Action: Make sure that node name &2 is valid for selected control action.

IAP0031

```
Failure opening IASPS file on node &1.
```

Cause: IASPS file was not found or was not able to be opened.

IAP0032

```
IASPS file on node &1 has no records.
```

Cause: IASPS file has no records. Make sure that the configuration process completed correctly.

IAP0033

```
Record read from IASPS file for selected image/path/step does not
contain correct data.
```

Cause: Expected amount of data read from IASPS file did not match required data length.

Action: Make sure that the configuration process (option #16) completed correctly for selected image. Check if there is an entry for this image in the IASP file.

IAP0034

```
Previous step is not configured. Previous step &4 not found in IASP
file for image &1 path &2.
```

Cause: Previous step &4 need to be configured first before using configuring step &3

Action: Make sure that the configuration process (option #16, or IASPCTL with option CONFIG) completed correctly for step &4 before attempting to configure step &3.

IAP0035

```
IASP status on node &1 could not be retrieved.
```

Action: Check joblog for details.

IAP0036

```
IASP &1 on node &2 with path &3 step &4 is not eligible for control
action &5.
```

Cause: Selected control action cannot be used with this path.

Action: Use a different control action with this path type.

IAP0037

```
&1 IOA(s) connected to IASP &2 and path &3 step &4 on node &5 are still
active.
```

Cause: IOAs for selected IASP &2 are still active.

Action: In order to complete this control action, the IOAs need to be deactivated (removed) first.

IAP0038

```
&1 IOA(s) connected to IASP &2 and path &3 step &4 on node &5 are
already deactivated (removed.
```

Cause: IOAs for selected IASP &1 are already deactivated (removed).

IAP0039

```
&1 IOA(s) connected to IASP &2 and path &3, step &4 on node &5 are
already activated (added).
```

Cause: IOAs for selected IASP &1 are already activated (added).

IAP0040

IOA(s) connected to IASP &1 and path &2 step &3 on node &4 are not in the same state.

Cause: IOAs for selected IASP &1 are not in the same state.

Action: Use SST to check the state of the IOAs that are connected to this IASP.

IAP0041

One or more of the IOA(s) for IASP &1 and path &2 step &3 on node &4 is not dedicated.

Cause: To be able to activate or deactivate IOAs connected to this IASP &1, it is required that the IOAs are NOT shared between ASPs.

Action: Use SST to check the state of the IOAs that are connected to this IASP. Change configuration of the IOAs and retry.

IAP0042

Could not find any IOA configured for IASP &1 and path &2 step &3 on node &4.

Cause: IOAs for selected IASP &1 could not be found.

Action: Make sure that the configuration was completed correctly. Use SST to check the state of the IOAs that are connected to this IASP.

IAP0043

Could not detect status of the IOAs configured for IASP &1 and path &2 step &3 on node &4.

Cause: Status IOAs for selected IASP &1 could not be retrieved.

Action: Make sure that the configuration was completed correctly. Use SST to check the state of the IOAs that are connected to this IASP.

IAP0044

IASP &1 has IOAs activated for other path &2 step &3 on node &4.

Cause: IASP &1 has IOAs active for paths that have same image and target node.

Action: Make sure that IASP &1 is only presented once to the node. Use SST to check the state of the IOAs that are connected to this IASP.

IAP0045

HA IASP Operation &1 failed. Selected node name &2 is equal to local node name.

Cause: Selected node name &2 that is going to act as the HA node is equal to local node name.

Action: Use a different node name for the HA node. Change input and try again.

IAP0046

```
The STM Server  job on the Primary node &2 is still active during HA
IASP operation &1.
```

Cause: The STM Server job on primary node &2 as defined in the IASP file is still active.

Action: Stop the STM Server job on the primary node &2 and try again.

IAP0047

```
HA IASP operation &1 failed.
```

Cause: Command &2 failed.

Action: Check joblog for details.

IAP0048

```
Disk management operation failed for IASP &1 on node &2.
```

Cause: Maybe a SST session is already active on the selected node.

Action: End all SST sessions, IBM i Access sessions and any other programs that have access to DASD management tools and try again.

IAP0049

```
STM Server on node &1 with port number &2 not ended.
```

Action: Check joblog for details.

IAP0050

```
STM Server with port number &1 ended.
```

IAP0051

```
Data queue not found.
```

Cause: API &1 failed with message id &2.

Action: Check joblog on remote node for details.

IAP0052

```
Data queue read failed.
```

Cause: API &1 returned no data.

Action: Check job log on remote node for details.

IAP0053

```
User space not created.
```

Cause: API QUSCRTUS failed with message id &1.

Action: Check job log on remote node for details.

IAP0054

```
User space pointer not resolved.
```

Cause: API &1 failed with message id &2.

Action: Check job log on remote node for details.

IAP0055

```
Spawn process failed.
```

Path to program : &1.

Cause: &2

Action: Check job log on remote node for details.

IAP0056

```
Command failed with message id &1.
```

Command: &2

Action: Check job log for details.

IAP0057

```
Spool file info not found.
```

Cause: API &1 failed with message id &2.

Action: Check joblog on remote node for details.

IAP0058

```
Remote command work file &1 open failed.
```

Cause: &2

Action: Check job log on remote node for details.

IAP0059

```
Spool file information not received.
```

Cause: Number of spool files received from STM server is 0.

Action: Check job log on remote node for details.

IAP0060

```
Remote command failed with message id &1.
Command: &4.
```

Action: Check file &2 on this node or check joblog for job &3 on remote node for details.

IAP0061

```
Job information not found.
```

Action: Check job log on remote node for details.

IAP0062

```
Failure opening PATH file on node &1.
```

Cause: PATH file was not found or was not able to be opened.

IAP0063

```
Path &1 was not found for the iasp image &2.
```

Cause: Path name could not be found in list.

Action: Enter appropriate path name or check if configuration completed correctly and try operation again.

IAP0064

```
Step number &1 was not found for the path &2 and image &3.
```

Cause: Step number could not be found in list.

Action: Enter appropriate step or check if configuration completed correctly and try operation again.

IAP0065

```
Snapshot name is not valid.
```

Action: Provide a valid snapshot name and try the operation again.

IAP0066

```
Snapshot generation value is not valid.
```

Action: Provide a valid value for the generation option and try the operation again

IAP0067

```
Snapshot time to live (TTL) value is not valid.
```

Action: Provide a valid time to live value and try the operation again.

IAP0068

```
New snapshot name is not valid.
```

Action: Provide a valid new snapshot name and try the operation again.'

IAP0069

```
Invalid date specified.
```

Cause: A date is not in the required format of MM/DD/YYYY, with reference to the current node date.

Action: Provide a date in the required format that is up to 400 days in the future from the current node date and try the operation again.

IAP0070

```
Remote command not allowed to run.
Command: &1.
```

Cause: Commands allowed to run are all commands included in the STM product library. Also the following IBM commands can be run through the STM server: PING, APING, DSPASPSTS, VRYCFG, VFYTCPCNN, and WRKCFGSTS.

Action: Either change the command or use the REXEC server to execute the unsupported commands

IAP9897

```
Command &1 failed.
Recovery.............: Check joblog for details
```

Cause: The command &1 failed to complete.

Action: Check the job log for cause.

IAP9898

```
Command &1 finished successfully.
```

Cause: Notification that the command &1 completed successfully.

Action: Informational message. No action is necessary.

GLOSSARY

This glossary contains terms related to business continuance operations and disk storage subsystems as they relate to Dell EMC SRDF/TimeFinder Manager for IBM i and Dell EMC arrays.

A

adapter Card that provides the physical interface between the director and disk devices, director and serial channels (Serial adapter).

ASPs Individual disks (storage units) or sets of disks reserved for particular objects (such as libraries). ASPs help organize data to limit the impact of storage-device failures and reduce recovery time. ASPs can be used to isolate those objects to facilitate backup and recovery.

B

backup node The partition or host that is the secondary owner for a particular IASP. The backup node has target devices connected to it.

BCV devices Standard array devices with special attributes that allow the devices to independently support applications and processes. BCVs are active production images that are logically or physically separate from the production volumes with no reliance on the production host, thus providing protection from physical or logical corruption. Once the BCV task is complete, the volume can be resynchronized with the production volume, reassigned to another production volume, or maintained "as is" for another task. See also "standard devices."

BCV mirror A standard device mirror (one of M2, M3, or M4) assigned to the BCV device upon establishing or re-establishing a BCV pair. See also "establish," "re-establish" and "BCV pair."

BCV pair Consists of a standard device and a BCV device attached together.

business continuance (BC) processes Processes that allow customers to access and manage instant copies of standard devices. See also "establish," "re-establish" and "Split."

business continuance volume (BCV) See "BCV devices."

C

cache Random access electronic storage used to retain frequently used data from disk for faster access by the channel.

cache slot Unit of cache equivalent to one track.

channel director — The component in the array subsystem that interfaces between the host channels and data storage. It transfers data between the channel and cache.

concurrent established BCV pair — The relationship that establishes two BCV devices as concurrent mirrors of a single standard device that allows two synchronized copies of the standard data to be created simultaneously.

consistency group — A *composite group* is a user-defined group of devices that can span multiple arrays and, if needed, provide consistency protection. When an SRDF composite group is enabled for consistency protection, it is known as an SRDF *consistency group*. *Consistency* means that the devices within the group act in unison to preserve dependent-write consistency of a database that may be distributed across multiple arrays or multiple SRDF groups within a single. (A *device group* can be used for consistency protection if the group of devices is within a single SRDF group in a single array.)

CRG (Cluster Resource group) — A Cluster resource group is a a set or grouping of cluster resources that are used to manage events that occur in a high availability environment.

D

DA — Disk Adapter. See "disk director."

data availability — Access to any and all user data by the application.

DEFAULT_POOL — A special Snap pool that exists by default. This pool contains SAVE devices that have not been assigned to any named pools but are available for Snap operations. SAVE devices in the DEFAULT_POOL are either enabled or disabled but have used tracks. See also "Save pool" and "Snap pool."

DATA device — A device that holds the actual storage space for the data written to a thin device. DATA devices are located in a thin pool.

define BCV pair — The process of identifying a BCV device and a standard device to be established.

delayed fast write — No existence of room in cache for the data presented by the write operation.

dependent-write I/O — An I/O that cannot be issued until a related predecessor I/O has completed. Most applications have embedded dependent-write logic to ensure data integrity in the event of a failure in the host or server processor, software, storage subsystem, or if an environmental power failure occurs.

dependent-write consistency — Dependent-write consistency is a data state where data integrity is guaranteed by dependent-write I/Os embedded in application logic. Database management systems are examples of applications that utilize the dependent-write consistency strategy.

destage — The asynchronous write of new or updated data from cache to disk device.

device — A uniquely addressable part of the array subsystem that consists of a set of access arms, the associated disk surfaces, and the electronic circuitry required to locate, read, and write data.

device address — A hexadecimal value that uniquely defines a physical I/O device on a channel path. A SCSI address consists of a Target ID and a Logical Unit Number (LUN).

device number — The value that logically identifies a disk device in a string.

Direct access storage device (DASD) — Most commonly known as a magnetic disk device.

director — The component in the array subsystem that allows the array to transfer data between the host channels and disk devices. See also "channel director" and "disk director."

disaster recovery — The process of restoring a previous copy of the data and applying logs or other necessary processes to that copy to bring it to a known point of consistency.

disaster recovery (DR) node — The node that takes over when the primary node fails in a three-node solution.

disk director — The component in the array subsystem that interfaces between cache and the disk devices.

disk pool — An auxiliary storage pool (ASP) of disks only. See also "ASPs."

DR node — See "disaster recovery (DR) node."

DSE pools — SRDF uses SRDF/A DSE pools to extend the cache space available for SRDF/A cycles by off loading some or all of the cycle data from cache to a pre-configured disk storage pool. Using SRDF/A DSE pools provides a safety net for running SRDF/A sessions. However, SRDF/A continues to operate normally with this feature disabled. See also "Snap pool."

DSS — Decision Support Systems. Database software used to support management decision making. Database data is extracted and analyzed, but not modified.

dual-Initiator — An array feature that automatically creates a backup data path to the disk devices serviced directly by a disk director if that disk director or the disk management hardware for those devices fails.

dynamic sparing — An array feature that automatically transfers data from a failing disk device to an available spare disk device without affecting data availability. This feature supports all non mirrored devices in the array subsystem.

dynamic SRDF pairing — The process of creating or deleting SRDF pairs while the array is in operation. Once created, these RDF pairs can be synchronized and managed the same as static RDF pairs that are paired during initial array configuration.

E

EA — ESCON Adapter.

establish — A Business Continuance process which assigns a BCV device as the next available mirror of a standard device.

established — The BCV pair condition where the BCV device and standard device are synchronized and functioning as an array mirror. A BCV pair is *established* by the BCV commands establish and re-establish.

F

fast-wide SCSI channel director — The component in the array subsystem that interfaces between host SCSI channels and data storage. It transfers data between the channel and cache.

FC — Fibre Channel. See "Fibre Channel director."

Fibre Channel director — The component in the array subsystem that interfaces between the host fibre channel interface and data storage. A Fibre Channel Director transfers data between the channel and cache.

G

gatekeeper — A small logical volume used to pass SCSI commands from a host to the array.

H

head and disk assembly (HDA) — A field replaceable unit in the array subsystem containing the disk and actuator.

hyper-volume extension — The ability to define more than one logical volume on a single physical disk device making use of its full formatted capacity. These logical volumes are user-selectable in size. The minimum volume size is one cylinder and the maximum size depends on the disk device capacity and the mode selected.

I

IASP — An independent disk pool, or independent auxiliary storage pool (ASP). An IASP is a collection of disks that can be brought online or taken offline independently from the remainder of the system's storage, which includes the system ASP, user ASPs, and other independent disk pools. An IASP can be either:

- Switchable among multiple systems in a clustered environment.
- Privately connected to a single server.

identifier (ID) — A sequence of bits or characters that identifies a program, device, controller, or system.

IFS — Integrated File System.

image — The ASP set that can be operated on simultaneously.

image types — There are several image types (UDFS IASP or an image containing primary and all secondary IASPs that belong to the same ASP group). See also "System image" and "UDFS."

IML — Initial microcode program loading.

incremental establish — A time-saving operation similar to an Establish. The source (R1) device copies to the target (R2) device only the new data that was updated on the source R1 device while the SRDF pair was split. Any changed tracks on the target (R2) device are also refreshed from the corresponding tracks on the source (R1) device. The R2 device is write disabled to the target host.

independent auxiliary storage pool — See "IASP."

instant split — A method of splitting a BCV that improves the performance of a typical split operation by performing a quick foreground BCV split, which reduces the time the application needs to be frozen and is shorter than using a regular Split.

I/O device — An addressable input/output unit, such as a disk device.

K

Kilobyte (K) — 1024 bytes.

L

least recently used algorithm (LRU) — The algorithm used to identify and make cache space available by removing the least recently used data.

logical partition — A subset of a single IBM i host that contains resources (processors, memory, and input/output devices). A logical partition operates as an independent system. When hardware requirements are met, multiple logical partitions can exist within a system.

logical volume — A user-addressable unit of storage. In the array subsystem, the user can define multiple logical volumes on a single physical disk device. Logical volumes can reside on an SRDF-enabled array, but do not participate in SRDF activity. Local volumes are typically protected either by local mirroring, RAID-S, or a dynamic spare.

LPAR — A function of the OS/400 licensed program that enables the creation of logical partitions.

M

media — The disk surface on which data is stored.

meta device — A logical device that spans multiple physical devices, which are concatenated to create larger devices. A meta device consists of a meta head and its member devices. The meta head is the first device in the meta device sequence and is responsible for receiving all incoming commands. It also identifies the entire meta device. When an incoming command for the metahead is processed, the array determines which meta device member should execute the command. Meta head devices can be added to a device group, however, a meta member cannot be added to a device group. This holds true for both standard and BCV devices. In an SRDF configuration, you can create meta devices for both primary and secondary volumes.

MII — Machine Initiated Interrupt.

Mirroring — The array maintains identical copies of a designated volume on separate disks. Each volume automatically updates during a write operation. If one disk device fails, the array automatically uses the other disk device(s).

Mirrored pair — A logical volume comprised of two or more physical devices with all data recorded on each device.

P

point of consistency A point in time to which data can be restored and recovered or restarted and maintain integrity for all data and applications.

primary LPAR The partition identifier (0) on an IBM i host.

primary node The partition or host that is the primary owner for a particular IASP.

private ASP A private independent ASP that is owned by a single system or a single logical partition. See also *ASPs*

promotion The process of moving data from a track on the disk device to cache. See also *stage*

protected BCV establish The process of moving all mirrors of locally-mirrored BCV devices to join the mirrors of a standard device.

Q

query A command that reports the state of all BCV devices in the system.

R

R1 See "source volume (R1)."

R2 See "target volume (R2)."

RA RLD Adapter. See "SRDF remote link director."

RAID Redundant Array of Independent Disks.

RDF/A DSE pool A collection of SAVE devices used for SRDF/A DSE. Also called RDF/A DSE Save Pool or RDF/A DSE Save Device Pool.

read hit Data requested by the read operation is in cache.

read miss Data requested by the read operation is not in cache.

re-establish A BC process which reassigns a BCV device as the next available mirror of the standard device with which it was previously paired. The BCV mirror is updated with the data that was written to the standard device during the period that the BCV pair was split. The data that was written to the BCV device during the split is overwritten by data from the standard device.

recreate A Snap session that is recreated on an existing virtual device (VDEV) to prepare to activate a new point-in-time image. The Recreate action is valid only when issued against sessions which have been previously activated. This feature is available with Enginuity 5876.

restore The process that reinstates a prior copy of the data.

rolling disaster A series of events that lead up to a complete disaster. For example, the loss of a communication link occurs prior to a site failure. Most disasters are rolling disasters; their duration may be only milliseconds or up to hours.

S

SA	SCSI Adapter. See "fast-wide SCSI channel director."
SAVE device	A device configured for use in TimeFinder/Snap operations and SRDF/A DSE. These devices provide polled physical storage space for storing pre-update images of the source device change tracks and new writes during a TimeFinder/Snap virtual copy session.
Save pool	A collection of SAVE devices. Also called Save Device pool.
scrubbing	The background process of reading, checking the error correction bits, and writing corrected data back to the source.
Snap device	See "virtual device" and "SAVE device."
Snap pool	A collection of SAVE devices used for Snap operation. Also called Snap Save pool or Snap Save Device pool (formerly known as SaveDev pool).
source image of an IASP	The set of the standard/R1 devices in a particular IASP.
source volume (R1)	A logical volume that is participating in SRDF operations. It resides in the *local* array. All CPUs attached to the array may access a source volume for read/write operations. All writes to this *primary* source volume are mirrored (copied to a *secondary* target volume) in another array, which can be remote. A source volume is not available for local mirroring or dynamic sparing operations.
Split	A Business Continuance process that removes the BCV mirror from the existing BCV pair and assigns the BCV mirror back to its original device address. The BCV device then holds an instant copy of the data from the standard device.
source image of IASP	A set of standard/R1 devices in a particular IASP. See also *ASPs*
SRDF	Symmetrix Remote Data Facility. SRDF consists of the microcode and hardware required to support remote mirroring.
SRDF/Asynchronous mode	A combination of hardware and software that extends the capability of SRDF to support Asynchronous Ordered Writes that ensures a consistent copy of data for remote replication over unlimited distances, for a variety of uses. SRDF/A inherits all the advantages of SRDF/Synchronous (SRDF/S).
SRDF/A DSE	SRDF/A Delta Set Extension
SRDF remote link director	The Remote Link Director (RLD) is a 2-port, serial-channel director microcode configured as the link between the two arrays in a Symmetrix Remote Data Facility (SRDF) configuration.
stage	The process of writing data from a disk device to cache. See "promotion" and "destage."
standard devices	Devices configured for normal array operation under a desired protection method (such as RAID 1, RAID 5, RAID 6, RAID-S, SRDF).

switchable IASP — An independent ASP that can switch between two separate IBM i hosts or separate LPARs. See also *ASPs*.

System ASP — An auxiliary storage pool consisting of system programs and system data. See also *ASPs*

System image — A combination of the system ASP and all user ASPs. SRDF/TimeFinder Manager references a System image as SYSTEM on its screens.

T

target image of an IASP — A set of BCV/R2 or Snap devices comprising a mirror copy of an IASP. See also *ASPs*

target volume (R2) — A Logical volume that is participating in SRDF operations. It resides in the *remote* array. This *secondary* target volume is paired with a *primary* source volume in the local array and receives all write data from its mirrored pair. This volume is not accessed by user applications during normal I/O operations. A target volume is not available for local mirroring or dynamic sparing operations.

TimeFinder — A business continuance solution that allows customers to use special devices (BCVs, Clones, and VDEVs) that contain copies of devices while the standard devices are on-line for regular array operation.

U

UDFS — See "user defined file system (UDFS)."

user defined file system (UDFS) — The user-defined file system that you create and manage. This file system resides on the auxiliary storage pool (ASP) or independent auxiliary storage pool (IASP) of your choice.

V

validate — The action that makes all tracks for a source volume valid on a target volume.

vary off — The process of making an IASP image unavailable for its intended (normal) use. All primary and secondary disk pools in the IASP vary off together.

vary on — The process of making an IASP image available for its intended (normal) use. All primary and secondary disk pools in the IASP vary on together.

virtual device — A host-accessible device containing track-level location information (pointers), which indicate where the copy session data is located in the physical storage. Virtual devices consume minimal physical disk storage, as they store only the address pointers to the data stored on the source device or a pool of SAVE devices. Using virtual devices, TimeFinder/Snap operations provide instant snapshot copies.

Virtual Provisioning — Commonly known as thin provisioning allows for storage to be allocated and accessed on demand from a pool of storage devices. Virtual Provisioning provides for the separation of the physical storage devices from the storage devices as perceived by the host.

volume — A general term referring to a storage device. In the array subsystem, a volume corresponds to single disk device.